5G RAN

# RedCap Feature Parameter Description

**Issue** Draft B

**Date** 2026-01-30

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# Contents



# 1 Change History

The following describes the specific changes in different issues.

## 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

| Change Description | Parameter Change | RAT | Base Station Model |
| --- | --- | --- | --- |
| Modified the mechanism of mapping between 5QI bearers and specific QCIs in "differentiated QoS management for RedCap UEs". For details, see **4.1.3.6 Differentiated QoS Management for RedCap UEs** and **4.2.2 Impacts**. | Added the **gNodeBAlgo**.***QciAlgoOptSw*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

- Editorial changes

  Revised descriptions of data preparation and command examples for "uplink-experience-based inter-frequency handover collaboration". For details, see **Data Preparation** and **Using MML Commands**.

  Revised descriptions of function switch for "data rate limitation in flexible experience differentiation". For details, see **4.3.2.3 Function Impacts**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

## 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 08 (2025-12-14).

- Technical changes

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added PDCCH symbol adaptation for RedCap BWP with the CD-SSB. For details, see:<br>• **4.1.3.11 PDCCH Symbol Adaptation for RedCap BWP with the CD-SSB (Low-Frequency TDD)**<br>• **4.2 Network Analysis**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameters: Added the **CD_BWP_PDCCH_SYM_ADAPTIVE_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added RedCap UE weight optimization. For details, see:<br>• **4.1.3.12 RedCap UE Weight Optimization (Low-Frequency TDD)**<br>• **4.2 Network Analysis**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Added the **gNBRedCapServIdenCfg**.***DlBigPktDataVolThld*** and **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** parameters.<br>Modified parameters:<br>• Added the **REDCAP_SRS_WEIGHT_SW** option to the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter.<br>• Added the **SRS_WEIGHT_UE_NUM_INCR_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added support for PUCCH CSI RB adjustment for the RedCap-dedicated BWP in NR TDD. For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.2.1 Benefits**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameters: Added compatibility of the **REDCAP_PUCCH_CSI_RB_ADJ_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter with low-frequency TDD. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added "CSI RB and format-1 RB reposition". For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.2.1 Benefits**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.3.4 Cells**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **PUCCH_RB_REPOSITION_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Optimized the maintenance and test for RedCap resources and the number of UEs in a specific network slice. For details, see **4.4.2 Activation Verification**. | None | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added an impact relationship between time-domain power saving BWP and RedCap Introduction. For details, see **4.3.2.3 Function Impacts**. | None | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Replaced the reserved-parameter bit that controls RedCap UE device-pipe identification optimization with a formal-parameter option. For details, see:<br>• **4.1.4.2 Networking or UE Compatibility Optimization**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Replaced the **RSVDSWPARAM2_BIT23** option of the **gNBRsvd**.***RsvdSwParam2*** parameter with the **REDCAP_SPEC_UE_IDENTIFY_OPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter.<br>• Modified parameter: Added the **REDCAP_SPEC_UE_IDENTIFY_OPT_SW** option to the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter.<br>• Added **RSVDSWPARAM2_BIT23** of the **gNBRsvd**.***RsvdSwParam2*** parameter to the disuse list. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added support for "scheduling in common PDCCH symbols for RedCap UEs" for BWP2 UEs. For details, see:<br>• **4.1.4.1 Resource Optimization**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Replaced the **RSVDSWPARAM1_BIT23** option of the **NRDUCellRsvdExt03**.***RsvdSwParam1*** parameter with the **BWP2_COM_PDCCH_SCH_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.<br>• Modified parameters: Added the **BWP2_COM_PDCCH_SCH_SW** option to the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.<br>• Added **RSVDSWPARAM1_BIT23** of the **NRDUCellRsvdExt03**.***RsvdSwParam1*** parameter to the disuse list. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added CORESET 0 offset policy configuration. For details, see:<br>• **4.1.4.1 Resource Optimization**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Added the **NRDUCellPdcchAlgo**.***Coreset0OffsetPol*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
| --- | --- | --- | --- |
| Added "SRS resource extension in a high-speed cell". For details, see:<br>• 4.1.4.1 Resource Optimization<br>• 4.4.1.1 Data Preparation<br>• 4.4.1.2 Using MML Commands | Modified parameter: Added the **SRS_RES_EXT_SW** option to the **NRDUCellHighSpeedMob**.*HighSpeedAlgoOptSw* parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added the **UeType** field to the **STR NRCELLSERVICEPARAMEFFECT** command. For details, see 5.5.2.1 Principles. | Modified MML commands: Added the **UeType** field to the **STR NRCELLSERVICEPARAMEFFECT** command. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added uplink-experience-based inter-frequency handover collaboration. For details, see:<br>• 5.4.1.1 Principles<br>• Data Preparation<br>• Using MML Commands | Modified parameter: Added the **UL_EXP_INTER_FREQ_HO_COLLAB_SW** option to the **NRCellRedCap**.*RedCapAlgoSwitch* parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added "BWP scheduling optimization for RedCap UEs". For details, see:<br>• **4.1.3.13 BWP Scheduling Optimization for RedCap UEs**<br>• **4.2.1 Benefits**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.3.2.3 Function Impacts**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **REDCAP_UE_BWP_SCH_OPT_SW** option to the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added impact relationships between RedCap Introduction and the following functions: "proactive avoidance of remote interference" and UL and DL Decoupling. For details, see **4.3.2.3 Function Impacts**. | None | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |
| Added an impact relationship between data rate limitation and RedCap Introduction. For details, see **4.3.2.3 Function Impacts**. | None | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added RedCap power aggregation. For details, see:<br>• **4.1.3.10 RedCap Power Aggregation (Low-Frequency TDD)**<br>• **4.2.1 Benefits**<br>• **4.3.2.3 Function Impacts**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Added the **NRDUCellPdcchAlgo**.***RedCapPwrBoostingCceThld*** parameter.<br>Modified parameter: Added the **REDCAP_POWER_AGGREGATION_SW** option to the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added RedCap BWP frequency-domain staggering. For details, see:<br>• **4.1.3.8 RedCap BWP Frequency-Domain Staggering (Low-Frequency TDD)**<br>• **4.2.1 Benefits**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.3.2.3 Function Impacts**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands**<br>• **5.1.3.2.3 Function Impacts**<br>• **6.6.3.2.3 Function Impacts** | Modified parameter: Added the **REDCAP_BWP_NON_OVERLAP_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added format-1 RB adaptation. For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **PUCCH_F1_RB_ADAPT_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
|---|---|---|---|
| Added CSI RB adaptation. For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **PUCCH_CSI_RB_ADAPT_SW** option to the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added format-1 SR resource optimization. For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.2.1 Benefits**<br>• **4.2.2 Impacts**<br>• **4.3.2.1 Prerequisite Functions**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **FORMAT1_SR_RES_OPT_SW** option to the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added the "uplink MU grouping and pairing avoidance for RedCap UEs" function. For details, see:<br>• **4.1.3.1 RedCap MU-MIMO**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option to the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter. | Low-frequency TDD | 3900 and 5900 series base stations (macro base stations) |

| Change Description | Parameter Change | RAT | Base Station Model |
| --- | --- | --- | --- |
| Added support for RedCap power saving functions of non-1RX RedCap UEs (including RedCap UE release preference, RedCap SSSG switching, RedCap UL skipping, and RedCap DRX adaptation) and support for these functions of specified UEs. For details, see:<br>• **5.3.7.1 Principles**<br>• **Data Preparation**<br>• **Using MML Commands**<br>• **5.3.8.1 Principles**<br>• **Data Preparation**<br>• **Using MML Commands**<br>• **5.3.9.1 Principles**<br>• **Data Preparation**<br>• **Using MML Commands**<br>• **5.3.10.1 Principles**<br>• **Data Preparation**<br>• **Using MML Commands** | Modified parameters:<br>• Added the **REDCAP_RELEASE_PREFER_EXT_SW** option to the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter.<br>• Added the **REDCAP_SSSG_SWITCH_EXT_SW** option to the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter.<br>• Added the **REDCAP_UPLINK_SKIP_EXT_SW** option to the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter.<br>• Added the **REDCAP_DRX_ADAPTATION_EXT_SW** option to the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter.<br>• Added the following options to the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter: **REDCAP_RELEASE_PREF_EXT_SW_ON**, **REDCAP_SSSG_SWITCH_EXT_SW_ON**, **REDCAP_UPLINK** | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

| Change Description | Parameter Change | RAT | Base Station Model |
| --- | --- | --- | --- |
| | **_SKIP_EXT_SW_ON**, and **REDCAP_DRX_ADAPT_EXT_SW_ON**. | | |
| Added an impact relationship between RedCap Introduction and RedCap DRX adaptation (for non-1RX RedCap UEs). For details, see **4.3.2.3 Function Impacts**. | None | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added aperiodic CSI-RS optimization in heavy-load scenarios. For details, see:<br>• **4.1.3.2 Multiple BWP1s for RedCap**<br>• **4.4.1.1 Data Preparation**<br>• **4.4.1.2 Using MML Commands** | Modified parameter: Added the **APER_CSIRS_FLEX_OPT_SW** option to the **NRDUCellCsirs**.***CsiRsExtSwitch*** parameter. | Low-frequency TDD | • 3900 and 5900 series base stations (macro base stations)<br>• DBS3900 LampSite and DBS5900 LampSite |

- Editorial changes
  - For NR TDD, added descriptions of the impact relationship between PDCCH resource allocation enhancement and RedCap power aggregation. For details, see **4.3.2.3 Function Impacts**.
  - For NR TDD, added descriptions of the network impacts of disabling "CD-SSB inclusion in the RedCap BWP0" in small-bandwidth scenarios. For details, see **4.2.2 Impacts**.
  - Added descriptions of support for blacklist control over RedCap UE release preference, RedCap UL skipping, and RedCap DRX adaptation by 1RX RedCap UEs. For details, see **5.3.7.1 Principles**, **5.3.9.1 Principles**, and **5.3.10.1 Principles**.
  - For NR TDD, added descriptions of the impact of false alarms on the **N.PRBUsed.UL.RedCap.Avg** counter. For details, see **4.2.2 Impacts**.
  - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

**NOTE**

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

### Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

## 2.2 Features in This Document

This document describes the following features.

| Feature ID | Feature Name | Chapter/Section |
|---|---|---|
| FOFD-070316 | RedCap Introduction | **4 RedCap Introduction** |
| FOFD-091206 | Premium RedCap | **5 Premium RedCap** |
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | **6 RedCap Downlink Experience Improvement (NR TDD)** |

## 2.3 Differences

### Differences Between NSA and SA

| Function Name | Difference | Chapter/Section |
|---|---|---|
| RedCap Introduction | Supported only in SA networking | **4 RedCap Introduction** |
| Premium RedCap | Supported only in SA networking | **5 Premium RedCap** |
| RedCap Downlink Experience Improvement (NR TDD) | Supported only in SA networking | **6 RedCap Downlink Experience Improvement (NR TDD)** |

## Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

| Function Name | Difference | Chapter/Section |
|---|---|---|
| RedCap Introduction | Supported only in low frequency bands | **4 RedCap Introduction** |
| Premium RedCap | Supported only in low frequency bands | **5 Premium RedCap** |
| RedCap Downlink Experience Improvement (NR TDD) | Supported only in low frequency bands | **6 RedCap Downlink Experience Improvement (NR TDD)** |

# 3 Overview

As one of the three major 5G scenarios, Massive Machine-Type Communications (mMTC) involves a large number of terminals that do not have high requirements on data traffic and rate. The capabilities of existing 5G UEs are far beyond the requirements in mMTC scenarios. However, high prices of 5G UE chips and modules hinder the industrial implementation of a massive number of connections. To solve this issue, 3GPP Release 17 introduces the reduced capability (RedCap) technology for medium- and high-rate IoT services, aiming to reduce UE costs by reducing the supported bandwidth, number of antennas, and modulation and demodulation capabilities.

**Table 3-1** lists the major differences between RedCap UEs and common NR UEs.

**Table 3-1** Differences between RedCap UEs and common NR UEs

| Item | RedCap UE | Common NR UE |
|---|---|---|
| Maximum bandwidth | 20 MHz | NR TDD: 100 MHz per cell |
| TX/RX mode | 1T1R/1T2R | 1T4R/1T2R/2T4R |
| Modulation scheme (corresponding to the maximum supported modulation order) | DL: 64QAM/256QAM<br>UL: 64QAM/256QAM | DL: 256QAM<br>UL: 256QAM |
| Support for carrier aggregation (CA) and dual connectivity (DC) | Not supported | Supported |

# 4 RedCap Introduction

## 4.1 Principles

Chips and modules of common NR UEs have high costs and have become obsolete for today's large-scale IoT deployments for massive connections. Therefore, RedCap UEs are introduced. RedCap UEs feature lower terminal complexity, while meeting the requirements of large-scale medium- and high-data-rate IoT services. RedCap Introduction is introduced to enable RedCap UEs to access the network. After the function takes effect, **RedCap UEs can access 5G networks.**

Similar to common NR UEs, RedCap Introduction involves basic functions on mobile networks, including basic signaling procedures and mobility management, as well as experience enhancement and scenario-specific optimization functions for RedCap capacity improvement and user experience improvement. Specifically,

- **4.1.1 Basic Signaling Procedures** describes differences between RedCap UEs and common NR UEs in initial access, inactivity management, and system information update.
- **4.1.2 Mobility Management** describes differences between RedCap UEs and common NR UEs in mobility management and optimization functions.
- **4.1.3 Experience Enhancement** describes RedCap experience enhancement functions.
- **4.1.4 Scenario-specific Optimization Functions** describes scenario-specific optimization functions, such as networking or UE compatibility optimization.

For details about involved functions, see **Table 4-1**.

**Table 4-1** Functions in RedCap Introduction (NR TDD)

| Category | Function Name | Section |
|---|---|---|
| Basic signaling procedures | RedCap UE initial access | **4.1.1.1 RedCap UE Initial Access** |
| | RedCap UE inactivity management | **4.1.1.2 RedCap UE Inactivity Management** |

| Category | Function Name | Section |
|---|---|---|
| | System information update through RRC reconfiguration | **4.1.1.3 System Information Update Through RRC Reconfiguration** |
| Mobility management | Idle-mode RRM measurement relaxation | **4.1.2.1 Idle-mode RRM Measurement Relaxation** |
| | NCD-SSB-based mobility | **4.1.2.2 NCD-SSB-based Mobility** |
| | RedCap handover policy control | **4.1.2.3 RedCap Handover Policy Control** |
| | Frequency selection for RedCap | **4.1.2.4 Frequency Selection for RedCap** |
| | Admission guarantee for emergency calls | **4.1.2.5 Admission Guarantee for Emergency Calls** |
| | QCI-specific handover priority configuration | **4.1.2.6 QCI-specific Handover Priority Configuration** |
| Experience enhancement | RedCap MU-MIMO | **4.1.3.1 RedCap MU-MIMO** |
| | Multiple BWP1s for RedCap | **4.1.3.2 Multiple BWP1s for RedCap** |
| | Uplink RedCap preallocation | **4.1.3.3 Uplink RedCap Preallocation** |
| | RedCap downlink scheduling policy | **4.1.3.4 RedCap Downlink Scheduling Policy** |
| | QCI-based RedCap DRX configuration | **4.1.3.5 QCI-based RedCap DRX Configuration** |
| | Differentiated QoS management for RedCap UEs | **4.1.3.6 Differentiated QoS Management for RedCap UEs** |
| | SSB interference avoidance for RedCap UEs | **4.1.3.7 SSB Interference Avoidance for RedCap UEs** |
| | RedCap BWP frequency-domain staggering | **4.1.3.8 RedCap BWP Frequency-Domain Staggering (Low-Frequency TDD)** |
| | Support for super uplink by RedCap | **4.1.3.9 Support for Super Uplink by RedCap** |

| Category | Function Name | Section |
| --- | --- | --- |
| | RedCap power aggregation | **4.1.3.10 RedCap Power Aggregation (Low-Frequency TDD)** |
| | PDCCH symbol adaptation for RedCap BWP with the CD-SSB | **4.1.3.11 PDCCH Symbol Adaptation for RedCap BWP with the CD-SSB (Low-Frequency TDD)** |
| | RedCap UE weight optimization | **4.1.3.12 RedCap UE Weight Optimization (Low-Frequency TDD)** |
| | BWP scheduling optimization for RedCap UEs | **4.1.3.13 BWP Scheduling Optimization for RedCap UEs** |
| Scenario-specific optimization functions | These functions are scenario-specific optimization functions, including resource optimization and networking or UE compatibility optimization, which are not listed in detail here. | **4.1.4 Scenario-specific Optimization Functions** |

When RedCap Introduction is enabled, the gNodeB configures a BWP1 with a maximum bandwidth of 20 MHz for RedCap UEs.

This function is controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter.

## 4.1.1 Basic Signaling Procedures

This section describes basic RedCap signaling procedures of functions including RedCap UE initial access, inactivity management, and system information update.

### 4.1.1.1 RedCap UE Initial Access

RedCap UEs are new to networks. Therefore, it is important to identify RedCap UEs and allocate resources related to initial access to them to ensure that they can successfully access the NR network. The resources include the BWP0, PRACH, PDCCH, CORESET 0, and CD-SSB.

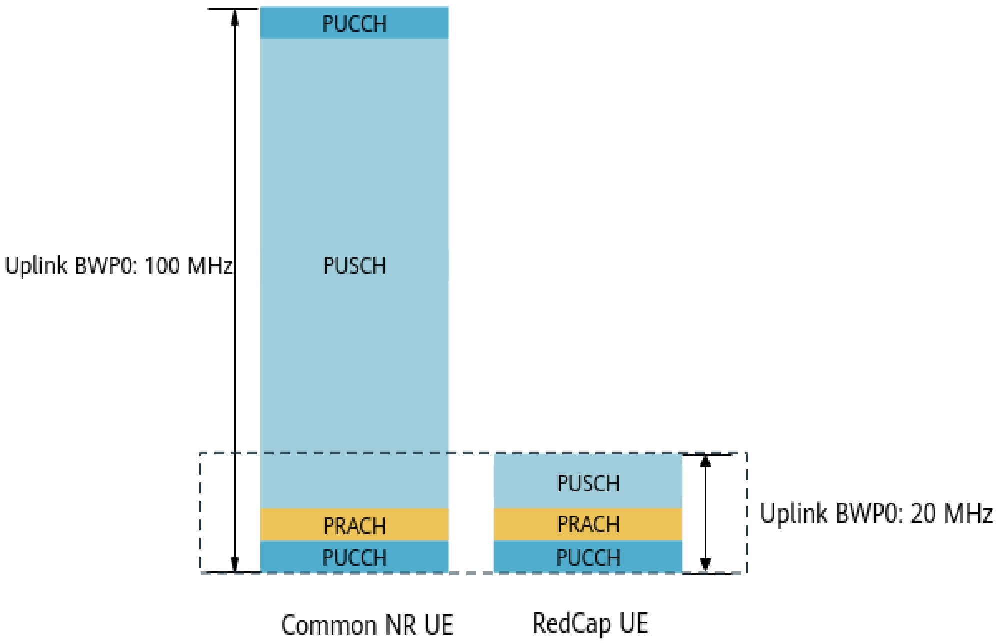
When a UE accesses an NR TDD cell, the gNodeB uses a scenario-specific method to identify whether the UE is a RedCap UE, as described in **Table 4-2**.

**Table 4-2** RedCap UE identification in NR TDD scenarios

| Scenario Classification | Specific Identification Method and Resource Allocation Related to Initial Access | Parameter |
|---|---|---|
| Scenario 1 | The gNodeB identifies whether a UE is a RedCap UE based on the LCID field in Msg3 and whether the reported UE capability contains "supportOfRedCap-r17". | **NRDUCell**.***UlBandwidth*** or **NRDUCellBwp**.***UlMinCarrierBw*** set to **20M** |
| Scenario 2 | • The common NR UE is scheduled in the 100 MHz BWP0. The RedCap UE is scheduled in the 20 MHz BWP0 and shares the PRACH and PUCCH resources within the 20 MHz bandwidth with common NR UEs.<br>• The gNodeB configures dedicated PRACH preamble resources for the RedCap UE. That is, the gNodeB identifies the RedCap UE based on Msg1 through independently reserved PRACH preamble resources.<br>Figure 4-1 shows the initial access of a RedCap UE and a common NR UE to an NR TDD cell. | • **NRDUCell**.***UlBandwidth*** and **NRDUCellBwp**.***UlMinCarrierBw*** set to **NOT_CONFIG** or a value greater than **20M**<br>• **NRDUCellRedCap**.***RedCapIdentifyStrategy*** set to **DEFAULT** |

**Figure 4-1** Example of initial access of a RedCap UE and a common NR UE to an NR TDD cell (in this example, a 100 MHz uplink BWP1 is configured and the RedCap uplink BWP0 is configured at the lower-frequency end of the cell bandwidth)

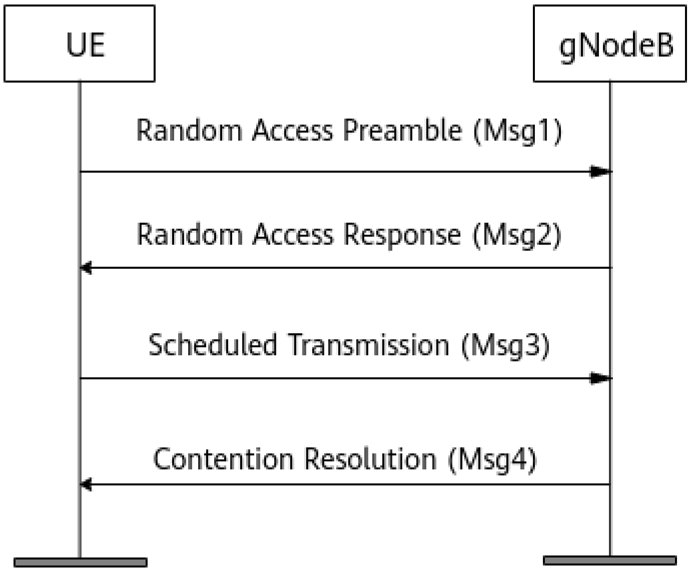


NOTE

For an NR TDD cell, when the RedCap BWP0 is not configured within the 20 MHz bandwidth at the lower-frequency end of the cell bandwidth and the **NRDUCell**.***PrachFreqStartPosition*** parameter is set to a value other than **65535**, the actual PRACH frequency-domain position may be inconsistent with the setting of the **NRDUCell**.***PrachFreqStartPosition*** parameter in certain scenarios. This is because the system requires that PRACH resources be configured in RedCap BWP0.

RedCap UE identification mentioned above is required only during contention-based random access, and other procedures for RedCap UEs during initial access are the same as those for common NR UEs, as shown in **Figure 4-2**.

**Figure 4-2** Random access procedure of RedCap UEs




### 4.1.1.2 RedCap UE Inactivity Management

**After detecting an inactive RedCap UE, the gNodeB performs inactivity management on this UE. This prevents the inactive RedCap UE from occupying system resources for a long time.** A RedCap UE becomes inactive because of no services or its disconnection from the gNodeB.

When detecting that a RedCap UE is in any of the following situations, the gNodeB considers it inactive:

- The gNodeB detects that the RedCap UE is in the signaling-only connection (no DRB) for a period longer than the length of **timer 1**.
- After a DRB is set up for the RedCap UE, the gNodeB detects that the UE does not transmit or receive any data (excluding MAC CEs) within the length of **timer 2**. The RedCap UE may have set up multiple DRBs, and each DRB is associated with a QCI. If different UE inactivity timer lengths are configured for these QCIs, the largest value takes effect.

**After considering the RedCap UE inactive, the gNodeB initiates a UE state transition or an RRC connection release procedure.** For RedCap UEs, the **RRC_INACTIVE connection management** function can be enabled.

- If this function is enabled, the gNodeB instructs the RedCap UE to transit from the RRC_CONNECTED state to RRC_INACTIVE state. For details about mobility management in inactive mode, see *SA Mobility Management in Inactive Mode*.

- If this function is disabled, the gNodeB sends a UE CONTEXT RELEASE REQUEST message carrying the release cause "User Inactivity" to the AMF and initiates an RRC connection release.

Table 4-3 describes the parameters related to the preceding function.

**Table 4-3** Functions related to RedCap UE Inactivity Management

| Item | Parameter |
|---|---|
| Timer 1 | **gNBConnStateTimer**.***SigUeNoNasMsgTransTmr*** |
| Timer 2 | **NRDUCellQciBearer**.***RedCapUeInactivityTimer*** |
| RRC_INACTIVE connection management | **REDCAP_RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter |

### 4.1.1.3 System Information Update Through RRC Reconfiguration

When a RedCap BWP1 does not include the CD-SSB, this BWP1 does not include CORESET 0, either. A RedCap UE on this BWP1 cannot monitor the system information update procedure in CORESET 0. SIB1 and SIB8 updates need to be delivered through RRC reconfiguration messages.

## 4.1.2 Mobility Management

This section describes functions related to RedCap mobility, including idle-mode RRM measurement relaxation, NCD-SSB-based mobility, and frequency selection.

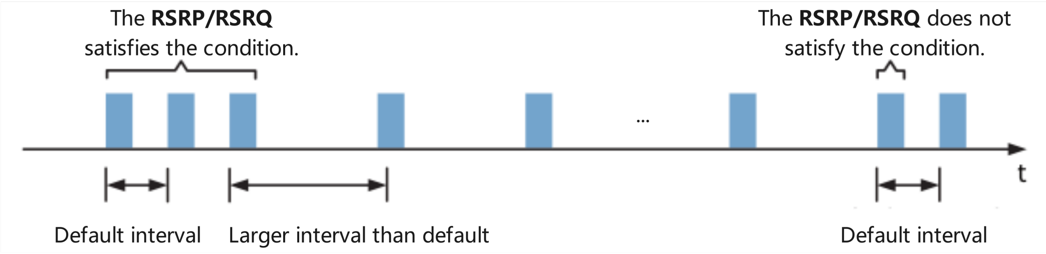
### 4.1.2.1 Idle-mode RRM Measurement Relaxation

In compliance with 3GPP specifications, RRC_IDLE UEs need to periodically measure neighboring cells for radio resource management (RRM) and the measurement results are used as the basis for cell reselection. When a UE moves at a low speed or the UE is stationary, the cell that the UE camps on does not change (that is, no cell reselection occurs). In this case, cell reselection is not urgent, and periodic RRM measurement is a waste of UE power. Therefore, when a UE moves at a low speed or the UE is stationary, the RRM measurement can be made less strict (for example, by increasing the RRM measurement interval) to reduce UE power consumption.

The power consumption of a UE is directly proportional to the RRM measurement period and measurement quantities (including SSB and cell- or carrier-level measurements). In addition, the power consumption also depends on the network measurement configuration. Each measurement result from the UE requires multiple measurement samples. 3GPP Release 17 proposes a new measurement method for RedCap UEs. As shown in Figure 4-3, **measurement relaxation on neighboring cells is triggered for RRC_IDLE/RRC_INACTIVE RedCap UEs running delay-insensitive services to measure at longer intervals. In this way, UE power consumption in RRM measurement is reduced.**

**Figure 4-3** Idle-mode RRM measurement relaxation




When both conditions for the measurement relaxation and requirements for the measurement relaxation duration are met, RedCap UEs in the RRC_IDLE/RRC_INACTIVE state can perform relaxed RRM measurements on neighboring cells.

- **Conditions for measurement relaxation:**
  - UE moving speed: For UEs moving at a low speed, relaxed measurements can be performed as long as the variation in measurement results on the serving cell within a certain period (that is, the difference between the UE-measured *Srxlev* and *Srxlev_RefStationary* of the serving cell within a period of time) is less than **threshold 1**.
    - *Srxlev* indicates the current *Srxlev* value (dB) of the serving cell.
    - *Srxlev_RefStationary* indicates the reference *Srxlev* value (dB) of the serving cell.
    - This threshold corresponds to the *s-SearchDeltaP-Stationary-r17* IE defined in 3GPP TS 38.331 V17.4.0.

    For details about the preceding fields and how this IE is used, see section 5.2.4.9.3 "Relaxed measurement criterion for a stationary RedCap UE" in 3GPP TS 38.304 V17.4.0.
  - UE location: For UEs not located at the cell edge, relaxed measurements can be performed as long as the *Srxlev* of the serving cell is higher than **threshold 2**. According to 3GPP TS 38.331 V17.4.0, this parameter must be set to a value less than or equal to **threshold 3** as well as **threshold 4**. Threshold 2[c] corresponds to the *s-SearchThresholdP2-r17* IE in SIB2 defined in 3GPP TS 38.331 V17.4.0. For details about how this IE is used, see section 5.2.4.9.4 "Relaxed measurement criterion for a stationary RedCap UE not at cell edge" in 3GPP TS 38.304 V17.4.0.
- **Measurement relaxation duration requirements**: After the UE receives the *relaxedMeasurement-r17* IE (indicating that RRM relaxation that complies with 3GPP Release 17 takes effect) in SIB2, the duration for which the RRM measurement relaxation conditions are met is longer than or equal to the length of **timer 3**.
- **Measurement relaxation methods**: The RRM measurement interval is increased, and the number of cells or carriers to be measured is reduced.

For details about how RRC_IDLE/RRC_INACTIVE UEs perform relaxed RRM measurements on neighboring cells, see section 4.2.2 "Requirements" in 3GPP TS 38.133 V17.4.0 and section 5.2.4.9 "Relaxed measurement" in 3GPP TS 38.304 V17.4.0.

**Table 4-4** describes the parameters related to the preceding function.

**Table 4-4** Parameters related to idle-mode RRM measurement relaxation

| Item | Parameter |
|---|---|
| Threshold 1 | **NRCellReselConfig**.***RrmRelaxLSUeRsrpDiffThld*** |
| Threshold 2 | **NRCellReselConfig**.***RrmRelaxNonEdgeRsrpThld*** |
| Threshold 3 | **NRCellReselConfig**.***IntraFreqMeasStartThld*** |
| Threshold 4 | **NRCellReselConfig**.***NonIntraFreqMeasRsrpThld*** |
| Timer 3 | **NRCellReselConfig**.***RrmRelaxLSUeJudgeTimer*** |

### 4.1.2.2 NCD-SSB-based Mobility

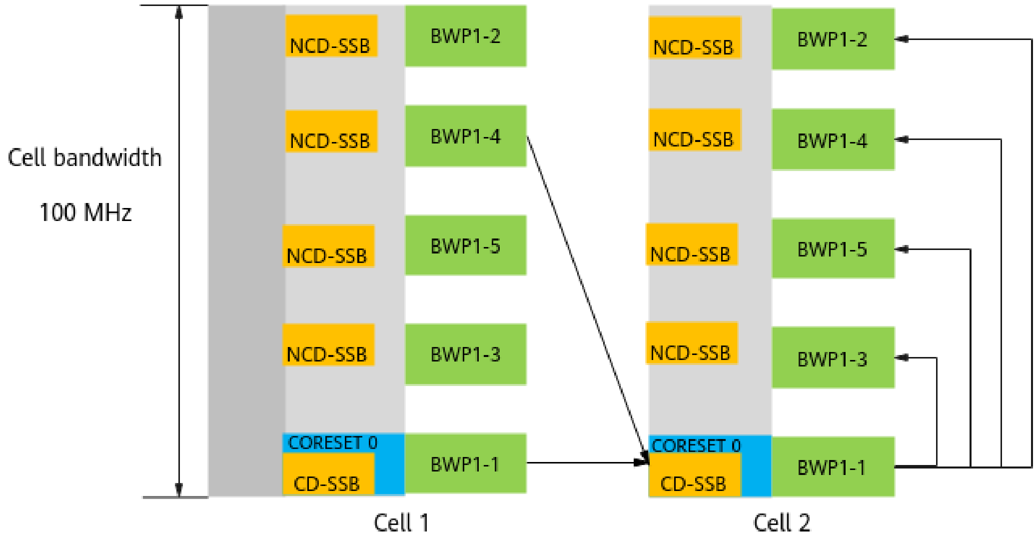
If RedCap UEs are concentrated on a BWP1 with the CD-SSB, resource congestion occurs. If a RedCap on a BWP1 without the CD-SSB needs to perform measurements, it needs to be switched to a BWP1 with the CD-SSB and switched back after measurement completion. This prolongs delay and decreases the capacity. One practice is the non-cell-defining synchronization signal and PBCH block (NCD-SSB), which is defined in 3GPP TS 38.331 V17.4.0. For a BWP1 without the CD-SSB, the NCD-SSB can be configured on the BWP1, so that RedCap UEs can also perform measurements and handovers based on the NCD-SSB. Accordingly, NCD-SSB-based mobility is introduced.

NCD-SSB-based mobility includes RedCap NCD-SSB-based mobility management and RedCap mobility optimization functions.

- NCD-SSB-based mobility management: When the number of activated RedCap BWP1s is greater than one, some BWP1s do not contain CD-SSBs. **To support RRM measurement and mobility measurement of RedCap UEs, BWP1s without CD-SSBs must contain NCD-SSBs**. RedCap UEs can perform measurements and handovers based on the NCD-SSB. As shown in **Figure 4-4**, a RedCap UE is handed over to BWP1-1 of cell 2 and then is switched between BWP1s. The mobility management procedure for RedCap UEs using BWP1s with CD-SSBs is the same as that for common NR UEs.

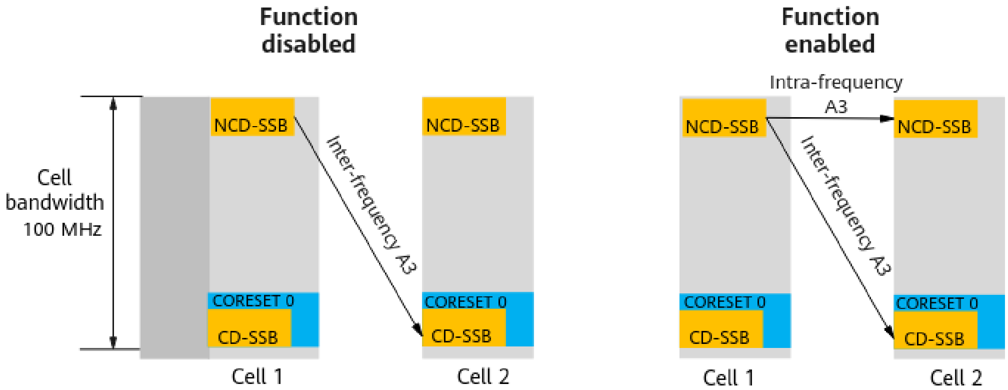
**Figure 4-4** NCD-SSB (using NR TDD 100 MHz as an example)




- Differentiated configuration of RedCap NCD-SSB-based mobility: When RedCap UEs on BWP1s with NCD-SSBs measure CD-SSBs of neighboring cells, differentiated setting of the A1 and A2 RSRP thresholds and A3 offset can be achieved by using the following parameters:
  - **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1RsrpThld***
  - **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld***
  - **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA3Offset***

  The **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1RsrpThld*** parameter value must be greater than the **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld*** parameter value.
- RedCap NCD-SSB measurement: RedCap UEs can perform intra-frequency A3 measurements on NCD-SSBs of neighboring cells, as shown in **Figure 4-5**. This reduces the probability of gap-assisted inter-frequency CD-SSB measurements and reduces the gap-assisted-measurement-induced decrease in average uplink and downlink throughputs of RedCap UEs. This function is controlled by the **REDCAP_NCD_SSB_MEAS_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter. This option is selected by default.

**Figure 4-5** NCD-SSB intra-frequency measurement (using NR TDD 100 MHz as an example)



- RedCap NCD-SSB SINR measurement: When co-channel interference exists between neighboring cells, it is recommended that this function be enabled. After this function takes effect, RedCap UEs can perform A1 and A2 measurements based on the NCD-SSB SINR triggering quantity, which involve the RedCap NCD-SSB A1 SINR threshold and RedCap NCD-SSB A2 SINR threshold, respectively.
  - RedCap NCD-SSB A1 SINR threshold: **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1SinrThld***
  - RedCap NCD-SSB A2 SINR threshold: **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2SinrThld***

  This function is controlled by the **REDCAP_NCD_SSB_SINR_MEAS_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter.

### 4.1.2.3 RedCap Handover Policy Control

**RedCap handover policy control refers to the process of determining the handover policy for RedCap/eRedCap UEs based on the *Redcap-Bcast-Information*/*eRedcap-Bcast-Information* IE or the cell RedCap barring state (specified by the NRDUCellRedCap.*CellBarredRedCap* parameter).** Based on specific handover scenarios, this function consists of Xn-based and intra-base-station handover control, NG-based and inter-RAT handover control, and control over handovers to external NR cells.

### Xn-based and Intra-Base-Station Handover Control

The Xn-based and intra-base-station handover control function only controls the outgoing handovers of RedCap UEs. (Incoming handovers are always allowed.)

- Common RedCap UEs (not performing emergency calls)

  After Xn-based and intra-base-station handover control takes effect for common RedCap UEs, the processing is as follows:
  - Xn-based handover: The gNodeB controls the outgoing handovers of RedCap UEs based on the bit values in the *Redcap-Bcast-Information* IE of the target cell, as described in **Table 4-5**. For details about the *Redcap-Bcast-Information* IE, see sections 8.4 "Global procedures" and 9.2.2.11 "Served Cell Information NR" in 3GPP TS 38.423 V18.2.0.

**Table 4-5** Xn-based handover policies for common RedCap UEs

| First Bit | Second Bit | Handover Processing |
|---|---|---|
| 0 | 0 | Outgoing handovers of 1RX and 2RX RedCap UEs are prohibited. |
| 0 | 1 | Outgoing handovers of 1RX RedCap UEs are prohibited, and those of 2RX RedCap UEs are allowed. |
| 1 | 0 | Outgoing handovers of 1RX RedCap UEs are allowed, and those of 2RX RedCap UEs are prohibited. |

| First Bit | Second Bit | Handover Processing |
|---|---|---|
| 1 | 1 | Outgoing handovers of 1RX and 2RX RedCap UEs are allowed. |

– Intra-base-station handover: The gNodeB controls the outgoing handovers of RedCap UEs based on the **NRDUCellRedCap**.***CellBarredRedCap*** parameter setting of the local cell, as described in **Table 4-6**.

**Table 4-6** Intra-base-station handover policies for common RedCap UEs

| Handover Policy | Handover Processing | Parameter Setting |
|---|---|---|
| Policy 1 | RedCap UEs are allowed to camp on the local cell, meaning that outgoing handovers are not performed. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **NOT_BARRED**. |
| Policy 2 | 1RX RedCap UEs are prohibited from camping on the local cell, meaning that outgoing handovers are required for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **1RX_BARRED**. |
| Policy 3 | 2RX RedCap UEs are prohibited from camping on the local cell, meaning that outgoing handovers are required for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **2RX_BARRED**. |
| Policy 4 | 1RX and 2RX RedCap UEs are prohibited from camping on the local cell, meaning that outgoing handovers are required for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **1RX_2RX_BARRED**. |

This function is controlled by the **XN_INTRA_GNB_HO_CTRL_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter. Incoming handovers of RedCap UEs are not controlled by this option. That is, incoming handovers of RedCap UEs are allowed regardless of the value of the *Redcap-Bcast-Information* IE of the local cell.

- For RedCap UEs performing emergency calls:
  – Outgoing handovers are allowed regardless of the bit values in the *Redcap-Bcast-Information* IE of the target cell.
  – Incoming handovers are allowed regardless of the bit values in the *Redcap-Bcast-Information* IE of the local cell.

## NG-based and Inter-RAT Handover Control

This function only controls incoming handover processing of RedCap UEs, because the *Redcap-Bcast-Information* IE of the peer base station cannot be perceived during NG-based handovers of such UEs.

- Common RedCap UEs (not performing emergency calls)

  After NG-based and inter-RAT handover control takes effect, the gNodeB controls the incoming handovers of RedCap UEs based on the **NRDUCellRedCap**.***CellBarredRedCap*** parameter setting in the local cell.

  **Table 4-7** NG-based and inter-RAT handover policies for common RedCap UEs

  | Handover Policy | Handover Processing | Parameter Setting |
  |---|---|---|
  | Policy 1 | RedCap UEs are allowed to camp on the local cell, meaning that incoming handovers are allowed. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **NOT_BARRED**. |
  | Policy 2 | 1RX RedCap UEs are prohibited from camping on the local cell, meaning that incoming handovers are prohibited for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **1RX_BARRED**. |
  | Policy 3 | 2RX RedCap UEs are prohibited from camping on the local cell, meaning that incoming handovers are prohibited for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **2RX_BARRED**. |
  | Policy 4 | 1RX and 2RX RedCap UEs are prohibited from camping on the local cell, meaning that incoming handovers are prohibited for such UEs. | The **NRDUCellRedCap**.***CellBarredRedCap*** parameter is set to **1RX_2RX_BARRED**. |

  This function is controlled by the **NG_INTER_RAT_HO_COMPAT_SW** option of the **NRCellRedCap.*RedCapAlgoSwitch*** parameter.

- For RedCap UEs performing emergency calls:

  Incoming handovers are allowed regardless of the **NRDUCellRedCap**.***CellBarredRedCap*** parameter setting in the local cell.

## Control over Handovers to External NR Cells

The handover policy for RedCap UEs can be configured in the neighbor relationships to determine the barring state of UE handovers to neighboring cells, as described in **Table 4-8**.

**Table 4-8** Handover policies for RedCap UEs

| Handover Policy | Handover Processing | Parameter Setting |
|---|---|---|
| Policy 1 | Handovers of 1RX RedCap UEs are prohibited. | The **REDCAP_1RX_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter is selected. |
| Policy 2 | Handovers of 2RX RedCap UEs are prohibited. | The **REDCAP_2RX_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter is selected. |

Control over handovers to external NR cells collaborates with Xn-based and intra-base-station handover control and NG-based and inter-RAT handover control as follows:

- Xn-based outgoing handover: The handover policy for common RedCap UEs is determined by the **REDCAP_1RX_NO_HO_SW** or **REDCAP_2RX_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter of external cells and the *Redcap-Bcast-Information* IE of the target cell. The UEs can be handed over only when both allow outgoing handovers.
- Intra-base-station handover: The handover policy for common RedCap UEs is determined by the **NRDUCellRedCap**.***CellBarredRedCap*** parameter.
- NG-based incoming handover: The handover policy for common RedCap UEs is determined by the **NRDUCellRedCap**.***CellBarredRedCap*** parameter.
- NG-based outgoing handover: The handover policy for common RedCap UEs is determined by the **REDCAP_1RX_NO_HO_SW** or **REDCAP_2RX_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter and the **REDCAP_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter. The UEs can be handed over only when both allow outgoing handovers.

### 4.1.2.4 Frequency Selection for RedCap

### Measurement Frequency Selection for RedCap

**To facilitate frequency filtering by the gNodeB for inter-frequency measurements by RedCap UEs, the measurement activation flag is configurable for intra-RAT neighboring frequencies**, as listed in **Table 4-9**.

**Table 4-9** Measurement frequency selection policies

| Category | Measurement Frequency Selection Policy | Parameter |
|---|---|---|
| Policy 1 | The gNodeB can include a frequency in the inter-frequency measurement configurations delivered to RedCap UEs. | **NRCellFreqRelation**.***RedCapMeasActvFlag*** set to **ACTIVE** |
| Policy 2 | The gNodeB does not include a frequency in the inter-frequency measurement configurations delivered to RedCap UEs. | **NRCellFreqRelation**.***RedCapMeasActvFlag*** set to **INACTIVE** |

NOTE

If the **NRCellFreqRelation**.***MeasActvFlag*** parameter is set to **INACTIVE** and the **AUTO_NR_FREQ_RELATION_SW** option of the **NRCELLALGOSWITCH**.***AnrSwitch*** parameter is deselected, the RedCap measurement activation flag does not take effect.

## Cell-Reselection Frequency Selection for RedCap

This function allows the configuration of the **redCapAccessAllowed-r17** IE in SIB4 to facilitate cell-reselection frequency selection in inter-frequency broadcast scenarios, as listed in **Table 4-10**.

**Table 4-10** Cell-reselection frequency selection policy

| Category | Cell-Reselection Frequency Selection | Parameter |
|---|---|---|
| Policy 1 | The **redCapAccessAllowed-r17** IE is set to **true**, indicating that cell reselections of RedCap UEs to this frequency are allowed. | **NRCellFreqRelation**.***RedCapAccessAllowedFlag*** set to **ALLOWED** |
| Policy 2 | The **redCapAccessAllowed-r17** IE is not contained, indicating that cell reselections of RedCap UEs to this frequency are not allowed. | **NRCellFreqRelation**.***RedCapAccessAllowedFlag*** set to **NOT_ALLOWED** |

## RedCap Frequency Priority Policy

The RedCap frequency priority policy is introduced to meet differentiated configuration requirements of customers and implement differentiated mobility processing for RedCap UEs. After this function takes effect, the gNodeB supports the configuration of dedicated gNodeB frequency priority group IDs for RedCap UEs based on the RFSP, QCI, or operator to implement customized preferential camping and handovers for RedCap UEs.

This function is set through the following parameters:

- RFSP-based differentiation: The **gNBRfspConfig**.***RedCapGnbFreqPriGroupId*** parameter is set to a value other than **255**.
- QCI-based differentiation: The **NRCellQciBearer**.***RedCapGnbFreqPriGroupId*** parameter is set to a value other than **255**.
- Operator-based differentiation: The **NRCellOpPolicy**.***RedCapGnbFreqPriGroupId*** parameter is set to a value other than **255**.

If these parameters are set to **255**, no camping and handover policies are customized for RedCap UEs based on the RFSP, QCI, or operator.

#### 4.1.2.5 Admission Guarantee for Emergency Calls

According to 3GPP TS 38.304 of Release 17, emergency calls of RedCap UEs are prohibited in a cell if the IE indicating the cell barring state of the cell is set to **not barred** while the IE indicating the RedCap-specific cell barring state of the cell is set to **barred**. **Table 4-11** lists the IEs related to the cell barring state.

**Table 4-11** IEs related to the cell barring state

| IE Type | IE | IE Value |
|---|---|---|
| Cell barring state | cellBarred | **barred** and **not barred** |
| RedCap-specific cell barring state (introduced in 3GPP Release 17) | cellBarredRedCap1RX<br>cellBarredRedCap2RX | **barred** and **not barred** |

According to 3GPP TS 38.304 of Release 18, SIB1 can include the *barringExemptEmergencyCall* IE. Inclusion of this IE means that in a cell with the IE indicating the cell barring state set to **not barred**, RedCap UEs can make emergency calls even if the IE indicating the RedCap-specific cell barring state is set to **barred**. In compliance with the protocol, admission guarantee for emergency calls has been introduced.

If this function is enabled, the *barringExemptEmergencyCall* IE in SIB1 sent from the gNodeB to UEs is set to **true**. If this function is disabled, SIB1 does not include this IE. For details about this IE, see section 5.3.1 "Cell status and cell reservations" in 3GPP TS 38.304 V18.2.0.

This function is controlled by the **EMC_ADMISSION_GUAR_SW** option of the **NRDUCellRac**.***RacAlgoSwitch*** parameter.

#### 4.1.2.6 QCI-specific Handover Priority Configuration

The QCI-specific handover priority configuration is introduced to meet differentiated configuration requirements of customers and implement differentiated mobility processing for RedCap UEs.

- If the **gNBQciBearer**.***QciPriorityForRedCapHo*** parameter is set to a value other than **255**, the value of this parameter is used as the QCI priority for handovers of RedCap UEs.
- If the **gNBQciBearer**.***QciPriorityForRedCapHo*** parameter is set to **255**, the priority indicated by the *priorityLevelQos* IE delivered by the core network is preferentially used as the QCI priority for handovers of RedCap UEs. If the core network does not deliver this IE, the value of the **gNBQciBearer**.***PriorityLevel*** parameter is used.

## 4.1.3 Experience Enhancement

This section describes the following functions related to RedCap experience improvement:

- RedCap MU-MIMO
- Multiple BWP1s for RedCap
- Uplink preallocation
- RedCap downlink scheduling policy
- QCI-based RedCap DRX configuration
- Differentiated QoS management for RedCap UEs
- SSB interference avoidance for RedCap UEs
- RedCap BWP frequency-domain staggering (low-frequency TDD)
- Support for Super Uplink by RedCap
- RedCap power aggregation (low-frequency TDD)
- PDCCH symbol adaptation for RedCap BWP with the CD-SSB (low-frequency TDD)
- RedCap UE weight optimization (low-frequency TDD)
- BWP scheduling optimization for RedCap UEs

### 4.1.3.1 RedCap MU-MIMO

RedCap multi-user MIMO (MU-MIMO) **implements spatial multiplexing for multiple RedCap UEs. It enables data transmission for the UEs using the same time-frequency resources, increasing system capacity and spectral efficiency in both uplink and downlink.** For details about MU-MIMO, see *MIMO (TDD)*.

Table 4-12 lists the MU-MIMO functions supported by RedCap UEs.

**Table 4-12** MU-MIMO functions supported by RedCap UEs

| Function | Function Switch | Description |
|---|---|---|
| Downlink MU-MIMO for RedCap UEs | **REDCAP_DL_MU_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | This function takes effect only when the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected. |

| Function | Function Switch | Description |
|---|---|---|
| Uplink MU-MIMO for RedCap UEs | **REDCAP_UL_MU_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | This function takes effect only when the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected. |

For NR TDD cells, to increase the pairing probability of RedCap UEs and the average uplink throughput of RedCap UEs, it is recommended that the following functions be enabled:

- Uplink MU grouping and pairing optimization for RedCap UEs: This function optimizes the uplink MU grouping and pairing procedure for RedCap UEs, increasing the average uplink throughput of RedCap UEs.

  This function is controlled by the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.
- Uplink MU grouping and pairing avoidance for RedCap UEs: In uplink scheduling, non-RedCap UEs with higher priorities avoid RedCap UEs. This increases the average uplink throughput of RedCap UEs.

  This function is controlled by the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter and takes effect only when the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter is selected.

For details about "uplink MU grouping and pairing optimization for RedCap UEs" and "uplink MU grouping and pairing avoidance for RedCap UEs", see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

### 4.1.3.2 Multiple BWP1s for RedCap

According to 3GPP specifications, RedCap UEs support a maximum bandwidth of 20 MHz. That is, after RedCap Introduction is enabled, one RedCap BWP1 is configured for RedCap UEs by default. When there are a large number of RedCap UEs or the traffic volume is heavy on the network, configuring only one RedCap BWP1 cannot meet the traffic volume requirements of RedCap UEs. In this context, Huawei introduces the "multiple BWP1s for RedCap" function. **If a cell bandwidth is greater than 20 MHz, this function allows an operator to configure multiple available BWP1s for RedCap UEs based on the load status. This increases the number of RedCap UEs supported by the cell to fulfill the traffic volume requirements of RedCap UEs.**

This function is controlled by the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter.

#### Supported Cell Bandwidth and Number of BWP1s Available for RedCap

For TDD cells:

**Multiple BWP1s for RedCap is only supported in cells with a bandwidth of 40 MHz or higher**.

**Table 4-13** lists the mapping between the number of BWP1s that can be used by RedCap UEs and the value of **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth***.

**Table 4-13** Mapping between the number of BWP1s available for RedCap UEs and the parameter value (NR TDD)

| Maximum Number of BWP1s Available for RedCap UEs (per Cell) | Value of NRDUCellRedCap.*RedCapUeMaxAvailBandwidth* |
|---|---|
| 1 | 20M |
| 2 | 40M |
| 3 | 60M |
| 4 | 80M |
| 5 | 100M |

NOTE

For NR TDD, after multiple BWP1s for RedCap takes effect, the number of BWP1s that take effect for RedCap is determined by the following factors: BBP type (which can be queried by running **DSP NRDUCELLREDCAP**), baseband mode (specified by **BBP**.***BBWS***), RAT resource ratio (specified by **BBP**.***SRT***), number of cells configured on the BBP, and BWP1 capability of the entire BBP. When the BWP1 capability of the entire BBP is insufficient, the number of BWP1s that actually take effect is less than the maximum number of BWP1s available for RedCap UEs mapped to the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter value. For example, assume that a single-mode BBP supports a maximum of 18 BWP1s. If the number of cells configured on the BBP is less than or equal to three, a maximum of five BWP1s can take effect for each cell. If the number of cells configured on the BBP is greater than three, a maximum of five BWP1s can take effect only in some of the cells.

## BWP1 Selection Policy for RedCap UEs

In a TDD cell, when RedCap UEs are configured with multiple available BWP1s, system information and paging message overheads on BWP1s with CD-SSBs in SSB multi-beam scenarios are large, resulting in poor user experience. To address this issue, preferential selection of BWP1s with NCD-SSBs for RedCap UEs is introduced. This function preferentially allocates UEs to BWP1 with the NCD-SSB based on the setting of the **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** parameter, improving user experience, as listed in **Table 4-14**.

**Table 4-14** BWP1 selection policy for RedCap UEs (NR TDD)

| Policy Category | BWP1 Selection Policy for RedCap UEs | Effective Conditions |
|---|---|---|
| Policy 1 | Newly admitted RedCap UEs are preferentially allocated to a BWP1 with the NCD-SSB. | The **NRDUCellRedCap**.***NcdSsbBwpPrefer-MaxUeNum*** parameter is set to a non-zero value and the number of UEs on a BWP1 with the NCD-SSB is less than this parameter value. |
| Policy 2 | RedCap UEs access the network evenly among BWP1s. | The **NRDUCellRedCap**.***NcdSsbBwpPrefer-MaxUeNum*** parameter is set to a non-zero value and the number of UEs on BWP1 with the NCD-SSB is greater than or equal to this parameter value. Alternatively, the **NRDUCellRedCap**.***NcdSsbBwpPrefer-MaxUeNum*** parameter is set to **0**. |

NOTE

The gNodeB also considers eMBB BWP2 UEs when balancing RedCap UEs among BWP1s.

## Scenario-based Optimization

After multiple BWP1s for RedCap is enabled, resource optimization functions described in **Table 4-15** can be used to further improve RedCap user experience in specific scenarios.

**Table 4-15** Scenario-based optimization

| RAT | Function Name | Function Description | Application Scenario | Parameter Setting |
|---|---|---|---|---|
| Low-frequency TDD | PUCCH CSI RB adjustment for the RedCap-dedicated BWP | This function optimizes the PUCCH resource allocation method. When there are a large number of RedCap UEs, the PUCCH CSI RB resources are adaptively adjusted on the RedCap-dedicated BWP based on the CSI resource usage.<br>RedCap-dedicated BWP refers to RedCap BWP1 that is not deployed at the band edges and does not contain the CD-SSB. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, and the number of UEs on the RedCap-dedicated BWP is greater than or equal to 100. | **REDCAP_PUCCH_CSI_RB_ADJ_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter |
| Low-frequency TDD | PUCCH format-1 RB resource control | This function optimizes the PUCCH resource allocation method and dynamically allocates PUCCH format-1 RB resources within a BBP. For details about PUCCH format-1 RB resource control, see *Channel Management* . | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**. | **FORMAT1_RB_RES_CTRL_SW** option of the **gNodeBAlgo**.***PucchSwitch*** parameter |
| Low-frequency TDD | Inter-TRP RB resource sharing policy | If this function is enabled, a multi-TRP mechanism is used for PUCCH RB resource sharing. For details about the inter-TRP RB resource sharing policy, see *Channel Management*. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**. | **INTER_TRP_RB_SHARE_POLICY_SW** option of the **NRDUCellPucch**.***PucchAlgoExtSwitch*** parameter |

| RAT | Function Name | Function Description | Application Scenario | Parameter Setting |
|---|---|---|---|---|
| Low-frequency TDD | RedCap BWP format-4 PUCCH configuration | In a RedCap BWP1 that is not located at the edge of the BWP0 in the cell, the number of PUCCH format 4 ACK RBs can be configured. These RBs are used to transmit HARQ-ACKs whose size is greater than two bits, thereby increasing the average downlink throughput of RedCap UEs performing large-packet services. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**. | **NRDUCellRedCap**.***PucchFormat4AckRbNum*** |
| Low-frequency TDD | CSI resource balancing | CSI resources for eMBB UEs are evenly allocated at both ends of the frequency band to ensure CSI resources for RedCap UEs. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**. | **CSI_RES_BALANCE_SW** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter |
| Low-frequency TDD | RedCap PUCCH resource guarantee | The PUCCH resource allocation mode in a cell is optimized so that RedCap UEs can use more PUCCH resources and more RedCap BWP1s can be activated. | A large number of RedCap BWPs are configured on the BBP. Specifically, the values of the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameters are greater than **20M** for multiple cells on the BBP. | **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter |
| Low-frequency TDD | Format-1 RB adaptation | With this function, the gNodeB adaptively adjusts the format-1 RB resources based on the format-1 resource usage on the RedCap BWP1. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M**. | **PUCCH_F1_RB_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter |

| RAT | Function Name | Function Description | Application Scenario | Parameter Setting |
|---|---|---|---|---|
| Low-frequency TDD | CSI RB adaptation | With this function, the gNodeB adaptively adjusts the CSI RB resources based on the CSI resource usage on the RedCap BWP1. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**. | **PUCCH_CSI_RB_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter |
| Low-frequency TDD | Format-1 SR resource optimization | With this function, 18 SR code channels are configured for each format-1 RB, and more format-1 SR resources can be used in the cell | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M**. | **FORMAT1_SR_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter |
| Low-frequency TDD | Aperiodic CSI-RS optimization in heavy-load scenarios | With this function, aperiodic CSI-RSs take effect even when the system is heavily loaded, fully utilizing aperiodic CSI-RS capabilities. For details about aperiodic CSI-RS optimization in heavy-load scenarios, see *Uplink Scheduling*. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M**. | **APER_CSIRS_FLEX_OPT_SW** option of the **NRDUCellCsirs**.***CsiRsExtSwitch*** parameter |
| Low-frequency TDD | CSI RB and format-1 RB reposition | The gNodeB adjusts the positions of the extended format-1 RBs to within the first activated RedCap BWP1 and adaptively adjusts the number of CSI RBs based on the CSI resource usage on the RedCap BWP. This increases the number of RedCap UEs supported on the first activated RedCap BWP1. | The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M**. | **PUCCH_RB_REPOSITION_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter |

#### 4.1.3.3 Uplink RedCap Preallocation

For RedCap UEs that access a cell through initial access, an incoming handover, or RRC reconfiguration, this function enables the base station to proactively schedule such a UE at a specified interval as long as there are remaining scheduling resources, regardless of whether the UE has sent a scheduling request indicator (SRI) to the base station. This shortens the waiting time that would otherwise be required between when a RedCap UE sends an SRI and when the UE obtains an uplink scheduling grant. This function is controlled by the following items:

- Function switch: **UL_REDCAP_PREALLOCATION_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter. This option is selected by default.
- Minimum preallocation interval (between two scheduling occasions for a UE): specified by the **NRDUCellPusch**.***MinPreallocationPeriod*** parameter
- Maximum number of UEs to be preallocated resources in a slot: specified by the **NRDUCellPusch**.***PreschUeNumUpperLimit*** parameter
- Amount of data to be scheduled in preallocation for a UE: specified by the **NRDUCellPusch**.***PreallocationSize*** parameter
- Type of RedCap UEs for which uplink preallocation takes effect: specified by the **NRDUCellRedCap**.***RedCapUlPreallocPolicy*** parameter

#### 4.1.3.4 RedCap Downlink Scheduling Policy

RedCap downlink scheduling policy is introduced to increase the average downlink throughput of RedCap UEs in SU-MIMO scenarios. **Table 4-16** describes the detailed policies.

**Table 4-16** RedCap downlink scheduling policy

| Downlink Scheduling Policy for RedCap UEs | Description | Parameter Setting |
| --- | --- | --- |
| Policy 1 | The base station adaptively selects type 0 or type 1 scheduling for RedCap UEs that support dynamic indication of PDSCH type 0 or 1 (the corresponding capability field is **dynamicSwitchRA-Type0-1-PDSCH**) based on the traffic volume of the UEs. In this way, type 1 scheduling takes effect for RedCap UEs running large-packet services, scheduling is performed at the RB level, the number of downlink RBs that can be scheduled increases, and the average downlink throughput of RedCap UEs increases in SU-MIMO scenarios. | Scenarios where all services of a single UE are carried on only one bearer: The **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter is deselected.<br>Scenarios where all services of a single UE are carried on multiple bearers: The **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter is deselected, and the **RSVDSWPARAM3_BIT30** option of the **NRDUCellRsvdExt01**.***RsvdSwParam3*** parameter is selected. |
| Policy 2 | RedCap UEs use the same scheduling policy as eMBB UEs. Type 0 scheduling takes effect by default for RedCap UEs running large-packet services. | Scenarios where all services of a single UE are carried on only one bearer: The **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter is selected.<br>Scenarios where all services of a single UE are carried on multiple bearers: The **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter is selected, or the **RSVDSWPARAM3_BIT30** option of the **NRDUCellRsvdExt01**.***RsvdSwParam3*** parameter is deselected. |

#### 4.1.3.5 QCI-based RedCap DRX Configuration

Quality of service (QoS) is a measure that helps define the level of service and provides an indication of customer satisfaction. The QoS class identifier (QCI) is used to identify the QoS class of a data radio bearer (DRB). QCI-based DRX parameter configuration allows different DRX parameter groups to be configured for different QCIs. The **NRCellQciBearer** and **NRDUCellQciBearer** MOs are used to bind a QCI to a DRX parameter group (configured using the **gNBDrxParamGroup** and **gNBDUDrxParamGroup** MOs).

QCI-based RedCap DRX configuration allows independent configuration of DRX parameter group IDs and DU DRX parameter group IDs for RedCap UEs. **Table 4-17** and **Table 4-18** list the function activation scenarios.

**Table 4-17** DRX activation scenarios (CU side)

| Scenario | Function Activation Status | Parameter Setting | |
|---|---|---|---|
| | | **NRCellQciBearer**.***RedCapDrxParamGroupId*** | **NRCellQciBearer**.***DrxParamGroupId*** |
| Scenario 1 | DRX does not take effect. | 255 | 255 |
| Scenario 2 | DRX takes effect based on the value of the **NRCellQciBearer**.***DrxParamGroupId*** parameter. | 255 | Not 255 |
| Scenario 3 | DRX takes effect based on the value of the **NRCellQciBearer**.***RedCapDrxParamGroupId*** parameter. | Not 255 | Any value |

**Table 4-18** DRX activation scenarios (DU side)

| Scenario | Function Activation Status | Parameter Setting | |
|---|---|---|---|
| | | **NRDUCellQciBearer**.***RedCapDuDrxParamGroupId*** | **NRDUCellQciBearer**.***DuDrxParamGroupId*** |
| Scenario 1 | DRX does not take effect. | 255 | 255 |
| Scenario 2 | DRX takes effect based on the value of the **NRDUCellQciBearer**.***DuDrxParamGroupId*** parameter. | 255 | Not 255 |

| Scenario | Function Activation Status | Parameter Setting | |
|---|---|---|---|
| | | **NRDUCellQciBearer**.***RedCapDuDrxParamGroupId*** | **NRDUCellQciBearer**.***DuDrxParamGroupId*** |
| Scenario 3 | DRX takes effect based on the value of the **NRDUCellQciBearer**.***RedCapDuDrxParamGroupId*** parameter. | Not 255 | Any value |

QCI-based RedCap DRX configuration takes effect on a UE only when all of the following conditions are met:

- The QCIs added to the **NRCellQciBearer** and **NRDUCellQciBearer** MOs exist in the **gNBQciBearer** MO.
- The scenario where the DRX does not take effect is not present on both the CU and DU sides.

A UE may have multiple DRBs and each DRB corresponds to a QCI. Different QCIs have different priorities (specified by the **gNBQciBearer**.***PriorityLevel*** parameter). After QCI-based RedCap DRX configuration takes effect on the UE, the UE uses the DRX parameter group bound with the QCI of the highest priority.

#### 4.1.3.6 Differentiated QoS Management for RedCap UEs

In SA networking, there is a one-to-one or many-to-one mapping between standard 5QIs and QCIs. When the 5QIs of RedCap and eMBB UEs are the same, RedCap and eMBB bearers cannot be mapped to different QCIs. As a result, differentiated scheduling and handovers cannot be performed for RedCap UEs. Differentiated QoS management for RedCap UEs is therefore introduced.

**This function allows specific QCIs to be configured for 5QI bearers of RedCap UEs to implement differentiated QoS configuration for RedCap UEs. This function can be used in differentiated scheduling and handovers for RedCap UEs.** For details about principles for 5QI bearers, see *QoS Management*.

This function uses the **gNBOp5qiConfig** MO to map the 5QI bearers of RedCap UEs to specific QCIs.

- If the **gNBOp5qiConfig**.***RedCapQci*** parameter is set to a value other than **65535**, the parameter value is a QCI actually used by RedCap UEs.
- If the **gNBOp5qiConfig**.***RedCapQci*** parameter is set to **65535**, the mapping mechanism is as follows:
  - If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is deselected, the value of the **gNBOp5qiConfig**.***Qci*** parameter is the QCI actually used by RedCap UEs.
  - If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is selected, the QCI actually used by RedCap UEs is determined based on the configuration of certain functions (The functions include SAI-based service experience

improvement. For details about the involved functions, see **4.3.2.3 Function Impacts** of "Differentiated QoS Management for RedCap UEs"). For details about SAI-based service experience improvement, see *SAI-based Service Experience Improvement*.

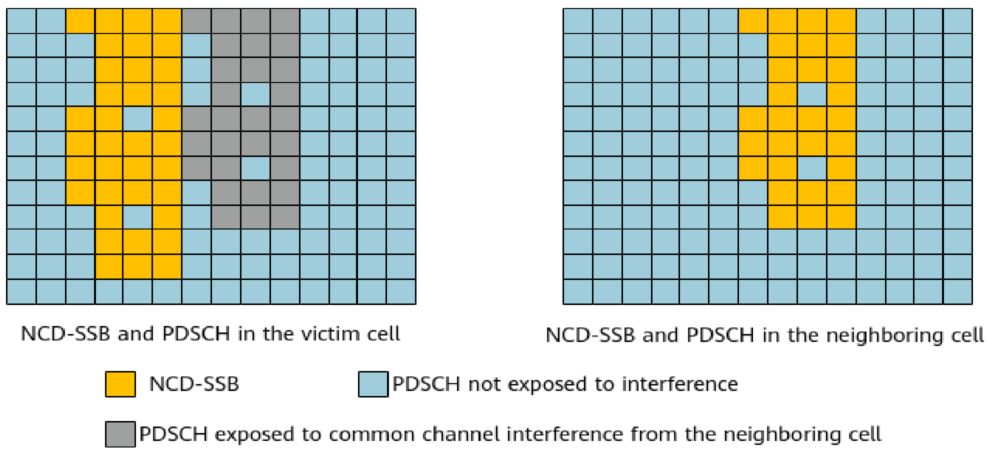
NOTE

- Extended QCIs specified by the **gNBOp5qiConfig.*Qci*** and **gNBOp5qiConfig.*RedCapQci*** parameters must be configured through the **NRCellQciBearer** and **NRDUCellQciBearer** MOs. If they are not configured, parameter settings of the **NRCellQciBearer** and **NRDUCellQciBearer** MOs corresponding to QCI 9 take effect. For details about principles for the settings of the **NRCellQciBearer** and **NRDUCellQciBearer** MOs, see *QoS Management*.
- If the **gNBOp5qiConfig**.***Nr5qi*** parameter is set to **2**, **65**, **66**, **67**, **69**, or **70**, the **gNBOp5qiConfig**.***RedCapQci*** parameter must be set to **65535**.
- The **gNBOp5qiConfig**.***Qci*** and **gNBOp5qiConfig**.***RedCapQci*** parameters cannot be both set to **65535**.

### 4.1.3.7 SSB Interference Avoidance for RedCap UEs

If the NCD-SSB frequency configuration or the number of NCD-SSB beams differs between adjacent cells, the time-frequency positions of NCD-SSBs cannot be completely aligned between the cells during data transmission. In this case, the NCD-SSBs of the neighboring cell interfere with the PDSCH of the local cell and therefore severely affect the downlink cell throughput, as shown in **Figure 4-6**. SSB interference avoidance for RedCap UEs is therefore introduced.

**Figure 4-6** Common channel interference




**With SSB interference avoidance for RedCap UEs enabled, the gNodeB checks whether the local cell experiences interference from the NCD-SSBs of neighboring cells. If the cell experiences the interference, the gNodeB performs downlink scheduling in the local cell in such a way that it avoids allocating resources in the time-frequency positions of the NCD-SSBs of the neighboring cells.** This function is recommended when there is interference from the NCD-SSBs of neighboring cells.

This function is controlled by the **REDCAP_SSB_INTRF_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter. As it is dependent on the identification and evaluation procedures of the common channel interference avoidance function and the related SSB interference avoidance functions, it takes effect only if at least one of the following functions is enabled:

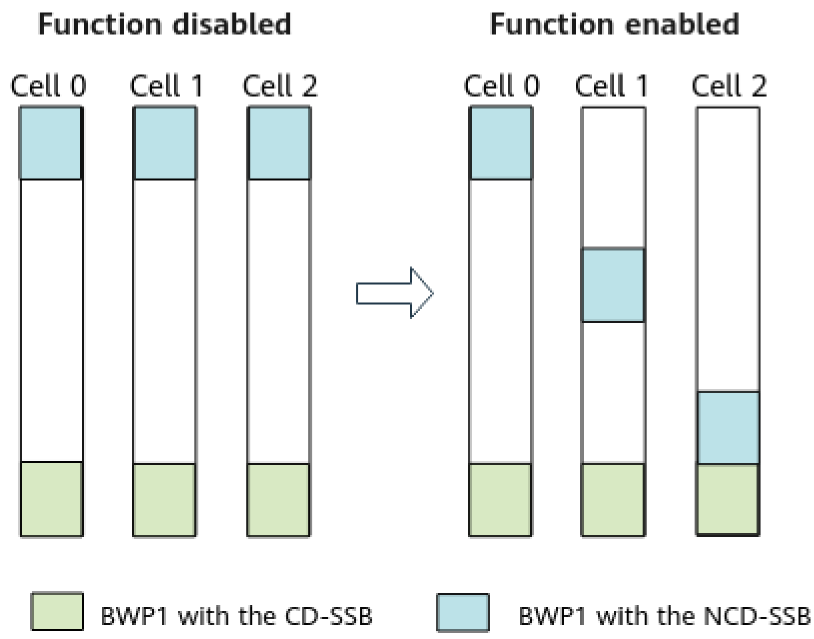
- Common channel interference avoidance: controlled by the **SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about this function, see *Interference Avoidance*.
- Inter-frequency SSB interference avoidance: controlled by the **INTER_FREQ_SSB_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about this function, see *3D Networking Experience Improvement*.
- Enhanced inter-frequency SSB interference avoidance: controlled by the **INTER_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about this function, see *3D Networking Experience Improvement*.
- Intra-frequency SSB interference avoidance: controlled by the **INTRA_FREQ_SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about this function, see *Interference Avoidance*.
- Enhanced intra-frequency SSB interference avoidance: controlled by the **INTRA_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about this function, see *3D Networking Experience Improvement*.

#### 4.1.3.8 RedCap BWP Frequency-Domain Staggering (Low-Frequency TDD)

If the RedCap BWP1s with the NCD-SSBs of the local cell and its neighboring cells are completely aligned in the frequency domain, the PUSCHs of these cells interfere with each other, affecting the average uplink cell throughput. To address this issue, **RedCap BWP frequency-domain staggering is introduced to stagger the frequency-domain positions of RedCap BWP1s with the NCD-SSBs among cells, reducing inter-cell PUSCH interference**, as shown in **Figure 4-7**.

**Figure 4-7** Inter-cell RedCap BWP1 interference avoidance




This function is controlled by the **REDCAP_BWP_NON_OVERLAP_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter.

### 4.1.3.9 Support for Super Uplink by RedCap

The RedCap Introduction function and super uplink can be enabled at the same time. **For RedCap UEs that support super uplink, the SUL spectrum can be used to improve their average uplink throughput.** The principles of super uplink to take effect on RedCap UEs are similar to those on common NR UEs. The difference is that only 20 MHz NR TDD carriers and 20 MHz SUL carriers take effect for RedCap UEs. For details about super uplink, see *Super Uplink*.

### 4.1.3.10 RedCap Power Aggregation (Low-Frequency TDD)

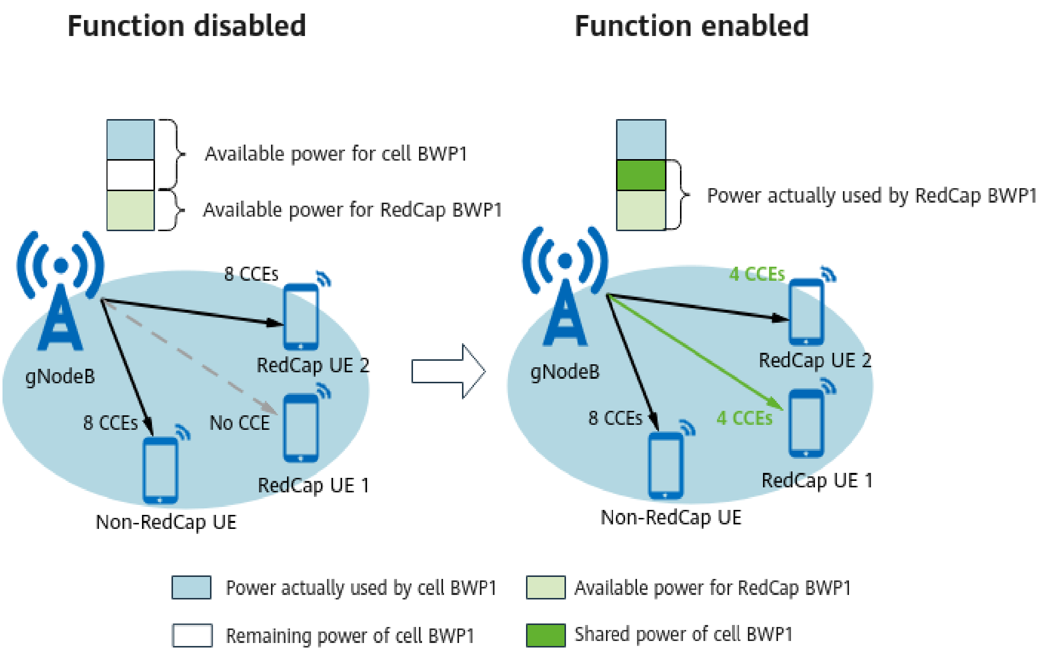
A RedCap BWP1, which has a small bandwidth (20 MHz), is more likely to be affected by CCE allocation failures caused by insufficient PDCCH CCE resources than the cell BWP1, which has a large bandwidth (up to 100 MHz for non-RedCap). RedCap power aggregation is introduced to increase the number of PDCCH CCEs that can be allocated to RedCap UEs and improve the uplink CCE allocation success rate of RedCap UEs.

**After this function is enabled, the gNodeB checks the uplink CCE usage in each TTI.If the uplink CCE usage is less than the CCE threshold for RedCap power aggregation, the gNodeB allows the remaining power on the BWP1 of the cell to be used by the RedCap BWP1 and reduces the PDCCH CCE aggregation level for RedCap UEs, thereby increasing the uplink CCE allocation success rate of RedCap UEs.**

**Figure 4-8** shows the principles of RedCap power aggregation.

**Figure 4-8** RedCap power aggregation




This function is controlled by the **REDCAP_POWER_AGGREGATION_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter. The CCE threshold for RedCap power aggregation is specified by the **NRDUCellPdcchAlgo**.***RedCapPwrBoostingCceThld*** parameter.

### 4.1.3.11 PDCCH Symbol Adaptation for RedCap BWP with the CD-SSB (Low-Frequency TDD)

RedCap UEs use small bandwidths. Small bandwidths are prone to insufficient CCE resources. This is especially true for RedCap BWP1 with the CD-SSB. Therefore, PDCCH symbol adaptation for RedCap BWP with the CD-SSB is introduced.

**The "PDCCH symbol adaptation for RedCap BWP with the CD-SSB" function allows RedCap BWP1 with the CD-SSB to be considered in PDCCH symbol number adaptation. If RedCap BWP1 with the CD-SSB meets the adaptation conditions, the number of PDCCH symbols can be scaled up or down.** For details about the conditions for triggering PDCCH symbol number adaptation, see "adaptive configuration" in "PDCCH Symbol Number Configuration" described in *Channel Management*.

This function is controlled by the **CD_BWP_PDCCH_SYM_ADAPTIVE_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter.

### 4.1.3.12 RedCap UE Weight Optimization (Low-Frequency TDD)

In TDD, commonly, UEs can send SRS signals using antenna switching so that the base station can obtain the full downlink channel state information based on uplink and downlink channel reciprocity. However, RedCap UEs can send SRS signals through only fixed antennas. Therefore, the base station cannot obtain full downlink channel state information, affecting downlink experience of RedCap UEs. To address this issue, RedCap UE weight optimization is introduced.

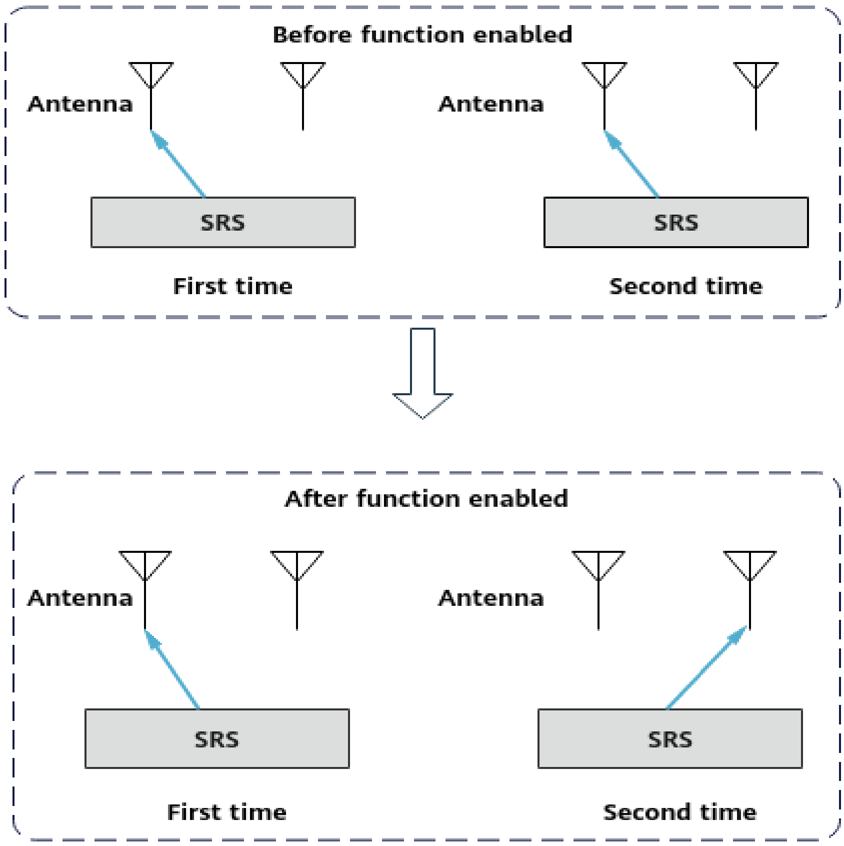
  

**After this function is enabled,the base station allocates 1T2R SRS resources to RedCap UEs that are identified as downlink large-packet UEs. In this case, the base station can receive SRS signals sent by these UEs using antenna switching and obtain full downlink channel state information. This improves the accuracy of channel estimation and downlink user experience of RedCap UEs.** Specifically,

- A RedCap UE supports antenna switching if the value of the reported *supportedSRS-TxPortSwitch* IE is **t1r2**.
- The UE is identified as a downlink large-packet UE if it meets the following condition at least five times within 10s: The downlink traffic volume of the UE in 1s is greater than or equal to the threshold specified by the **gNBRedCapServIdenCfg**.***DlBigPktDataVolThld*** parameter in the **gNBRedCapServIdenCfg** MO associated with the **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** parameter.

Figure 4-9 shows how this function works.

**Figure 4-9** RedCap UE weight optimization




RedCap UE weight optimization is controlled by the **REDCAP_SRS_WEIGHT_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. This function is not applicable to FWA users. For details about FWA user identification, see *FWA*.

In addition, to increase the number of RedCap UEs for which RedCap UE weight optimization takes effect, the function of increasing the number of RedCap UEs using SRS-based weights is introduced. This function is controlled by the

**SRS_WEIGHT_UE_NUM_INCR_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter.

NOTE

When the **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** parameter is set to **255**, the traffic volume threshold for RedCap UEs running downlink large-packet services is 1 Mbit/s.

#### 4.1.3.13 BWP Scheduling Optimization for RedCap UEs

In RedCap hybrid service scenarios, UEs are unevenly distributed on different RedCap BWP1s. As a result, the experience of some RedCap UEs is poor. To address this issue, BWP scheduling optimization for RedCap UEs is introduced. After this function takes effect, **RedCap UEs in the RRC_CONNECTED state are preferentially scheduled on light-load RedCap BWP1s to achieve load balancing.**

This function is controlled by the **REDCAP_UE_BWP_SCH_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.

## 4.1.4 Scenario-specific Optimization Functions

This section describes scenario-specific RedCap optimization functions, including resource optimization and "networking or UE compatibility optimization".

### 4.1.4.1 Resource Optimization

### Optimization in the Downlink

**RedCap PDCCH symbol number adjustment (low-frequency TDD):**

In an NR TDD cell, RedCap PDCCH symbol number adjustment is supported. This function provides the following policies:

- The number of PDCCH symbols occupied by RedCap UEs equals the number of PDCCH symbols in the cell, which improves the coverage for RedCap UEs.
- The number of PDCCH symbols occupied by RedCap UEs is adaptively adjusted based on the cell bandwidth and the number of PDCCH symbols in the cell, which improves the capacity for RedCap UEs.

This function is controlled by the **REDCAP_PDCCH_SYM_NUM_ADJ_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

**Scheduling in common PDCCH symbols for RedCap UEs (low-frequency TDD):**

For an NR TDD cell, when RedCap BWP1 contains CORESET 0, common signaling such as system broadcast messages and paging messages in CORESET 0 consumes a large number of resources, which leads to a decrease in the number of available PDCCH CCE resources for RedCap UEs. In this case, the average uplink and downlink throughputs of UEs on RedCap BWP1 are affected. Scheduling in common PDCCH symbols for RedCap UEs is introduced to increase the average uplink and downlink throughputs of RedCap UEs. After this function takes effect, the base station decreases the PDCCH CCE aggregation level for common signaling such as system broadcast messages and paging messages to 4, and configures PDCCH UE-specific search space (USS) for RedCap UEs in CORESET 0 so that RedCap UEs can use resources in common PDCCH symbols for data transmission. This increases the average uplink and downlink throughputs of RedCap UEs on the RedCap BWP1s with CORESET 0.

For non-BWP2 UEs, this function is controlled by the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch***. For BWP2 UEs, this function is controlled by both the **REDCAP_COMM_PDCCH_SYM_SCH_SW** of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter and **BWP2_COM_PDCCH_SCH_SW** options of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.

It is recommended that the **UL_DL_COMM_CCE_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter also be selected when this function is enabled.

**Configuration of multiple PDCCH search spaces (low-frequency TDD):**

For an NR TDD cell in a 5G ToB scenario, when the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter (specifying the number of symbols occupied by the PDCCH) is set to **2SYM** or **3SYM** and the network coverage is favorable (that is, the average PDCCH aggregation level is low), it is recommended that configuration of multiple PDCCH search spaces be enabled. This function is controlled by the **MULTI_SEARCH_SPACE_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter. For details about configuration of multiple PDCCH search spaces, see *Channel Management*.

**CORESET 0 offset policy configuration (low-frequency TDD):**

For an NR TDD cell, if RedCap BWP0 is located to either end of cell BWP1 in the frequency domain, RedCap BWP0 cannot directly reuse the PUCCH resources of the cell. In this case, PUCCH resources need to be independently configured for RedCap UEs. As a result, extra resource overhead is generated, affecting uplink performance. In this context, CORESET 0 offset policy configuration is introduced. The offset of the frequency-domain position of CORESET 0 relative to the center frequency of the SSB is adjusted to make CORESET 0 located as close as possible to either cell bandwidth edge in the frequency domain.

This function is enabled if the **NRDUCellPdcchAlgo**.***Coreset0OffsetPol*** parameter is set to **EDGE_FIRST**. For details about CORESET 0 offset policy configuration, see *Channel Management*.

**RedCap PDCCH resource avoidance (low-frequency TDD):**

The available bandwidth for RedCap UEs is small, and therefore RedCap UEs are more prone to CCE insufficiency than eMBB UEs. With this function, PDCCH resources are allocated to eMBB UEs in a way that avoids possible PDCCH positions allocated to RedCap UEs, increasing the downlink CCE allocation success rate for RedCap UEs.

This function is controlled by the **REDCAP_PDCCH_AVOID_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter.

**RedCap downlink scheduling optimization factor:**

For cells in high loads in terms of RedCap UEs and non-RedCap UEs, the RedCap downlink scheduling optimization factor can be used to adjust the average downlink throughputs of both types of UEs.

The RedCap downlink scheduling optimization factor is specified by the **NRDUCellRedCap**.***DlSchOptFactor*** parameter. If this parameter is set to a value other than **255**, a larger value of this parameter results in a higher average downlink throughput of RedCap UEs and a lower average downlink throughput of non-RedCap UEs.

**RedCap RLC retransmission guarantee:**

It is recommended that RedCap RLC retransmission guarantee be enabled if service drops occur due to abnormal RLC retransmissions for RedCap UEs. After this function takes effect, RedCap UEs can complete RLC retransmissions more promptly, reducing the risk of service drops caused by abnormal RLC retransmissions.

This function is controlled by the **REDCAP_RLC_RETRANS_GUAR_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

## Optimization in the Uplink

**SR code channel reservation for RedCap (low-frequency TDD):**

For an NR TDD cell, SR code channel reservation for RedCap can be enabled to ensure sufficient SR code channel resources for RedCap UEs. After this function takes effect, the base station reserves a certain number of SR code channels for RedCap UEs and dynamically adjusts the number of reserved SR code channels based on the number of RedCap UEs.

This function is controlled by the **REDCAP_SR_CODE_RSV_SW** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter.

**BWP2 PUCCH frequency-domain position optimization (low-frequency TDD):**

For an NR TDD cell, it is recommended that BWP2 PUCCH frequency-domain position optimization be enabled when the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected and the BWP2 is configured at the cell bandwidth edges. After this function takes effect, BWP2 PUCCH resources are configured at the upper and lower edges of the cell bandwidth to increase the number of consecutive RBs available for the PUSCH in the cell. This increases the average uplink throughput of RedCap UEs on the RedCap BWP1 that overlaps the BWP2 in the frequency domain.

This function is controlled by the **BWP2_PUCCH_POS_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter.

**SRS resource allocation policy for RedCap UEs in a high-speed cell (low-frequency TDD):**

For an NR TDD high-speed cell, the SRS allocation policy for RedCap UEs is now configurable. After this function takes effect, a higher level results in more SRS resources allocated to RedCap UEs and more RedCap UEs that can be served.

This function is controlled by the **NRDUCellHighSpeedMob**.***RedCapUeSrsResAllocPol*** parameter (a larger number following Level indicates a higher level). This function takes effect only in low-frequency TDD high-speed cells.

**SRS resource extension in a high-speed cell (low-frequency TDD):**

For an NR TDD high-speed cell, SRS resource extension is now supported. After this function takes effect, the base station allocates more SRS resources to UEs and more RedCap UEs can be served.

This function is controlled by the **SRS_RES_EXT_SW** option of the **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** parameter and takes effect only in high-speed cells. For details about SRS resource extension in a high-speed cell, see *High Speed Mobility*.

When this function is enabled, it is recommended that DCCH type 1 scheduling (controlled by the **DCCH_TYPE1_SCH_SW** option of the **NRDUCellDlSchRes**.***DlSchResAlgoSw*** parameter) also be enabled, to improve the downlink scheduling resource usage and support scheduling of more UEs. For details about DCCH type 1 scheduling, see *Downlink Scheduling*.

### 4.1.4.2 Networking or UE Compatibility Optimization

**PUCCH interference avoidance (low-frequency TDD):**

For an NR TDD cell, PUCCH interference avoidance can be enabled when there is co-channel interference from LTE to the NR PUSCH/PDSCH or external interference.

This function is controlled by the **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter. For details about PUCCH interference avoidance, see *Interference Avoidance*.

**CD-SSB inclusion in the RedCap BWP0 (low-frequency TDD):**

For an NR TDD cell, CD-SSB inclusion in the RedCap BWP0 can be enabled if RRC connection reestablishments, resynchronizations, or service drops occur due to UE incompatibility with RedCap BWP0s that do not include CORESET 0. After this function takes effect, the base station adjusts the RedCap BWP0 allocation policy to ensure that the RedCap BWP0s contain the CD-SSB.

This function is controlled by the **REDCAP_BWP0_INCLUDE_CD_SSB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

**PDCCH CSS configuration for RedCap UEs**:

PDCCH CSS configuration for RedCap UEs can be enabled when there are RedCap UEs that require that downlink RedCap BWP1s must contain PDCCH common search space (CSS). After this function takes effect, the base station configures CSS resources for RedCap BWP1s that contain part of CORESET 0 or do not contain CORESET 0.

This function is controlled by the **REDCAP_PDCCH_CSS_CONFIG_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

**Prohibition of RedCap UE handovers to external NR cells:**

Prohibition of RedCap UE handovers to external NR cells can be enabled when handover-related counters deteriorate because the RedCap deployment status of neighboring cells fails to be obtained through the Xn interface and RedCap UEs are handed over to cells where RedCap is not enabled. After this function takes effect, RedCap UEs are prohibited from being handed over to cells that do not support RedCap.

This function is controlled by the **REDCAP_NO_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter.

**NCD-SSB SIB1 compatibility:**

The NCD-SSB SIB1 compatibility function is introduced to prevent UEs that use the NCD-SSB from failing to receive the other system information (OSI) after they are switched to the RRC_IDLE or RRC_INACTIVE state. With this function, SIB1 sent on the NCD-SSB contains the *si-SchedulingInfo* IE, which indicates the OSI for scheduling.

This function is controlled by the **NCD_SSB_SIB1_COMPAT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter.

**PRACH allocation direction optimization (low-frequency TDD):**

PRACH allocation direction optimization is introduced to ensure that UEs performing strong verification on PRACH positions can access the network. With this function, the gNodeB always allocates PRACH resources from the lower end of RedCap BWP0. This ensures that UEs with compatibility issues can access the network and increases the incoming handover success rate.

This function is controlled by the **PRACH_ALLOC_DIR_OPT_SW** option of the **NRDUCellPrach**.***PrachAlgoExtSwitch*** parameter.

**RedCap UE device-pipe identification optimization**

RedCap UE device-pipe identification optimization is introduced to increase the proportion of RedCap UEs identified as capable of device-pipe synergy. Assume that this function takes effect. If a RedCap UE carries the Huawei Private Signaling (HPS) capability flag of the source cell but the base station does not receive the HPS capability reported by the UE, the base station can send RRC HPS DL TRANSFER messages to such UEs in subsequent handovers. This type of message is used to indicate that the cell supports device-pipe synergy.

This function is controlled by the **REDCAP_SPEC_UE_IDENTIFY_OPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter.

# 4.2 Network Analysis

## 4.2.1 Benefits

The benefits of functions in RedCap Introduction are described in **Table 4-19**. For the definitions of the involved indicators, see **4.4.3 Network Monitoring**.

**Table 4-19** Benefits of functions in RedCap Introduction (NR TDD)

| Category | Function Name | Benefits |
|---|---|---|
| Basic functions | RedCap Introduction | • Improved spectral efficiency<br>The existing 5G network can be upgraded and evolved to support new user types and service types. In this way, the existing spectrum resources of operators are fully utilized, improving spectrum utilization.<br>• Support for the access of a large number of UEs in IoT scenarios<br>This function applies to low-rate and low-activity services, allowing a large number of users to access the 5G network.<br>• Reduced NR UE cost<br>RedCap UEs with low costs can access 5G networks. |
| Mobility management | RedCap NCD-SSB measurement | If intra-frequency NCD-SSBs exist in neighboring cells, the average uplink and downlink throughputs of RedCap UEs may increase. |
| | RedCap NCD-SSB SINR measurement | For cells with co-channel interference, this function enables RedCap UEs to promptly complete SINR-based measurements and initiate handovers, decreasing the service drop rate. |
| Experience Enhancement | RedCap MU-MIMO | RedCap UEs support MU-MIMO to increase the average UE throughput and average cell throughput. |

| Category | Function Name | Benefits |
| --- | --- | --- |
| | Multiple BWP1s for RedCap | When there are a large number of RedCap UEs or the traffic volume is heavy on the network, configuring only one BWP1 cannot meet the traffic volume requirements of RedCap UEs. In this case, configuring multiple BWP1s increases the number of RedCap UEs supported by a cell and meets the traffic volume requirements of RedCap UEs.<br>Scenario-based optimization:<br>● PUCCH CSI RB adjustment for the RedCap-dedicated BWP: When a large number of UEs access the RedCap-dedicated BWP (RedCap BWP1 that is not deployed at the band edges and does not contain the CD-SSB), more PUCCH resources are used and the average downlink UE throughput may increase.<br>● CSI resource balancing: The average uplink throughput of RedCap UEs increases when the **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected in a moderately or heavily loaded cell.<br>● RedCap PUCCH resource guarantee: The number of RedCap BWP1s that can be activated in a cell increases because RedCap UEs can use more PUCCH resources.<br>● Format-1 SR resource optimization: With this function, more format-1 SR resources are available in the cell, and therefore more UEs can access the cell.<br>● CSI RB and format-1 RB reposition: More RedCap UEs can access the first activated RedCap BWP1. |
| | Uplink RedCap preallocation | The uplink user-perceived delay of RedCap UEs is shortened. |
| | RedCap downlink scheduling policy | In SU-MIMO scenarios, when type 1 scheduling takes effect on RedCap UEs running large-packet services, the average downlink throughput of RedCap UEs increases in the full-buffer traffic model. |
| | SSB interference avoidance for RedCap UEs | Avoidance of the NCD-SSB interference from neighboring cells increases the average downlink throughput of cell-edge UEs when the serving cell is exposed to strong NCD-SSB interference from neighboring cells. |

| Category | Function Name | Benefits |
| --- | --- | --- |
| | RedCap BWP frequency-domain staggering | With this function, the frequency-domain positions of RedCap BWP1s with the NCD-SSBs among different cells are staggered, reducing inter-cell PUSCH interference and increasing the average uplink throughput of RedCap UEs. |
| | Support for super uplink by RedCap | For RedCap UEs that support super uplink, the SUL spectrum can be used to improve the average uplink UE throughput. |
| | RedCap power aggregation | This function increases the number of PDCCH CCEs that can be allocated to RedCap UEs and improves the uplink CCE allocation success rate of RedCap UEs and average uplink UE throughput. |
| | PDCCH symbol adaptation for RedCap BWP with the CD-SSB | The number of PDCCH symbols in the RedCap BWP with the CD-SSB increases, and more UEs can be scheduled. |
| | RedCap UE weight optimization | The average downlink throughput of RedCap UEs may increase. |
| | BWP scheduling optimization for RedCap UEs | The average uplink and downlink throughputs of RedCap UEs may increase. |
| Scenario-specific optimization functions—Resource optimization (downlink) | RedCap PDCCH symbol number adjustment | If the number of PDCCH symbols in a cell is greater than 1, this function increases the CCE allocation success rate for RedCap UEs at the cell center or a medium distance from the cell center. |
| | Scheduling in common PDCCH symbols for RedCap UEs | As the CCE allocation success rate increases, the average uplink and downlink throughputs of UEs on the RedCap BWP1s with CORESET 0 increase. |
| | RedCap PDCCH resource avoidance | PDCCH resources are allocated to eMBB UEs in a way that avoids possible PDCCH positions allocated to RedCap UEs. When the avoidance conditions are met, this function increases the downlink CCE allocation success rate for UEs and average downlink throughput for RedCap UEs. |

| Category | Function Name | Benefits |
|---|---|---|
| | RedCap RLC retransmission guarantee | If abnormal RedCap RLC retransmissions occur, the service drop rate of RedCap UEs in the cell decreases. |
| Scenario-specific optimization functions—Resource optimization (uplink) | SR code channel reservation for RedCap | The number of RedCap UEs supported by a cell increases. When there are a large number of UEs in RRC_CONNECTED mode in a cell, the average uplink/downlink throughput of RedCap UEs may increase. |
| | BWP2 PUCCH frequency-domain position optimization | If the BWP2 is configured at the cell bandwidth edges, after this function takes effect, the average uplink cell throughput increases because the number of consecutive RBs available for the PUSCH increases. In addition, the average uplink throughput of RedCap UEs on the RedCap BWP1 that overlaps the BWP2 increases. |
| | SRS resource allocation policy for RedCap UEs in a high-speed cell | A higher level specified by the **NRDUCellHighSpeedMob**.***RedCapUeSrsResAlloc-Pol*** parameter results in more SRS resources allocated to RedCap UEs and more RedCap UEs that can be served. |
| Networking or UE compatibility optimization | CD-SSB inclusion in the RedCap BWP0 | If there are RedCap UEs that support only the BWP0s with CD-SSBs in the cell, the RRC reestablishment success rate and RRC resynchronization success rate increase and the service drop rate decreases. |
| | PDCCH CSS configuration for RedCap UEs: | If there are RedCap UEs that require that downlink RedCap BWP1s must contain PDCCH CSS, the cell service drop rate decreases. |
| | Prohibition of RedCap UE handovers to external NR cells | If the RedCap deployment status of neighboring cells fails to be obtained through the Xn interface and neighboring cells have low handover success rates of RedCap UEs, it is recommended that this function be enabled. This function prevents handover attempts to neighboring cells with low handover success rates of RedCap UEs, thereby increasing the handover success rate of the serving cell. |
| | PRACH allocation direction optimization | The incoming handover success rate increases when there are UEs performing strong verification on PRACH positions on the network. |

| Category | Function Name | Benefits |
| --- | --- | --- |
| | RedCap UE device-pipe identification optimization | In handover scenarios, this function increases the proportion of RedCap UEs identified as capable of device-pipe synergy. |

## 4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

The network impacts of functions in RedCap Introduction are described in **Table 4-20**. For the definitions of the involved indicators, see **4.4.3 Network Monitoring**.

**Table 4-20** Network impacts of functions in RedCap Introduction (NR TDD)

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| Basic functions | RedCap Introduction | RedCap Introduction may have the following network impacts:<br>● The gNodeB configures certain PRACH preambles dedicated to RedCap UEs. As a result, the contention-based and non-contention-based random access success rates decrease, the access delay of common NR UEs increases, and the value of the **N.RA.Contention.SSBBeam.Preamble.Avail.Num** counter may increase.<br>● PRACH resources need to be configured within the RedCap BWP0. In contiguous cell networking scenarios, if the PUSCH and PRACH interfere with each other due to inconsistent frequency-domain positions of RedCap BWP0s between cells, the network performance is affected. (1) For cells experiencing PUSCH interference, the uplink initial block error rate (IBLER) slightly increases, and the average uplink MCS index slightly decreases. As a result, the uplink cell throughput slightly decreases and the PDCCH CCE usage slightly increases. (2) For cells experiencing PRACH interference, the number of false alarms increases, and the contention-based and non-contention-based random access success rates decrease.<br>● As RedCap-related configuration information is added to SIB1, the number of scheduled RBs increases, which increases the downlink PRB usage and SRB PRB usage, decreases the downlink spectral efficiency, and increases the average downlink cell power. In addition, as dedicated BWP0s are configured for RedCap UEs, the MCS index selected for SIB1 increases and the downlink coverage performance deteriorates.<br>● The PUCCH resource allocation mode changes, which may cause fluctuations in the number of DTXs. As a result, the PDCCH CCE aggregation level and PDCCH CCE usage fluctuate. In addition, as the PUCCH frequency hopping is changed to disabled, the values of the following counters fluctuate: **N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7**.<br>● RedCap Introduction optimizes the SRS resource allocation mode for UEs using small-bandwidth |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | | resources, such as RedCap and BWP2 UEs, to reduce interference between neighboring cells and improve user experience.<br>● When both RedCap Introduction and the power saving BWP function take effect, RedCap UEs and UEs in the power saving BWP state share the 20 MHz bandwidth of the cell. As a result, the actual resources available to UEs in the power saving BWP state are affected and the value of the **N.Bwp2.Dur.Total** counter decreases.<br>● When both the RedCap Introduction and UE bandwidth adaptation functions are enabled, cell-related KPIs deteriorate. For details about the network impact, see *Scalable Bandwidth*. It is recommended that the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter be set to **2SYM** to double CCE resources for common NR UEs. This prevents CCE resources from being insufficient for common NR UEs.<br>● Once the multi-BWP1 PDCCH transmission and enhanced multi-BWP1 PDCCH transmission functions are enabled at the same time, values of the CCE-related indicators fluctuate. For details about these two functions, see *Channel Management*.<br>● Both RedCap UEs and common NR UEs access the base station. However, as the radio performance of RedCap UEs is lower than that of common NR UEs, the following network impacts may occur:<br>– The maximum number of RRC_CONNECTED UEs supported in the cell includes both RedCap UEs and common NR UEs. After RedCap UEs access a cell, the allowed number of common NR UEs in RRC_CONNECTED mode decreases.<br>– After RedCap UEs access a cell, the contention-based random access success rate fluctuates because the access-related counters related to RedCap Introduction are measured in the original contention-based random access counters.<br>– The average data transmission delay in the cell increases. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | | – The uplink and downlink IBLERs, and uplink and downlink RBLERs increase.<br>– The uplink/downlink CCE allocation success rate in the cell decreases.<br>– The total service drop rate in the cell increases.<br>– The average numbers of layers and RBs paired for downlink MIMO decrease.<br>– The average cell throughput and average UE throughput decrease.<br>– The cell spectral efficiency decreases.<br>– The average uplink and downlink RB usages increase because the number of UEs in RRC_CONNECTED mode in the cell increases.<br>– The counters related to RedCap Introduction are measured in the original counters (such as access- and throughput-related counters), causing value fluctuation of the original counters.<br>● In high-speed cells, SRS resource allocation works in a different way, which results in fewer eMBB UEs for which code division multiplexing (CDM) is preferentially prevented from taking effect. As a result, the overall service drop rate of the cells slightly increases in high load scenarios.<br>● Assume that the PUCCH experiences interference and the cell is configured with an interference range. If RedCap UEs suffer from interference, the cell performance may deteriorate or RedCap UEs may fail to access the cell. To avoid this impact, it is recommended that the PUCCH interference avoidance function be enabled (by selecting the **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter). In this way, the PUCCH is transmitted out of the interference range. However, the number of activated RedCap BWP1s may be less than the number of BWP1s corresponding to the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter value. For details about PUCCH interference avoidance, see *Interference Avoidance*.<br>● If no RedCap UE accesses the cell, the average number of uplink PRBs used by RedCap UEs, |

| Category | Function Name | Network Impacts |
|---|---|---|
| | | indicated by **N.PRBUsed.UL.RedCap.Avg**, may be non-zero due to false alarms. |
| Mobility management | RedCap NCD-SSB SINR measurement | Gap-assisted measurements are more likely to be initiated for RedCap UEs, and the average uplink and downlink throughputs of RedCap UEs fluctuate during the measurements. |
| | RedCap handover policy control | ● Xn-based and intra-base-station handover control: The number of handovers decreases, and the service drop rate may increase.<br>● NG-based and inter-RAT handover control: The incoming handover success rate decreases.<br>● Control over handovers to external NR cells: For neighbor relationships with external cells with a low 1RX/2RX RedCap UE handover success rate, this function prevents attempts of handovers to these external cells and the incurred handover failures, improving the handover success rate of the local cell. For neighbor relationships with external cells with a high 1RX RedCap UE handover success rate, the prohibition of handovers increases the service drop rate. |
| | Measurement frequency selection for RedCap | When the **NRCellFreqRelation**.***RedCapMeasActvFlag*** parameter is set to **ACTIVE**, the number of inter-frequency measurements increases. When the **NRCellFreqRelation**.***RedCapMeasActvFlag*** parameter is set to **INACTIVE**, the number of inter-frequency measurements decreases. |
| | Cell-reselection frequency selection for RedCap | The value of the **NRCellFreqRelation**.***RedCapAccessAllowedFlag*** parameter affects the RedCap UE distribution model. If this parameter is set to **ALLOWED**, the average number of RedCap UEs in RRC_CONNECTED mode served by cells on the frequency increases. If this parameter is set to **NOT_ALLOWED**, this average number decreases. |
| Experience enhancement | RedCap MU-MIMO | ● MU-MIMO for RedCap UEs has the same network impacts as MU-MIMO for common NR UEs. For details, see *MIMO (TDD)*.<br>● When UEs are densely distributed or scheduling resources are insufficient, the average downlink UE throughput may decrease after RedCap PDSCH MU-MIMO is enabled. |

| Category | Function Name | Network Impacts |
|---|---|---|
| | Multiple BWP1s for RedCap | • When multiple BWP1s are configured for RedCap UEs, the service experience of RedCap UEs improves because RedCap UEs can use more network resources. However, the service experience of eMBB UEs deteriorates. As a result, the overall network KPIs fluctuate. The impact depends on the number of BWP1s and the traffic volume on the live network.<br>• NCD-SSB resources need to be configured in certain BWP1s, which consumes extra resources and has the following impacts on the network:<br>– The number of available PDSCH RBs decreases, and both the average downlink cell and UE throughputs decrease.<br>– The average number of used downlink PRBs and the downlink RB usage increase.<br>– The downlink IBLER, downlink retransmission ratio, and average downlink MCS index fluctuate.<br>The impact depends on the number of BWP1s and the traffic volume on the live network.<br>• As PUCCH resources need to be configured in multiple BWP1s, the number of available PUSCH RBs decreases and PUSCH RB resources for scheduling become discontinuous, which may have the following impacts on the network:<br>– The average uplink cell throughput and average uplink UE throughput decrease.<br>– The average number of used uplink PRBs and the uplink RB usage fluctuate.<br>– The uplink IBLER, uplink retransmission ratio, and average uplink MCS index fluctuate.<br>The impact depends on the number of BWP1s and the traffic volume on the live network.<br>• When the distribution of RedCap UEs on multiple BWP1s changes, the following indicators will fluctuate: average uplink throughput of RedCap UEs, average downlink throughput of RedCap UEs, uplink CCE allocation success rate of UEs, downlink CCE allocation success rate of UEs, uplink RB usage, and downlink RB usage.<br>• Preferential selection of BWP1s with NCD-SSBs for RedCap UEs |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | | – If the **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** parameter is set to a larger value, there will be more RedCap UEs that preferentially access BWP1 with the NCD-SSB. If the **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** parameter is set to a smaller value, there will be fewer such UEs.<br>– When the number of RedCap UEs in a cell is small and the **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** parameter is set to a large value, RedCap UEs are mostly served by BWPs with NCD-SSBs. As a result, the uplink and downlink throughputs of RedCap UEs decrease.<br>Scenario-based optimization:<br>● PUCCH CSI RB adjustment for the RedCap-dedicated BWP: When a large number of UEs access the RedCap-dedicated BWP (RedCap BWP1 that is not deployed at the band edges and does not contain the CD-SSB), more PUCCH resources are used, fewer PUSCH resources are available, and the average uplink UE throughput may decrease.<br>● RedCap BWP format-4 PUCCH configuration: When there are RedCap UEs performing downlink large-packet services in a cell and the **NRDUCellRedCap**.***PucchFormat4AckRbNum*** parameter is set to a non-zero value, the following network impacts occur:<br>– A larger value of this parameter results in higher average downlink throughput of RedCap UEs but a higher probability of decreasing the average downlink throughput of eMBB UEs and a change in the overall average downlink UE throughput in the cell. In addition, the available uplink scheduling resources become fewer, and thereby the overall average uplink UE throughput in the cell becomes lower.<br>– A smaller value of this parameter results in lower average downlink throughput of RedCap UEs but a higher probability of increasing the average downlink throughput of eMBB UEs and a change in the overall average downlink UE throughput in the cell. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | | In addition, the available uplink scheduling resources become more, and thereby the overall average uplink UE throughput in the cell becomes higher.<br>● RedCap PUCCH resource guarantee: As PUCCH resources in a cell are preferentially allocated on RedCap BWP1, the average uplink and downlink cell throughputs fluctuate. In addition, the number of RedCap BWP1s that can be activated in a cell increases because RedCap UEs can use more PUCCH resources. The specific impact is the same as that of the "multiple BWP1s for RedCap" function.<br>● Format-1 RB adaptation: When a large number of UEs access a cell, the PUCCH resources increase, the number of admitted RedCap UEs may increase, the available PUSCH resources decrease, and the average uplink UE throughput may decrease.<br>● CSI RB adaptation: When a large number of UEs access a cell, more PUCCH resources can be used for CSI reporting, the available PUSCH resources become fewer, the uplink UE throughput decreases, and the downlink UE throughput increases.<br>● format-1 SR resource optimization: When the number of UEs in a cell increases, the probability of HARQ resource conflicts increases. As a result, the number of downlink scheduling times decreases, and the average downlink UE throughput and average downlink cell throughput may decrease.<br>● CSI RB and format-1 RB reposition: When a large number of UEs access a cell, more PUCCH resources are used for CSI reporting, the available PUSCH resources become fewer, the uplink UE throughput may decrease, and the downlink UE throughput may increase. |
| | Uplink RedCap preallocation | The number of uplink scheduling resources increases, and the average cell throughput decreases. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | RedCap downlink scheduling policy | In SU-MIMO scenarios, when type 1 scheduling takes effect on RedCap UEs running large-packet services, the following impacts occur:<br>● The average downlink throughput of RedCap UEs may decrease in the burst traffic model.<br>● The average downlink cell throughput may decrease because the number of resources scheduled for eMBB UEs decreases.<br>● As the scheduling mechanism (RB scheduling granularity and number of scheduling occasions) for RedCap UEs changes, the values of scheduling-related indicators such as indicators related to downlink RBs, CCEs, MCS indexes, and service delay fluctuate. |
| | SSB interference avoidance for RedCap UEs | The number of available time-frequency resources for downlink scheduling in the cell decreases. The proportion by which the time-frequency resources available for downlink scheduling are reduced is related to the cell bandwidth and the number of NCD-SSB resources in neighboring cells. When the serving cell is exposed to weak NCD-SSB interference from neighboring cells, the average downlink UE throughput decreases. |

| Category | Function Name | Network Impacts |
|---|---|---|
| | RedCap BWP frequency-domain staggering | After RedCap BWP frequency-domain staggering takes effect, the following are true:<br>● The values of the following indicators may decrease:<br>– Average throughput of eMBB UEs<br>– Average uplink MCS index<br>– Average number of available PRBs on the PUSCH and average number of actually available PRBs on the PUSCH<br>● The values of the following indicators may fluctuate:<br>– Uplink and downlink IBLERs<br>– Uplink and downlink RB usages<br>– Uplink CCE allocation success rate<br>– Average number of layers paired for uplink MIMO, average number of RBs paired for uplink MIMO, and total number of RBs paired for uplink MU-MIMO<br>– Average uplink interference (average interference and noise received on each uplink PRB)<br>● The number of available uplink PRBs within the BWP may increase. |
| | RedCap power aggregation | After RedCap power aggregation takes effect, the uplink CCE aggregation level and PDCCH CCE usage may decrease, and the uplink RB usage fluctuates.<br>A smaller CCE threshold for RedCap power aggregation (specified by the **NRDUCellPdcchAlgo**.***RedCapPwrBoostingCceThld*** parameter) results in a lower probability that RedCap power aggregation takes effect and a lower probability that the average uplink throughput of RedCap UEs increases. A larger threshold results in a higher probability that RedCap power aggregation takes effect, and a higher probability that the average uplink throughput of RedCap UEs increases and the average uplink throughput of non-RedCap UEs decreases. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | PDCCH symbol adaptation for RedCap BWP with the CD-SSB | The number of PDSCH symbols decreases due to an increase in the number of PDCCH symbols. As a result, the average cell throughput may fluctuate. |
| | RedCap UE weight optimization | After RedCap UE weight optimization takes effect, the following are true:<br>● The average uplink and downlink throughputs of eMBB UEs may decrease.<br>● The average uplink throughput of RedCap UEs may fluctuate.<br>● The downlink RB usage and value of service delay-related indicators may decrease. The average uplink MCS index, average downlink MCS index, average downlink rank, uplink IBLER, uplink RBLER, downlink IBLER, downlink RBLER, and values of downlink MU-related indicators may fluctuate. |
| | BWP scheduling optimization for RedCap UEs | The number of RRC reconfigurations increases, and the service drop rate in a cell may increase. |
| Scenario-specific optimization functions—Resource optimization (downlink) | RedCap PDCCH symbol number adjustment | If this function is disabled, network performance is not affected. If this function is enabled for a cell where the number of PDCCH symbols is greater than one, the CCE allocation success rate may decrease for RedCap UEs at the cell edge. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | Scheduling in common PDCCH symbols for RedCap UEs | ● The average downlink cell throughput may decrease because the number of PDCCH resources on the eMBB BWP1 decreases.<br>● The CCE aggregation level of the common PDCCH for system information and paging messages is decreased to 4, which has the following impacts:<br>– The coverage for common signaling shrinks. As a result, the number of UEs that can access the cell and the paging success rate may decrease.<br>– The number of used PDCCH CCEs, the downlink CCE aggregation level, and the PDCCH CCE usage decrease.<br>● The uplink and downlink BLERs, duration in which the BWP2 takes effect, and total service drop rate of the cell fluctuate due to changes in the uplink and downlink RedCap traffic volumes.<br>● In heavy-load RedCap scenarios, the traffic volumes, RB usage, and CCE usage increase in both the uplink and downlink for RedCap and the average MCS index and average rank decrease in both the uplink and downlink for RedCap.<br>● In medium- and heavy-load eMBB scenarios, the common PDCCH resource allocation mode is modified. As a result, the number of PDCCH symbols for BWP2 UEs decreases from two to one, and the duration in which BWP2 takes effect may decrease.<br>● The change in the duration in which BWP2 takes effect leads to the change in RB positions for PUCCH SINR measurement. As a result, the values of the following counters fluctuate due to the interference difference between different RBs in the frequency domain: **N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7**.<br>● In SUL and NR FDD co-deployment mode (controlled by the **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter), UE scheduling behavior changes, which has the following impacts:<br>– The uplink and downlink BLERs increase.<br>– The service delay increases. |

| Category | Function Name | Network Impacts |
| --- | --- | --- |
| | | – The values of the counters related to the following aspects fluctuate: average uplink and downlink MCS indexes, average uplink and downlink ranks, handover success rate, total service drop rate, and number of RRC connection reestablishments.<br>– The average cell throughput and average UE throughput decrease in both the uplink and downlink. In addition, the uplink rank decreases. |
| | RedCap downlink scheduling optimization factor | If the **NRDUCellRedCap**.***DlSchOptFactor*** parameter is set to a value other than **255**, a larger value of this parameter results in a higher average downlink throughput of RedCap UEs and a lower average downlink throughput of non-RedCap UEs. |
| | RedCap RLC retransmission guarantee | When a cell is heavily loaded in terms of the UE number, the average uplink and downlink throughputs and random access success rate of eMBB UEs decrease, and the service drop rate of eMBB UEs may increase. |
| Scenario-specific optimization functions—Resource optimization (uplink) | SR code channel reservation for RedCap | This function may have the following impacts on the network:<br>● When a cell serves a large number of RRC_CONNECTED UEs, the average uplink and downlink throughputs of eMBB UEs decrease and those of RedCap UEs decrease. As a result, the average uplink and downlink throughputs of UEs fluctuate.<br>● The number of RRC reconfiguration messages increases, the PDCCH aggregation level slightly increases, and the uplink and downlink IBLERs slightly increase. |
| | BWP2 PUCCH frequency-domain position optimization | When the BWP2 is configured at the cell bandwidth edges and at least two RedCap BWP1s are configured in the cell, the average uplink throughput of RedCap UEs in the second RedCap BWP1 decreases. |
| | SRS resource allocation policy for RedCap UEs in a high-speed cell | Setting the **NRDUCellHighSpeedMob**.***RedCapUeSrsResAllocPol*** parameter to **LEVEL2** or **LEVEL3** may increase the service drop rate of non-RedCap UEs (this occurs for the **LEVEL2** setting only when the cell is heavily loaded). |

| Category | Function Name | Network Impacts |
|---|---|---|
| Networking or UE compatibility optimization | CD-SSB inclusion in the RedCap BWP0 | If CD-SSBs are not configured at the cell edge, this function has the following impacts on the network:<br>● The average uplink and downlink UE throughputs decrease.<br>● The average uplink cell throughput decreases and the average downlink cell throughput fluctuates.<br>● The uplink and downlink RB usages decrease, and the average number of available PUSCH PRBs may decrease.<br>● The CCE allocation failure rate increases.<br>● The random access success rate fluctuates.<br>If CD-SSB inclusion in the RedCap BWP0 is disabled in small-bandwidth scenarios (the cell bandwidth is less than 60 MHz) and SSBs are configured in the middle of the cell frequency band, the BWP0 bandwidth of the cell may be compressed, affecting the access-related and scheduling-related indicators of non-RedCap UEs. |
| | PDCCH CSS configuration for RedCap UEs | The CCE allocation failure rate may increase. |
| | Prohibition of RedCap UE handovers to external NR cells: | If the RedCap deployment status of neighboring cells fails to be obtained through the Xn interface and success rate of handovers of RedCap UEs to the neighboring cells are high, this function leads to an increase in the service drop rate of the serving cell because handovers to these neighboring cells are prohibited. |
| | NCD-SSB SIB1 compatibility | For RedCap UEs using NCD-SSBs, the number of RRC connection reconfigurations over the air interface may increase. After the UEs transit from the RRC_CONNECTED state to the RRC_IDLE or RRC_INACTIVE state, the UEs can receive the OSI broadcast in the cell. |
| | PRACH allocation direction optimization | In medium- and heavy-load scenarios, the average uplink UE throughput and average uplink cell throughput may decrease, and the cell access success rate fluctuates. |

# 4.3 Requirements

## 4.3.1 Licenses

SSB interference avoidance for RedCap UEs is a basic function, and there are no license requirements for this function.

The following are feature and capacity license requirements for other functions in RedCap Introduction.

**Table 4-21** Feature licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-070316 | RedCap Introduction | NR0S00RCID00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

**Table 4-22** Capacity licenses

| Model | Description | Sales Unit |
|---|---|---|
| NR0S0RCBWP00 | RedCap BWP Hardware License (NR) | per BWP |

NOTE

The "RedCap BWP Hardware License" license is consumed based on the number of BWP1s specified by the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter.

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

## 4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 4.3.2.1 Prerequisite Functions

RedCap Introduction:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Full-bandwidth initial BWP configuration | **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter | *5G Networking and Signaling* | For an NR TDD cell, the RedCap Introduction function depends on the full-bandwidth initial BWP configuration function. |
| Low-frequency TDD | UE-characteristic-based SRS period adaptation | **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter | *Channel Management* | For an NR TDD low-speed cell (with the **NRDUCell**.***HighSpeedFlag*** parameter set to **LOW_SPEED**), the RedCap Introduction function depends on the UE-characteristic-based SRS period adaptation function. |
| Low-frequency TDD | Long PUCCH format | **NRDUCellPucch**.***StructureType*** | *Channel Management* | For an NR TDD cell, the RedCap Introduction function can be enabled only when the long PUCCH format is configured for the cell (by setting **NRDUCellPucch**.***StructureType*** to **LONG_STRUCTURE**). |
| Low-frequency TDD | UE-specific PDCCH Aggregation Level | **NRDUCellPdcch**.***UeSpecificPdcchAggLvl*** | None | For an NR TDD cell, to prevent CCE resource insufficiency, the **NRDUCellPdcch**.***UeSpecificPdcchAggLvl*** parameter must be set to **ADAPTIVE** when RedCap Introduction is enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | SRS period and CDM adaptation | **SRS_PERIOD_AND_CDM_ADAPT_SW** option of the **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** parameter | *High Speed Mobility* | For an NR TDD high-speed cell (with the **NRDUCell**.***HighSpeedFlag*** parameter set to **HIGH_SPEED**), the RedCap Introduction function depends on the SRS period and CDM adaptation function. |
| Low-frequency TDD | Minimum uplink/downlink carrier bandwidth | **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw*** | *Scalable Bandwidth* | When the cell bandwidth is 30MHz, RedCap Introduction requires that the parameters specifying the minimum uplink and downlink carrier bandwidths be set to **20M**. |

RedCap UE inactivity management:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UE inactivity management depends on the RedCap Introduction function. |

RedCap MU-MIMO:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PDSCH MU-MIMO | **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter | *MIMO (TDD)* | Downlink MU-MIMO for RedCap UEs (controlled by the **REDCAP_DL_MU_SW** option of the **NRDUCellRedCap**.***ReDCapAlgoSwitch*** parameter) takes effect only when the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected. |
| Low-frequency TDD | PUSCH MU-MIMO | **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter | *MIMO (TDD)* | Uplink MU-MIMO for RedCap UEs (controlled by the **REDCAP_UL_MU_SW** option of the **NRDUCellRedCap**.***ReDCapAlgoSwitch*** parameter) takes effect only when the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected. |

Multiple BWP1s for RedCap:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Multiple BWP1s for RedCap depends on the RedCap Introduction function. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Uplink bandwidth/ Downlink bandwidth | **NRDUCell**.***UlBandwidth*** and **NRDUCell**.***DlBandwidth*** | None | For an NR TDD cell, the value of **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** cannot be greater than the values of **NRDUCell**.***UlBandwidth*** and **NRDUCell**.***DlBandwidth***. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Max SSB Power Offset | **NRDUCellTrpBeam**.***MaxSsbPwrOffset*** | *Power Control* | For an NR TDD cell, the **NRDUCellTrpBeam**.***MaxSsbPwrOffset*** parameter for static power control must be set to a value less than or equal to **4** if the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter is selected and any of the following conditions is met:<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_100M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **80M** or **100M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_90M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **80M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_80M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | parameter is set to **60M** or **80M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_70M** or **CELL_BW_60M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **60M**. |

Preferential selection of BWP1s with NCD-SSBs for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** set to a value greater than **20M** | *RedCap* | For an NR TDD cell, preferential selection of BWP1s with NCD-SSBs for RedCap UEs (with the **NRDUCellRedCap**.***NcdSsbBwpPreferMax-UeNum*** parameter set to a value other than **0**) requires that the value of the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter be greater than **20M**. |

PUCCH CSI RB adjustment for the RedCap-dedicated BWP:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | PUCCH CSI RB adjustment for the RedCap-dedicated BWP can take effect only when RedCap Introduction is enabled. |

RedCap BWP format-4 PUCCH configuration:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap BWP format-4 PUCCH configuration requires that RedCap Introduction be enabled. |

CSI resource balancing:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | CSI resource balancing can be enabled only when RedCap Introduction is enabled. |

RedCap PUCCH resource guarantee:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap PUCCH resource guarantee can be enabled only when RedCap Introduction is enabled. |

Format-1 RB adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Format-1 RB adaptation can be enabled only when both RedCap Introduction and PUCCH RB adaptation are enabled. |
| Low-frequency TDD | PUCCH RB adaptation | **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | *Channel Management* | |

CSI RB adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | CSI RB adaptation can be enabled only when both RedCap Introduction and PUCCH RB adaptation are enabled. |
| Low-frequency TDD | PUCCH RB adaptation | **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | *Channel Management* | |

Format-1 SR resource optimization:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Format-1 SR resource optimization can be enabled only when RedCap Introduction is enabled. |

QCI-specific handover priority configuration:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | QCI-specific handover priority configuration can take effect only when RedCap Introduction is enabled. |

QCI-based RedCap DRX configuration:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | DRX | **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter | *DRX* | QCI-based RedCap DRX configuration requires DRX to be enabled. |

Uplink RedCap preallocation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Uplink basic preallocation | **UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter | *Uplink Scheduling* | Uplink RedCap preallocation requires at least one of the following options to be selected:<br>• **UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter<br>• **UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter |
| Low-frequency TDD | Uplink smart preallocation | **UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter | *Uplink Scheduling* | |

SSB interference avoidance for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Common channel interference avoidance | **SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter | *Interference Avoidance* | SSB interference avoidance for RedCap UEs requires at least one of the following options to be selected:<br>• **SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter<br>• **INTER_FREQ_SSB_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter<br>• **INTER_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter<br>• **INTRA_FREQ_SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter<br>• **INTRA_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter |
| Low-frequency TDD | Inter-frequency SSB interference avoidance | **INTER_FREQ_SSB_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter | *3D Networking Experience Improvement* | |
| Low-frequency TDD | Enhanced inter-frequency SSB interference avoidance | **INTER_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter | *3D Networking Experience Improvement* | |
| Low-frequency TDD | Intra-frequency SSB interference avoidance | **INTRA_FREQ_SSB_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter | *Interference Avoidance* | |
| Low-frequency TDD | Enhanced intra-frequency SSB interference avoidance | **INTRA_FREQ_SSB_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter | *3D Networking Experience Improvement* | |

RedCap BWP frequency-domain staggering:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap BWP frequency-domain staggering can be enabled only when RedCap Introduction is enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** set to a value greater than **20M** | *RedCap* | RedCap BWP frequency-domain staggering can be enabled when any of the following conditions is met:<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_100M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **40M**, **60M**, **80M**, or **100M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_90M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **40M**, **60M**, or **80M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_80M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | set to **40M**, **60M**, or **80M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_70M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **40M** or **60M**.<br>● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_60M**, and the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **40M** or **60M**. |

PDCCH symbol adaptation for RedCap BWP with the CD-SSB:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | PDCCH symbol adaptation for RedCap BWP with the CD-SSB requires that RedCap Introduction be enabled. |

CSI RB and format-1 RB reposition:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | CSI RB and format-1 RB reposition requires RedCap Introduction to be enabled. |
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | *RedCap* | CSI RB and format-1 RB reposition requires the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter to be set to **20M**. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | CSI measurement optimization | **CSI_MEAS_OPT_SW** option of the **NRDUCellCsirs**.***CsiRsExtSwitch*** parameter | *Channel Management* | If "CSI RB and format-1 RB reposition " is enabled in any of the following scenarios, aperiodic CSI-RS resources may be insufficient for UEs in non-overlapping areas, and therefore CSI-RS measurement may fail. Therefore, in any of the following scenarios, "CSI RB and format-1 RB reposition" depends on CSI-RS measurement optimization. Adaptive adjustment of periodic CSI-RS resources ensures that UEs in non-overlapping areas can perform CSI-RS measurement.<br>● The **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter is selected.<br>● The **NRDUCellAlgoSwitch**.***VmimoSwitch*** parameter is set to **ON**.<br>● The **RF_COMBINING_NOISE_SUPPR_SW** option of the **NRDUCellUlSch**.***RfCombSwitch*** parameter is selected.<br>● The **DISTRIBUTED_AN** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | **T_PMI_WT_OPT_SW** option of the **NRDUCellAlgoSwitch**.***BeamOptSwitch*** parameter is selected. |

RedCap UE weight optimization:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UE weight optimization requires that RedCap Introduction be enabled. |
| Low-frequency TDD | SRS interference coordination based on self-contained slots | **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | *Channel Management* | RedCap UE weight optimization requires that SRS interference coordination based on self-contained slots be enabled. |
| Low-frequency TDD | Scenario-specific SRS resource allocation policy | **NRDUCellSrs**.***ScenarioSrsResAllocPol*** | None | RedCap UE weight optimization requires that the **NRDUCellSrs**.***ScenarioSrsResAllocPol*** parameter be set to **CAPC_PRI**. |

BWP scheduling optimization for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | BWP scheduling optimization for RedCap UEs requires that RedCap Introduction be enabled. |

Scheduling in common PDCCH symbols for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Scheduling in common PDCCH symbols for RedCap UEs depends on RedCap Introduction. |
| Low-frequency TDD | Common Control Resource RB Number | **NRDUCellCoreset**.***CommonCtrlResRbNum*** | None | Scheduling in common PDCCH symbols for RedCap UEs requires that the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter be set to **RB48** or **RB96**. |

RedCap PDCCH resource avoidance:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap PDCCH resource avoidance requires that the RedCap Introduction function be enabled. |

SR code channel reservation for RedCap:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | SR code channel reservation for RedCap requires that the RedCap Introduction function be enabled. |

BWP2 PUCCH frequency-domain position optimization:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | BWP2 PUCCH frequency-domain position optimization requires that the RedCap Introduction function be enabled. |
| Low-frequency TDD | Power saving BWP | **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | BWP2 PUCCH frequency-domain position optimization requires that the power saving BWP function be enabled. |

CD-SSB inclusion in the RedCap BWP0:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | CD-SSB inclusion in the RedCap BWP0 requires that the RedCap Introduction function be enabled. |

PDCCH CSS configuration for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | PDCCH CSS configuration for RedCap UEs requires that the RedCap Introduction function be enabled. |

Prohibition of RedCap UE handovers to external NR cells:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Prohibition of RedCap UE handovers to external NR cells requires that the RedCap Introduction function be enabled. |

PRACH allocation direction optimization:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | PRACH allocation direction optimization requires RedCap Introduction to be enabled. |

RedCap UE device-pipe identification optimization:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Device-pipe identification | **SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop**.***SpecUeCooperationSwitch*** parameter | *Device-Pipe Synergy* | RedCap UE device-pipe identification optimization requires that device-pipe identification be enabled. |
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UE device-pipe identification optimization requires that RedCap Introduction be enabled. |

### 4.3.2.2 Mutually Exclusive Functions

RedCap Introduction:

- Basic uplink functions:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PUCCH channel management | **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter<br>**NRDUCellPucch**.***StructureType*** set to **LONG_STRUCTURE** | *Channel Management* | If RedCap Introduction is enabled, the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected, and the **NRDUCellPucch**.***StructureType*** parameter is set to **LONG_STRUCTURE** for an NR TDD cell, then the value sum of the following parameters cannot exceed 32: **NRDUCellPucch**.***Format1RbNum***, **NRDUCellPucch**.***CsiDedicatedRbNum***, **NRDUCellPucch**.***Format3RbNum***, **NRDUCellPucch**.***Format4RbNum***, and **NRDUCellPucch**.***Format4CsiDedicatedRbNum***. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PUCCH RB adaptation | **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | *Channel Management* | After RedCap Introduction is enabled, the sum of the values of parameters **NRDUCellPucch.*Format1RbNum***, **NRDUCellPucch.*CsiDedicatedRbNum***, **NRDUCellPucch.*Format3RbNum***, **NRDUCellPucch.*Format4RbNum***, and **NRDUCellPucch.*Format4CsiDedicatedRbNum*** cannot exceed 32 for an NR TDD cell that meets all of the following conditions:<br>● The **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnlIntrfAvoidSwitch*** parameter is selected.<br>● The **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB48** or **RB96**.<br>● The **NRDUCellBwp**.***UlMinCarrierBw*** parameter is set to **NOT_CONFIG** |
| Low-frequency TDD | PUCCH interference avoidance | ● **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** parameter<br>● **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** set to **UL_MANUAL** | *Interference Avoidance* | |
| Low-frequency TDD | PUSCH interference avoidance | ● **UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent**.***IntrfOptimizationMethod*** parameter<br>● **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** set to **UL_MANUAL** | *Interference Avoidance* | |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Common Control Resource RB Number | **NRDUCellCoreset**.***CommonCtrlResRbNum*** | None | or a value greater than **20M**.<br>● Any of the following conditions is met:<br>– The **NRDUCell**.***FrequencyBand*** parameter is set to **N79**, PUSCH interference avoidance is enabled, and the SSB center frequency position is greater than RB23.<br>– The **NRDUCell**.***FrequencyBand*** parameter is set to **N38**, **N40**, **N41**, **N77**, or **N78**, PUSCH interference avoidance is enabled, and the SSB center frequency position is greater than RB35. |
| Low-frequency TDD | Uplink Minimum Carrier Bandwidth | **NRDUCellBwp**.***UlMinCarrierBw*** | *Scalable Bandwidth* | |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Format3 RB Number | **NRDUCellPucch**.***Format3RbNum*** | None | For an NR TDD cell, the **NRDUCellPucch**.***Format3RbNum*** parameter cannot be set to **RB2** when the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter is selected and all the following conditions are met:<br>● The **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL**.<br>● The **NRDUCell**.***UlBandwidth*** parameter is set to **CELL_BW_60M**, **CELL_BW_70M**, **CELL_BW_80M**, **CELL_BW_90M**, or **CELL_BW_100M**.<br>● The **NRDUCellBwp**.***UlMinCarrierBw*** parameter is set to **NOT_CONFIG** or a value greater than **20M**.<br>● The **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter is deselected. |

● Basic downlink functions:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Periodic TRS transmission | **NRDUCellCsirs**.***TrsPeriod*** | *Channel Management* | For an NR TDD cell, when the **NRDUCellCsirs.*TrsPeriod*** parameter is set to **MS0** or **MS10**, the RedCap Introduction function cannot be enabled. |
| Low-frequency TDD | DC Component rate matching | **DC_COMPONENT_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter | *Downlink Scheduling* | In an NR TDD cell, RedCap Introduction and DC component rate matching cannot be enabled at the same time. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Rate-matching-pattern-configuration-based PDCCH rate matching | **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter | *Downlink Scheduling* | For an NR TDD cell, when RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) is enabled, the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter can be selected only if any of the following conditions is met:<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M** and the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value greater than **4**.<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M**, the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value less than **4**, and the **NRDUCellPdcch**.***PdcchAggLevel1Strategy*** parameter is set to **CONFIG**.<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | parameter is set to **40M**, the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value less than **4**, and the **NRDUCellPdcch**.***PdcchAggLevel1Strategy*** parameter is set to **CONFIG**.<br>● The **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value less than **4** and the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to a value other than **RB24**. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Common Control Resource RB Number | **NRDUCellCoreset**.***CommonCtrlResRbNum*** | None | For an NR TDD cell, when the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter is selected, the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter must be set to **RB48** or **RB96** if any of the following conditions is met:<br>● The **NRDUCellBwp**.***DlMinCarrierBw*** parameter is set to **NOT_CONFIG**, and the **NRDUCell**.***DlBandwidth*** parameter is not set to **CELL_BW_20M**.<br>● The **NRDUCellBwp**.***DlMinCarrierBw*** parameter is set to a value greater than **20M**. |

- Other functions:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | RB resource control | **NRDUCellRes**.***DlMinRbStartPos*** and **NRDUCellRes**.***DlRbMinRatio*** | *Network Slicing*<br>*URLLC* | When RedCap Introduction is enabled for an NR TDD cell and the **NRDUCellRes**.***ResUsageObject*** parameter is set to **NETWORK_SLICE_GROUP** or **HIGH_RELIABILITY**:<br>● If the **NRDUCellRes.*DlMinRbStartPos*** parameter is set to a value other than **65535**, the **NRDUCellRes.*DlRbMinRatio*** parameter cannot be set to a value greater than **10**.<br>● If the **NRDUCellRes.*UlMinRbStartPos*** parameter is set to a value other than **65535**, the **NRDUCellRes.*UlRbMinRatio*** parameter cannot be set to a value greater than **10**. |
| Low-frequency TDD | Staggered PRACH frequency-domain resource allocation | **PRACH_FREQ_POS_STAGGER_SW** option of the **NRDUCellPrach**.***PrachAlgoExtSwitch*** parameter | *Random Access Control* | For an NR TDD cell, RedCap Introduction and the staggered PRACH frequency-domain resource allocation function cannot be enabled at the same time. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Power saving BWP | **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | When both RedCap Introduction and the power saving BWP function are enabled, the **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter cannot be set to a value other than **RB51** for an NR TDD cell. |

Multiple BWP1s for RedCap:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Differentiated eMBB QoS | **EMBB_QOS_DIFF_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter | None | For an NR TDD cell, if differentiated eMBB QoS is enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M** or **40M**. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | For an NR TDD hyper cell, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to **20M**, **40M**, or **60M** if RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) is enabled and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected. At least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** | *Cell Combination* | For an NR TDD combined cell in basic transmission mode, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **DPS_MODE** | *Cell Combination* | For an NR TDD combined cell in DPS transmission mode, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected. At least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Joint Transmission Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE** | *Cell Combination* | For an NR TDD cell with Joint Transmission Cell Combination enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected. At least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Narrowband TRS rate matching | **NARROW_BAND_TRS_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter | None | If narrowband TRS rate matching is enabled and the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter is selected, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**. |
| Low-frequency TDD | UE bandwidth adaptation | **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter | *Scalable Bandwidth* | In the case of an NR TDD cell:<br>When the UE bandwidth adaptation function is enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to a value less than or equal to the **NRDUCellBwp**.***UlMinCarrierBw*** parameter value. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Rate-matching-pattern-configuration-based PDCCH rate matching | **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter | *Downlink Scheduling* | For an NR TDD cell, when RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) is enabled, the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter can be selected only if any of the following conditions is met:<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M** and the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value greater than **4**.<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M**, the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value less than **4**, and the **NRDUCellPdcch**.***PdcchAggLevel1Strategy*** parameter is set to **CONFIG**.<br>● The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | parameter is set to **40M**, the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to a value less than **4**, and the **NRDUCellPdcch**.***PdcchAggLevel1Strategy*** parameter is set to **CONFIG**. |
| Low-frequency TDD | 80 MHz UE access | **80M_UE_ACCESS_SW** option of the **NRDUCellUeCoop**.***UEAccessSwitch*** parameter | None | For an NR TDD cell, if RedCap Introduction and 80 MHz UE access are both enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter cannot be set to a value greater than **80M**. |
| Low-frequency TDD | Bi-Carrier | **BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | None | For an NR TDD cell, when Bi-Carrier and RedCap Introduction are both enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter cannot be set to a value greater than **60M**. |
| Low-frequency TDD | BWP frequency-domain adaptation for RedCap UEs | **BWP_FREQ_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | If the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **100M**, BWP frequency-domain adaptation for RedCap UEs must not be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PLLC primary cell | **PLLC_PRIMARY_CELL_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | None | When PLLC primary cell is enabled, multiple BWP1s for RedCap cannot be enabled. In other words, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to **20M**. |

RedCap BWP format-4 PUCCH configuration:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Differentiated eMBB QoS | **EMBB_QOS_DIFF_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter | None | If the **EMBB_QOS_DIFF_SW** option is selected and the **NRDUCell**.***SlotAssignment*** parameter is set to **7_3_DDDSUDDSUU**, the **NRDUCellRedCap**.***PucchFormat4AckRbNum*** parameter must be set to **RB0**. |

RedCap PUCCH resource guarantee:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Baseband resource mutual aid | **NRDUCellTrp**.***BbResMutualAidSw*** | None | RedCap PUCCH resource guarantee and baseband resource mutual aid cannot be both enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | For an NR TDD hyper cell, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to **20M**, **40M**, or **60M** if RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) is enabled and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected. At least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

RedCap PDCCH symbol number adjustment:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Scheduling in common PDCCH symbols for RedCap UEs | **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | For an NR TDD cell, if the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB96**, "scheduling in common PDCCH symbols for RedCap UEs" and RedCap PDCCH symbol number adjustment cannot be both enabled. |

Scheduling in common PDCCH symbols for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | PDCCH symbol number configuration | **NRDUCellPdcch**.***OccupiedSymbolNum*** | *Channel Management* | For an NR TDD cell, the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter can only be set to **2SYM** or **3SYM** if the following conditions are met: (1) The **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB96**. (2) The **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter is deselected. (3) Scheduling in common PDCCH symbols for RedCap UEs is enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Configuration of multiple PDCCH search spaces | **MULTI_SEARCH_SPACE_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter | *Channel Management* | For an NR TDD cell, if the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB96**, "scheduling in common PDCCH symbols for RedCap UEs" and configuration of multiple PDCCH search spaces cannot be both enabled. |
| Low-frequency TDD | RedCap PDCCH symbol number adjustment | **REDCAP_PDCCH_SYM_NUM_ADJ_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | For an NR TDD cell, if the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB96**, "scheduling in common PDCCH symbols for RedCap UEs" and RedCap PDCCH symbol number adjustment cannot be both enabled. |
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** | *Cell Combination* | If an NR TDD cell is a combined cell in basic transmission mode, scheduling in common PDCCH symbols for RedCap UEs cannot be enabled. |
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | If an NR TDD cell is a hyper cell, scheduling in common PDCCH symbols for RedCap UEs cannot be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PDCCH multi-symbol resource pool | **MULTI_SYM_CORESET_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter | *Channel Management* | For an NR TDD cell, if PDCCH multi-symbol resource pool is enabled, scheduling in common PDCCH symbols for RedCap UEs cannot be enabled. |
| Low-frequency TDD | Smart PDCCH symbol allocation | **PDCCH_SYM_SMART_ALLOC_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter | *Massive MIMO Uplink Boosting (Low-Frequency TDD)* | For an NR TDD cell, "smart PDCCH symbol allocation" and "scheduling in common PDCCH symbols for RedCap UEs" cannot be both enabled. |

### 4.3.2.3 Function Impacts

RedCap Introduction:

- Basic functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Uplink basic preallocation | **UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter | *Uplink Scheduling* | When RedCap Introduction is enabled, uplink basic preallocation does not take effect on RedCap UEs. The **UL_REDCAP_PREALLOCATION_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter must also be selected so that uplink basic preallocation takes effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Uplink smart preallocation | **UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter | *Uplink Scheduling* | When RedCap Introduction is enabled, uplink smart preallocation does not take effect on RedCap UEs. The **UL_REDCAP_PREALLOCATION_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter must also be selected so that uplink smart preallocation takes effect. |
| Low-frequency TDD | Multi-DCI retransmission adaptation | **MULTI_DCI_RETRANS_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***MultiDciAlgoSwitch*** parameter | None | RedCap UEs do not support multi-DCI retransmission adaptation. |
| Low-frequency TDD | Common PUCCH RB resource optimization | **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | None | If common PUCCH RB resource optimization is enabled and then RedCap Introduction is enabled, the average uplink UE throughput decreases and the average downlink UE throughput fluctuates due to a change in the PUCCH allocation mode. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | PDCCH power control for CEUs | **NRDUCellPdcchAlgo**.***PdcchEdgePwrCtrlPolicy*** set to **BASED_ON_AGG_LVL** or **BASED_ON_CSI_RPT** | *Power Control* | When PDCCH power control for CEUs and RedCap Introduction are both enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the uplink and downlink CCE allocation success rates as well as the average uplink and downlink throughputs of RedCap UEs. |
| Low-frequency TDD | Smart PDCCH symbol allocation | **PDCCH_SYM_SMART_ALLOC_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter | *Massive MIMO Uplink Boosting (Low-Frequency TDD)* | When smart PDCCH symbol allocation is enabled in scenarios where RedCap Introduction takes effect, the CCE allocation success rate may increase for RedCap UEs at the cell center or at a medium distance from the cell center, and may decrease for RedCap UEs at the cell edge. As a result, the average uplink and downlink UE throughputs may fluctuate. |

- Functions related to carrier management

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Experience-based smart carrier selection | **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter | *Multi-Frequency Smart Selection* | Experience-based smart carrier selection does not take effect on RedCap UEs. |
| Low-frequency TDD | Multi-frequency beam coordination | **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | *Multi-Frequency Smart Selection* | When multi-frequency beam coordination takes effect, the gNodeB does not select RedCap UEs. |
| Low-frequency TDD | Latency-based optimal carrier selection | **DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter | *Multi-Frequency Smart Selection* | Latency-based optimal carrier selection does not take effect for RedCap UEs. |

- Other functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | MT access control | **MT_RRC_CONN_SWITCH** option of the **NRDUCellConnMgmt**.***IabCtrlSwitch*** parameter | None | RedCap UEs do not support MT access control. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Energy-efficiency-based inter-frequency handover | **ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter | *Energy Conservation and Emission Reduction* | Energy-efficiency-based inter-frequency handover does not take effect on RedCap UEs. |
| Low-frequency TDD | Intra-base-station DL CoMP | **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | Intra-base-station DL CoMP does not take effect on RedCap UEs. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | RB resource control | • **NRDUCellRes**.***DlMinRbStartPos*** and **NRDUCellRes**.***DlRbMinRatio***<br>• **NRDUCellRes**.***UlMinRbStartPos*** and **NRDUCellRes**.***UlRbMinRatio*** | *Network Slicing* | If the uplink slice RB resources (indicated by a product of **NRDUCellRes**.***UlRbMaxRatio*** or **NRDUCellRes**.***UlRbMinRatio*** and the cell bandwidth) are greater than 15 MHz, slice UEs may consume excessive uplink scheduling resources. As a result, RedCap UEs outside the slice group may fail to be scheduled in a timely manner due to insufficient uplink scheduling resources. If the downlink slice RB resources (indicated by a product of **NRDUCellRes**.***DlRbMaxRatio*** or **NRDUCellRes**.***DlRbMinRatio*** and the cell bandwidth) are greater than 15 MHz, slice UEs may consume excessive downlink scheduling resources. As a result, RedCap UEs outside the slice group may fail to be scheduled in a timely manner due to insufficient downlink scheduling resources. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Precise MU-MIMO evaluation | **PRECISE_MUMIMO_EVAL_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | When RedCap UEs are paired for MU-MIMO, the benefits of precise MU-MIMO evaluation slightly decrease because the power usage of MU-MIMO UEs may decrease. |
| Low-frequency TDD | Symbol power saving | **NRDUCellAlgoSwitch**.***SymbolShutdownSwitch*** | *Energy Conservation and Emission Reduction* | When RedCap Introduction and symbol power saving take effect at the same time, the value of the **N.PowerSaving.SymbolShutdown.Dur** counter (indicating the symbol power saving duration) may decrease. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** | *Cell Combination* | For an NR TDD combined cell in basic transmission mode, if the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected and the **NRDUCellPucch**.***Format3RbNum*** parameter is set to **RB2**, the downlink user-perceived rates of RedCap UEs decrease. To prevent this issue, it is recommended that the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter be selected, the **NRDUCellPucch**.***Format3RbNum*** parameter be set to **RB4**, or the **NRDUCellMultiTrp**.***DataTransMode*** parameter be set to **DPS_MODE**. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink resource isolation | **DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter | *FWA* | After downlink resource isolation is enabled, the number of downlink scheduling opportunities for RedCap FWA users further decreases. (For the definition of such users, see *FWA*.) As a result, the average downlink throughput of FWA users decreases more significantly. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Power saving BWP | **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | Power saving BWP does not take effect on RedCap UEs.<br>When both RedCap Introduction and the power saving BWP function take effect, RedCap UEs and UEs in the power saving BWP state share the 20 MHz bandwidth of the cell. As a result, the actual resources available to UEs in the power saving BWP state are affected and the value of the **N.Bwp2.Dur.Total** counter decreases.<br>When the RedCap load is heavy in a cell, RedCap UEs occupy SRS resources of BWP2 UEs. As a result, the number of BWP2 UEs decreases. In this case, it is recommended that SRS resource reservation for BWP2 UEs in RedCap scenarios be enabled (by selecting the **BWP2_SRS_RSV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter) to increase SRS resources allocated to BWP2 UEs and improve the energy saving effect. For details about the SRS resource reservation for BWP2 UEs in RedCap scenarios function, |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | see *UE Power Saving*. |
| Low-frequency TDD | Time-domain power saving BWP | **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | • Time-domain power saving BWP does not take effect on RedCap UEs.<br>• When time-domain power saving BWP takes effect, the scheduling of RedCap UEs on BWP1 is affected, and RedCap user experience fluctuates. |
| Low-frequency TDD | RedCap DRX adaptation | • **REDCAP_DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter<br>• **REDCAP_DRX_ADAPTATION_EXT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter | *RedCap* | When RedCap UEs are using BWP1 with the NCD-SSB, RedCap DRX adaptation does not take effect on these RedCap UEs. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Inter-cell interference avoidance | **INTER_CELL_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | Inter-cell interference avoidance does not take effect on RedCap UEs. |
| Low-frequency TDD | Data rate limitation in flexible experience differentiation | **FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter | *Service-differentiated Experience Guarantee* | Data rate limitation in flexible experience differentiation does not take effect on RedCap UEs. |
| Low-frequency TDD | Proactive avoidance of remote interference | **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled. |
| Low-frequency TDD | UL and DL Decoupling | **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** | *UL and DL Decoupling* | |

RedCap UE inactivity management:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap UE release preference | **REDCAP_RELEASE_PREFERENCE_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter or **REDCAP_RELEASE_PREFER_EXT_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When the RRC_INACTIVE connection management function takes effect, RedCap UE release preference allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state. |

RedCap NCD-SSB measurement:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap BWP frequency-domain staggering | **REDCAP_BWP_NON_OVERLAP_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | When both RedCap BWP frequency-domain staggering and RedCap NCD-SSB measurement are enabled, the probability that intra-frequency NCD-SSBs are measured decreases. If intra-frequency NCD-SSBs cannot be measured, gap-assisted inter-frequency CD-SSB measurement is initiated, which decreases the average uplink and downlink throughputs of RedCap UEs. You can reduce the value of **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld*** to reduce the impact. |

Multiple BWP1s for RedCap:

- Functions related to interference reduction

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink multi-BWP interference avoidance | **DL_BWP_HYBRID_INTRF_RANDOM_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | For an NR TDD cell, when both downlink multi-BWP interference avoidance and multiple BWP1s for RedCap (with the **NRDUCellRedCap.*RedCapUeMaxAvail-Bandwidth*** parameter set to a value other than **20M**) are enabled and multiple RedCap UEs access the network through different BWP1s, the downlink interference on each BWP1 may change. As a result, the performance of RedCap UEs may be affected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | External interference avoidance | • **UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** parameter<br>• **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** set to **UL_MANUAL** | *Interference Avoidance* | For an NR TDD cell, the number of activated RedCap BWP1s may be less than the number of BWP1s corresponding to the **NRDUCellRedCap**.***RedCapUeMaxAvail-Bandwidth*** parameter value when all of the following conditions are met:<br>• PUCCH interference avoidance (controlled by the **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.*CommChnlntrfAvoidSwitch*** parameter) is enabled.<br>• External interference avoidance is enabled.<br>• At least one **NRDUCellIntrfBand** MO instance has been configured for the cell (that is, the command output of **LST NRDUCELLINTRFBAND** shows that there is at least one interference band). |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Start RB avoidance | **START_RB_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When multiple BWP1s for RedCap is in effect, start RB avoidance does not take effect. |

- Other functions

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | PUCCH RB adaptation | **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter | *Channel Management* | PUCCH RB adaptation takes effect only on RedCap UEs using BWP1s at cell bandwidth edges. |
| Low-frequency TDD | Virtual Grid-based Multi-Frequency Coordination | **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter | *Virtual Grid-based Multi-Frequency Coordination* | During virtual grid model building and maintenance, data of RedCap UEs using BWP1s with NCD-SSBs is not collected, and virtual grid models are not used for these UEs, either. |
| Low-frequency TDD | UE-level MRO | **UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter | *MRO* | UE-level MRO does not take effect on RedCap UEs using BWP1s with NCD-SSBs. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | CSI reporting period adaptation | **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter | *Channel Management* | When CSI reporting period adaptation is enabled:<br>• For a TDD cell, CSI reporting period adaptation takes effect on RedCap UEs when the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value less than or equal to **40M**. In other scenarios, the CSI reporting period is always 320 slots. |
| Low-frequency TDD | Bandwidth-based power aggregation | **BANDWIDTH_BASED_POWER_AGG_SW** option of the **NRDUCellChnPwr**.***ChnPwrAlgoSwitch*** parameter | *Energy Conservation and Emission Reduction* | For a TDD cell, when RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) is enabled and the **NRDUCellRedCap**.***RedCapUeMaxAvail-Bandwidth*** parameter is set to a value other than **20M**, bandwidth-based power aggregation does not take effect after being enabled, if the adjusted bandwidth does not cover all the RedCap BWP1s. The purpose is to ensure that UEs on all the RedCap BWP1s can perform services properly. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Dynamic RF channel shutdown | **RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter | *Energy Conservation and Emission Reduction* | For a TDD cell, when dynamic RF channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, dynamic RF channel shutdown does not take effect when SSB power aggregation is insufficient. |
| Low-frequency TDD | RF channel intelligent shutdown | **RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | For a TDD cell, when RF channel intelligent shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, RF channel intelligent shutdown does not take effect when SSB power aggregation is insufficient. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | TTI-level channel shutdown | **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | For a TDD cell, when TTI-level channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, TTI-level channel shutdown does not take effect when SSB power aggregation is insufficient. |
| Low-frequency TDD | LNR coordinated channel shutdown | • NR: **MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter<br>• LTE: **CellRfShutdown**.***MultiRatJointChnShutdownSw*** | *Multi-RAT Coordinated Energy Saving* | For a TDD cell, when LNR coordinated channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, LNR coordinated channel shutdown does not take effect when SSB power aggregation is insufficient. |
| Low-frequency TDD | LTE TDD and NR spectrum sharing | **LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter | None | When LTE TDD and NR spectrum sharing is enabled, the total bandwidth of the maximum number of BWP1s that can be activated for RedCap UEs does not exceed the NR-dedicated spectrum range. |

Preferential selection of BWP1s with NCD-SSBs for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | UE-number-based BWP1 adaptation | **USER_NUM_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | Assume that preferential selection of BWP1s with NCD-SSBs for RedCap UEs and any of the following functions are enabled: UE-number-based BWP1 adaptation and PRB-and-CCE-resource-based BWP1 adaptation. The base station reserves at least one RedCap BWP1 with the NCD-SSB because the UEs preferentially access the BWP1s with NCD-SSBs. |
| Low-frequency TDD | PRB-and-CCE-resource-based BWP1 adaptation | **RESOURCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | |

QCI-specific handover priority configuration:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | RedCap frequency priority policy | **NRCellQciBearer**.***RedCapGnbFreqPriGroupId*** set to a value other than **255** | *RedCap* | If both QCI-specific handover priority configuration and RedCap frequency priority policy take effect, the QCI priority is preferentially determined by the **gNBQciBearer**.***QciPriorityForRedCapHo*** parameter. If this parameter is set to **255**, the QCI priority will then be determined by the **gNBQciBearer**.***QciPriorityForHo*** parameter. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Service-based inter-frequency handover | **SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter | *SA Mobility Management in Connected Mode* | If both QCI-specific handover priority configuration and service-based inter-frequency handover take effect, the QCI priority is preferentially determined by the **gNBQciBearer**.***QciPriorityForRedCapHo*** parameter. If this parameter is set to **255**, the QCI priority will then be determined by the **gNBQciBearer**.***QciPriorityForHo*** parameter. |

RedCap downlink scheduling policy:

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | FWA RedCap downlink type 1 scheduling | **FWA_TYPE1_4RB_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | The FWA RedCap downlink type 1 scheduling function is on a per four RB basis. Therefore, after this function takes effect for RedCap UEs, the downlink scheduling policy for RedCap UEs will not take effect (downlink type 1 scheduling on a per one RB basis). |

Differentiated QoS management for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | SAI-based service experience improvement | **gNBSchAdaptToTfcMdl**.***SchAdaptToTfcModelPol*** set to **SAI** | *SAI-based Service Experience Improvement* | If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is deselected and the corresponding **gNBOp5qiConfig** MO is configured for a bearer, the equivalent QCI configured for the bearer for SAI-based service experience improvement (specified by **gNBServExpGroup**.***EquivalentQci***) and SAI-based service experience improvement do not take effect.<br>If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is selected and a 5QI-specific bearer has been configured with a QCI specified by **gNBOp5qiConfig**.***RedCapQci*** for "differentiated QoS management for RedCap UEs", the following are true:<br>● If the **gNBOp5qiConfig**.***RedCapQci*** parameter is set to **65535**, SAI-based service experience improvement does not take effect on non-RedCap UEs.<br>● If the **gNBOp5qiConfig**.***RedCapQci*** |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | parameter is set to a value other than **65535**, the following are true:<br>– If the **gNBOp5qiConfig**.***Qci*** parameter is set to **65535**, SAI-based service experience improvement does not take effect on RedCap UEs.<br>– If the **gNBOp5qiConfig**.***Qci*** parameter is set to a value other than **65535**, SAI-based service experience improvement does not take effect on all UEs. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Service identification-based assurance | **gNBSchAdaptToTfcMdl**.***SchAdaptToTfcModelPol*** set to **TAI** | *Device-Pipe Synergy* | If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is deselected and the corresponding **gNBOp5qiConfig** MO is configured for a bearer, the equivalent QCI configured for the bearer for service identification-based assurance (specified by **gNBServExpGroup**.***EquivalentQci***) and service identification-based assurance do not take effect.<br>If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo**.***QciAlgoOptSw*** parameter is selected and a 5QI-specific bearer has been configured with a QCI specified by **gNBOp5qiConfig**.***RedCapQci*** for "differentiated QoS management for RedCap UEs", the following are true:<br>● If the **gNBOp5qiConfig**.***RedCapQci*** parameter is set to **65535**, service identification-based assurance does not take effect on non-RedCap UEs.<br>● If the **gNBOp5qiConfig**.***RedCapQci*** |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | parameter is set to a value other than **65535**, the following are true:<br>– If the **gNBOp5qiConfig**.***Qci*** parameter is set to **65535**, service identification-based assurance does not take effect on RedCap UEs.<br>– If the **gNBOp5qiConfig**.***Qci*** parameter is set to a value other than **65535**, service identification-based assurance does not take effect on all UEs. |

RedCap BWP frequency-domain staggering:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | BWP frequency-domain adaptation for RedCap UEs | **BWP_FREQ_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When RedCap BWP frequency-domain staggering and "BWP frequency-domain adaptation for RedCap UEs" are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, RedCap BWP frequency-domain adaptation takes effect based on the new activation sequence of RedCap BWP1s. |
| Low-frequency TDD | UE-number-based BWP1 adaptation | **USER_NUM_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When both RedCap BWP frequency-domain staggering and UE-number-based BWP1 adaptation are enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, UE-number-based BWP1 adaptation takes effect based on the new activation sequence of RedCap BWP1s. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PRB-and-CCE-resource-based BWP1 adaptation | **RESOURCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When RedCap BWP frequency-domain staggering and PRB-and-CCE-resource-based BWP1 adaptation are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, PRB-and-CCE-resource-based BWP1 adaptation takes effect based on the new activation sequence of RedCap BWP1s. |
| Low-frequency TDD | RedCap NCD-SSB measurement | **REDCAP_NCD_SSB_MEAS_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When both RedCap BWP frequency-domain staggering and RedCap NCD-SSB measurement are enabled, the probability that intra-frequency NCD-SSBs are measured decreases. If intra-frequency NCD-SSBs cannot be measured, gap-assisted inter-frequency CD-SSB measurement is initiated, which decreases the average uplink and downlink throughputs of RedCap UEs. You can reduce the value of **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld*** to reduce the impact. |

RedCap power aggregation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PDCCH power control for CEUs | **NRDUCellPdcchAlgo**.***PdcchEdgePwrCtrlPolicy*** set to **BASED_ON_AGG_LVL** or **BASED_ON_CSI_RPT** | *Power Control* | When PDCCH power control for CEUs and RedCap power aggregation are both enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the CCE allocation success rate and may reduce overall benefits. |
| Low-frequency TDD | PDCCH aggregation level reduction | **PDCCH_AGG_LVL_COMPR_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | When PDCCH aggregation level reduction and RedCap power aggregation are both enabled, the overall benefits provided by these two functions decrease. |
| Low-frequency TDD | PDCCH resource allocation enhancement | **PDCCH_RES_ALLOC_ENH_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter | *Channel Management* | When PDCCH resource allocation enhancement and RedCap power aggregation are both enabled, RedCap power aggregation does not take effect. |

BWP scheduling optimization for RedCap UEs:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | QCI-specific UE-number-based periodic BWP load balancing | **QCI_UE_NUM_BASED_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter and **QCI_UE_NUM_BASED_PRDC_BLB_SW** option of the **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** parameter | *RedCap* | When both BWP scheduling optimization for RedCap UEs and QCI-specific UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect. |
| Low-frequency TDD | UE-number-based periodic BWP load balancing | **UE_NUM_BASED_PRDC_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | When both BWP scheduling optimization for RedCap UEs and UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect. |

RedCap PDCCH symbol number adjustment:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Configuration of multiple PDCCH search spaces | **MULTI_SEARCH_SPACE_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter | *Channel Management* | When configuration of multiple PDCCH search spaces is enabled, RedCap PDCCH symbol number adjustment does not take effect. |

CD-SSB inclusion in the RedCap BWP0:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Policy configuration for PUCCH resource sharing with the PUSCH | **NRDUCellPusch**.***PuschSharePucchResPol*** set to a value other than **NOT_SHARED** | None | For an NR TDD cell, when CD-SSB inclusion in RedCap BWP0 is enabled and the **NRDUCellPusch**.***PuschSharePucchResPol*** parameter is set to a value other than **NOT_SHARED**, the average uplink throughput of RedCap UEs may decrease. |

## 4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

For NR TDD: All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

**NOTE**

When the multiple BWP1s for RedCap function takes effect on a UBBPfw series board, a maximum of two BWP1s are available for RedCap.

### RF Modules

No requirements

## 4.3.4 Cells

In the case of an NR TDD cell:

- RedCap Introduction:
  - The **NRDUCell**.***DuplexMode*** parameter must be set to a value other than **CELL_SDL**.

- Independent BWP0 can be configured for RedCap UEs only in cells with a bandwidth of 40 MHz or higher. In other cell bandwidth scenarios, RedCap UEs and common NR UEs share BWP0.
- The value of **NRDUCellPrach**.***PrachConfigurationIndex*** corresponding to **NRDUCellSulRelation**.***SulNrDuCellId*** can only be **13**, **14**, **15**, **17**, **18**, **21**, or **65535**, and the value of the corresponding **NRDUCell**.***CellRadius*** parameter cannot exceed **14500** if: (1) **NRDUCellSulRelation**.***SulType*** is set to **FIX_SUL**; (2) RedCap Introduction is enabled for the cell corresponding to **NRDUCellSulRelation**.***SulNrDuCellId***; and (3) the cell specified by **NRDUCellSulRelation**.***NrDuCellId*** meets one of the following conditions:
  - **NRDUCellBwp**.***DlMinCarrierBw*** is set to **NOT_CONFIG** or a value greater than **20M**, **NRDUCell**.***FrequencyBand*** is set to **N79**, **NRDUCellCoreset**.***CommonCtrlResRbNum*** is set to **RB48** or **RB96**, and the center frequency position is greater than RB23. For details about how to calculate the SSB frequency-domain position, see *Cell Management*.
  - **NRDUCellBwp**.***DlMinCarrierBw*** is set to **NOT_CONFIG** or a value greater than **20M**, **NRDUCell**.***FrequencyBand*** is set to **N38**, **N40**, **N41**, **N77**, or **N78**, **NRDUCellCoreset**.***CommonCtrlResRbNum*** is set to **RB48** or **RB96**, and the center frequency position is less than RB35. For details about how to calculate the SSB frequency-domain position, see *Cell Management*.

- Multiple BWP1s for RedCap: The cell bandwidth must be greater than or equal to 40 MHz.
- RedCap PUCCH resource guarantee:
  - If the **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** is selected for a cell served by a base station, the **NRDUCellTrp**.***BasebandEqmId*** parameter must be set to a value other than **255** for all intra-base-station cells with the **NRCell**.***CellActiveState*** parameter being **CELL_ACTIVE**.
  - RedCap PUCCH resource guarantee must be enabled or disabled for all cells on the BBP serving an activated cell (with the **NRCell**.***CellActiveState*** parameter being **CELL_ACTIVE**).
  - The **NRDUCellTrp**.***SecondaryBasebandEqmId*** parameter must be set to **255**.
- Other functions: None

### 4.3.5 Others

UEs must support the supportOfRedCap-r17 capability defined in 3GPP Release 17.

## 4.4 Operation and Maintenance

### 4.4.1 Data Configuration

#### 4.4.1.1 Data Preparation

The following tables describe the parameters used for function activation.

**Table 4-23** Parameters used for activation (RedCap Introduction)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | If RedCap UEs need to access the network, select the **REDCAP_SW** option of this parameter. |

**Table 4-24** Parameters used for activation (parameters that need to be configured when RedCap Introduction is enabled)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| PDCCH Algorithm Enhancement Switch | **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** | For an NR TDD cell, as the number of PDCCH symbols and the uplink-downlink CCE resource ratio need to be adaptively adjusted and the initial CCE aggregation level selection needs to be optimized in multiple BWP1s for the RedCap Introduction function, it is recommended that multi-BWP1 PDCCH transmission[a] be enabled by selecting the **PDCCH_MULTI_BWP1_ALGO_SW** option. |
| PDCCH Algorithm Enhancement Switch | **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** | For an NR TDD cell, as the number of PDCCH symbols and the uplink-downlink CCE resource ratio need to be adaptively adjusted and the initial CCE aggregation level selection needs to be optimized in multiple BWP1s for the RedCap Introduction function, it is recommended that enhanced multi-BWP1 PDCCH transmission[a] be enabled by selecting the **PDCCH_MULTI_BWP1_ALGO_ENH_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| PDCCH Algorithm Switch | **NRDUCellPdcch**.***PdcchAlgoSwitch*** | For an NR TDD cell, when the RedCap Introduction function is enabled and the RedCap BWP1 bandwidth is 20 MHz, CCE resources will be limited. Therefore, it is recommended that PDCCH uplink and downlink CCE sharing be enabled by selecting the **UL_DL_CCE_SHARING_SW**.<br>For details about PDCCH uplink and downlink CCE sharing, see *Channel Management*. |
| SRS Algorithm Extension Switch | **NRDUCellSrs**.***SrsAlgoExtSwitch*** | For an NR TDD cell, after the RedCap Introduction function is enabled, SRS resources need to be allocated to BWP0s with different bandwidths. Therefore, it is recommended that BWP0 SRS resource allocation be enabled by selecting the **BWP0_SRS_ALLOC_SW** option. |
| a: Once the multi-BWP1 PDCCH transmission and enhanced multi-BWP1 PDCCH transmission functions are enabled at the same time, values of the CCE-related indicators may fluctuate. For details about these two functions, see *Channel Management*. | | |

**Table 4-25** Parameters used for activation (basic signaling procedures)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap UE Inactivity Timer | **NRDUCellQciBearer**.***RedCapUeInactivityTimer*** | Set this parameter based on the network plan. |
| Sig UE No NAS Msg Trans Timer | **gNBConnStateTimer**.***SigUeNoNasMsgTransTmr*** | Use the recommended value. |
| INACTIVE Strategy Switch | **NRCellAlgoSwitch**.***InactiveStrategySwitch*** | It is recommended that the **REDCAP_RRC_INACTIVE_SWITCH** option be selected when RRC_INACTIVE connection management is required for RedCap UEs. |

**Table 4-26** Parameters used for activation (mobility management)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RRM Meas Relax Low-Speed UE Judge Timer | **NRCellReselConfig**.***RrmRelaxLSUeJudgeTimer*** | Use the recommended value. |
| RRM Meas Relax Low-Speed UE RSRP Diff Thld | **NRCellReselConfig**.***RrmRelaxLSUeRsrpDiffThld*** | Use the recommended value. |
| RRM Meas Relax Non-Edge UE RSRP Thld | **NRCellReselConfig**.***RrmRelaxNonEdgeRsrpThld*** | Use the recommended value.<br>According to 3GPP TS 38.331 V17.4.0, this parameter must be set to a value less than or equal to the value of the **NRCellReselConfig**.***IntraFreqMeasStartThld*** parameter as well as the value of the **NRCellReselConfig**.***NonIntraFreqMeasRsrpThld*** parameter. |
| RedCap NCD-SSB A1 RSRP Thld | **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1RsrpThld*** | Use the recommended value.<br>The **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1RsrpThld*** parameter value must be greater than the **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld*** parameter value. |
| RedCap NCD-SSB A2 RSRP Thld | **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2RsrpThld*** | Use the recommended value. |
| RedCap NCD-SSB A3 Offset | **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA3Offset*** | Use the recommended value. |
| RedCap Algorithm Switch | **NRCellRedCap**.***RedCapAlgoSwitch*** | Select the **REDCAP_NCD_SSB_MEAS_SW** option.<br>This option is selected by default, and it is recommended that this option be selected. |
| RedCap Algorithm Switch | **NRCellRedCap**.***RedCapAlgoSwitch***. | It is recommended that the **REDCAP_NCD_SSB_SINR_MEAS_SW** option be selected when the cell suffers co-channel interference. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap NCD-SSB A1 SINR Thld | **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1SinrThld*** | Use the recommended value.<br>The **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA1SinrThld*** parameter value must be greater than the **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2SinrThld*** parameter value. |
| RedCap NCD-SSB A2 SINR Thld | **NRCellInterFHoMeaGrp**.***RedCapNcdSsbA2SinrThld*** | Use the recommended value. |
| RedCap Algorithm Switch | **NRCellRedCap**.***RedCapAlgoSwitch*** | If the RedCap handover policy needs to be adjusted based on the **Redcap-Bcast-Information** IE or the RedCap barring status (specified by the **NRDUCellRedCap**.***CellBarredRedCap*** parameter) in the link, select the **XN_INTRA_GNB_HO_CTRL_SW** option for Xn-based handovers and intra-base-station handovers and the **NG_INTER_RAT_HO_COMPAT_SW** option for NG-based handovers and inter-RAT handovers. |
| Cell Barred for RedCap | **NRDUCellRedCap**.***CellBarredRedCap*** | Set this parameter based on the network plan. |
| Mobility Function Indication | **NRExternalNCell**.***MobilityFunInd*** | Select the **REDCAP_1RX_NO_HO_SW** and **REDCAP_2RX_NO_HO_SW** options if 1RX and 2RX RedCap UE handovers to external NR cells are prohibited. |
| RedCap Measurement Activation Flag | **NRCellFreqRelation**.***RedCapMeasActvFlag*** | Set this parameter based on the network plan. |
| RedCap Access Allowed Flag | **NRCellFreqRelation**.***RedCapAccessAllowed-Flag*** | Set this parameter based on the network plan. |
| RedCap gNodeB Frequency Priority Group ID | **gNBRfspConfig**.***RedCapGnbFreqPriGroupId*** | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap gNodeB Frequency Priority Group ID | **NRCellQciBearer**.***RedCapGnbFreqPriGroupId*** | Set this parameter based on the network plan. |
| RedCap gNodeB Frequency Priority Group ID | **NRCellOpPolicy**.***RedCapGnbFreqPriGroupId*** | Set this parameter based on the network plan. |
| RA Control Algorithm Switch | **NRDUCellRac**.***RacAlgoSwitch*** | Select the **EMC_ADMISSION_GUAR_SW** option when there are RedCap UEs performing emergency calls on the network. |
| RedCap QCI Priority for Handover | **gNBQciBearer**.***QciPriorityForRedCapHo*** | Set this parameter based on the network plan. |

**Table 4-27** Parameters used for activation (experience enhancement)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **REDCAP_DL_MU_SW** option be selected when the cell is heavily loaded. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **REDCAP_UL_MU_SW** option be selected when the cell is heavily loaded. |
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** | Select the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of this parameter if uplink MU grouping and pairing optimization for RedCap UEs is required. |
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** | Select the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option of this parameter if uplink MU grouping and pairing avoidance for RedCap UEs is required. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap UE Maximum Available Bandwidth | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | Set this parameter based on the network plan. |
| NCD-SSB BWP-preferred Max UE Number | **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** | In common scenarios, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, the recommended value is **1**. To achieve the optimal gains in certain scenarios, you can adjust the configuration based on the specific application scenario and traffic model. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | If the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M** and the number of UEs on the RedCap-dedicated BWP (for details about its definition, see **4.1.3.2 Multiple BWP1s for RedCap**) is greater than or equal to 100, select the **REDCAP_PUCCH_CSI_RB_ADJ_SW** option. |
| PUCCH Switch | **gNodeBAlgo**.***PucchSwitch*** | For a TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, it is recommended that the **FORMAT1_RB_RES_CTRL_SW** option be selected. |
| PUCCH Algorithm Extension Switch | **NRDUCellPucch**.***PucchAlgoExtSwitch*** | For a TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, it is recommended that the **INTER_TRP_RB_SHARE_POLICY_SW** option be selected. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| PUCCH Format4 ACK RB Number | **NRDUCellRedCap**.***PucchFormat4AckRbNum*** | For a TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **40M** and there are multiple RedCap UEs performing downlink large-packet services, the value **RB1** is recommended, which increases the average downlink throughput of RedCap UEs performing large-packet services. |
| CSI Resource Algo Switch | **NRDUCellPucch**.***CsiResoureAlgoSwitch*** | For an NR TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, it is recommended that the **CSI_RES_BALANCE_SW** option be selected. |
| PUCCH Algorithm Switch | **NRDUCellPucch**.***PucchAlgoSwitch*** | For an NR TDD cell, if a large number of RedCap BWPs are configured on the BBP serving this cell (that is, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M** for multiple cells on this BBP), it is recommended that the **REDCAP_PUCCH_RES_GUAR_SW** option be selected. |
| PUCCH Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | For an NR TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M**, it is recommended that the **PUCCH_F1_RB_ADAPT_SW** option be selected. |
| PUCCH Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | For an NR TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M**, it is recommended that the **PUCCH_CSI_RB_ADAPT_SW** option be selected. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| PUCCH Algorithm Switch | **NRDUCellPucch**.***PucchAlgoSwitch*** | For an NR TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M**, it is recommended that the **FORMAT1_SR_RES_OPT_SW** option be selected. |
| CSI-RS Extend Switch | **NRDUCellCsirs**.***CsiRsExtSwitch*** | If the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than or equal to **20M** for a heavy-load cell, it is recommended that the **APER_CSIRS_FLEX_OPT_SW** option be selected. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | For an NR TDD cell, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **20M**, it is recommended that the **PUCCH_RB_REPOSITION_SW** option be selected. If BWP2 is configured in a cell, it is recommended that the **FORMAT1_SR_CHN_ALLOC_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter and the **PUCCH_F1_RB_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter be selected. |
| Uplink Preallocation Switch | **NRDUCellPusch**.***UlPreallocationSwitch*** | Select the **UL_REDCAP_PREALLOCATION_SW** option of this parameter.<br>This option is selected by default, and it is recommended that this option be selected. |
| Minimum Preallocation Period | **NRDUCellPusch**.***MinPreallocationPeriod*** | Retain the default value. |
| Prescheduled UE Num Upper Limit | **NRDUCellPusch**.***PreschUeNumUpperLimit*** | Retain the default value. |
| Preallocation Size | **NRDUCellPusch**.***PreallocationSize*** | Use the recommended value. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap UL Preallocation Policy | **NRDUCellRedCap**.***RedCapUlPreallocPolicy*** | Use the recommended value. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | Retain the default value of the **REDCAP_DL_SCH_POL_SW** option. |
| RedCap DU DRX Parameter Group ID | **NRDUCellQciBearer**.***RedCapDuDrxParamGroupId*** | Set this parameter based on the network plan.<br>This parameter specifies the ID of a DRX parameter group for RedCap UEs. If this parameter is set to a value in the range of 0 to 65, valid DRX parameters are configured for the specified QCI. If this parameter is set to **255**, the DRX configurations for RedCap UEs are the same as those for common UEs; that is, the parameter group specified by **NRDUCellQciBearer.*DuDrxParamGroupId*** is used. |
| RedCap UE DRX Parameter Group ID | **NRCellQciBearer**.***RedCapDrxParamGroupId*** | Set this parameter based on the network plan.<br>This parameter specifies the ID of a DRX parameter group for RedCap UEs. If this parameter is set to a value in the range of 0 to 65, valid DRX parameters are configured for the specified QCI. If this parameter is set to **255**, the DRX configurations for RedCap UEs are the same as those for common UEs; that is, the parameter group specified by **NRCellQciBearer.*DrxParamGroupId*** is used. |
| Operator 5QI Configure Index | **gNBOp5qiConfig.*Op5qiConfigIndex*** | Set this parameter based on the network plan. |
| Operator ID | **gNBOp5qiConfig.*OperatorId*** | Set this parameter to the same value as **gNBNetworkSlice.*OperatorId***. |
| Slice Service Type | **gNBOp5qiConfig.*SliceServiceType*** | Set this parameter to the same value as **gNBNetworkSlice.*SliceServiceType***. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Slice Differentiator | **gNBOp5qiConfig.*SliceDifferentiator*** | Set this parameter to the same value as **gNBNetworkSlice.*SliceDifferentiator***. |
| NR 5G QoS Indicator | **gNBOp5qiConfig.*Nr5qi*** | Set this parameter based on the network plan. |
| QoS Class Identifier | **gNBOp5qiConfig.*Qci*** | Set this parameter based on the network plan. |
| RedCap QoS Class Identifier | **gNBOp5qiConfig.*RedCapQci*** | Set this parameter based on the network plan. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | Set the **REDCAP_SSB_INTRF_AVOID_SW** option based on the network plan. |
| PDCCH Algorithm Extension Switch | **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** | It is recommended that the **REDCAP_POWER_AGGREGATION_SW** option be selected for an NR TDD cell. |
| RedCap Power Aggregation CCE Threshold | **NRDUCellPdcchAlgo**.***RedCapPwrBoostingCceThld*** | For an NR TDD cell, use the recommended value. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | It is recommended that the **REDCAP_BWP_NON_OVERLAP_SW** option be selected for an NR TDD cell on a light- or medium-load network where multiple cells offer contiguous coverage. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | You are advised to select the **CD_BWP_PDCCH_SYM_ADAPTIVE_SW** option when CCE resources are insufficient. |
| Service Differentiation Algorithm Switch | **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** | For an NR TDD cell, select the **REDCAP_SRS_WEIGHT_SW** option if RedCap UE weight optimization is required. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| UL Big Packet Data Volume Threshold | **gNBRedCapServIdenCfg**.***DlBigPktDataVolThld*** | For an NR TDD cell, use the default value if RedCap UE weight optimization is required.<br>The combination of the value of the **gNBRedCapServIdenCfg**.***UlBigPktDataVolThld*** and **gNBRedCapServIdenCfg**.***DlBigPktDataVolThld*** parameter in the same MO specified by the **gNBRedCapServIdenCfg**.***ServParamGroupId*** parameter must be unique. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | For an NR TDD cell, select the **SRS_WEIGHT_UE_NUM_INCR_SW** option if to increase the number of RedCap UEs for which RedCap UE weight optimization takes effect is required. |
| DL Exp SRS Weight Parameter Group ID | **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** | Set this parameter based on the network plan.<br>If the **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** parameter is set to a value other than **255**, the value of the **NRDUCellRedCap**.***DlExpSrsWeightParamGrpId*** parameter must be the same as that of the **gNBRedCapServIdenCfg**.***ServParamGroupId*** parameter in a **gNBRedCapServIdenCfg** MO. |
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** | Select the **REDCAP_UE_BWP_SCH_OPT_SW** option if the loads on different RedCap BWP1s in a cell are unbalanced. |

**Table 4-28** Parameters used for activation (scenario-specific optimization functions — resource optimization (downlink))

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **REDCAP_PDCCH_SYM_NUM_ADJ_SW** option be deselected. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | Select the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of this parameter when the network coverage is good (that is, the average PDCCH CCE aggregation level is low). |
| BWP Power Saving Switch | **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** | Select the **BWP2_COM_PDCCH_SCH_SW** option of this parameter when the network coverage is good (that is, the average PDCCH CCE aggregation level is low) and there are UEs using BWP2. |
| PDCCH Algorithm Switch | **NRDUCellPdcch**.***PdcchAlgoSwitch*** | It is recommended that the **UL_DL_COMM_CCE_SHARING_SW** option be selected when scheduling in common PDCCH symbols for RedCap UEs is enabled. |
| PDCCH Algorithm Enhancement Switch | **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** | For an NR TDD cell in a 5G ToB scenario, when the number of symbols occupied by the PDCCH is always two or three and the network coverage is favorable (that is, the average PDCCH aggregation level is low), it is recommended that the **MULTI_SEARCH_SPACE_SW** option be selected. |
| CORESET0 Offset Policy | **NRDUCellPdcchAlgo**.***Coreset0OffsetPol*** | Set this parameter to **EDGE_FIRST** if RedCap BWP0 is located to either end of cell BWP1 in the frequency domain. |
| PDCCH Algorithm Extension Switch | **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** | For an NR TDD cell, it is recommended that the **REDCAP_PDCCH_AVOID_SW** option be selected if CCE resources are insufficient. |
| Downlink Scheduling Optimization Factor | **NRDUCellRedCap**.***DlSchOptFactor*** | Set this parameter based on the network plan. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **REDCAP_RLC_RETRANS_GUAR_SW** option be selected if service drops occur due to abnormal RLC retransmissions for RedCap UEs. |

**Table 4-29** Parameters used for activation (scenario-specific optimization functions — resource optimization (uplink))

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| SR Resource Algo Switch | **NRDUCellPucch**.***SrResoureAlgoSwitch*** | It is recommended that the **REDCAP_SR_CODE_RSV_SW** option be selected. |
| PUCCH Algorithm Switch | **NRDUCellPucch**.***PucchAlgoSwitch*** | For an NR TDD cell, it is recommended that the **BWP2_PUCCH_POS_OPT_SW** option be selected when the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected and BWP2 is configured at the cell bandwidth edges. |
| RedCap UE SRS Resource Alloc Policy | **NRDUCellHighSpeedMob**.***RedCapUeSrsResAllocPol*** | If a high-speed NR TDD cell needs to support the access of more RedCap UEs, set this parameter to **LEVEL2** or **LEVEL3**. |
| High Speed Algorithm Optimization Switch | **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** | If a high-speed NR TDD cell needs to support the access of more RedCap UEs, select the **SRS_RES_EXT_SW** option. |
| DL Scheduling Resource Algorithm Switch | **NRDUCellDlSchRes**.***DlSchResAlgoSw*** | If a high-speed NR TDD cell needs to support the access of more RedCap UEs, select the **DCCH_TYPE1_SCH_SW** option. |

**Table 4-30** Parameters used for activation (scenario-specific optimization functions — networking or UE compatibility optimization)

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Common Channel Interference Avoid Switch | **NRDUCellAlgoSwitch**.***CommChnIntrfAvoidSwitch*** | It is recommended that the **PUCCH_INTRF_AVOID_SW** option be selected when there is co-channel interference from LTE to the NR PUSCH/PDSCH or external interference. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | For an NR TDD cell, when the SSB frequency is configured in the middle of the frequency band, the RedCap BWP0 does not include CORESET 0. In this case, it is recommended that the **REDCAP_BWP0_INCLUDE_CD_SSB_SW** option be selected. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | If there are RedCap UEs that require that downlink RedCap BWP1s must contain PDCCH CSS, it is recommended that the **REDCAP_PDCCH_CSS_CONFIG_SW** option be selected. |
| Mobility Function Indication | **NRExternalNCell**.***MobilityFunInd*** | It is recommended that the **REDCAP_NO_HO_SW** option be selected when handover-related counters deteriorate because the RedCap deployment status of neighboring cells fails to be obtained through the Xn interface and RedCap UEs are handed over to cells where RedCap is not enabled. |
| Compatibility Algorithm Switch | **gNodeBParam**.***CompatibilityAlgoSwitch*** | It is recommended that the **NCD_SSB_SIB1_COMPAT_SW** option be selected. This option is selected by default. |
| PRACH Algorithm Extension Switch | **NRDUCellPrach**.***PrachAlgoExtSwitch*** | It is recommended that the **PRACH_ALLOC_DIR_OPT_SW** option be selected under the guidance of Huawei engineers if there are UEs performing strong verification on PRACH positions on the network. |
| Mobility Algorithm Switch | **gNBMobilityComm-Param**.***MobilityAlgoSwitch*** | To increase the proportion of RedCap UEs identified as capable of device-pipe synergy in handover scenarios, select the **REDCAP_SPEC_UE_IDENTIFY_OPT_SW** option. |

### 4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

Before applying mobility management in inactive mode to RedCap UEs, the RAN notification area ID of the local cell must be configured. For details, see the MML configuration examples for mobility management in inactive mode in *SA Mobility Management in Inactive Mode*.

Before applying QCI-based RedCap DRX configuration to RedCap UEs, ensure that the corresponding QCIs have been configured. For details about QCI configuration, see the MML command examples in *QoS Management* .

- RedCap Introduction

```
//Enabling RedCap Introduction
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_SW-1;
//For an NR TDD cell, enabling multi-BWP1 PDCCH transmission
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-1;
//For an NR TDD cell, enabling enhanced multi-BWP1 PDCCH transmission
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_ENH_SW-1;
//For an NR TDD cell, turning on UL_DL_CCE_SHARING_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
//For an NR TDD cell, enabling BWP0 SRS resource allocation
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=BWP0_SRS_ALLOC_SW-1;
```

- Basic signaling procedures

```

//Changing the UE inactivity timer length to 10s, which should be adjusted based on the network
plan (This command takes QCI 9 as an example.)
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, RedCapUeInactivityTimer=10;
//Modifying the configuration of the gNodeB connection status timer with Sig UE No NAS Msg Trans
Timer set to 8
MOD GNBCONNSTATETIMER: SigUeNoNasMsgTransTmr=8;
//Enabling RRC_INACTIVE connection management
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=REDCAP_RRC_INACTIVE_SWITCH-1;
```

- Mobility management

```
//Configuring the low-speed UE decision timer, low-speed UE RSRP difference threshold, and non-
edge UE decision RSRP threshold for RRM measurement relaxation
MOD NRCELLRESELCONFIG: NrCellId=0, RrmRelaxLSUeJudgeTimer=5S,
RrmRelaxLSUeRsrpDiffThld=6DB, RrmRelaxNonEdgeRsrpThld=7;
//Setting the RedCapNcdSsbA1RsrpThld, RedCapNcdSsbA2RsrpThld, and RedCapNcdSsbA3Offset
parameters
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
RedCapNcdSsbA1RsrpThld=-96, RedCapNcdSsbA2RsrpThld=-100, RedCapNcdSsbA3Offset=2;
//Turning on REDCAP_NCD_SSB_MEAS_SW
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_NCD_SSB_MEAS_SW-1;
//(Optional) Turning on REDCAP_NCD_SSB_SINR_MEAS_SW when the cell suffers co-channel
interference and the SINRs of UEs are low
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_NCD_SSB_SINR_MEAS_SW-1;
//Setting the RedCapNcdSsbA1SinrThld and RedCapNcdSsbA2SinrThld parameters
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
RedCapNcdSsbA1SinrThld=-6, RedCapNcdSsbA2SinrThld=-9;
//(Optional) Selecting the XN_INTRA_GNB_HO_CTRL_SW option for Xn-based handovers and intra-
base-station handovers and the NG_INTER_RAT_HO_COMPAT_SW option for NG-based handovers
and inter-RAT handovers and setting the NRDUCellRedCap.CellBarredRedCap parameter if the
handover policy for RedCap UEs needs to be adjusted based on the Redcap-Bcast-Information IE or
the setting of the CellBarredRedCap parameter
MOD NRCELLREDCAP: NrCellId=0,
RedCapAlgoSwitch=XN_INTRA_GNB_HO_CTRL_SW-1&NG_INTER_RAT_HO_COMPAT_SW-1;
```

```
MOD NRDUCELLREDCAP: NrDuCellId=0, CellBarredRedCap=NOT_BARRED;
//(Optional) Turning on REDCAP_1RX_NO_HO_SW and REDCAP_2RX_NO_HO_SW if 1RX and 2RX RedCap UE handovers to external NR cells are prohibited
ADD NREXTERNALNCELL: Mcc="460", Mnc="00", gNBId=1, CellId=0, PhysicalCellId=0, Tac=0, SsbFreqPos=8816;
MOD NREXTERNALNCELL: Mcc="460", Mnc="00", gNBId=1, CellId=0, Tac=0, SsbFreqPos=8816, MobilityFunInd=REDCAP_1RX_NO_HO_SW-1&REDCAP_2RX_NO_HO_SW-1;
//Setting the RedCapMeasActvFlag parameter
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77, RedCapMeasActvFlag=ACTIVE;
//Setting the RedCapAccessAllowedFlag parameter (using NR TDD as an example)
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77, RedCapAccessAllowedFlag=ALLOWED;
//Running the following base-station-level command to configure a frequency priority group and NR frequency in the group (The following uses NR FDD as an example. If no gNodeB frequency priority group has been added, run the ADD GNBFREQPRIORITYGROUP command. Otherwise, run the MOD GNBFREQPRIORITYGROUP command.)
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1, RatType=NR, SsbFreqPos=2000;
MOD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1, RatType=NR, SsbFreqPos=2000;
//Running the following base-station-level commands to configure RFSP-, QCI-, and operator-specific RedCap frequency priorities
//Method 1: Configuring RFSP-specific RedCap frequency priorities (If no gNodeB RFSP configuration has been added, run the ADD GNBRFSPCONFIG command. Otherwise, run the MOD GNBRFSPCONFIG command.)
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=2, RedCapGnbFreqPriGroupId=1;
MOD GNBRFSPCONFIG: OperatorId=0, RfspIndex=2, RedCapGnbFreqPriGroupId=1;
//Method 2: Configuring QCI-specific RedCap frequency priorities (If no QCI-specific bearer has been added for the NR cell, run the ADD NRCELLQCIBEARER command. Otherwise, run the MOD NRCELLQCIBEARER command.)
ADD NRCELLQCIBEARER: NrCellId=0, Qci=128, RedCapGnbFreqPriGroupId=1;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=128, RedCapGnbFreqPriGroupId=1;
//Method 3: Configuring operator-specific RedCap frequency priorities (If no operator-specific policy has been added for the NR cell, run the ADD NRCELLOPPOLICY command. Otherwise, run the MOD NRCELLOPPOLICY command.)
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=0, RedCapGnbFreqPriGroupId=1;
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=0, RedCapGnbFreqPriGroupId=1;
//(Optional) Turning on EMC_ADMISSION_GUAR_SW if there are RedCap UEs performing emergency calls on the network
MOD NRDUCELLRAC: NrDuCellId=0, RacAlgoSwitch=EMC_ADMISSION_GUAR_SW-1;
//Configuring the QCI priority for handovers of RedCap UEs (using QCI 128 and priority 21 as an example)
MOD GNBQCIBEARER: Qci=128, QciPriorityForRedCapHo=21;
```

- Experience enhancement

```
//Enabling RedCap MU-MIMO
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_DL_MU_SW-1&REDCAP_UL_MU_SW-1, RedCapSchAlgoSwitch=REDCAP_UL_MU_GRP_PAIR_OPT_SW-1&REDCAP_UL_MU_GRP_PAIR_AVOID_SW-1;
//Configuring the maximum available bandwidth for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUeMaxAvailBandwidth=40M;
//(Optional) Setting the NcdSsbBwpPreferMaxUeNum parameter when service experience of RedCap UEs needs to be improved in SSB multi-beam scenarios with a small number of RedCap UEs (This command takes the value 1 of NcdSsbBwpPreferMaxUeNum as an example.)
MOD NRDUCELLREDCAP: NrDuCellId=0, NcdSsbBwpPreferMaxUeNum=1;
//(Optional)(Recommended) For an NR TDD cell, turning on FORMAT1_RB_RES_CTRL_SW if the maximum bandwidth available to RedCap UEs is greater than 20 MHz
MOD GNODEBALGO: PucchSwitch=FORMAT1_RB_RES_CTRL_SW-1;
//(Optional)(Recommended) For an NR TDD cell, turning on INTER_TRP_RB_SHARE_POLICY_SW if the maximum bandwidth available to RedCap UEs is greater than 20 MHz
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-1;
//(Optional) Turning on REDCAP_PUCCH_CSI_RB_ADJ_SW if the maximum bandwidth available to RedCap UEs is greater than 20 MHz and the number of UEs on the RedCap-dedicated BWP is greater than or equal to 100
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=REDCAP_PUCCH_CSI_RB_ADJ_SW-1;
//(Optional) For an NR TDD cell, setting this parameter to RB1 if the maximum bandwidth available to RedCap UEs is greater than 40 MHz and there are multiple RedCap UEs performing downlink large-packet services
```

```
MOD NRDUCELLREDCAP: NrDuCellId=0, PucchFormat4AckRbNum=RB1;
//(Optional)(Recommended) For an NR TDD cell, turning on CSI_RES_BALANCE_SW if the maximum bandwidth available to RedCap UEs is greater than 20 MHz
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_RES_BALANCE_SW-1;
//(Optional)(Recommended) For an NR TDD cell, turning on REDCAP_PUCCH_RES_GUAR_SW if a large number of RedCap BWPs are configured on a BBP serving this cell (that is, the maximum bandwidth available to RedCap UEs is greater than 20 MHz for multiple cells on this BBP)
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=REDCAP_PUCCH_RES_GUAR_SW-1;
//(Optional)(Recommended) For an NR TDD cell, turning on PUCCH_F1_RB_ADAPT_SW if the maximum bandwidth available to RedCap UEs is greater than or equal to 20 MHz
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_F1_RB_ADAPT_SW-1;
//(Optional)(Recommended) For an NR TDD cell, turning on PUCCH_CSI_RB_ADAPT_SW if the maximum bandwidth available to RedCap UEs is greater than 20 MHz
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_CSI_RB_ADAPT_SW-1;
//(Optional)(Recommended) For an NR TDD cell, turning on FORMAT1_SR_RES_OPT_SW if the maximum bandwidth available to RedCap UEs is greater than or equal to 20 MHz
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_SR_RES_OPT_SW-1;
//(Optional)(Recommended) Turning on APER_CSIRS_FLEX_OPT_SW if the maximum bandwidth available to RedCap UEs is greater than or equal to 20 MHz and the cell is heavily loaded
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=APER_CSIRS_FLEX_OPT_SW-1;
//(Optional) (Recommended) For an NR TDD cell, turning on PUCCH_RB_REPOSITION_SW if the maximum bandwidth available to RedCap UEs is equal to 20 MHz
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_RB_REPOSITION_SW-1;
//(Optional) For an NR TDD cell, turning on FORMAT1_SR_CHN_ALLOC_OPT_SW and PUCCH_F1_RB_ADAPT_SW if BWP2 is configured in the cell and PUCCH_RB_REPOSITION_SW is turned on
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_SR_CHN_ALLOC_OPT_SW-1;
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_F1_RB_ADAPT_SW-1;
//Turning on UL_REDCAP_PREALLOCATION_SW
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_REDCAP_PREALLOCATION_SW-1;
//Configuring the minimum preallocation period
MOD NRDUCELLPUSCH: NrDuCellId=0, MinPreallocationPeriod=5;
//Configuring the maximum number of UEs in the preallocation queue
MOD NRDUCELLPUSCH: NrDuCellId=0, PreschUeNumUpperLimit=2;
//Configuring the amount of UE data to be scheduled for uplink preallocation
MOD NRDUCELLPUSCH: NrDuCellId=0, PreallocationSize=100;
//Configuring the uplink preallocation policy for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUlPreallocPolicy=2RX_REDCAP;
//Modifying the downlink scheduling policy for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=REDCAP_DL_SCH_POL_SW-0;
//Adding a DRX parameter group for a long DRX cycle
ADD GNBDRXPARAMGROUP: DrxParamGroupId=2, LongCycle=MS320;
ADD GNBDUDRXPARAMGROUP: DuDrxParamGroupId=2, OnDurationTimer=MS10, DrxInactivityTimer=MS100, DrxRetransTimer=SL16;
//Binding the DRX parameter group to the QCI (QCI 100 is used as an example)
MOD NRCELLQCIBEARER: NrCellId=0, Qci=100, RedCapDrxParamGroupId=2;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, RedCapDuDrxParamGroupId=2;
//Adding gNodeB-level QCI configurations for bearers
ADD GNBQCIBEARER: Qci=201;
//Configuring QoS parameter mapping for network slice-level 5QIs
ADD GNBOP5QICONFIG: Op5qiConfigIndex=1, Nr5qi=8, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0, Qci=65535, RedCapQci=201;
//Adding QCI bearer configurations for NRCellQciBearer and NRDUCellQciBearer MOs
ADD NRCELLQCIBEARER: NrCellId=0, Qci=201, AmPdcpParamGroupId=5, UmPdcpParamGroupId=1;
ADD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=201, MacParamGroupId=9, AmRlcParamGroupId=3, UmRlcParamGroupId=0;
//(Optional) Enabling SSB interference avoidance for RedCap UEs if the interference from the NCD-SSBs of neighboring cells to the PDSCH of the serving cell needs to be reduced
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_SSB_INTRF_AVOID_SW-1;
//(Optional) For an NR TDD cell, turning on REDCAP_BWP_NON_OVERLAP_SW when the network load is light and multiple cells on the network offer contiguous coverage
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=REDCAP_BWP_NON_OVERLAP_SW-1;
//For an NR TDD cell, turning on REDCAP_POWER_AGGREGATION_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=REDCAP_POWER_AGGREGATION_SW-1;
//For an NR TDD cell, setting RedCapPwrBoostingCceThld
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, RedCapPwrBoostingCceThld=50;
//(Optional) For an NR TDD cell, turning on CD_BWP_PDCCH_SYM_ADAPTIVE_SW when CCE resources are insufficient
MOD NRDUCELLREDCAP: NrDuCellId=0,
```

```
RedCapAlgoExtSwitch=CD_BWP_PDCCH_SYM_ADAPTIVE_SW-1;
//(Optional) For an NR TDD cell, if RedCap UE weight optimization is required, turning on
REDCAP_SRS_WEIGHT_SW (using QCI 8 as an example) and configuring DlBigPktDataVolThld
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=8,
ServiceDiffAlgoSwitch=REDCAP_SRS_WEIGHT_SW-1;
MOD GNBREDCAPSERVIDENCFG: ServParamGroupId=0, DlBigPktDataVolThld=1M;
MOD NRDUCELLREDCAP: NrDuCellId=0, DlExpSrsWeightParamGrpId=0;
//(Optional) For an NR TDD cell, turning on SRS_WEIGHT_UE_NUM_INCR_SW if the function of
increasing the number of RedCap UEs for which RedCap UE weight optimization takes effect is
required
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=SRS_WEIGHT_UE_NUM_INCR_SW-1;
//(Optional) Turning on REDCAP_UE_BWP_SCH_OPT_SW if the loads on different RedCap BWP1s in a
cell are unbalanced.
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=REDCAP_UE_BWP_SCH_OPT_SW-1;
```

- Scenario-specific optimization functions — Resource optimization (downlink)

```
//(Optional) For an NR TDD cell, turning off REDCAP_PDCCH_SYM_NUM_ADJ_SW if the number of
PDCCH symbols occupied by RedCap UEs is expected to be equal to the number of PDCCH symbols in
the cell
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_PDCCH_SYM_NUM_ADJ_SW-0;
//(Optional) For an NR TDD cell, turning on REDCAP_COMM_PDCCH_SYM_SCH_SW and
UL_DL_COMM_CCE_SHARING_SW when the network coverage is good (that is, the average PDCCH
CCE aggregation level is low)
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_COMM_PDCCH_SYM_SCH_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_COMM_CCE_SHARING_SW-1;
//(Optional) For an NR TDD cell, turning on BWP2_COM_PDCCH_SCH_SW when the network
coverage is good (that is, the average PDCCH CCE aggregation level is low) and there are UEs using
BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, BwpPwrSavingSw=BWP2_COM_PDCCH_SCH_SW-1;
//(Optional) (Recommended) For an NR TDD cell in a 5G ToB scenario, turnning on
MULTI_SEARCH_SPACE_SW when the number of symbols occupied by the PDCCH is always two or
three and the network coverage is favorable (that is, the average PDCCH aggregation level is low)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SEARCH_SPACE_SW-1;
//(Optional) For an NR TDD cell, enabling CORESET 0 offset policy configuration if RedCap BWP0 is
located to either end of cell BWP1 in the frequency domain
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, Coreset0OffsetPol=EDGE_FIRST;
//(Optional)(Recommended) For an NR TDD cell, turning on REDCAP_PDCCH_AVOID_SW if CCE
resources are insufficient
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=REDCAP_PDCCH_AVOID_SW-1;
//(Optional) Setting the DlSchOptFactor parameter (The value 50 is used as an example. Set this
parameter based on the network plan.) to reduce the impact of RedCap UEs on non-RedCap UEs and
ensure the average downlink throughput of non-RedCap UEs in a heavily loaded cell serving both
RedCap UEs and non-RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, DlSchOptFactor=50;
//(Optional) Enabling RedCap RLC retransmission guarantee if service drops occur due to abnormal
RLC retransmissions for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_RLC_RETRANS_GUAR_SW-1;
```

- Scenario-specific optimization functions — Resource optimization (uplink)

```
//For an NR TDD cell, enabling SR code channel reservation for RedCap
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=REDCAP_SR_CODE_RSV_SW-1;
//(Optional) For an NR TDD cell, turning on BWP2_PUCCH_POS_OPT_SW when the BWP2_SWITCH
option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected and BWP2 is configured
at the cell bandwidth edges
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=BWP2_PUCCH_POS_OPT_SW-1;
//(Optional) For a high-speed NR TDD cell, setting the RedCapUeSrsResAllocPol parameter (The value
LEVEL2 is used as an example.) to allow more RedCap UEs to access the cell
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, RedCapUeSrsResAllocPol=LEVEL2;
//(Optional) Turning on SRS_RES_EXT_SW if a high-speed NR TDD cell needs to support the access of
more RedCap UEs
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=SRS_RES_EXT_SW-1;
//(Optional) Turning on DCCH_TYPE1_SCH_SW if a high-speed NR TDD cell needs to support the
access of more RedCap UEs
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DlSchResAlgoSw=DCCH_TYPE1_SCH_SW-1;
```

- Scenario-specific optimization functions — Networking or UE compatibility optimization

```
//(Optional) For an NR TDD cell, enabling PUCCH interference avoidance when there is co-channel
interference from LTE to the NR PUSCH/PDSCH or external interference
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=101,
CommChnIntrfAvoidSwitch=PUCCH_INTRF_AVOID_SW-1;
//(Optional) For an NR TDD cell, when the SSB frequency is configured in the middle of the frequency
band, the RedCap BWP0 does not include CORESET 0. In this case, enable CD-SSB inclusion in the
RedCap BWP0.
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_BWP0_INCLUDE_CD_SSB_SW-1;
//(Optional) Enabling PDCCH CSS configuration for RedCap UEs if there are RedCap UEs that require
that downlink RedCap BWP1s must contain PDCCH CSS
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_PDCCH_CSS_CONFIG_SW-1;
//(Optional) Enabling prohibition of RedCap UE handovers to external NR cells when handover-
related counters deteriorate because the RedCap deployment status of neighboring cells fails to be
obtained through the Xn interface and RedCap UEs are handed over to cells where RedCap is not
enabled (using NR TDD as an example)
MOD NREXTERNALNCELL: Mcc="460", Mnc="00", gNBId=1, CellId=0, Tac=1, SsbFreqPos=8816,
MobilityFunInd=REDCAP_NO_HO_SW-1;
//Turning on NCD_SSB_SIB1_COMPAT_SW
MOD GNODEBPARAM: CompatibilityAlgoSwitch=NCD_SSB_SIB1_COMPAT_SW-1;
//(Optional)(Recommended) Turning on PRACH_ALLOC_DIR_OPT_SW under the guidance of Huawei
engineers if there are UEs performing strong verification on PRACH positions on the network
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=PRACH_ALLOC_DIR_OPT_SW-1;
//(Optional) Enabling RedCap UE device-pipe identification optimization to increase the proportion of
RedCap UEs identified as capable of device-pipe synergy in handover scenarios
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=REDCAP_SPEC_UE_IDENTIFY_OPT_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Basic signaling procedures

```
//Disabling RRC_INACTIVE connection management
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=REDCAP_RRC_INACTIVE_SWITCH-0;
```

- Mobility management

```
//(Optional) Turning off REDCAP_1RX_NO_HO_SW and REDCAP_2RX_NO_HO_SW
MOD NREXTERNALNCELL: Mcc="460", Mnc="00", gNBId=1, CellId=0, Tac=0, SsbFreqPos=8816,
MobilityFunInd=REDCAP_1RX_NO_HO_SW-0&REDCAP_2RX_NO_HO_SW-0;
//(Optional) Turning off XN_INTRA_GNB_HO_CTRL_SW for Xn-based handovers and intra-base-
station handovers and NG_INTER_RAT_HO_COMPAT_SW for NG-based handovers and inter-RAT
handovers
MOD NRCELLREDCAP: NrCellId=0,
RedCapAlgoSwitch=XN_INTRA_GNB_HO_CTRL_SW-0&NG_INTER_RAT_HO_COMPAT_SW-0;
//(Optional) Turning off EMC_ADMISSION_GUAR_SW
MOD NRDUCELLRAC: NrDuCellId=0, RacAlgoSwitch=EMC_ADMISSION_GUAR_SW-0;
//(Optional) Turning off REDCAP_NCD_SSB_MEAS_SW
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_NCD_SSB_MEAS_SW-0;
//(Optional) Turning off REDCAP_NCD_SSB_SINR_MEAS_SW
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_NCD_SSB_SINR_MEAS_SW-0;
```

- Experience enhancement

```
//(Optional) Turning off REDCAP_UE_BWP_SCH_OPT_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=REDCAP_UE_BWP_SCH_OPT_SW-0;
//(Optional) For an NR TDD cell, turning off SRS_WEIGHT_UE_NUM_INCR_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=SRS_WEIGHT_UE_NUM_INCR_SW-0;
//(Optional) For an NR TDD cell, turning off REDCAP_SRS_WEIGHT_SW
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=8,
ServiceDiffAlgoSwitch=REDCAP_SRS_WEIGHT_SW-0;
//(Optional) For an NR TDD cell, turning off CD_BWP_PDCCH_SYM_ADAPTIVE_SW
MOD NRDUCELLREDCAP: NrDuCellId=0,
RedCapAlgoExtSwitch=CD_BWP_PDCCH_SYM_ADAPTIVE_SW-0;
//(Optional) For an NR TDD cell, turning off REDCAP_BWP_NON_OVERLAP_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=REDCAP_BWP_NON_OVERLAP_SW-0;
//For an NR TDD cell, turning off REDCAP_POWER_AGGREGATION_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=REDCAP_POWER_AGGREGATION_SW-0;
//(Optional) Disabling SSB interference avoidance for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_SSB_INTRF_AVOID_SW-0;
//Modifying the downlink scheduling policy for RedCap UEs
```

```
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=REDCAP_DL_SCH_POL_SW-1;
//Turning off UL_REDCAP_PREALLOCATION_SW
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_REDCAP_PREALLOCATION_SW-0;
//(Optional) For an NR TDD cell, turning off PUCCH_RB_REPOSITION_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_RB_REPOSITION_SW-0;
//(Optional) Turning off APER_CSIRS_FLEX_OPT_SW
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsExtSwitch=APER_CSIRS_FLEX_OPT_SW-0;
//(Optional) For an NR TDD cell, turning off FORMAT1_SR_RES_OPT_SW
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=FORMAT1_SR_RES_OPT_SW-0;
//(Optional) For an NR TDD cell, turning off PUCCH_CSI_RB_ADAPT_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_CSI_RB_ADAPT_SW-0;
//(Optional) For an NR TDD cell, turning off PUCCH_F1_RB_ADAPT_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=PUCCH_F1_RB_ADAPT_SW-0;
//(Optional) For an NR TDD cell, turning off REDCAP_PUCCH_RES_GUAR_SW
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=REDCAP_PUCCH_RES_GUAR_SW-0;
//(Optional) For an NR TDD cell, turning off CSI_RES_BALANCE_SW
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=CSI_RES_BALANCE_SW-0;
//(Optional) Turning off REDCAP_PUCCH_CSI_RB_ADJ_SW
MOD NRDUCELLREDCAP: NrDuCellId=0,RedCapAlgoExtSwitch=REDCAP_PUCCH_CSI_RB_ADJ_SW-0;
//(Optional) For an NR TDD cell, turning off INTER_TRP_RB_SHARE_POLICY_SW
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=INTER_TRP_RB_SHARE_POLICY_SW-0;
//(Optional) For an NR TDD cell, turning off FORMAT1_RB_RES_CTRL_SW
MOD GNODEBALGO: PucchSwitch=FORMAT1_RB_RES_CTRL_SW-0;
//Restoring the maximum bandwidth available for RedCap UEs to the default value
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUeMaxAvailBandwidth=20M;
//Disabling RedCap MU-MIMO
MOD NRDUCELLREDCAP: NrDuCellId=0,
RedCapAlgoSwitch=REDCAP_DL_MU_SW-0&REDCAP_UL_MU_SW-0,
RedCapSchAlgoSwitch=REDCAP_UL_MU_GRP_PAIR_OPT_SW-0&REDCAP_UL_MU_GRP_PAIR_AVOID_S
W-0;
```

- Scenario-specific optimization functions — Resource optimization (downlink)

```
//(Optional) Disabling RedCap RLC retransmission guarantee
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_RLC_RETRANS_GUAR_SW-0;
//(Optional) For an NR TDD cell, turning off REDCAP_PDCCH_AVOID_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=REDCAP_PDCCH_AVOID_SW-0;
//For an NR TDD cell, disabling CORESET 0 offset policy configuration
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, Coreset0OffsetPol=DEFAULT;
//(Optional) For an NR TDD cell, turning off MULTI_SEARCH_SPACE_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=MULTI_SEARCH_SPACE_SW-0;
//(Optional) For an NR TDD cell, turning off BWP2_COM_PDCCH_SCH_SW
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, BwpPwrSavingSw=BWP2_COM_PDCCH_SCH_SW-0;
//(Optional) For an NR TDD cell, turning off REDCAP_COMM_PDCCH_SYM_SCH_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_COMM_PDCCH_SYM_SCH_SW-0;
//(Optional) For an NR TDD cell, enabling RedCap PDCCH symbol number adjustment if the number
of PDCCH symbols occupied by RedCap UEs is expected to be adaptively adjusted based on the cell
bandwidth and the number of PDCCH symbols in the cell
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_PDCCH_SYM_NUM_ADJ_SW-1;
```

- Scenario-specific optimization functions — Resource optimization (uplink)

```
//(Optional) Turning off DCCH_TYPE1_SCH_SW
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DlSchResAlgoSw=DCCH_TYPE1_SCH_SW-0;
//(Optional) Turning off SRS_RES_EXT_SW
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, HighSpeedAlgoOptSw=SRS_RES_EXT_SW-0;
//(Optional) For an NR TDD cell, restoring the default SRS resource allocation policy for RedCap UEs
MOD NRDUCELLHIGHSPEEDMOB: NrDuCellId=0, RedCapUeSrsResAllocPol=LEVEL1;
//(Optional) For an NR TDD cell, turning off BWP2_PUCCH_POS_OPT_SW
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=BWP2_PUCCH_POS_OPT_SW-0;
//For an NR TDD cell, disabling SR code channel reservation for RedCap
MOD NRDUCELLPUCCH: NrDuCellId=0, SrResoureAlgoSwitch=REDCAP_SR_CODE_RSV_SW-0;
```

- Scenario-specific optimization functions — Networking or UE compatibility optimization

```
//(Optional) Disabling RedCap UE device-pipe identification optimization
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=REDCAP_SPEC_UE_IDENTIFY_OPT_SW-0;
//(Optional) Turning off PRACH_ALLOC_DIR_OPT_SW
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=PRACH_ALLOC_DIR_OPT_SW-0;
//Turning off NCD_SSB_SIB1_COMPAT_SW
MOD GNODEBPARAM: CompatibilityAlgoSwitch=NCD_SSB_SIB1_COMPAT_SW-0;
//(Optional) Disabling prohibition of RedCap UE handovers to external NR cells (using NR TDD as an
```

```
example)
MOD NREXTERNALNCELL: Mcc="460", Mnc="00", gNBId=1, CellId=0, Tac=1, SsbFreqPos=8816,
MobilityFunInd=REDCAP_NO_HO_SW-0;
//(Optional) Disabling PDCCH CSS configuration for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_PDCCH_CSS_CONFIG_SW-0;
//(Optional) For an NR TDD cell, disabling CD-SSB inclusion in the RedCap BWP0
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_BWP0_INCLUDE_CD_SSB_SW-0;
//(Optional) For an NR TDD cell, disabling PUCCH interference avoidance
MOD NRDUCELLALGOSWITCH: NrDuCellId=101,
CommChnIntrfAvoidSwitch=PUCCH_INTRF_AVOID_SW-0;
```

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```
//For an NR TDD cell, disabling BWP0 SRS resource allocation
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=BWP0_SRS_ALLOC_SW-0;
//For an NR TDD cell, turning off UL_DL_CCE_SHARING_SW
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-0;
//For an NR TDD cell, disabling enhanced multi-BWP1 PDCCH transmission
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_ENH_SW-0;
//For an NR TDD cell, disabling multi-BWP1 PDCCH transmission
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_MULTI_BWP1_ALGO_SW-0;
//Disabling the RedCap UE access function
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_SW-0;
```

#### 4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 4.4.2 Activation Verification

**RedCap Introduction:**

Observe the following counters to check whether the RedCap Introduction function has taken effect. If the value of any counter in the table is not 0, the RedCap Introduction function has taken effect.

#### Cell User Quantity Measurement

| Counter ID | Counter Name |
| --- | --- |
| 1911837919 | **N.User.RRCConn.RedCap.Avg** |
| 1911840354 | **N.User.RRCConn.RedCap.Max** |

#### Intra-Cell Network Slice User Number Measurement

| Counter ID | Counter Name |
| --- | --- |
| 1914453772 | **N.User.RedCap.RRCConn.Avg.Ns** |
| 1914453773 | **N.User.RedCap.RRCConn.Max.Ns** |

### Throughput and Data Volume Measurement

| Counter ID | Counter Name |
|---|---|
| 1911837921 | **N.ThpVol.DL.RedCap** |
| 1911837930 | **N.ThpVol.UL.RedCap** |
| 1911840378 | **N.ThpVol.UE.UL.SmallPkt.RedCap** |
| 1911840380 | **N.ThpTime.UE.UL.RmvSmallPkt.RedCap** |
| 1911840382 | **N.ThpVol.DL.LastSlot.RedCap** |
| 1911840387 | **N.ThpTime.DL.RmvLastSlot.RedCap** |

### RRC Setup Measurement

| Counter ID | Counter Name |
|---|---|
| 1911837922 | **N.RRC.SetupReq.Succ.RedCap** |
| 1911837923 | **N.RRC.SetupReq.Att.RedCap** |
| 1911837924 | **N.RRC.SetupReq.Msg.RedCap** |
| 1911844975 | **N.RRC.SetupSucc.ReEst.RedCap** |
| 1911844976 | **N.RRC.Setup.ReEst.RedCap** |

### RRC Reestablishment Measurement

| Counter ID | Counter Name |
|---|---|
| 1911844981 | **N.RRC.ReEst.Att.RedCap** |
| 1911844982 | **N.RRC.ReEst.Succ.RedCap** |
| 1911844983 | **N.RRC.NoCntxReEst.Att.RedCap** |
| 1911844984 | **N.RRC.NoCntxReEst.Succ.RedCap** |

### QoS Flow Measurement

| Counter ID | Counter Name |
|---|---|
| 1911837925 | **N.QosFlow.NormRel.RedCap** |
| 1911837926 | **N.QosFlow.AbnormRel.RedCap** |
| 1911837927 | **N.QosFlow.Est.Succ.RedCap** |

| Counter ID | Counter Name |
|---|---|
| 1911837928 | **N.QosFlow.Est.Att.RedCap** |
| 1911844939 | **N.QosFlow.FailEst.AMF.Redcap** |
| 1911844940 | **N.QosFlow.FailEst.RNL.RedCap** |
| 1911844941 | **N.QosFlow.FailEst.TNL.RedCap** |
| 1911844978 | **N.QosFlow.AbnormRel.RedCap.RNL** |
| 1911844979 | **N.QosFlow.AbnormRel.RedCap.TNL** |
| 1911844980 | **N.QosFlow.AbnormRel.RedCap.AMF** |

### UE Context Setup Measurement

| Counter ID | Counter Name |
|---|---|
| 1911844936 | **N.NGSig.ConnEst.Att.RedCap** |
| 1911844937 | **N.NGSig.ConnEst.Succ.RedCap** |

### RRC_INACTIVE Measurement

| Counter ID | Counter Name |
|---|---|
| 1911840365 | **N.RRC.ResumeReq.Succ.RedCap** |
| 1911840366 | **N.RRC.ResumeReq.Att.RedCap** |
| 1911840367 | **N.RRC.ResumeReq.Msg.RedCap** |

### Outgoing Intra-RAT/Inter-RAT Handover Measurement

| Counter ID | Counter Name |
|---|---|
| 1911840355 | **N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut.RedCap** |
| 1911840356 | **N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut.RedCap** |
| 1911840357 | **N.HO.IntraFreq.IntragNB.ExecSuccOut.RedCap** |
| 1911840368 | **N.HO.IntraFreq.Ng.IntergNB.ExecAttOut.RedCap** |

| Counter ID | Counter Name |
|---|---|
| 1911840369 | **N.HO.IntraFreq.IntragNB.ExecAttOut.RedCap** |
| 1911840370 | **N.HO.IntraFreq.Xn.IntergNB.ExecAttOut.RedCap** |
| 1911840383 | **N.HO.IntraFreq.IntragNB.PrepAttOut.RedCap** |
| 1911840384 | **N.HO.IntraFreq.Xn.IntergNB.PrepAttOut.RedCap** |
| 1911840385 | **N.HO.IntraFreq.Ng.IntergNB.PrepAttOut.RedCap** |
| 1911840371 | **N.HO.InterFreq.IntragNB.ExecSuccOut.RedCap** |
| 1911840372 | **N.HO.InterFreq.Xn.IntergNB.ExecSuccOut.RedCap** |
| 1911840373 | **N.HO.InterFreq.Ng.IntergNB.ExecSuccOut.RedCap** |
| 1911840358 | **N.HO.InterFreq.IntragNB.ExecAttOut.RedCap** |
| 1911840359 | **N.HO.InterFreq.Xn.IntergNB.ExecAttOut.RedCap** |
| 1911840360 | **N.HO.InterFreq.Ng.IntergNB.ExecAttOut.RedCap** |
| 1911840388 | **N.HO.InterFreq.IntragNB.PrepAttOut.RedCap** |
| 1911840389 | **N.HO.InterFreq.Xn.IntergNB.PrepAttOut.RedCap** |
| 1911840390 | **N.HO.InterFreq.Ng.IntergNB.PrepAttOut.RedCap** |
| 1911840363 | **N.HO.InterRAT.N2E.ExecSuccOut.RedCap** |
| 1911840364 | **N.HO.InterRAT.N2E.ExecAttOut.RedCap** |
| 1911840386 | **N.HO.InterRAT.N2E.PrepAttOut.RedCap** |
| 1911844977 | **N.RRCRedirection.N2E.Redcap** |

## PDCP Measurement

| Counter ID | Counter Name |
|---|---|
| 1911840352 | **N.PDCP.Vol.UL.TrfSDU.Tx.RedCap** |
| 1911840361 | **N.PDCP.Vol.DL.TrfSDU.Rx.RedCap** |

## PRB Measurement

| Counter ID | Counter Name |
|---|---|
| 1911840374 | **N.PRBUsed.DL.RedCap.Avg** |
| 1911840376 | **N.PRBUsed.UL.RedCap.Avg** |
| 1911849700 | **N.PRB.UL.RedCap.BWP.Avail.Avg** |
| 1911849702 | **N.PRB.DL.RedCap.BWP.Avail.Avg** |
| 1911849704 | **N.PRBUsed.UL.RedCap.BWP.Avg** |
| 1911849706 | **N.PRBUsed.DL.RedCap.BWP.Avg** |
| 1911853896 | **N.PRBUsed.UL.RedCap.PLMN** |
| 1911853894 | **N.PRBUsed.DL.RedCap.PLMN** |
| 1911853890 | **N.CCE.RedCap.Used.Avg** |
| 1911853892 | **N.CCE.RedCap.Avail.Avg** |

NOTE

The preceding counters are measured for RedCap UEs. .

**Idle-mode RRM measurement relaxation:**

Observe the signaling trace results to check whether the idle-mode RRM measurement relaxation function has taken effect.

**Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the SIB2 message. If SIB2 contains the **relaxedMeasurement-r17** IE, idle-mode RRM measurement relaxation has taken effect. Otherwise, idle-mode RRM measurement relaxation has not taken effect.

**----End**

**RedCap handover policy control:**

Observe the counters described in Table 4-31 to check whether RedCap handover policy control has taken effect.

If the value of any of these counters decreases, this function has taken effect.

**Table 4-31** Counters for observing the performance of RedCap handover policy control

| Counter ID | Counter Name |
|---|---|
| 1911840383 | **N.HO.IntraFreq.IntragNB.PrepAttOut.RedCap** |
| 1911840384 | **N.HO.IntraFreq.Xn.IntergNB.PrepAttOut.RedCap** |
| 1911840385 | **N.HO.IntraFreq.Ng.IntergNB.PrepAttOut.RedCap** |
| 1911840388 | **N.HO.InterFreq.IntragNB.PrepAttOut.RedCap** |
| 1911840389 | **N.HO.InterFreq.Xn.IntergNB.PrepAttOut.RedCap** |
| 1911840390 | **N.HO.InterFreq.Ng.IntergNB.PrepAttOut.RedCap** |
| 1911840386 | **N.HO.InterRAT.N2E.PrepAttOut.RedCap** |

**Multiple BWP1s for RedCap:**

- Using Counters

  Observe the following counters to check whether multiple BWP1s for RedCap has taken effect.

  For an NR TDD cell, if the value of any counter in the table is greater than **51**, multiple BWP1s for RedCap has taken effect.

  | Counter ID | Counter Name |
  |---|---|
  | 1911840391 | **N.PRB.UL.RedCap.Avail.Avg** |
  | 1911840531 | **N.PRB.DL.RedCap.Avail.Avg** |

- Using MML Commands

  Run the **DSP NRDUCELLREDCAP** command to check the value of **RedCap UE Active Bandwidth**. If the value is greater than **20M**, multiple BWP1s for RedCap has taken effect.

**SSB interference avoidance for RedCap UEs:**

After enabling SSB interference avoidance for RedCap UEs, use the drive test tool to check whether the number of RBs allocated to UEs with full-buffer services decreases. If it does, no scheduling has occurred in the local cell at the time-frequency positions where neighboring cells transmit NCD-SSBs but the local cell does not. This function has taken effect.

NOTE

The number of NCD-SSB beams is indicated by the *ssb-PositionInBurst* field in the ServingCellConfigCommonSIB message. For details, see 3GPP TS 38.331 V17.4.0.

**Uplink RedCap preallocation:**

After uplink RedCap preallocation is enabled, if the average number of PRBs used by uplink preallocation UEs (indicated by **N.PRB.UL.Prealloc.Used.Avg**) increases, this function has taken effect.

### 4.4.3 Network Monitoring

After RedCap Introduction is enabled, network monitoring involves the indicators listed in **Table 4-32**.

**Table 4-32** Indicators involved after RedCap Introduction is enabled

| Indicator | Formula |
| --- | --- |
| Service delay | (**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**)/1000 + (**N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**)/ 1000 |
| Uplink IBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100% |
| Downlink IBLER | (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100% |

| Indicator | Formula |
| --- | --- |
| Uplink RBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100% |
| Downlink RBLER | (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100% |
| Service drop rate of RedCap UEs | **N.QosFlow.AbnormRel.RedCap**/(**N.QosFlow.AbnormRel.RedCap** + **N.QosFlow.NormRel.RedCap**) × 100% |
| Service drop rate of a cell | **N.QosFlow.AbnormRel**/(**N.QosFlow.AbnormRel** + **N.QosFlow.NormRel**) × 100% |
| Uplink CCE allocation success rate | ((**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**) × 100% |
| Downlink CCE allocation success rate | ((**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**) × 100% |
| Contention-based random access success rate | (**N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att**) × 100% |
| Non-contention-based random access success rate | (**N.RA.Dedicated.Msg3**/**N.RA.Dedicated.Att**) × 100% |
| SRB PRB usage | (**N.PRB.DL.Used.Avg** - **N.PRB.DL.DrbUsed.Avg**)/**N.PRB.DL.Avail.Avg** |

| Indicator | Formula |
|---|---|
| Downlink spectral efficiency [kbits/(TTI x PRB)] | **N.ThpVol.DL.Cell**/ (**N.PRB.DL.DrbUsed.Avg** x **N.DL.PDSCH.Tti.Num**) |
| Average number of layers paired for downlink MIMO | **N.ChMeas.MIMO.DL.Pair.Layer**/ **N.ChMeas.MIMO.DL.Pair.PRB** |
| Average number of layers paired for uplink MIMO | **N.ChMeas.MIMO.UL.Pair.Layer**/ **N.ChMeas.MIMO.UL.Pair.PRB** |
| Average number of RBs paired for downlink MIMO | (1 x **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + 2 × **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + 3 × **N.ChMeas.MIMO.DL.MuPairing.3Layers.RB** + 4 × **N.ChMeas.MIMO.DL.MuPairing.4Layers.RB** + 5 × **N.ChMeas.MIMO.DL.MuPairing.5Layers.RB** + 6 × **N.ChMeas.MIMO.DL.MuPairing.6Layers.RB** + 7 × **N.ChMeas.MIMO.DL.MuPairing.7Layers.RB** + 8 × **N.ChMeas.MIMO.DL.MuPairing.8Layers.RB**)/ (**N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.3Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.4Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.5Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.6Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.7Layers.RB** + **N.ChMeas.MIMO.DL.MuPairing.8Layers.RB**) |

| Indicator | Formula |
| --- | --- |
| Average number of RBs paired for uplink MIMO | (1 × **N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + 2 × **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + 3 × **N.ChMeas.MIMO.UL.MuPairing.3Layers.RB** + 4 × **N.ChMeas.MIMO.UL.MuPairing.4Layers.RB** + 5 × **N.ChMeas.MIMO.UL.MuPairing.5Layers.RB** + 6 × **N.ChMeas.MIMO.UL.MuPairing.6Layers.RB** + 7 × **N.ChMeas.MIMO.UL.MuPairing.7Layers.RB** + 8 × **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**)/(**N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.3Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.4Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.5Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.6Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.7Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**) |
| Total number of RBs paired for uplink MU-MIMO | **N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.3Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.4Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.5Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.6Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.7Layers.RB** + **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB** |

| Indicator | Formula |
|---|---|
| Average uplink cell throughput | **Cell Uplink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average uplink UE throughput | **User Uplink Average Throughput (DU)** |
| Average downlink UE throughput | **User Downlink Average Throughput (DU)** |
| Average uplink throughput of RedCap UEs | (**N.ThpVol.UL.RedCap** – **N.ThpVol.UE.UL.SmallPkt.RedCap**)/ **N.ThpTime.UE.UL.RmvSmallPkt.RedCap** |
| Average downlink throughput of RedCap UEs | (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/ **N.ThpTime.DL.RmvLastSlot.RedCap** |
| Uplink RB usage | **Uplink Resource Block Utilizing Rate (DU)** |
| Downlink RB usage | **Downlink Resource Block Utilizing Rate (DU)** |
| Average number of used uplink PRBs | **N.PRB.UL.Used.Avg** |
| Average number of used downlink PRBs | **N.PRB.DL.Used.Avg** |
| Average number of PDCCH CCEs used for common DCI | **N.CCE.CommDCI.Used.Avg** |
| Average number of available uplink PUSCH PRBs | **N.PRB.UL.PUSCH.Avail.Avg** |
| Average number of actually available PRBs for the PUSCH | **N.PRB.UL.PUSCH.Actual.Avail.Avg** |
| Number of available uplink PRBs within the BWP | **N.PRB.UL.RedCap.BWP.Avail.Avg** |

| Indicator | Formula |
|---|---|
| Uplink retransmission ratio | (**N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB** + **N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**) × 100% |
| Downlink retransmission ratio | (**N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**) × 100% |
| Average uplink MCS index | (0 × **N.ChMeas.PUSCH.MCS.0** + 1 × **N.ChMeas.PUSCH.MCS.1** + 2 × **N.ChMeas.PUSCH.MCS.2** + 3 × **N.ChMeas.PUSCH.MCS.3** + ... + 27 × **N.ChMeas.PUSCH.MCS.27** + 28 × **N.ChMeas.PUSCH.MCS.28**)/(**N.ChMeas.PUSCH.MCS.0** + **N.ChMeas.PUSCH.MCS.1** + **N.ChMeas.PUSCH.MCS.2** + **N.ChMeas.PUSCH.MCS.3** + ... + **N.ChMeas.PUSCH.MCS.27** + **N.ChMeas.PUSCH.MCS.28**) |

| Indicator | Formula |
|---|---|
| Average downlink MCS index | (0 x **N.ChMeas.PDSCH.MCS.0** + 1 x **N.ChMeas.PDSCH.MCS.1** + 2 x **N.ChMeas.PDSCH.MCS.2** + 3 x **N.ChMeas.PDSCH.MCS.3** + ... + 27 × **N.ChMeas.PDSCH.MCS.27** + 28 × **N.ChMeas.PDSCH.MCS.28**)/(**N.ChMeas.PDSCH.MCS.0** + **N.ChMeas.PDSCH.MCS.1** + **N.ChMeas.PDSCH.MCS.2** + **N.ChMeas.PDSCH.MCS.3** + ... + **N.ChMeas.PDSCH.MCS.27** + **N.ChMeas.PDSCH.MCS.28**) |
| Average downlink cell power | **N.DL.Pwr.Avg** |
| Number of RRC connection reestablishments | **N.RRC.ReEst.Att** |
| Average number of used PDCCH CCEs | **N.CCE.Used.Avg** |
| PDCCH CCE usage | **N.CCE.Used.Avg**/**N.CCE.Avail.Avg** × 100% |
| Uplink CCE aggregation level | (**N.CCE.UL.AggLvl16Num** × 16+ **N.CCE.UL.AggLvl8Num** × 8+ **N.CCE.UL.AggLvl4Num** × 4+ **N.CCE.UL.AggLvl2Num** × 2+ **N.CCE.UL.AggLvl1Num**)/(**N.CCE.UL.AggLvl16Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl1Num**) |
| Downlink CCE aggregation level | (**N.CCE.DL.AggLvl16Num** × 16 + **N.CCE.DL.AggLvl8Num** × 8 + **N.CCE.DL.AggLvl4Num** × 4 + **N.CCE.DL.AggLvl2Num** × 2+ **N.CCE.DL.AggLvl1Num**)/(**N.CCE.DL.AggLvl16Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl1Num**) |
| Average uplink rank | SUM(***k*** **x N.PUSCH.InitTbUl.Rank.*k***)/SUM(**N.PUSCH.InitTbUl.Rank.*k***)<br>***k*** = 1 to 2 |
| Average downlink rank | SUM(***k*** **x N.PDSCH.InitTbDl.Rank.*k***)/SUM(**N.PDSCH.InitTbDl.Rank.*k***)<br>***k*** = 1 to 4 |

| Indicator | Formula |
| --- | --- |
| Duration in which the BWP2 takes effect | **N.Bwp2.Dur.Total** |
| Average interference and noise received on each uplink PRB | **N.UL.NI.Avg** |

NOTE

The downlink spectral efficiency can be expressed as bits per second per Hz by using the following formula: **N.ThpVol.DL.Cell**/(**N.PRB.DL.DrbUsed.Avg** x **N.DL.PDSCH.Tti.Num** x 1000 x 2000/360000). The idea behind this formula is as follows: 1000 is introduced for a kbit-to-bit conversion, because the unit of the **N.ThpVol.DL.Cell** counter is kbit. 2000 is introduced for a TTI-to-second conversion, because one second has 2000 TTIs (1s = 1000 ms, and 1 TTI = 0.5 ms). 360000 is introduced for a PRB-to-Hz conversion, because one PRB spans 360,000 Hz in the frequency domain (1 PRB = 12 subcarriers, and 1 subcarrier = 30,000 Hz).

# 5 Premium RedCap

Compared with RedCap Introduction, Premium RedCap offers enhanced performance in terms of capacity, coverage, energy saving, mobility management, and BWP load balancing, as listed in **Table 5-1**.

**Table 5-1** TDD-related Premium RedCap function overview

| Category | Function Name | Section |
|---|---|---|
| Capacity enhancement | UE-number-based BWP1 adaptation | **UE-Number-based BWP1 Adaptation** |
| | PRB-and-CCE-resource-based BWP1 adaptation | **PRB-and-CCE-Resource-based BWP1 Adaptation** |
| Coverage enhancement | Dynamic uplink slot aggregation | **5.2.1.1 Dynamic Uplink Slot Aggregation (Trial Function in NR TDD)** |
| | Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD) | **5.2.1.2 Power Class 2 based on Device-Pipe Synergy for RedCap UEs (Low-Frequency TDD)** |
| RedCap Power Saving | Inactive RA-SDT (trial) | **5.3.1 Inactive RA-SDT (Trial)** |
| | Assistance TRS | **5.3.2 Assistance TRS** |
| | Connected-mode RRM measurement relaxation | **5.3.3 Connected-mode RRM Measurement Relaxation** |
| | RedCap flexible DRX configuration | **5.3.4 RedCap Flexible DRX Configuration** |
| | RLM relaxation | **5.3.5 RLM Relaxation** |
| | eDRX in idle mode (trial) | **5.3.6 eDRX in Idle Mode (Trial)** |

| Category | Function Name | Section |
| --- | --- | --- |
| | RedCap UE release preference | **5.3.7 RedCap UE Release Preference** |
| | RedCap SSSG switching | **5.3.8 RedCap SSSG Switching** |
| | RedCap UL skipping | **5.3.9 RedCap UL Skipping** |
| | RedCap DRX adaptation | **5.3.10 RedCap DRX Adaptation** |
| RedCap mobility management enhancement | Uplink-experience-based inter-frequency handover | **5.4.1 Uplink-Experience-based Inter-Frequency Handover** |
| BWP load balancing | UE-number-based periodic BWP load balancing | **5.5.1 UE-Number-based Periodic BWP Load Balancing** |
| | QCI-specific UE-number-based periodic BWP load balancing | **5.5.2 QCI-specific UE-Number-based Periodic BWP Load Balancing** |

# 5.1 Capacity Enhancement

## 5.1.1 Principles

When "multiple BWP1s for RedCap" enabled, a fixed number of RedCap BWPs is set. When factors such as the number of UEs and load change, this can lead to a decrease in overall cell throughput. Therefore, capacity enhancement functions are introduced to address this issue. Capacity enhancement consists of the following functions:

- UE-number-based BWP1 adaptation
- PRB-and-CCE-resource-based BWP1 adaptation

### UE-Number-based BWP1 Adaptation

In 5G to customer (5GtoC) scenarios, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M** for a cell, it is recommended that the **USER_NUM_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter be selected. In this case, **the gNodeB adaptively adjusts the number of activated BWP1s for RedCap UEs based on the relative load relationship between RedCap UEs and common NR UEs, and selects the BWP1 that serves the smallest number of UEs for RedCap UE access to achieve the optimal total cell traffic volume**.

- A new BWP1 will be activated when all of the following conditions are met:
  - Total number of RRC_CONNECTED UEs on all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell ≥ **NRDUCellRedCap**.***RedCapUeNumThld*** x 1.1
  - Proportion of non-RedCap UEs in a cell (equal to "Number of non-RedCap UEs in the cell/Number of RRC_CONNECTED UEs supported by the cell") < **NRDUCellRedCap**.***CellMediumLoadUeRatioThld***
  - Total bandwidth of all activated RedCap BWP1s in the cell < Configured maximum available bandwidth for RedCap UEs (specified by **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth***)
- An activated BWP1 will be deleted when either of the following conditions is met throughout a period of time specified by **NRDUCellRedCap**.***BwpShrinkWaitingTime***:
  - Both of the following sub-conditions are met: (1) Total number of RRC_CONNECTED UEs on all activated RedCap BWP1s in a cell/(Number of activated RedCap BWP1s in the cell – 1) < **NRDUCellRedCap**.***RedCapUeNumThld***; (2) Total bandwidth of all activated RedCap BWP1s in the cell > 20 MHz
  - Both of the following sub-conditions are met: (1) Proportion of non-RedCap UEs in the cell ≥ Sum of **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** and a specific hysteresis (a preset value); (2) Total bandwidth of all activated RedCap BWP1s in the cell > 20 MHz

## PRB-and-CCE-Resource-based BWP1 Adaptation

In 5GtoC scenarios, if the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M** for a cell, it is recommended that the **RESOURCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter be selected. In this case, the gNodeB performs the following:

In NR TDD scenarios, **the gNodeB adaptively adjusts the number of activated RedCap BWP1s based on the PRB usage and PDCCH CCE failure rate on BWP1s available for RedCap UEs, and selects a BWP1 for RedCap UE access based on the average number of RRC_CONNECTED UEs, PRB usage, and PDCCH CCE failure rate of all activated RedCap BWP1s, to balance the RedCap UE load among BWP1s**.

The gNodeB follows the following rules when adaptively adjusting the number of activated RedCap BWP1s:

- A new BWP1 will be activated when all of the following conditions are met:
  - One of the following sub-conditions is met: (1) Total uplink PRB usage of all activated RedCap BWP1s in a cell/Number of activated RedCap BWP1s in the cell > Uplink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpUlPrbRatioThld***); (2) Total downlink PRB usage of all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell > Downlink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpDlPrbRatioThld***); (3) Total PDCCH CCE failure rate of all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell > PDCCH CCE failure rate threshold (specified by **NRDUCellRedCap**.***RedCapBwpCceFailRatioThld***)

  - Total bandwidth of all activated RedCap BWP1s in the cell < Configured maximum available bandwidth for RedCap UEs (specified by **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth***)
- An activated BWP1 will be deleted when all of the following conditions are met throughout a period of time specified by **NRDUCellRedCap**.***BwpShrinkWaitingTime***:
  - Total uplink PRB usage of all activated RedCap BWP1s in the cell/ (Number of activated RedCap BWP1s in the cell – 1) < Uplink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpUlPrbRatioThld***)
  - Total downlink PRB usage of all activated RedCap BWP1s in the cell/ (Number of activated RedCap BWP1s in the cell – 1) < Downlink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpDlPrbRatioThld***)
  - Total PDCCH CCE failure rate of all activated RedCap BWP1s in the cell/ (Number of activated RedCap BWP1s in the cell – 1) < PDCCH CCE failure rate threshold (specified by **NRDUCellRedCap**.***RedCapBwpCceFailRatioThld***)
  - Total bandwidth of all activated RedCap BWP1s in the cell > 20 MHz

When UE-number-based BWP1 adaptation is enabled together with PRB-and-CCE-resource-based BWP1 adaptation, the gNodeB comprehensively evaluates the average number of RRC_CONNECTED RedCap UEs, as well as the PRB usage and PDCCH CCE failure rate on BWP1s available for RedCap UEs, and adaptively adjusts the number of activated RedCap BWP1s.

- A new BWP1 will be activated when all of the following conditions are met:
  - One of the following sub-conditions is met: (1) Total number of RRC_CONNECTED UEs on all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell ≥ **NRDUCellRedCap**.***RedCapUeNumThld*** x 1.1; (2) Total uplink PRB usage of all activated RedCap BWP1s in a cell/Number of activated RedCap BWP1s in the cell > Uplink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpUlPrbRatioThld***); (3) Total downlink PRB usage of all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell > Downlink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpDlPrbRatioThld***); (4) Total PDCCH CCE failure rate of all activated RedCap BWP1s in the cell/Number of activated RedCap BWP1s in the cell > PDCCH CCE failure rate threshold (specified by **NRDUCellRedCap**.***RedCapBwpCceFailRatioThld***)
  - Proportion of non-RedCap UEs in the cell < **NRDUCellRedCap**.***CellMediumLoadUeRatioThld***
  - Total bandwidth of all activated RedCap BWP1s in the cell < Configured maximum available bandwidth for RedCap UEs (specified by **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth***)
- An activated BWP1 will be deleted when either of the following conditions is met throughout a period of time specified by **NRDUCellRedCap**.***BwpShrinkWaitingTime***:
  - All of the following sub-conditions are met: (1) Total number of RRC_CONNECTED UEs on all activated RedCap BWP1s in the cell/ (Number of activated RedCap BWP1s in the cell – 1) < **NRDUCellRedCap**.***RedCapUeNumThld***;(2) Total uplink PRB usage of all

activated RedCap BWP1s in the cell/(Number of activated RedCap BWP1s in the cell – 1) < Uplink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpUlPrbRatioThld***); (3) Total downlink PRB usage of all activated RedCap BWP1s in the cell/(Number of activated RedCap BWP1s in the cell – 1) < Downlink PRB usage threshold (specified by **NRDUCellRedCap**.***RedCapBwpDlPrbRatioThld***), (4) Total PDCCH CCE failure rate of all activated RedCap BWP1s in the cell/(Number of activated RedCap BWP1s in the cell – 1) < PDCCH CCE failure rate threshold (specified by **NRDUCellRedCap**.***RedCapBwpCceFailRatioThld***); (5) Total bandwidth of all activated RedCap BWP1s in the cell > 20 MHz

– Both of the following sub-conditions are met: (1) Proportion of non-RedCap UEs in the cell ≥ Sum of **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** and a specific hysteresis (a preset value); (2) Total bandwidth of all activated RedCap BWP1s in the cell > 20 MHz

**NOTE**

The average number of RRC_CONNECTED UEs, PRB usage, and PDCCH CCE failure rate of activated BWP1s cover RedCap UEs and non-RedCap UEs.

## 5.1.2 Network Analysis

### 5.1.2.1 Benefits

- UE-number-based BWP1 adaptation: After this function takes effect, the number of BWP1s can be adaptively allocated based on RedCap service requirements. When the conditions for activating a new BWP1 are met, the new BWP1 can be activated to provide more network resources for RedCap UEs, improving service experience of RedCap UEs. When the conditions for deleting an activated BWP1 are met, the activated BWP1 can be deleted to provide more network resources for eMBB UEs, thereby improving service experience of eMBB UEs.
- PRB-and-CCE-resource-based BWP1 adaptation:

  After PRB-and-CCE-resource-based BWP1 adaptation takes effect, the number of BWP1s can be adaptively allocated based on the CCE failure rate and uplink and downlink PRB usages of BWP1s used by RedCap UEs. When the conditions for activating a new BWP1 are met, the new BWP1 can be activated to provide more network resources for RedCap UEs, improving service experience of RedCap UEs. When the conditions for deleting an activated BWP1 are met, the activated BWP1 can be deleted to provide more network resources for eMBB UEs, thereby improving service experience of eMBB UEs.
- Configurable threshold of the proportion of UEs with medium load in the cell: When the proportion of non-RedCap UEs in a cell is less than the **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** parameter value and other BWP1 expansion conditions are met, RedCap UE throughput increases. When the proportion of non-redcap UEs in a cell is greater than or equal to the sum of the **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** parameter value and a specific hysteresis and the number of activated BWP1s in the cell is greater than 1, the expanded BWP1s need to be reduced, improving eMBB UE throughput performance.

#### 5.1.2.2 Impacts

The impacts between this function and other functions are described in **5.1.3.2.3 Function Impacts**.

- UE-number-based BWP1 adaptation: When the conditions for activating a new BWP1 are met, the new BWP1 can be activated to provide more network resources for RedCap UEs. In this case, network resources available for eMBB UEs decrease, and therefore service experience of eMBB UEs deteriorates. When the conditions for deleting an activated BWP1 are met, the activated BWP1 can be deleted to provide more network resources for eMBB UEs. In this case, network resources available for RedCap UEs decrease, and therefore service experience of RedCap UEs deteriorates.
- PRB-and-CCE-resource-based BWP1 adaptation:
  - When the conditions for activating a new BWP1 are met, the new BWP1 can be activated to provide more network resources for RedCap UEs. In this case, network resources available for eMBB UEs decrease, and therefore service experience of eMBB UEs deteriorates. When the conditions for deleting an activated BWP1 are met, the activated BWP1 can be deleted to provide more network resources for eMBB UEs. In this case, network resources available for RedCap UEs decrease, and therefore service experience of RedCap UEs deteriorates.
- When the proportion of non-RedCap UEs in a cell is less than the **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** parameter value and other BWP1 expansion conditions are met, eMBB UE throughput decreases. When the proportion of non-redcap UEs in a cell is greater than or equal to the sum of the **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** parameter value and a specific hysteresis and the number of activated BWP1s in the cell is greater than 1, the expanded BWP1s need to be reduced, decreasing RedCap UE throughput.
- A larger value of the **NRDUCellRedCap**.***BwpShrinkWaitingTime*** parameter indicates a longer waiting time for RedCap BWP shrink, and results in less frequent BWP shrinking and a lower probability of service drops caused by RedCap UE reconfigurations after shrinking. A smaller value of this parameter produces the opposite effects.

### 5.1.3 Requirements

#### 5.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.1.3.2.1 Prerequisite Functions

UE-number-based BWP1 adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | UE-number-based BWP1 adaptation can be enabled only when RedCap Introduction is enabled. |
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | *RedCap* | UE-number-based BWP1 adaptation can be enabled only when multiple BWP1s for RedCap is enabled (by setting the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter to a value greater than **20M**). |

PRB-and-CCE-resource-based BWP1 adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | PRB-and-CCE-resource-based BWP1 adaptation can be enabled only when RedCap Introduction is enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | *RedCap* | PRB-and-CCE-resource-based BWP1 adaptation can be enabled only when multiple BWP1s for RedCap is enabled (by setting the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter to a value greater than **20M**). |

#### 5.1.3.2.2 Mutually Exclusive Functions

UE-number-based BWP1 adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | In a high-speed NR TDD cell, if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter), UE-number-based BWP1 adaptation cannot be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | For an NR TDD hyper cell, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected and at least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACT** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | **V_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **DPS_MODE** | *Cell Combination* | For an NR TDD combined cell in DPS transmission mode, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected, and at least one of the **USER_NUM_BASED_BWP_ACTV_SW** and |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Joint Transmission Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE** | *Cell Combination* | For an NR TDD cell with Joint Transmission Cell Combination enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected, and at least one of the **USER_NUM_BASED_BWP_AC** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | **TV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

PRB-and-CCE-resource-based BWP1 adaptation:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | In a high-speed NR TDD cell, if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter), PRB-and-CCE-resource-based BWP1 adaptation cannot be enabled. |
| Low-frequency TDD | UE-number-based periodic BWP load balancing | **UE_NUM_BASED_PRDC_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | UE-number-based periodic BWP load balancing cannot be enabled together with PRB-and-CCE-resource-based BWP1 adaptation. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | For an NR TDD hyper cell, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected and at least one of the **USER_NUM_BASED_BWP_ACTV_SW** and **RESOURCE_BASED_BWP_ACT** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | **V_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **DPS_MODE** | *Cell Combination* | For an NR TDD combined cell in DPS transmission mode, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected, and at least one of the **USER_NUM_BASED_BWP_ACTV_SW** and |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| | | | | **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Joint Transmission Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE** | *Cell Combination* | For an NR TDD cell with Joint Transmission Cell Combination enabled, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter can only be set to **20M**, **40M**, or **60M** if RedCap Introduction is enabled (by selecting the **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter) and any of the following conditions is met:<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected.<br>● The **REDCAP_PUCCH_RES_GUAR_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is selected, and at least one of the **USER_NUM_BASED_BWP_AC** |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | **TV_SW** and **RESOURCE_BASED_BWP_ACTV_SW** options of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected. |

#### 5.1.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Preferential selection of BWP1s with NCD-SSBs for RedCap UEs | **NRDUCellRedCap**.***NcdSsbBwpPreferMaxUeNum*** set to a value other than **0** | *RedCap* | Assume that preferential selection of BWP1s with NCD-SSBs for RedCap UEs and any of the following functions are enabled: UE-number-based BWP1 adaptation and PRB-and-CCE-resource-based BWP1 adaptation. The base station reserves at least one RedCap BWP1 with the NCD-SSB because the UEs preferentially access the BWP1s with NCD-SSBs. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | QCI-specific UE-number-based periodic BWP load balancing | **QCI_UE_NUM_BASED_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter and **QCI_UE_NUM_BASED_PRDC_BLB_SW** option of the **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** parameter | *RedCap* | For an NR TDD cell, the base station reserves at least one RedCap BWP1 with the NCD-SSB when the **gNBDURedCapBlbGrp**.***DLBwpLbWeightFactor*** parameter is set to **0** and any of the following functions is enabled: UE-number-based BWP1 adaptation and PRB-and-CCE-resource-based BWP1 adaptation. This is because RedCap UEs running services of the corresponding 5QI are not expected to access the CD-SSB BWP. |
| Low-frequency TDD | RedCap BWP frequency-domain staggering | **REDCAP_BWP_NON_OVERLAP_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | When RedCap BWP frequency-domain staggering and PRB-and-CCE-resource-based BWP1 adaptation or PRB-and-CCE-resource-based BWP1 adaptation are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, PRB-and-CCE-resource-based BWP1 adaptation or PRB-and-CCE-resource-based BWP1 adaptation takes effect based on the new activation sequence of RedCap BWP1s. |

#### 5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 5.1.3.4 Cells

In NR TDD, the cell bandwidth must be greater than or equal to 40 MHz.

#### 5.1.3.5 Others

None

## 5.1.4 Operation and Maintenance

### 5.1.4.1 Data Configuration

#### 5.1.4.1.1 Data Preparation

**Table 5-2** describes the parameters used for function activation.

**Table 5-2** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap UE Maximum Available Bandwidth | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | When the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to a value greater than **20M** for a cell:<br>● It is recommended that the **USER_NUM_BASED_BWP_ACTV_SW** option of this parameter be selected so that UE-number-based BWP1 adaptation can take effect.<br>● In NR TDD, it is recommended that the **RESOURCE_BASED_BWP_ACTV_SW** option of this parameter be selected when the single-UE traffic volume is large and two BWP1s are configured, so that PRB-and-CCE-resource-based BWP1 adaptation can take effect.<br>In 5G to business (5GtoB) scenarios, the number of RedCap UEs fluctuates slightly, and services are stable. Therefore, these functions are not recommended. |
| RedCap UE Number Threshold | **NRDUCellRedCap**.***RedCapUeNumThld*** | Set this parameter based on the network plan. |
| Cell Medium Load UE Ratio Threshold | **NRDUCellRedCap**.***CellMediumLoadUeRatioThld*** | Use the recommended value. |
| BWP Shrink Waiting Time | **NRDUCellRedCap**.***BwpShrinkWaitingTime*** | Use the recommended value. |

#### 5.1.4.1.2 Using MML Commands

Before using MML commands, refer to **5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Configuring the maximum available bandwidth for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUeMaxAvailBandwidth=40M;
//Enabling UE-number-based BWP1 adaptation
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=USER_NUM_BASED_BWP_ACTV_SW-1;
//Enabling PRB-and-CCE-resource-based BWP1 adaptation for an NR TDD cell
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=RESOURCE_BASED_BWP_ACTV_SW-1;
//Setting the RedCap UE number threshold for an NR TDD cell
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUeNumThld=40;
//Setting CellMediumLoadUeRatioThld
MOD NRDUCELLREDCAP: NrDuCellId=0, CellMediumLoadUeRatioThld=100;
//Configuring the BWP Shrink Waiting Time
MOD NRDUCELLREDCAP: NrDuCellId=0, BwpShrinkWaitingTime=10MIN;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling PRB-and-CCE-resource-based BWP1 adaptation for an NR TDD cell
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=RESOURCE_BASED_BWP_ACTV_SW-0;
//Disabling UE-number-based BWP1 adaptation
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=USER_NUM_BASED_BWP_ACTV_SW-0;
//Restoring the maximum bandwidth available for RedCap UEs to the default value
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapUeMaxAvailBandwidth=20M;
```

### 5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.1.4.2 Activation Verification

Observe the following counters to check whether the RedCap capacity enhancement function has taken effect.

For an NR TDD cell, if the value of any counter in the following table is greater than **51**, the RedCap capacity enhancement function has taken effect.

| Counter ID | Counter Name |
|---|---|
| 1911840391 | **N.PRB.UL.RedCap.Avail.Avg** |
| 1911840531 | **N.PRB.DL.RedCap.Avail.Avg** |

## 5.1.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **4.2.2 Impacts**.

# 5.2 Coverage Enhancement

## 5.2.1 Principles

Coverage enhancement includes the following functions, which improve PUSCH coverage and maximum transmit power of specific UEs, thereby improving uplink experience of RedCap UEs at the cell edge.

- Dynamic uplink slot aggregation: This function adaptively adjusts the number of aggregated uplink slots based on radio channel quality, to improve PUSCH coverage.
- Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD): The base station can set the maximum uplink transmit power to 26 dBm for RedCap UEs that support the function of power class 2 based on device-pipe synergy, to improve the uplink coverage and user experience of RedCap UEs. This function is supported only in TDD cells.

Coverage enhancement is controlled by the **REDCAP_UL_COVERAGE_ENH_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter.

### 5.2.1.1 Dynamic Uplink Slot Aggregation (Trial Function in NR TDD)

Dynamic uplink slot aggregation enables **a RedCap UE to use different HARQ redundancy versions to transmit the same data block in multiple consecutive uplink slots, which are bound and treated as a single resource unit, in one HARQ retransmission corresponding to an ACK or NACK**. This function together with HARQ retransmission combining improves the transmission efficiency and reliability over the air interface.

A RedCap UE enters the dynamic uplink slot aggregation state when all of the following conditions are met:

- The RedCap UE is not running a voice service.
- The RedCap UE is not running a URLLC service.
- The RedCap UE is identified as one under weak coverage. That is, the SINR for uplink scheduling of the RedCap UE, which is measured in a measurement period of 10 ms, is less than or equal to the value of **NRDUCellRedCap**.***UlSlotAggSinrThld*** for three consecutive times.

After dynamic uplink slot aggregation takes effect for a RedCap UE under weak coverage, the base station instructs the UE to adaptively adjust the number of aggregated uplink slots through the downlink control information (DCI) message based on the SRS SINR of the RedCap UE.

For a TDD cell, the number of aggregated uplink slots can be 1 or 2.

A RedCap UE exits the dynamic uplink slot aggregation state when any of the following conditions is met:

- The RedCap UE is running a voice service.
- The RedCap UE is running a URLLC service.

- The SINR for uplink scheduling of the RedCap UE, which is measured in a measurement period of 10 ms, is greater than the sum of **NRDUCellRedCap**.***UlSlotAggSinrThld*** and 5 dB for three consecutive times.

When dynamic uplink slot aggregation is enabled, you are advised to enable the functions listed in **Table 5-3**.

**Table 5-3** Functions that are recommended to be used together with dynamic uplink slot aggregation

| RAT | Function Name | Function Description | Parameter |
|---|---|---|---|
| Low-frequency TDD | Uplink DMRS SINR correction | The base station performs power correction on the DMRS SINR before the SINR is used in MCS selection and scheduling. In this way, the base station can more accurately trace UE channel fluctuation and predict the SINR adjustment value at the next scheduling moment. | **UL_DMRS_SINR_CORRECTION_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter |
| Low-frequency TDD | PUSCH-DTX-based aggregation level adaptation | To prevent a failure to select an appropriate CCE aggregation level due to missing DCI detection, you can enable PUSCH-DTX-based aggregation level adaptation. After this function is enabled, the gNodeB dynamically adjusts the estimated PDCCH SINR based on PUSCH DTX information to select a more appropriate CCE aggregation level. | **PUSCH_DTX_AGG_LVL_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter |

Dynamic uplink slot aggregation is a trial function in NR TDD.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

#### 5.2.1.2 Power Class 2 based on Device-Pipe Synergy for RedCap UEs (Low-Frequency TDD)

Because the maximum uplink transmit power of RedCap UEs is only 23 dBm, the uplink coverage may be limited due to limited uplink power in certain scenarios. To improve the uplink coverage and user experience of RedCap UEs, power class 2 based on device-pipe synergy for RedCap UEs is introduced. This function enables the base station to set the maximum uplink transmit power to 26 dBm for RedCap UEs that support the function of power class 2 based on device-pipe synergy. This function is controlled by the **REDCAP_COORD_PC2_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter.

## 5.2.2 Network Analysis

### 5.2.2.1 Benefits

The coverage enhancement function increases the average uplink throughput of UEs under weak coverage.

The benefits of each sub-function are as follows:

- Dynamic uplink slot aggregation: This function together with HARQ retransmission combining provides the following benefits:
  - Increased MCS index used for scheduling
  - Increased TB size used for RedCap UE scheduling under weak coverage
  - Reduced proportion of RLC header overhead
  - Improved transmission efficiency and reliability over the air interface
- Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD): In weak coverage scenarios, this function increases the uplink throughput of RedCap UEs that support power class 2 based on device-pipe synergy.

### 5.2.2.2 Impacts

The impacts between this function and other functions are described in **5.2.3.2.3 Function Impacts**.

- Once the coverage enhancement function takes effect, more RedCap UEs at the cell edge or under weak coverage may access the cell, causing the following network impacts:
  - The number of online RedCap UEs in the cell increases.
  - The uplink data traffic volume in the cell increases.
  - Cell-level KPIs such as the following indicators related to access, handovers, and service drops deteriorate: the RRC setup success rate,

average uplink and downlink UE throughputs, and average uplink and downlink cell throughputs.

- Dynamic uplink slot aggregation: Once this function takes effect, the average uplink UE throughput may fluctuate significantly, and the uplink initial block error rate (IBLER) and residual block error rate (RBLER) decrease. In extreme cell edge or weak coverage scenarios, it is possible that the coverage gains and the uplink user-perceived rate decrease due to the increase in missing DCI detection.
- Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD): None

For the definitions of related indicators, see **5.2.4.3 Network Monitoring**.

## 5.2.3 Requirements

### 5.2.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.2.3.2.1 Prerequisite Functions

Coverage enhancement:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | The capacity enhancement function depends on RedCap Introduction. |

#### 5.2.3.2.2 Mutually Exclusive Functions

Coverage enhancement:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell.*HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | Coverage enhancement and High-speed Railway Superior Experience cannot be enabled at the same time. |
| Low-frequency TDD | Cell Combination | **NRDUCell.*NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | Coverage enhancement and Cell Combination cannot be enabled at the same time. |
| Low-frequency TDD | Light-load time sequence optimization | **LIGHT_LOAD_TIME_SEQ_OPT_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter | None | Coverage enhancement and light-load time sequence optimization cannot be enabled at the same time. |

#### 5.2.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Uplink PAMC | **UL_PAMC_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter | *Uplink Scheduling* | Uplink PAMC does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Intra-base-station UL CoMP | **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | For an NR TDD cell, intra-base-station UL CoMP does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |
| Low-frequency TDD | PUSCH MU-MIMO | **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter | *MIMO (TDD)* | RedCap UEs with dynamic uplink slot aggregation in effect do not participate in PUSCH MU-MIMO. |
| Low-frequency TDD | Super uplink | **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter | *Super Uplink* | For an NR TDD cell, super uplink does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |
| Low-frequency TDD | Uplink DCI extension-based scheduling | **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter | *Uplink Scheduling* | For an NR TDD cell, uplink DCI extension-based scheduling does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |
| Low-frequency TDD | VoNR uplink slot aggregation | **VONR_UL_SLOT_AGGREGATION_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter | None | For an NR TDD cell, dynamic uplink slot aggregation does not take effect on RedCap UEs with VoNR uplink slot aggregation in effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Slot aggregation | **SLOT_AGGREGATION_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter | None | For an NR TDD cell, dynamic uplink slot aggregation does not take effect on RedCap UEs with slot aggregation in effect. |
| Low-frequency TDD | UL skipping | • **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter<br>• **UPLINK_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter | *UE Power Saving* | For an NR TDD cell, UL skipping does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |
| Low-frequency TDD | Flexible spectrum access | **FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter | *Uplink Flexible Spectrum Scheduling* | For an NR TDD cell, flexible spectrum access does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect. |

### 5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

#### Boards

Dynamic uplink slot aggregation: All NR-capable main control boards and baseband processing units, except the UBBPfw, support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD): All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

#### RF Modules

No requirements

### 5.2.3.4 Cells

- Coverage enhancement

  For NR TDD, coverage enhancement is supported only in cells for which the **NRDUCell**.***SlotAssignment*** parameter is set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**,.
- Dynamic uplink slot aggregation

  For NR TDD, dynamic uplink slot aggregation is supported only in cells for which the **NRDUCell**.***SlotAssignment*** parameter is set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.

### 5.2.3.5 Others

Dynamic uplink slot aggregation: UEs must support pusch-RepetitionTypeA-r16 or pusch-RepetitionTypeA-v16c0 defined in 3GPP Release 16.

Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD): RedCaps UEs must support device-pipe synergy.

## 5.2.4 Operation and Maintenance

### 5.2.4.1 Data Configuration

#### 5.2.4.1.1 Data Preparation

Table 5-4 describes the parameters used for function activation.

**Table 5-4** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_UL_COVERAGE_ENH_SW | Select this option if coverage enhancement is required. |
| UL Slot Aggregation SINR Threshold | **NRDUCellRedCap**.***UlSlotAggSinrThld*** | None | Use the recommended value. |
| UL DMRS Switch | **NRDUCellDmrs**.***UlDmrsSwitch*** | DMRS_POSTSINR_CORRECTION_SW | It is recommended that this option be selected when dynamic uplink slot aggregation is enabled. |
| Uplink AMC Performance Switch | **NRDUCellUlAmc**.***UlAmcPerformanceSw*** | UL_DMRS_SINR_CORRECTION_SW | It is recommended that this option be selected when dynamic uplink slot aggregation is enabled. |
| PDCCH Algorithm Enhancement Switch | **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** | PUSCH_DTX_AGG_LVL_ADAPT_SW | It is recommended that this option be selected when dynamic uplink slot aggregation is enabled. This is to avoid the unavailability of an appropriate CCE aggregation level due to DCI detection failures. |
| UL Scheduling Algorithm Switch | **NRDUCellUlSch**.***UlSchAlgoSwitch*** | UL_SLOT_AGG_ULCOMP_DECOUPLE_SW | Select this option if both dynamic uplink slot aggregation and UL CoMP need to take effect on RedCap UEs. |
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_COORD_PC2_SW | Select this option if power class 2 based on device-pipe synergy for RedCap UEs is required. |

### 5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling coverage enhancement
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_UL_COVERAGE_ENH_SW-1;
//Enabling DMRS PostSINR measurement correction
MOD NRDUCELLDMRS: NrDuCellId=0, UlDmrsSwitch=DMRS_POSTSINR_CORRECTION_SW-1;
//Enabling uplink DMRS SINR correction
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=UL_DMRS_SINR_CORRECTION_SW-1;
//Setting the SINR threshold for uplink slot aggregation
MOD NRDUCELLREDCAP: NrDuCellId=0, UlSlotAggSinrThld=38;
//Enabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-1;
//For an NR TDD cell, enabling power class 2 based on device-pipe synergy for RedCap UEs
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_COORD_PC2_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//For an NR TDD cell, disabling power class 2 based on device-pipe synergy for RedCap UEs
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_COORD_PC2_SW-0;
//Disabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-0;
//Disabling DMRS PostSINR measurement correction
MOD NRDUCELLDMRS: NrDuCellId=0, UlDmrsSwitch=DMRS_POSTSINR_CORRECTION_SW-0;
//Disabling uplink DMRS SINR correction
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=UL_DMRS_SINR_CORRECTION_SW-0;
//Disabling coverage enhancement
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_UL_COVERAGE_ENH_SW-0;
```

### 5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.2.4.2 Activation Verification

Dynamic uplink slot aggregation:

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. Check the **PUSCH-TimeDomainResourceAllocation-r16** signaling fields in **pusch-Config > pusch-TimeDomainAllocationListDCI-0-1-r16** and the **numberOfRepetitions-r16** IE in these signaling fields. If there are at least two different values of the **numberOfRepetitions-r16** IE in all **PUSCH-TimeDomainResourceAllocation-r16**

signaling fields, it indicates that dynamic uplink slot aggregation has taken effect. Otherwise, it indicates that this function has not taken effect.

**----End**

Power class 2 based on device-pipe synergy for RedCap UEs (low-frequency TDD):

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** If the value of **p-Max** is **26** in the **p-NR-FR1** field of the PhysicalCellGroupConfig message sent from the base station to the RedCap UE, power class 2 based on device-pipe synergy for RedCap UEs has been enabled.

**----End**

### 5.2.4.3 Network Monitoring

- Uplink data traffic volume of a cell: **N.PDCP.Vol.UL.TrfSDU.Tx** or **N.ThpVol.UL**
- RRC connection setup success rate = **RRC Setup Success Rate (CU)**
- Average uplink UE throughput = **User Uplink Average Throughput (DU)**
- Average downlink UE throughput = **User Downlink Average Throughput (DU)**
- Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
- UL IBLER = (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
- UL RBLER = (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
- Uplink user-perceived rate = (**N.ThpVol.UL.Edge** – **N.ThpVol.UE.UL.SmallPkt.Edge**)/**N.ThpTime.UE.UL.RmvSmallPkt.Edge**

Observe the uplink RLC single-UE throughput: Use the real-time monitoring items on the MAE-Access to observe the fluctuation of the uplink RLC single-UE throughput after the function is enabled. To compare the uplink RLC throughput before and after the function is enabled, choose **User Performance Monitoring** > **User Common Monitoring** > **Uplink RLC Throughput** on the MAE-Access.

## 5.3 RedCap Power Saving

The base station uses the RedCap Power Saving function to further reduce power consumption of RedCap UEs. This function is controlled by the

**REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter. It includes the following subfunctions:

- Inactive RA-SDT (trial)
- Assistance TRS
- Connected-mode RRM measurement relaxation
- RedCap flexible DRX configuration
- RLM relaxation
- eDRX in idle mode (trial)
- RedCap UE release preference
- RedCap SSSG switching
- RedCap UL skipping
- RedCap DRX adaptation

## 5.3.1 Inactive RA-SDT (Trial)

### 5.3.1.1 Principles

When a UE is running services with a small data volume, excessive power is consumed if it stays in the RRC_CONNECTED state for a long time. To address this issue, 3GPP Release 17 introduces the inactive small data transmission (SDT) function. This function **allows RRC_INACTIVE UEs to perform services with a small amount of data. This function increases the duration in which UEs stay in states other than RRC_CONNECTED and reduces power waste induced by signaling transmission and PDCCH monitoring, to achieve UE power saving**.

3GPP Release 17 support two SDT modes:

- Random access-based SDT (RA-SDT)
- Configured grant-based SDT (CG-SDT)

Currently, Huawei base stations implement the inactive SDT function in RA-SDT mode.

Inactive RA-SDT is controlled by the **REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter. When the inactive RA-SDT function is enabled, the gNodeB delivers the IEs required for enabling inactive RA-SDT in the RMSI. In this case, inactive RA-SDT can take effect on UEs capable of RA-SDT. When a UE in the RRC_INACTIVE state needs to send uplink data packets and the signal quality and to-be-transmitted data volume of the UE in the serving cell meets the triggering conditions specified using **NRDUCellSelConfig**.***RaSdtRsrpThld*** and **NRDUCellSelConfig**.***RaSdtDataVolumeThld***, respectively, are met, initial access and data transmission can be performed through the RA-SDT procedure. In addition, to obtain the UE power saving benefits and reduce the duration of UEs staying in RRC_CONNECTED mode, the gNodeB sets the inactivity timer length to 1s for UEs that access the network through the SDT procedure. If the UE has no uplink or downlink data to transmit within 1s, it is released to the RRC_INACTIVE state.

where

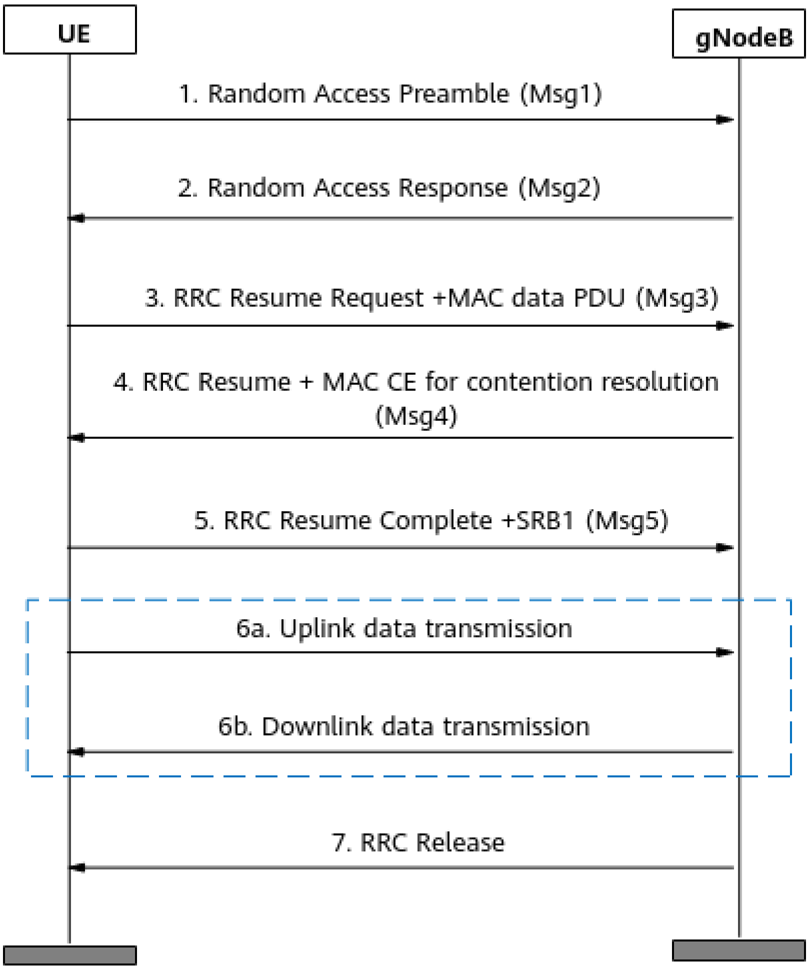
  

- The IEs required by inactive RA-SDT are as follows:
  - featureCombinationPreamblesList-r17: contains configurations for the base station to divide RA preambles for SDT.
  - sdt-ConfigCommon-r17: includes the following configurations:
    - sdt-RSRP-Threshold-r17: indicates the RSRP threshold for UEs to initiate RA through the SDT process. It is specified by the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter. When uplink data arrives, an RA-SDT-capable UE can send the data packet through the RA-SDT process only if the downlink RSRP is greater than this threshold.
    - sdt-DataVolumeThreshold-r17: indicates the data volume threshold for UEs to initiate RA through the SDT process. It is specified by the **NRDUCellSelConfig**.***RaSdtDataVolumeThld*** parameter. When uplink data arrives, an RA-SDT-capable UE can send the data packet through the RA-SDT process only if the data volume is less than this threshold.
    - t319a-r17: indicates the duration of the T319a timer.

  For details about the *featureCombinationPreamblesList-r17* and *sdt-ConfigCommon-r17* IEs, see 3GPP TS 38.331 V17.4.0.
- The signal quality and to-be-transmitted data volume of the UE in the serving cell must meet the following two conditions:
  - The RSRP of the UE is greater than the value of **NRDUCellSelConfig**.***RaSdtRsrpThld***.
  - The uplink data volume of the UE is less than or equal to the value of **NRDUCellSelConfig**.***RaSdtDataVolumeThld***.
- When releasing a RedCap UE supporting the inactive RA-SDT function to the RRC_INACTIVE state, the gNodeB includes **SuspendConfig->sdt-Config-r17** in the RRC Release message sent to the UE. In addition, **sdt-DRB-List** in the *sdt-Config-r17* IE contains the DRBs that are allowed to transmit uplink data through the SDT procedure.

For an RRC_INACTIVE RedCap UE, if the RRC Release message used to release the UE and enable the UE to enter the RRC_INACTIVE state contains the configuration of the SDT DRBs set by the gNodeB, the inactive RA-SDT procedure, shown in **Figure 5-1**, is performed when uplink data arrives.

**Figure 5-1** Procedure for inactive RA-SDT




In the procedure for this function, note the following:

1. The UE can select the RA preamble for SDT allocated by the gNodeB to initiate RA, and starts the T319a timer when initializing the RA-SDT procedure.
2. The gNodeB performs the RA response.
3. After the UE receives the random access response (RAR) from the gNodeB in response to the preamble for SDT, it sends the RRC Resume Request message and uplink data to the gNodeB through Msg3. The **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter specifies the MCS index used for Msg3 scheduling in the RA-SDT procedure.
   - A smaller value of the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter results in a smaller volume of uplink data that can be contained in Msg3, a higher probability of successful Msg3 demodulation during RA-SDT, and a lower requirement for the signal quality (RSRP value of the UE) of the serving cell of the UE.
   - A larger value of the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter results in a larger volume of uplink data that can be contained in Msg3, a lower probability of successful Msg3 demodulation during RA-SDT, and a higher requirement for the signal quality of the serving cell of the UE.

Therefore, when setting the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter, you are advised to take the setting of the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter into consideration. **Table 5-5** describes the configuration requirements for **NRDUCellSelConfig**.***RaSdtRsrpThld*** when the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter is set to different values in different interference scenarios.

**Table 5-5** RA-SDT RSRP threshold configuration requirements

| Scenario | Value of NRDUCellPusch.*RaSdtMsg3Mcs* | Value of NRDUCellSelConfig.*RaSdtRsrpThld* |
|---|---|---|
| Medium- and low-interference scenario (–120 dBm ≤ **N.UL.NI.Avg** < –100 dBm) | MCS1 to MCS3 | ≥ –110 dBm |
| | MCS4 to MCS6 | ≥ –107 dBm |
| | MCS7 to MCS9 | ≥ –102 dBm |
| | MCS10 to MCS11 | ≥ –98 dBm |
| High-interference scenario (**N.UL.NI.Avg** ≥ –100 dBm) | MCS1 to MCS3 | ≥ –100 dBm |
| | MCS4 to MCS6 | ≥ –95 dBm |
| | MCS7 to MCS9 | ≥ –90 dBm |
| | MCS10 to MCS11 | ≥ –85 dBm |

4. The gNodeB delivers an RRC Resume message and a MAC CE for contention resolution.
5. The UE responds with an RRC Resume Complete message and SRB1.
6. Data is transmitted in the uplink and downlink.
7. After data transmission is complete, the gNodeB sends an RRC Release message to release the UE and enable the UE to enter the RRC_INACTIVE state.

Before the T319a timer expires:

- If the UE receives an RRC Resume, RRC Setup, or RRC Release message from the gNodeB, the UE stops the T319a timer.
- If the T319a timer expires, the UE switches to the RRC_IDLE state.

Inactive RA-SDT is a trial function.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

### 5.3.1.2 Network Analysis

#### 5.3.1.2.1 Benefits

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

#### 5.3.1.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

Inactive RA-SDT has the following network impacts:

- After inactive RA-SDT is enabled, the contention-based random access success rate fluctuates because the access-related counters related to inactive RA-SDT are measured in the original contention-based random access counters.
- When inactive RA-SDT is enabled and uplink interference is strong on the live network, the **N.RA.SDT.Contention.Att** counter will not be zero due to PRACH false alarms, even if there are no RedCap UEs on the network.
- In different interference scenarios, if the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter is set to a value less than the required value listed in **Table 5-5**, Msg3 demodulation may fail and RedCap UEs may fail to access the network.
- When the random access load is heavy and the inactive RA-SDT function is enabled, as the number of available PRACH preambles decreases for UEs (including RedCap UEs and common NR UEs) on which inactive RA-SDT does not take effect, the probability of contention conflicts increases, the access delay may increase for UEs on which inactive RA-SDT does not take effect, and the contention-based random access success rate (indicated by **N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att** x 100%) may decrease.

For details about counters related to inactive RA-SDT, see **5.3.1.4.3 Network Monitoring**.

### 5.3.1.3 Requirements

#### 5.3.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

**Prerequisite Functions**

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | To enable inactive RA-SDT for RedCap UEs, RedCap Introduction must be enabled. |
| RRC_INACTIVE connection management | **REDCAP_RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter | *RedCap* | To enable inactive RA-SDT for RedCap UEs, RRC_INACTIVE connection management must be enabled. |
| RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap Power Saving must be enabled for inactive RA-SDT to take effect on RedCap UEs. |

**Mutually Exclusive Functions**

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Extended Cell Range (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **14500** and less than or equal to **60000** | *Extended Cell Range* | When Extended Cell Range(TDD) is enabled, the inactive RA-SDT function cannot be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Extended Cell Range of 90 km (TDD) | **NRDUCell**.***CellRadius*** set to a value greater than **60000** and less than or equal to **90000** | *Extended Cell Range* | When Extended Cell Range of 90 km (TDD) is enabled, the inactive RA-SDT function cannot be enabled. |
| Low-frequency TDD | PRACH configuration index | **NRDUCellPrach**.***PrachConfiguration-nIndex*** | None | For an NR TDD cell, when the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter is set to **0**, **1**, **2**, or **200**, the inactive RA-SDT function cannot be enabled.<br>For an NR TDD cell, when the **NRDUCell**.***CellRadius*** parameter is set to a value greater than or equal to **4610** and the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter is set to **195**, **202**, **206**, or **210**, the inactive RA-SDT function cannot be enabled. |
| Low-frequency TDD | IAB deployment | **NRDUCell**.***DeployPosition*** | None | For an NR TDD cell, when the **NRDUCell**.***DeployPosition*** parameter is set to **DEPLOY_IAB**, the inactive RA-SDT function cannot be enabled. |
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** | *High Speed Mobility* | For an NR TDD cell, when the **NRDUCell**.***CellRadius*** parameter is set to a value greater than or equal to **4610** and the **NRDUCell**.***HighSpeedFlag*** parameter is set to **HIGH_SPEED**, the inactive RA-SDT function cannot be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Slot configuration | **NRDUCell**.***SlotAssignment*** | *Standards Compliance* | For an NR TDD cell, when the **NRDUCell**.***CellRadius*** parameter is set to a value greater than or equal to **4610** and the **NRDUCell**.***SlotAssignment*** parameter is set to **4_1_DDDSU** or **1_1_DSDSDSDSDS**, the inactive RA-SDT function cannot be enabled. |
| Low-frequency TDD | Inactive RA-SDT | **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter | *UE Power Saving* | When the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter is selected, the inactive RA-SDT function cannot be enabled. |

## Function Impacts

None

### 5.3.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 5.3.1.3.4 Cells

The **NRDUCell**.***DuplexMode*** parameter must be set to a value other than **CELL_SUL**.

#### 5.3.1.3.5 Others

UEs must support RA-SDT-r17 defined in 3GPP Release 17. In addition, if the BWP0 used by a RedCap UE in a cell does not include the CD-SSB, this UE must also support ncd-SSB-forRedCapInitialBWP-SDT-r17.

## 5.3.1.4 Operation and Maintenance

### 5.3.1.4.1 Data Configuration

### Data Preparation

Table 5-6 describes the parameters used for function activation.

**Table 5-6** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | Select the **REDCAP_ENERGY_SAVING_SW** option of this parameter if inactive RA-SDT is required. |
| RA-SDT Maximum Preamble Number | **NRDUCellPrach**.***RaSdtPreambleMaxNum*** | Set this parameter based on the network plan. |
| RA-SDT RSRP Threshold | **NRDUCellSelConfig**.***RaSdtRsrpThld*** | For details about the configurations in different interference scenarios, see Table 5-5. |
| RA-SDT Data Volume Threshold | **NRDUCellSelConfig**.***RaSdtDataVolumeThld*** | Use the recommended value. |
| RA-SDT MSG3 MCS | **NRDUCellPusch**.***RaSdtMsg3Mcs*** | For details about the configurations in different interference scenarios, see Table 5-5. |
| INACTIVE Strategy Switch | **NRCellAlgoSwitch**.***InactiveStrategySwitch*** | Select the **REDCAP_RRC_INACTIVE_RA_SDT_SW** option of this parameter if inactive RA-SDT is required. |

### Using MML Commands

Before using MML commands, refer to **5.3.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Enabling the RRC INACTIVE connection management and inactive RA-SDT functions
MOD NRCELLALGOSWITCH: NrCellId=0,
InactiveStrategySwitch=REDCAP_RRC_INACTIVE_SWITCH-1&REDCAP_RRC_INACTIVE_RA_SDT_SW-1;
//Configuring the maximum number of preambles that can be allocated for RA-SDT
MOD NRDUCELLPRACH: NrDuCellId=0, RaSdtPreambleMaxNum=1;
//Configuring the RSRP threshold and data volume threshold for RA-SDT
MOD NRDUCELLSELCONFIG: NrDuCellId=0, RaSdtRsrpThld=-100, RaSdtDataVolumeThld=BYTE100;
//Configuring the MCS index for RA-SDT Msg3
MOD NRDUCELLPUSCH: NrDuCellId=0, RaSdtMsg3Mcs=MCS7;
```

### Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the RRC INACTIVE connection management and inactive RA-SDT functions
MOD NRCELLALGOSWITCH: NrCellId=0,
InactiveStrategySwitch=REDCAP_RRC_INACTIVE_SWITCH-0&REDCAP_RRC_INACTIVE_RA_SDT_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 5.3.1.4.2 Activation Verification

### Using Counters

Observe the following counters to check whether inactive RA-SDT has taken effect. If the values of all the counters are not zero, this function has taken effect.

| Counter ID | Counter Name |
|---|---|
| 1911840400 | **N.RA.SDT.Contention.Resp** |

| Counter ID | Counter Name |
|---|---|
| 1911840401 | **N.RA.SDT.Contention.Msg3** |
| 1911840402 | **N.RA.SDT.Contention.Att** |
| 1911840403 | **N.RA.SDT.Contention.Resolution.Succ** |

### Tracing Signaling

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** Check that RRC_Release message is displayed in the result of **Uu Interface Trace**.

If the RRC_Release message contains **SuspendConfig->sdt-Config-r17**, it indicates that the inactive RA-SDT function has taken effect. Otherwise, it indicates that the inactive RA-SDT function has not taken effect.

**----End**

#### 5.3.1.4.3 Network Monitoring

The indicators listed in **5.3.1.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

## 5.3.2 Assistance TRS

### 5.3.2.1 Principles

Tracking reference signal (TRS) for idle/inactive UE is a UE power saving technology introduced in 3GPP Release 17 and is referred to as assistance TRS. This function **enables the gNodeB to broadcast TRS information configured for UEs in connected mode to RRC_IDLE or RRC_INACTIVE UEs through SIB17, assisting such UEs in time and frequency synchronization.** It offers the following benefits:

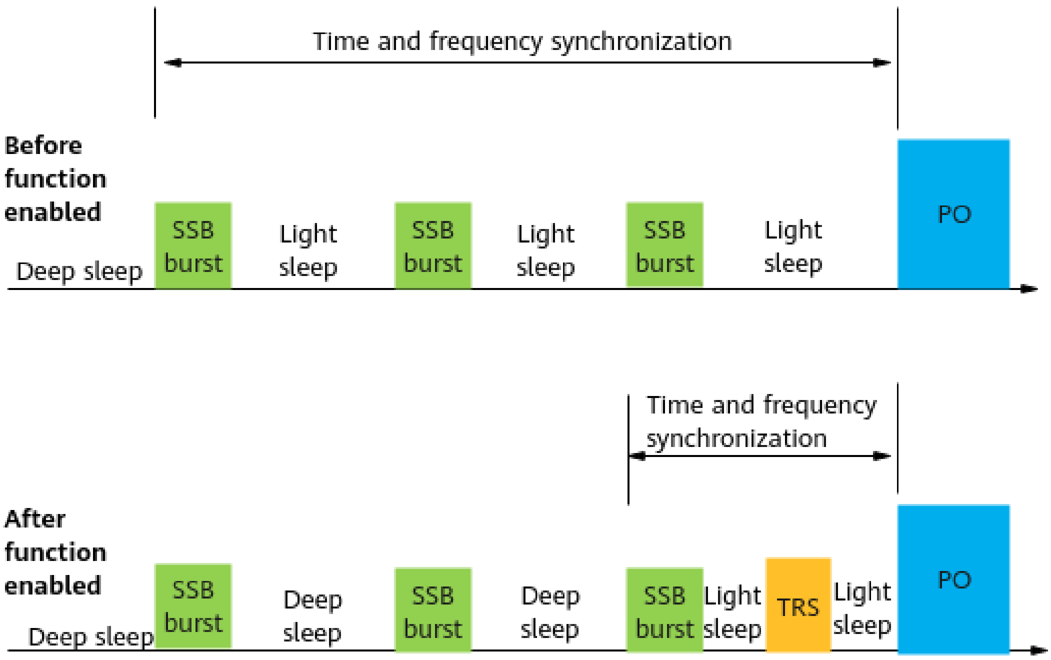
- This function reduces the number of SSBs that need to be received before paging message reception and reduces the time and power consumption for UEs to wake up in advance for time and frequency synchronization.
- This function multiplexes TRSs of RRC_CONNECTED UEs to reduce extra resource consumption of the gNodeB.

Assistance TRS is controlled by the **ASSISTANCE_TRS_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter. **Figure 5-2** illustrates the timelines of an RRC_IDLE or RRC_INACTIVE UE before and after the assistance TRS function is enabled.

- Before assistance TRS is enabled, a cell-edge UE needs to wake up and receive three SSB bursts for time and frequency synchronization before receiving the paging message in the PO.

- After assistance TRS is enabled, the UE needs to receive only one SSB burst and one TRS before receiving the paging message.

**Figure 5-2** Assistance TRS




## 5.3.2.2 Network Analysis

### 5.3.2.2.1 Benefits

This function provides the following benefits:

- This function reduces the number of SSBs that need to be received before paging message reception and reduces the time for UEs to wake up in advance for time and frequency synchronization. This reduces the power consumption of the communications modules of the UEs. The reduction in power consumption can be observed and evaluated on the UE side.
- This function multiplexes TRSs of RRC_CONNECTED UEs to reduce extra resource consumption of the gNodeB.

### 5.3.2.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

This function has no impact on the network.

## 5.3.2.3 Requirements

#### 5.3.2.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap Introduction must be enabled for assistance TRS to take effect on RedCap UEs. |
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap Power Saving must be enabled for assistance TRS to take effect on RedCap UEs. |
| Low-frequency TDD | Periodic TRS transmission | **NRDUCellCsirs**.***TrsPeriod*** | *Channel Management* | **NRDUCellCsirs**.***TrsPeriod*** must be set to a value other than **MS0** for assistance TRS to take effect. |

### Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Cell Combination | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** | *Cell Combination* | Assistance TRS and Cell Combination cannot be enabled at the same time. |
| Low-frequency TDD | Hyper Cell | **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL** | *Hyper Cell* | Assistance TRS and Hyper Cell cannot be enabled at the same time. |

### Function Impacts

None

### 5.3.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

### 5.3.2.3.4 Others

None

## 5.3.2.4 Operation and Maintenance

### 5.3.2.4.1 Data Configuration

## Data Preparation

Table 5-7 describes the parameters used for function activation.

**Table 5-7** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option when the assistance TRS function needs to be enabled. |
| Idle UE Power Saving Switch | **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** | ASSISTANCE_TRS_SW | Select this option when the assistance TRS function needs to be enabled. |
| SIB Type | **gNBSibConfig**.***SibType*** | None | Set this parameter based on site requirements. |
| SIB Transmission Policy | **gNBSibConfig**.***SibTransPolicy*** | None | Set this parameter to **BROADCAST**. |
| SIB Transmission Period | **gNBSibConfig**.***SibTransPeriod*** | None | Set this parameter based on site requirements. |

NOTE

For a system information configuration ID, if the minimum system information transmission period is 320 ms, a maximum of four system information types can be configured. If the minimum system information transmission period is 640 ms, a maximum of eight system information types can be configured.

## Using MML Commands

Before using MML commands, refer to **5.3.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

- If a cell uses the default system information configuration (that is, a set of system information configuration whose ID is 0), a new set of system information configuration needs to be created and bound to the cell when the default system information configuration cannot meet the service requirements. This happens because the default system information configuration cannot be modified.

- If a cell uses a non-default system information configuration (for example, a set of system information configuration whose ID is 1), the SIB17 configuration must be added to activate assistance TRS. In this case, the system information transmission period is prolonged and it takes a longer time for UEs to obtain system information compared with the default system information configuration.

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Adding gNodeB system information configurations
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_2, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_3, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_4, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_5, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_17, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
//Modifying the system information configuration of a cell
MOD NRDUCELL: NrDuCellId=0, SibConfigId=1;
//Enabling the assistance TRS function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=ASSISTANCE_TRS_SW-1;
```

## Deactivation Command Examples

```
//Disabling the assistance TRS function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=ASSISTANCE_TRS_SW-0;
//Modifying the system information configuration of a cell
MOD NRDUCELL: NrDuCellId=0, SibConfigId=0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.2.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

**Step 2** In the navigation tree on the left of the displayed **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** Observe the tracing result on the **Uu Interface Trace** tab page. If SIB17 is displayed, assistance TRS has taken effect. Otherwise, assistance TRS has not taken effect.

**----End**

#### 5.3.2.4.3 Network Monitoring

This function is used to reduce UE power consumption and does not need to be monitored separately on the network side.

## 5.3.3 Connected-mode RRM Measurement Relaxation

### 5.3.3.1 Principles

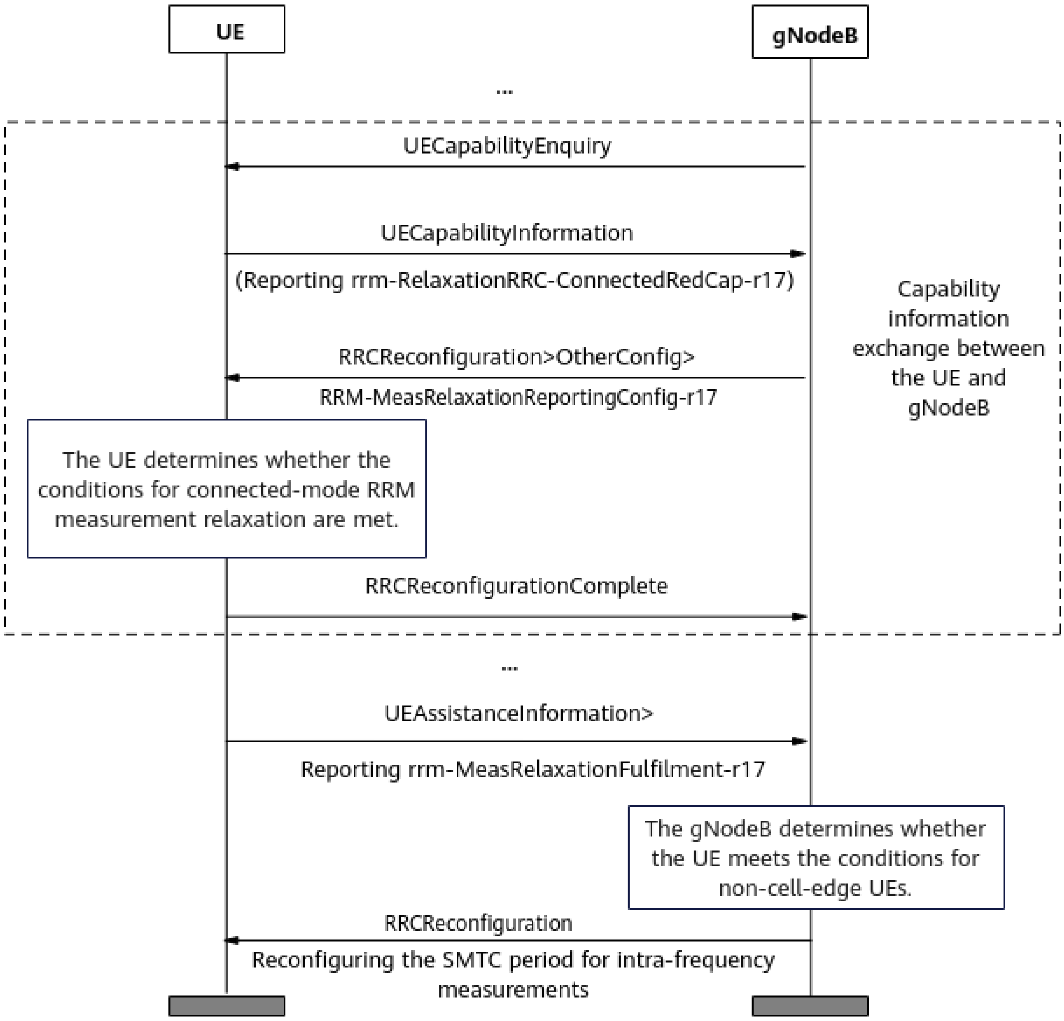
RRM measurements are used to meet user mobility experience requirements and implement seamless mobility management. In compliance with 3GPP specifications, UEs need to periodically measure neighboring cells for radio resource management (RRM) and use the measurement results as the basis for cell reselection or handover. When a UE moves at a low speed or the UE is stationary, the cell that the UE camps on does not change (that is, no cell reselection or handover occurs). In this case, cell reselection or handover is not urgent, and periodic RRM measurement is a waste of UE power. Therefore, when a UE moves at a low speed or the UE is stationary, the RRM measurement can be made less strict (for example, by increasing the RRM measurement interval) to reduce UE power consumption.

3GPP Release 17 allows RedCap UEs to reduce power consumption by idle-mode RRM measurement relaxation (for details, see **4.1.2.1 Idle-mode RRM Measurement Relaxation**) and connected-mode RRM measurement relaxation. Idle-mode RRM measurement relaxation does not require UE capability reporting of RRM measurement relaxation. Connected-mode RRM measurement relaxation depends on the UE capability rrm-RelaxationRRC-ConnectedRedCap-r17.

The connected-mode RRM measurement relaxation function is controlled by the **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter.

Connected-mode RRM measurement relaxation depends on the UAI procedure, as shown in **Figure 5-3**.

**Figure 5-3** Procedure for connected-mode RRM measurement relaxation




In the procedure for this function, note the following:

1. A RedCap UE and gNodeB exchange capability information.

   After an RRC connection is set up between a RedCap UE and base station, if the gNodeB does not obtain necessary UE capabilities, the gNodeB queries UE capabilities as follows:

   a. The gNodeB sends a UECapabilityEnquiry message to the RedCap UE.
   b. The RedCap UE responds to the gNodeB with a UECapabilityInformation message. The **rrm-RelaxationRRC-ConnectedRedCap-r17** field in the message indicates whether the UE supports connected-mode RRM measurement relaxation.
   c. If the RedCap UE supports connected-mode RRM measurement relaxation, the gNodeB delivers an RRCReconfiguration message carrying the *rrm-MeasRelaxationReportingConfig-r17* IE to the UE so that the UE can determine whether the conditions for connected-mode RRM measurement relaxation are met.
   d. The UE responds to the gNodeB with a RRCReconfigurationComplete message.

2. Based on the values of s-SearchDeltaP-Stationary-r17 and t-SearchDeltaP-Stationary-r17, the RedCap UE determines whether the conditions for connected-mode RRM measurement relaxation are met. If the conditions are met, the UE sends a UEAssistanceInformation message to the gNodeB.
   – If the difference between the UE-measured *SS-RSRP* and UE-measured *SS-RSRP_RefStationaryConnected* of the serving cell is less than **NRCellReselConfig**.***RrmRelaxLSUeRsrpDiffThld***, the UE meets the conditions for RRM measurement relaxation for low-speed UEs. The threshold corresponds to the *s-SearchDeltaP-Stationary-r17* IE defined in 3GPP TS 38.331 V17.4.0. where
     - *SS-RSRP* indicates the current SSB RSRP value (dB) of the serving cell measured by the UE.
     - *SS-RSRP_RefStationaryConnected* indicates the reference SSB RSRP value (dB) of the serving cell.

     For details about the preceding fields and how to use the IEs, see section 5.7.4.4 "Relaxed measurement criterion for a stationary RedCap UE" in 3GPP TS 38.331 V17.4.0.
   – If a UE continuously meets the conditions for triggering RRM measurement relaxation for low-speed UEs for a period of time longer than **NRCellReselConfig**.***RrmRelaxLSUeJudgeTimer***, the gNodeB considers that the UE can perform relaxed RRM measurements. The timer corresponds to the *t-SearchDeltaP-Stationary-r17* IE defined in 3GPP TS 38.331 V17.4.0.
3. If the value of rrm-MeasRelaxationFulfilment-r17 in the reported UEAssistanceInformation message is **TRUE**, the gNodeB determines whether to allow connected-mode RRM measurement relaxation to take effect on the UE based on the SRS SINR of the UE and the threshold for connected-mode RRM measurement relaxation (**NRCellRedCap**.***ConnMeasRelaxSinrThld***).
   – For a RedCap UE reporting that it meets the conditions for connected-mode RRM measurement relaxation, the gNodeB performs measurements every second. If the three consecutive measured uplink SINRs of the RedCap UE are greater than or equal to the threshold for connected-mode RRM measurement relaxation plus 3 dB, the gNodeB allows connected-mode RRM measurement relaxation to take effect on the UE.
   – For a RedCap UE in the connected-mode RRM measurement relaxation state, the gNodeB performs measurements every second. If the three consecutive measured uplink SINRs of the RedCap UE are less than the threshold for connected-mode RRM measurement relaxation minus 3 dB, the gNodeB instructs the UE to exit connected-mode RRM measurement relaxation.
4. For a RedCap UE that meets the conditions for connected-mode RRM measurement relaxation, the SMTC period (**NRDUCell**.***SmtcPeriod***) for intra-frequency measurements (including measurements related to events A1, A2, and A3) delivered to the UE is prolonged to 160 ms.

### 5.3.3.2 Network Analysis

#### 5.3.3.2.1 Benefits

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

#### 5.3.3.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

This function has no impact on the network.

### 5.3.3.3 Requirements

#### 5.3.3.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap Introduction must be enabled for connected-mode RRM measurement relaxation to take effect on RedCap UEs. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap Power Saving must be enabled for connected-mode RRM measurement relaxation to take effect on RedCap UEs. |

#### Mutually Exclusive Functions

None

#### Function Impacts

None

#### 5.3.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 5.3.3.3.4 Others

For connected-mode RRM measurement relaxation, UEs must support the rrm-RelaxationRRC-ConnectedRedCap-r17 capability defined in 3GPP Release 17.

### 5.3.3.4 Operation and Maintenance

#### 5.3.3.4.1 Data Configuration

## Data Preparation

Table 5-8 describes the parameters used for function activation.

**Table 5-8** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellFeatureSw**.*UeCapbBasedFunSw* | Select the **REDCAP_ENERGY_SAVING_SW** option of this parameter if connected mode RRM measurement relaxation is required. |
| Connected Measure Relaxation SINR Threshold | **NRCellRedCap**.*ConnMeasRelaxSinrThld* | Use the recommended value. |

## Using MML Commands

Before using MML commands, refer to 5.3.3.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling the connected-mode RRM measurement relaxation function
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Setting the SINR threshold for the connected-mode RRM measurement relaxation function
MOD NRCELLREDCAP: NrCellId=0, ConnMeasRelaxSinrThld=4;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the connected-mode RRM measurement relaxation function
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

## Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 5.3.3.4.2 Activation Verification

## Using Counters

Observe the following counter to check whether the connected-mode RRM measurement relaxation function has taken effect. If the counter value is not 0, it indicates that the connected-mode RRM measurement relaxation function has taken effect.

| Counter ID | Counter Name |
|---|---|
| 1911840393 | **N.User.RRCConn.RRM.Relax.RedCap.Avg** |

## Tracing Signaling

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the RRC_RECFG message contains **rrm-MeasRelaxationReportingConfig-r17**, connected-mode RRM measurement relaxation has been enabled.

**----End**

#### 5.3.3.4.3 Network Monitoring

This function is used to reduce UE power consumption and does not need to be monitored separately on the network side.

## 5.3.4 RedCap Flexible DRX Configuration

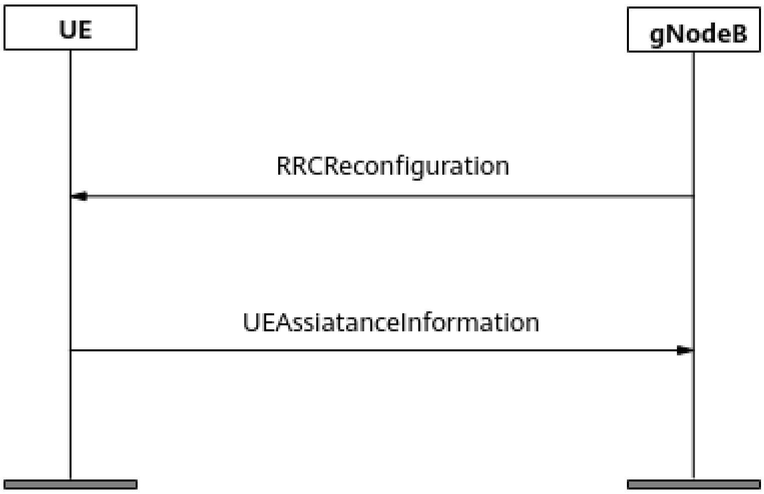
### 5.3.4.1 Principles

The DRX mechanism is a key solution for UE power saving. On the LTE network, base stations send DRX parameters to UEs over RRCReconfiguration messages, without considering the assistance information related to DRX configuration sent by the UEs. To meet the differentiated requirements of different UEs for experience and power saving effects, the NR network supports RedCap flexible DRX configuration.

**The base station can refer to the assistance information (UE-recommended configurations) reported by UEs for optimizing network resource utilization and reducing UE power consumption.**

When RedCap flexible DRX configuration is enabled, the base station configures the *drx-PreferenceConfig* IE in the RRCReconfiguration message for RedCap UEs that support **drx-Preference-r16**. UEs that meet relevant conditions provide the base station with assistance information related to the DRX configuration through the UAI procedure. After verifying that the assistance information is valid, the base station delivers new DRX configurations to the UEs through RRCReconfiguration messages, as shown in **Figure 5-4**.

**Figure 5-4** UE Assistance Information




To prevent RRC reconfiguration due to UEs' frequent reporting of the *drx-Preference-r16* IE, the base station sends the setting of the T346a timer (specified by the **NRCellUeTimer**.***DrxPreferenceProhibitTmr*** parameter) to UEs through the *drx-PreferenceProhibitTimer* IE. UEs then modify their local T346a timer settings accordingly and do not report *drx-Preference-r16* to the base station before the timer expires. For details about the meanings and mechanisms of the *drx-Preference-r16*, *drx-PreferenceConfig*, and *drx-PreferenceProhibitTimer* IEs, see 3GPP TS 38.331 V17.4.0.

RedCap flexible DRX configuration is controlled by the **REDCAP_DRX_FLEX_CONFIG_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter. This function **takes effect only on UEs that either newly access a cell or are newly handed over to it**.

## 5.3.4.2 Network Analysis

### 5.3.4.2.1 Benefits

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

### 5.3.4.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

After RedCap flexible DRX configuration takes effect, the base station performs RRC reconfiguration for UEs based on the DRX-configuration-related assistance information reported by the UEs. The average uplink or downlink UE throughput changes when the following DRX IE settings change:

- preferredDRX-LongCycle-r16

- preferredDRX-InactivityTimer-r16
- preferredDRX-ShortCycle-r16
- preferredDRX-ShortCycleTimer-r16

Different settings of the **NRCellUeTimer**.***DrxPreferenceProhibitTmr*** parameter have different impacts on the network:

- A smaller value of this parameter results in more frequent reporting of assistance information about DRX Preference by UEs, a larger number of RRC reconfiguration messages sent by the gNodeB, and a possible increase in the service drop rate.
- A larger value of this parameter results in less frequent reporting of assistance information about DRX Preference by UEs, a smaller number of RRC reconfiguration messages sent by the gNodeB, and reduced timeliness in UE responses.

To obtain UE power saving gains, the following parameter settings are recommended on the UE side:

- preferredDRX-LongCycle-r16 = ms320
- preferredDRX-InactivityTimer-r16 = ms20

### 5.3.4.3 Requirements

#### 5.3.4.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

## Prerequisite Functions

| Function Name | Function Switch | Reference | Description |
|---|---|---|---|
| DRX | **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter | *DRX* | RedCap flexible DRX configuration requires that DRX be enabled. |
| RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap flexible DRX configuration requires that RedCap Introduction be enabled. |
| RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap flexible DRX configuration requires that RedCap Introduction be enabled. |

## Mutually Exclusive Functions

None

## Function Impacts

None

### 5.3.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 5.3.4.3.4 Others

RedCap UEs that support the **drx-Preference-r16** capability are available.

### 5.3.4.4 Operation and Maintenance

#### 5.3.4.4.1 Data Configuration

### Data Preparation

Table 5-9 describes the parameters used for function activation.

QCI-specific DRX parameter settings require that bearers with specified QCIs have been set up for UEs. Therefore, before using this function, ensure that bearers of relevant QCIs have been set up. For details, see *QoS Management*.

**Table 5-9** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if UAI-based DRX configuration is required for RedCap UEs. |
| UE Assistance Information Switch | **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** | REDCAP_DRX_FLEX_CONFIG_SW | Select this option if RedCap flexible DRX configuration is required for RedCap UEs. |

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |
| DRX Preference Prohibit Timer | **NRCellUeTimer**.***DrxPreferenceProhibitTmr*** | None | A smaller value of this parameter results in more frequent reporting of assistance information about DRX Preference by UEs, a larger number of RRC reconfiguration messages sent by the gNodeB, and a possible increase in the service drop rate. A larger value of this parameter results in less frequent reporting of assistance information about DRX Preference by UEs, a smaller number of RRC reconfigurations, and reduced timeliness in UE responses.<br>This parameter corresponds to the *drx-PreferenceProhibitTimer-r16* IE, which is defined in 3GPP TS 38.331. The 0s and 0.5s timer values are not available, in order to prevent frequent UAI reporting from affecting UE performance. You are advised to use the recommended value. |

### Using MML Commands

Before using MML commands, refer to **5.3.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Enabling the DRX function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=BASIC_DRX_SW-1;
//Enabling the RedCap flexible DRX configuration function
MOD NRCELLALGOSWITCH: NrCellId=0, UeAssistanceInfoSwitch=REDCAP_DRX_FLEX_CONFIG_SW-1;
//Setting the DRX Preference Prohibit Timer
 MOD NRCELLUETIMER: NrCellId=0, DrxPreferenceProhibitTmr=S30;
```

### Deactivation Command Examples

```
//Disabling the RedCap flexible DRX configuration function
MOD NRCELLALGOSWITCH: NrCellId=0, UeAssistanceInfoSwitch=REDCAP_DRX_FLEX_CONFIG_SW-0;
//Disabling the DRX function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, NrDuCellDrxAlgoSwitch=BASIC_DRX_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.4.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the RRC_RECFG message contains **OtherConfig->drx-PreferenceConfig-r16**, RedCap flexible DRX configuration has taken effect.

**----End**

### 5.3.4.4.3 Network Monitoring

This function is used to reduce UE power consumption and does not need to be monitored separately on the network side.

## 5.3.5 RLM Relaxation

### 5.3.5.1 Principles

Radio link monitoring (RLM) relaxation is introduced in 3GPP Release 17. With this function, the base station delivers the IEs required for enabling RLM relaxation to RLM-relaxation-capable RedCap UEs. The UEs determine whether to perform RLM relaxation based on the standards in 3GPP Release 17. If the conditions for enabling RLM relaxation are met, UE power consumed by RLM measurements can be reduced by methods such as prolonging the RLM measurement interval.

This function is controlled by the **RLM_RELAX_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter. When this option is selected, RedCap UEs monitor the downlink radio link quality based on the reference signal for RLM (RLM-RS) in the out-of-synchronization evaluation period. When the physical layer of a RedCap UE detects that the downlink radio link quality is poor, it will report an out-of-synchronization indication to upper layers. If the number of consecutive out-of-synchronization indications received by the UE from the physical layer exceeds the value of **NRDUCellUeTimerConst**.***N310***, the RedCap UE starts the timer T310 (specified by **NRDUCellUeTimerConst**.***T310***).

- If any of the following occurs before the timer expires, the timer is stopped: (1) The UE detects that the physical layer fault has been rectified. (2) The UE receives an RRCReconfiguration message that carries the *reconfigurationWithSync* IE. (3) The UE receives a MobilityFromNRCommand message. (4) The UE initiates an RRC reestablishment procedure.
- Otherwise, the RedCap UE determines that a radio link failure occurs and triggers an RRC connection reestablishment procedure.

The 3GPP R17-defined conditions for UEs to perform RLM relaxation are as follows:

- The UE determines whether it meets the low-speed mobility conditions based on the s-SearchDeltaP-Connected-r17 and t-SearchDeltaP-Connected-r17 fields in the **lowMobilityEvaluationConnected-r17** IE in the RRC Reconfiguration message.
- The UE determines whether the channel quality of the serving cell is good based on the **goodServingCellEvaluationRLM-r17** IE.

For details about the RLM relaxation decision procedure for RedCap UEs, see 3GPP TS 38.331 V17.4.0.

The **NRDUCellUePwrSaving**.***RlmRelaxGoodServCellEvaOfs*** parameter specifies the value of **goodServingCellEvaluationRLM-r17**. When the downlink channel quality of the serving cell is greater than the sum of Q_in and the value of this parameter, RedCap UEs meet the conditions for high quality of the serving cell based on which RLM relaxation can take effect.

### 5.3.5.2 Network Analysis

#### 5.3.5.2.1 Benefits

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

#### 5.3.5.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

This function has no impact on the network.

### 5.3.5.3 Requirements

#### 5.3.5.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

## Prerequisite Functions

| Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- |
| RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RLM relaxation requires that RedCap Introduction be enabled. |
| RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RLM relaxation requires that RedCap Power Saving be enabled. |

## Mutually Exclusive Functions

None

## Function Impacts

None

#### 5.3.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

### 5.3.5.3.4 Others

UEs must support the rlm-Relaxation-r17 capability defined in 3GPP Release 17.

## 5.3.5.4 Operation and Maintenance

### 5.3.5.4.1 Data Configuration

## Data Preparation

**Table 5-10** describes the parameters used for function activation.

**Table 5-10** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if RLM relaxation is required. |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | RLM_RELAX_SW | Select this option if RLM relaxation is required. |
| RLM Relax Good Serving Cell Evaluation Offset | **NRDUCellUePwrSaving**.***RlmRelaxGoodServCellEvaOfs*** | None | Use the recommended value. |
| Low-Speed UE RSRP Diff Thld | **NRDUCellRedCap**.***LsUeRsrpDiffThld*** | None | Use the recommended value. |
| Low-Speed UE Evaluation Timer | **NRDUCellRedCap**.***LsUeEvalTimer*** | None | Use the recommended value. |

### Using MML Commands

Before using MML commands, refer to **5.3.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Enabling RLM relaxation
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=RLM_RELAX_SW-1;
//Setting the RlmRelaxGoodServCellEvaOfs parameter
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, RlmRelaxGoodServCellEvaOfs=DB4;
//Setting the LsUeRsrpDiffThld and LsUeEvalTimer parameters
MOD NRDUCELLREDCAP: NrDuCellId=0, LsUeRsrpDiffThld=6DB, LsUeEvalTimer=5S;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling RLM relaxation
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=RLM_RELAX_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.5.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

**Step 2** In the navigation tree on the left of the displayed **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** Check that RRC_RECFG signaling is displayed in the Uu interface tracing result.

If the **s-SearchDeltaP-Connected-r17** and **t-SearchDeltaP-Connected-r17** fields in the **lowMobilityEvaluationConnected-r17** IE of the RRC_RECFG message meet the conditions for low-speed mobility and the **goodServingCellEvaluationRLM-**

**r17** IE indicates that the channel quality of the serving cell is high, RLM relaxation has taken effect. Otherwise, RLM relaxation has not taken effect.

**----End**

#### 5.3.5.4.3 Network Monitoring

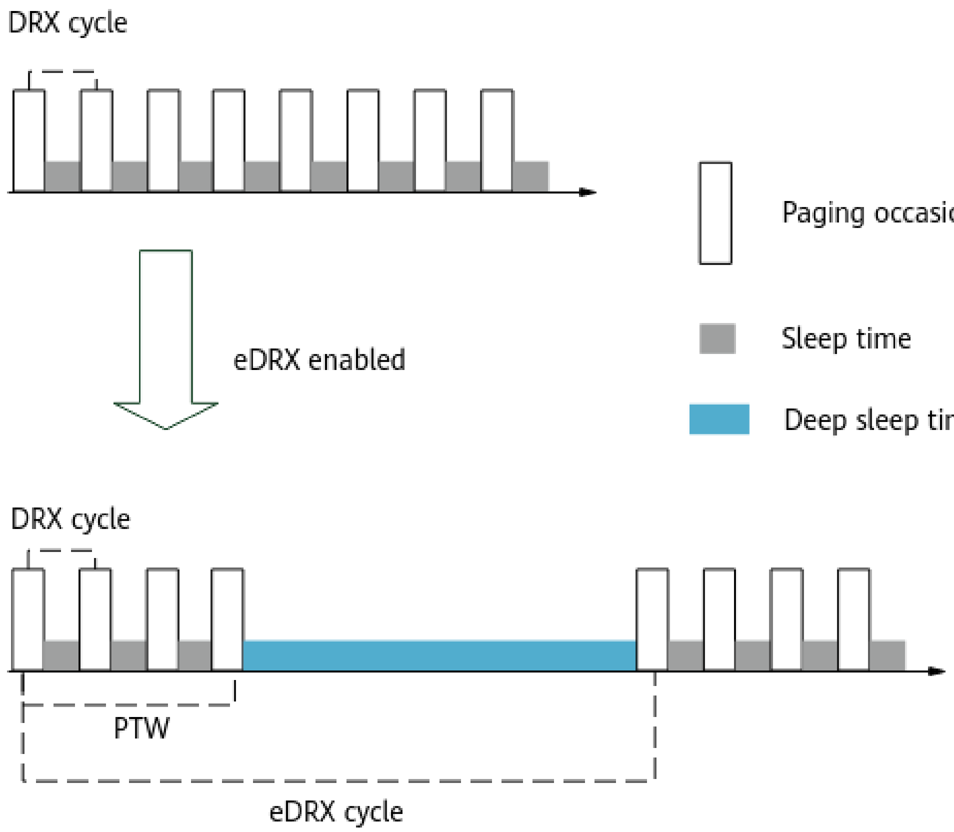
This function is used to reduce UE power consumption and does not need to be monitored separately on the network side.

## 5.3.6 eDRX in Idle Mode (Trial)

### 5.3.6.1 Principles

A UE in idle mode intermittently receives paging messages to save power. This mechanism is described in section 7.4 "Paging in extended DRX" in 3GPP TS 38.304 V17.0.0 and illustrated in **Figure 5-5**. **eDRX in idle mode enables UEs to stay in a long, deep sleep period during an eDRX cycle. The UEs monitor paging messages only during the paging time window (PTW), and the paging monitoring method is the same as that in DRX mode.** For details about DRX, see *DRX*. Other paging procedures in idle mode remain unchanged. For details, see *5G Networking and Signaling*.

**Figure 5-5** eDRX mechanism

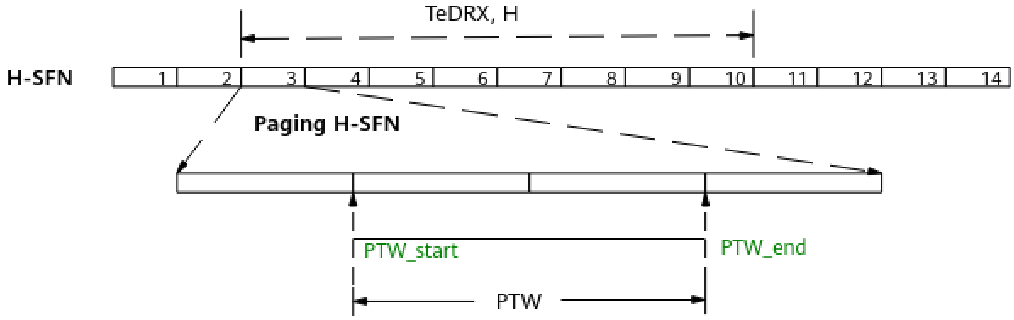



For NR, the DRX cycle is specified by the **NRDUCellPagingConfig**.***DefaultPagingCycle*** parameter, with the default value of

**rf128**, indicating 1.28s. The eDRX cycle length and PTW size are negotiated by the UE and core network. The unit of eDRX cycle length is hyper system frame number (H-SFN). One H-SFN is equal to 10.24s (1024 SFNs). The maximum eDRX cycle length is 2.9 hours (1024 H-SFNs). The PTW size is an integer multiple of 1.28s, and the maximum size is 40.96s (= 1.28s x 32).

The start and end positions of a PTW (PTW_start and PTW_end) are calculated by the gNodeB based on the length of the eDRX cycle ($T_{eDRX,\ H}$) and other related parameters. The gNodeB delivers paging messages and the UE monitors the paging messages during the PTW. **Figure 5-6** shows an example.

**Figure 5-6** Start and end positions of a PTW




When the system information of a cell changes and the eDRX cycle is not aligned with the system information update period, the system information for eDRX UEs will not be updated in time. In this case, the system information modification period (specified by the **NRDUCellSelConfig**.***SiModificationPeriod*** parameter) can be set to reduce the risk of delayed system information updates. If the **NRDUCellSelConfig**.***SiModificationPeriod*** parameter is set to **NOT_CONFIG**, the system information modification period is twice the DRX cycle. The default paging cycle for eDRX UEs is specified by the **NRDUCellPagingConfig**.***SiModEdrxPagingTime*** parameter.

eDRX in idle mode is controlled by the **IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter. This function can take effect on eDRX-capable UEs regardless of whether the UEs are RedCap UEs or not.

eDRX in idle mode is a trial function.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

### 5.3.6.2 Network Analysis

#### 5.3.6.2.1 Benefits

eDRX in idle mode helps UEs save power and prolong standby time by reducing the time UEs spend in monitoring paging messages. eDRX in idle mode is recommended when operators require power saving for eDRX-capable UEs and the core network supports eDRX in idle mode.

#### 5.3.6.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

After eDRX in idle mode takes effect, the paging cycle of UEs is prolonged, UEs cannot respond to paging messages in a timely manner, and the values of the following performance counters related to paging, PDCCH CCE, RRC connection setup, and random access may decrease.

- **N.Paging.NG.Rx**
- **N.Paging.NG.Rx.PLMN**
- **N.Paging.UU.Ue.Num**
- **N.Paging.PEI.UU.Ue.Num**
- **N.Paging.NG.Rx.Discard**
- **N.Paging.NG.Rx.Discard.PLMN**
- **N.CCE.CommDCI.Used.Avg**
- **N.RRC.SetupReq.Att**
- **N.RRC.SetupReq.Att.RedCap**
- **N.RRC.SetupReq.Succ**
- **N.RRC.SetupReq.Succ.RedCap**
- **N.RA.Contention.Att**
- **N.RA.Contention.Resp**
- **N.RA.Contention.Msg3**
- **N.RA.Contention.Resolution.Succ**

When paging messages are concentrated in the paging time window (PTW), the values of the following paging-related counters may increase.

- **N.Paging.Dis.PchCong**
- **N.Paging.Ng.Dis.PchCong**

- **N.Paging.Ng.Dis.PchCong.PLMN**

### 5.3.6.3 Requirements

#### 5.3.6.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

**Prerequisite Functions**

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | eDRX in idle mode requires that RedCap Introduction be enabled. |
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | eDRX in idle mode requires that RedCap Power Saving be enabled. |

### Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Dynamic adjustment of the number of PFs | **PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | None | eDRX in idle mode and dynamic adjustment of the number of PFs cannot be both enabled. |
| Low-frequency TDD | Remote interference adaptive avoidance | **RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter | *Remote Interference Management (Low-Frequency TDD)* | eDRX in idle mode and remote interference adaptive avoidance cannot be both enabled. |
| Low-frequency TDD | Paging capacity mode | **NRDUCellPagingConfig**.***PagingCapacityMode*** | *5G Networking and Signaling* | eDRX in idle mode cannot be enabled when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**. |

### Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RF channel intelligent shutdown | **RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When the setting of the **RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter changes, the beam coverage area may change. As a result, UEs with eDRX in idle mode in effect cannot receive paging messages for a maximum of three hours. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | TTI-level channel shutdown | **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter | *Energy Conservation and Emission Reduction* | When the setting of the **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter changes, the beam coverage area may change. As a result, UEs with eDRX in idle mode in effect cannot receive paging messages for a maximum of three hours. |
| Low-frequency TDD | 3D coverage pattern | **NRDUCellTrpBeam**.***CoverageScenario*** | *Beam Management* | When the setting of the **NRDUCellTrpBeam**.***CoverageScenario*** parameter changes, UEs with eDRX in idle mode in effect may not obtain the latest beam configuration information in a timely manner. As a result, they cannot receive paging messages for a maximum of three hours. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PEI | **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter | *UE Power Saving* | When the setting of the **PEI_SW** or **PAGING_SUBGROUP_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter changes, UEs with eDRX in idle mode in effect may not obtain the latest paging configuration information in a timely manner. As a result, they cannot receive paging messages for a maximum of three hours. |

#### 5.3.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units, except the UBBPfw, support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

#### 5.3.6.3.4 Others

Core network requirements: Core networks must comply with 3GPP Release 17 or later and support eDRX.

UE requirements:UEs must support eDRX.

### 5.3.6.4 Operation and Maintenance

#### 5.3.6.4.1 Data Configuration

##### Data Preparation

Table 5-11 describes the parameters used for function activation.

**Table 5-11** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if eDRX in idle mode is required. |
| Idle UE Power Saving Switch | **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** | IDLE_MODE_EDRX_SW | Select this option if eDRX in idle mode is required. |
| System Information Modification Period | **NRDUCellSelConfig**.***SiModificationPeriod*** | None | It is recommended that this parameter be set based on the network plan if system information is not updated in a timely manner for eDRX UEs because the eDRX cycle is not aligned with the system information update period.<br>If this parameter is set to a value other than **NOT_CONFIG**, the value must be twice, four times, eight times, or 16 times the value of **NRDUCellPagingConfig**.***DefaultPagingCycle***. |
| System Information Modify eDRX Paging Time | **NRDUCellPagingConfig**.***SiModEdrxPagingTime*** | None | Set this parameter based on the network plan. This parameter can be set to a value other than **NOT_CONFIG** only when the **IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter is selected. |

## Using MML Commands

Before using MML commands, refer to **5.3.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Enabling eDRX in idle mode
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=IDLE_MODE_EDRX_SW-1;
//(Optional) Setting the system information modification period if system information is not updated in a timely manner for eDRX UEs because the eDRX cycle is not aligned with the system information update period (using the value RF64 as an example)
MOD NRDUCELLSELCONFIG: NrDuCellId=0, SiModificationPeriod=RF256;
//(Optional) Setting the SiModEdrxPagingTime parameter (using the value HFQUARTER as an example)
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, SiModEdrxPagingTime=HFQUARTER;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling eDRX in idle mode
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=IDLE_MODE_EDRX_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

## Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.6.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RMSI message. If the RMSI message contains **eDRX-AllowedIdle-r17**, eDRX in idle mode has taken effect. Otherwise, eDRX in idle mode has not taken effect.

**----End**

#### 5.3.6.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **5.3.6.2.2 Impacts**.

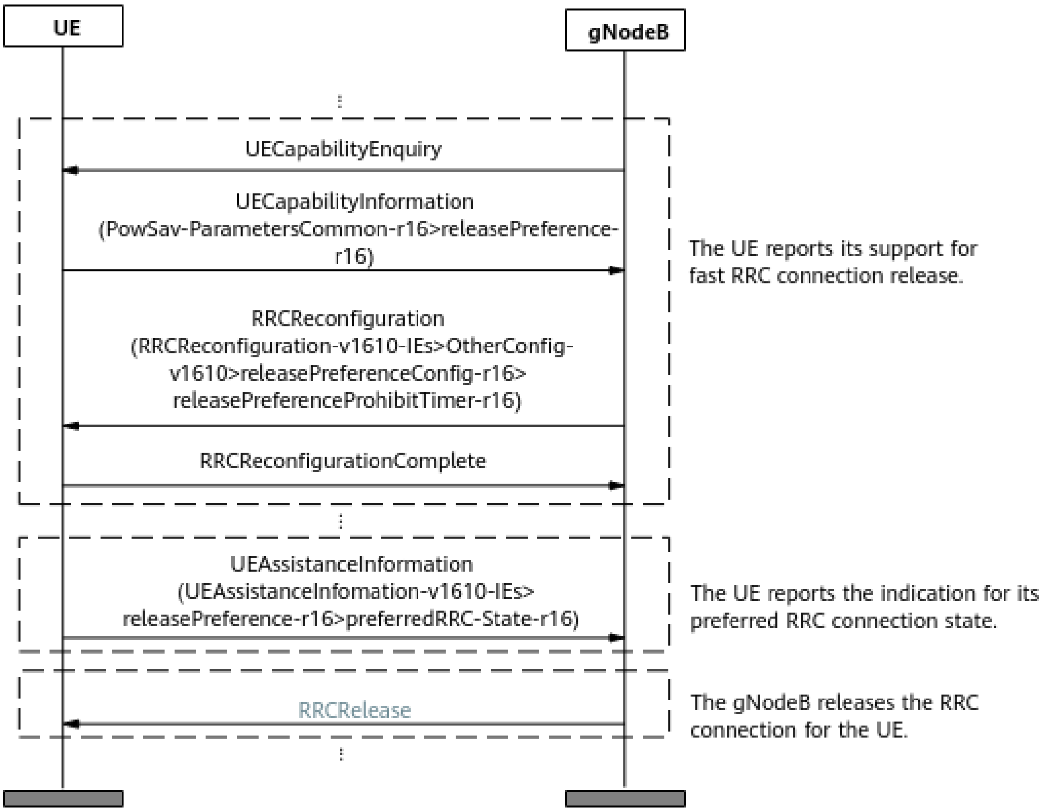
## 5.3.7 RedCap UE Release Preference

### 5.3.7.1 Principles

When a UE has no data to transmit, it is a waste of power for the UE to remain in the RRC_CONNECTED state. To reduce UE power consumption, the RedCap UE release preference function is introduced.

**Figure 5-7** shows the procedure for this function.

**Figure 5-7** Procedure for RedCap UE release preference




This function is implemented as follows:

1. The UE and base station exchange capability information.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.

   b. The UE sends a UECapabilityInformation message to the base station. The **releasePreference-r16** field in the message indicates that the RedCap UE supports fast RRC connection release.

c. The base station sends an RRCReconfiguration message containing the **releasePreferenceProhibitTimer-r16** field to the UE to configure the RRC connection release indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report RRC connection release indication messages. This prevents the UE from frequently reporting the indication.

d. The UE returns an RRCReconfigurationComplete message to the base station.

2. The UE proactively reports the indication for its preferred RRC connection state.

   After the RedCap UE release preference function is enabled, when detecting that the RRC connection can be released, the UE does not need to wait for the inactivity timer on the base station side to expire. Instead, the UE sends a UEAssistanceInformation message to the base station, requesting for an RRC connection release and a switch to the RRC_IDLE or RRC_INACTIVE state. For details about RRC connection states, see *SA Mobility Management in Idle Mode* or *SA Mobility Management in Inactive Mode*.

   The preferred RRC connection state indications proactively reported by the UE include **idle**, **inactive**, **connected**, and **outOfConnected**. **outOfConnected** indicates that the UE expects to exit the RRC_CONNECTED state and the base station determines whether to switch the UE to the RRC_IDLE or RRC_INACTIVE state.

3. Upon receiving this message, the base station releases the RRC connection for the UE, thereby reducing UE power consumption.

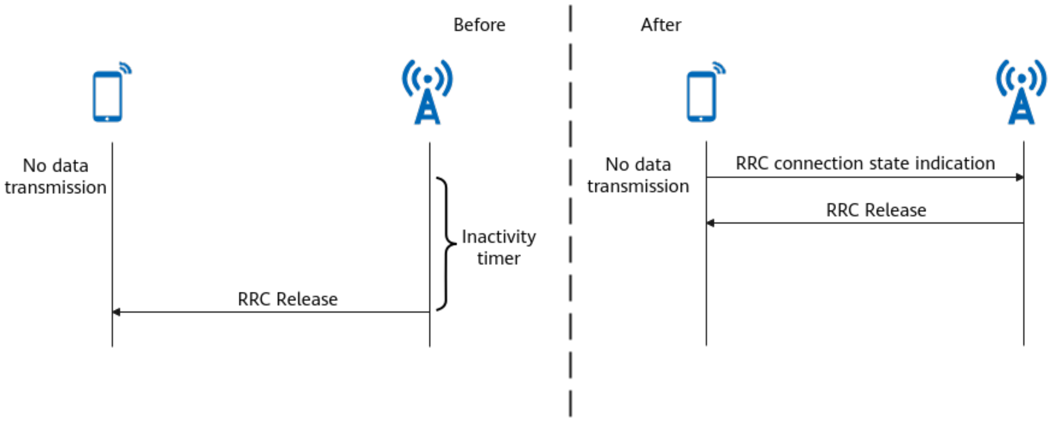
   **If a UE sends a UEAssistanceInformation message to the base station, requesting to change from the RRC_CONNECTED state to the RRC_INACTIVE state, but the conditions for the UE to enter the RRC_INACTIVE state are not met, then the base station switches the UE to the RRC_IDLE state. If the conditions are met, the base station switches the UE to the RRC_INACTIVE state.**

   If the UE is performing positioning or VoNR services, the base station ignores the RRC connection state indication of the UE and does not deliver an RRC connection release message.

**Figure 5-8** illustrates the difference in the mechanisms used before and after the RedCap UE release preference function is enabled.

**Figure 5-8** Comparison of the mechanisms used before and after the RedCap UE release preference function is enabled




This function is controlled by the following parameters:

- For 1RX RedCap UEs, this function is controlled by the **REDCAP_RELEASE_PREFERENCE_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter. To disable this function for specified 1RX RedCap UEs, select the **UE_RELEASE_PREFERENCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter. For details about the blacklist mechanism, see "fast RRC connection release" in *UE Power Saving*.
- For non-1RX RedCap UEs, this function is controlled by the **REDCAP_RELEASE_PREFER_EXT_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter. If whitelist control over RedCap UE release preference is required for specified non-1RX RedCap UEs, select the **REDCAP_RELEASE_PREF_EXT_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter.

## 5.3.7.2 Network Analysis

### 5.3.7.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

### 5.3.7.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

This function has the following impacts on the network:

- The numbers of times that the RRC_IDLE and RRC_INACTIVE states are entered, and the RRC connection is resumed may increase. The number of RRC_CONNECTED UEs may decrease. The number of UE context setup attempts and the number of successful UE context setups may increase.
- The number of RRC connection setup times may increase because the number of times UEs enter the RRC_IDLE state increases.

- In scenarios with ongoing services, the delay may increase if a UE initiates a fast RRC connection release. For details about the impact of the RRC_INACTIVE state on the network, see *SA Mobility Management in Inactive Mode*. The specific impact is closely related to the RRC connection release policy of the UEs.
- If UEs frequently report messages for fast RRC connection release, the CPU load of the main control board may increase.

The involved indicators are as follows:

- Counters related to RRC connection setups: **N.RRC.SetupReq.Att**, **N.RRC.Setup**, and **N.RRC.SetupReq.Succ**.
- Counters related to the number of RRC connection resumes: **N.RRC.ResumeReq.Att**, **N.RRC.Resume**, and **N.RRC.ResumeReq.Succ**
- Counters related to the number of UEs in the RRC_CONNECTED state: **N.User.RRCConn.Max**, **N.User.RRCConn.Avg**, and **N.User.NsaDc.PSCell.Avg**
- Number of UE context setup attempts in a cell: **N.UECntx.Est.Att**
- Number of successful UE context setups in a cell: **N.UECntx.Est.Succ**
- Delay: Packet Delay in DU = **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
- Board CPU usage: **VS.BBUBoard.CPULoad.Mean** and **VS.BBUBoard.CPULoad.Max**

### 5.3.7.3 Requirements

#### 5.3.7.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UE release preference requires RedCap Introduction. |
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UE release preference requires RedCap Power Saving. |

### Mutually Exclusive Functions

None

### Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RRC_INACTIVE connection management | **REDCAP_RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter | *RedCap* | When the RRC_INACTIVE connection management function is in effect, RedCap UE release preference allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state. |

#### 5.3.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 5.3.7.3.4 Others

RedCap UE release preference requires that the UEs be RedCap UEs that support *releasePreference-r16*.

### 5.3.7.4 Operation and Maintenance

#### 5.3.7.4.1 Data Configuration

### Data Preparation

**Table 5-12** describes the parameters used for function activation.

**Table 5-12** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
| --- | --- | --- | --- |
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if the RedCap UE release preference function is required for RedCap UEs. |
| UE Power Saving Switch | **NRCellAlgoSwitch**.***UePwrSavingSwitch*** | REDCAP_RELEASE_PREFERENCE_SW | Select this option if the RedCap UE release preference function is required for 1RX RedCap UEs. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| RedCap Algorithm Switch | **NRCellRedCap**.***RedCapAlgoSwitch*** | REDCAP_RELEASE_PREFER_EXT_SW | Select this option if the RedCap UE release preference function is required for non-1RX RedCap UEs. |
| Whitelist Control Extension Switch | **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** | REDCAP_RELEASE_PREF_EXT_SW_ON | Select this option if the RedCap UE release preference function is required for specified non-1RX RedCap UEs. |

## Using MML Commands

Before using MML commands, refer to **5.3.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Enabling the RedCap UE release preference function
MOD NRCELLALGOSWITCH: NrCellId=0, UePwrSavingSwitch=REDCAP_RELEASE_PREFERENCE_SW-1;
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_RELEASE_PREFER_EXT_SW-1;
//(Optional) Enabling whitelist control over RedCap UE release preference if RedCap UE release preference
is required for specified non-1RX RedCap UEs (UE information needs to be added based on live network
conditions.)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL;
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_RELEASE_PREF_EXT_SW_ON-1;
```

## Deactivation Command Examples

```
//(Optional) Disabling RedCap UE release preference for specified non-1RX RedCap UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_RELEASE_PREF_EXT_SW_ON-0;
//Disabling the RedCap UE release preference function
MOD NRCELLALGOSWITCH: NrCellId=0, UePwrSavingSwitch=REDCAP_RELEASE_PREFERENCE_SW-0;
MOD NRCELLREDCAP: NrCellId=0, RedCapAlgoSwitch=REDCAP_RELEASE_PREFER_EXT_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

## Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 5.3.7.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the RRC_RECFG message contains the *releasePreferenceConfig* IE and the value of **releasePreferenceProhibitTimer-r16** in this IE is not 0, the RedCap UE release preference function has taken effect. Otherwise, this function has not taken effect.

**----End**

#### 5.3.7.4.3 Network Monitoring

The indicators listed in **5.3.7.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

## 5.3.8 RedCap SSSG Switching

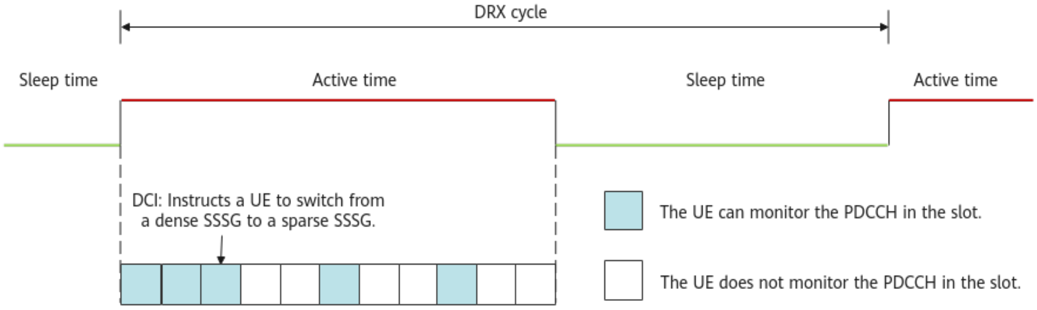
### 5.3.8.1 Principles

When DRX takes effect, UEs do not monitor the PDCCH during the DRX sleep time. This mechanism significantly reduces the power consumption of UEs in the RRC_CONNECTED state by minimizing the frequency of PDCCH monitoring. However, in the DRX active time, a UE monitors the PDCCH in each downlink slot. In heartbeat/sparse packet scenarios, if the base station does not send a PDCCH scheduling indication, monitoring the PDCCH still causes a waste of UE power.

To further reduce UE power, 3GPP Release 17 proposed the RedCap search space set group (SSSG) switching function. A base station supports multiple search space sets to form different SSSGs. PDCCH monitoring densities of the search space may vary with the SSSG. The PDCCH monitoring density of a dense SSSG is high, and that of a sparse SSSG is low. **For a RedCap UE that supports SSSG switching, the base station can instruct the UE to switch from a dense SSSG to a sparse SSSG by using the DCI. This reduces PDCCH monitoring occasions and saves power.**

RedCap SSSG switching can also be used in non-DRX scenarios. However, it is recommended that DRX be enabled preferentially because DRX provides more power saving gains. If further power saving is required, enable this function to reduce UE power consumption in the DRX active time. **Figure 5-9** shows the working principles of RedCap SSSG switching during DRX operation.

**Figure 5-9** RedCap SSSG switching in DRX mode




This function is implemented as follows:

1. The base station determines whether the UE supports SSSG switching.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The UE may indicate, by using a **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17** field, that the UE supports SSSG switching.

2. The base station indicates SSSGs through an RRC reconfiguration message.

   For UEs that support SSSG switching, the **searchSpaceGroupIDList-r17** field is delivered through the RRC reconfiguration message to indicate the SSSGs that can be used by the UE.

3. The base station instructs, by using the DCI, the UE to switch between SSSGs.

   Based on the service status, the base station indicates to the UE which SSSG in the preceding SSSGs should be activated through the **PDCCH monitoring adaptation indication** field in the DCI, completing the SSSG switching. For details about the switching conditions and methods, see **Switching from a Dense SSSG to a Sparse SSSG** and **Switching from a Sparse SSSG to a Dense SSSG**.

## Switching from a Dense SSSG to a Sparse SSSG

The base station determines whether the UE meets the following conditions every 1s. If the UE meets all of the following conditions, the base station instructs the UE to switch from a dense SSSG to a sparse SSSG.

- The downlink RLC throughput of the UE is lower than threshold A, and the downlink RLC buffer data volume is less than threshold B.
- The uplink RLC throughput of the UE is lower than threshold C, and the uplink BSR-indicated data volume is less than threshold D.

**Table 5-13** describes how the thresholds related to RedCap UEs are calculated.

**Table 5-13** Thresholds for switching from a dense SSSG to a sparse SSSG

| Threshold | Meaning | Calculation Method for RedCap UEs |
| --- | --- | --- |
| Threshold A | Threshold of downlink throughput for switching from a dense SSSG to a sparse SSSG | Specified by the **NRDUCellUePwrSaving**.***RedCapSssgSwThptThld*** parameter |
| Threshold B | Threshold of downlink traffic volume for switching from a dense SSSG to a sparse SSSG | min{**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.3, Downlink throughput estimated based on radio channel quality x **NRDUCellUePwrSaving**.***RedCapSssgSwVolThldSlotNum***} |
| Threshold C | Threshold of uplink throughput for switching from a dense SSSG to a sparse SSSG | Specified by the **NRDUCellUePwrSaving**.***RedCapSssgSwThptThld*** parameter |
| Threshold D | Threshold of uplink traffic volume for switching from a dense SSSG to a sparse SSSG | TDD: Uplink throughput estimated based on radio channel quality x **NRDUCellUePwrSaving**.***RedCapSssgSwVolThldSlotNum*** x 0.25 |

## Switching from a Sparse SSSG to a Dense SSSG

The base station performs determination at a granularity of ms. If the UE meets either of the following conditions, the base station instructs the UE to switch from a sparse SSSG to a dense SSSG.

- The downlink RLC buffer data volume of a UE is greater than or equal to threshold E.
- The uplink BSR-indicated data volume of a UE is greater than or equal to threshold F.

Table 5-14 describes how the thresholds related to RedCap UEs are calculated.

**Table 5-14** Threshold for switching from a sparse SSSG to a dense SSSG

| Threshold | Meaning | Calculation Method for RedCap UEs |
| --- | --- | --- |
| Threshold E | Threshold of downlink traffic volume for switching from a sparse SSSG to a dense SSSG | min{**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.4, Downlink throughput estimated based on radio channel quality x **NRDUCellUePwrSaving**.***RedCapSssgSwVolThldSlotNum*** x 4} |

| Threshold | Meaning | Calculation Method for RedCap UEs |
| --- | --- | --- |
| Threshold F | Threshold of uplink traffic volume for switching from a sparse SSSG to a dense SSSG | TDD: Uplink throughput estimated based on radio channel quality x **NRDUCellUePwrSaving**.***RedCapSssgSwVolThldSlotNum*** |

This function is controlled by the following parameters:

- For 1RX RedCap UEs, this function is controlled by the **REDCAP_SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter.
- For non-1RX RedCap UEs, this function is controlled by the **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter. If whitelist control over RedCap SSSG switching is required for specified non-1RX RedCap UEs, select the **REDCAP_SSSG_SWITCH_EXT_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter.

The level involved in SSSG switching can be configured using the **NRDUCellUePwrSaving**.***SssgSwitchingLevel*** parameter. Compared with **LEVEL0**, **LEVEL1** indicates that the SSSG is more sparse and the power saving effect is better.

### 5.3.8.2 Network Analysis

#### 5.3.8.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

#### 5.3.8.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

Enabling RedCap SSSG switching may have the following impacts:

- The average uplink and downlink cell throughputs and the average uplink and downlink UE throughputs decrease.
- Cell edge uplink and downlink throughputs may decrease.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The delay increases.
- The CCE usage decreases.
- The uplink and downlink CCE allocation success rates decrease.
- The QoS flow service drop rate of all services increases.
- The values of the counters related to the uplink and downlink RB usages may increase or decrease.

- The RRC connection setup success rate fluctuates.

The involved indicators are as follows:

| Counter Description | Formula |
| --- | --- |
| Average uplink cell throughput | **Cell Uplink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average uplink UE throughput | **User Uplink Average Throughput (DU)** |
| Average downlink UE throughput | **User Downlink Average Throughput (DU)** |
| Cell edge uplink cell throughput | (**N.ThpVol.UL.CellOverlap** – **N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap** |
| Cell edge downlink cell throughput | (**N.ThpVol.DL.CellOverlap** – **N.ThpVol.DL.LastSlot.CellOverlap**)/ **N.ThpTime.DL.RmvLastSlot.CellOverlap** |
| Uplink IBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink IBLER | (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |

| Counter Description | Formula |
|---|---|
| Uplink RBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink RBLER | (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |
| Average transmission delay of the first packet | **N.Traffic.DL.RlcFirstPktDelay.Time**/ **N.Qos.DL.RlcFirstPktDelay.Num** |
| Average air interface processing delay of downlink data packets | **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** |
| Average RLC processing delay of downlink data packets | **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num** |
| Average gNodeB DU processing delay of downlink data packets | **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num** |
| CCE usage | **N.CCE.Used.Avg/N.CCE.Avail.Avg** |
| Uplink CCE allocation success rate | ((**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**) / **N.CCE.UL.AllocReq.Num**) × 100% |
| Downlink CCE allocation success rate | ((**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**) / **N.CCE.DL.AllocReq.Num**) × 100% |
| Service drop rate | **Service Call Drop Rate (CU)** |

| Counter Description | Formula |
| --- | --- |
| Downlink RB usage in a cell | **Downlink Resource Block Utilizing Rate (DU)** |
| Uplink RB usage in a cell | **Uplink Resource Block Utilizing Rate (DU)** |
| PDSCH DRB PRB usage | **N.PRB.DL.DrbUsed.Avg**/ **N.PRB.DL.Avail.Avg** |
| PUSCH DRB PRB usage | **N.PRB.UL.DrbUsed.Avg**/ **N.PRB.UL.PUSCH.Avail.Avg** |
| Average PUSCH PRB usage | **N.PRB.PUSCH.Used.Avg**/ **N.PRB.UL.PUSCH.Avail.Avg** |
| RRC connection setup success rate | **RRC Setup Success Rate (CU)** |

**NOTE**

Throughput-related counters for UEs in overlapping areas are measured only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter is selected. In addition, the **NRCellComp**.***OverlappingRsrpOffset*** parameter needs to be configured, and its value must be the same as the value of either **NRCellComp**.***UlCompRsrpOffset*** or **NRCellComp**.***DlCompRsrpOffset***.

## 5.3.8.3 Requirements

### 5.3.8.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- |
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.3.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

## Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap SSSG switching depends on RedCap Introduction. |
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap SSSG switching requires that RedCap Power Saving be enabled. |

## Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | IAB deployment | **NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB** | None | If **NRDUCell**.***DeployPosition*** is set to **DEPLOY_IAB**, RedCap SSSG switching must be disabled. |

## Function Impacts

Coordination-related functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Co-MM | **INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter | *Co-MM (Low-Frequency TDD)* | UEs with Co-MM in effect do not switch from a dense SSSG to a sparse one. For Co-MM to take effect for a UE with a sparse SSSG in effect, the UE must switch to a dense SSSG. |
| Low-frequency TDD | Intra-base-station UL CoMP | **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | UEs with intra-base-station UL CoMP in effect do not switch from a dense SSSG to a sparse one. For intra-base-station UL CoMP to take effect for a UE with a sparse SSSG in effect, the UE must switch to a dense SSSG. |
| Low-frequency TDD | Intra-base-station DL CoMP | **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter | *CoMP* | UEs with intra-base-station DL CoMP in effect do not switch from a dense SSSG to a sparse one. For intra-base-station DL CoMP to take effect for a UE with a sparse SSSG in effect, the UE must switch to a dense SSSG. |

URLLC-related functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Low latency and high reliability | **HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter | *URLLC* | UEs with low latency and high reliability in effect do not switch to a sparse SSSG. |

Other functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Super uplink | **FLEX_SUPER_UPLINK_SW** and **INTER_GNB_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter | *Super Uplink* | UEs with super uplink in effect do not switch to a sparse SSSG. |
| Low-frequency TDD | Basic VoNR functions | **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter | *VoNR* | UEs running VoNR services do not switch to a sparse SSSG. |
| Low-frequency TDD | FWA | **gNBQciBearer**.***ServiceType*** set to **FWA_INTERNET**, **FWA_VIDEO**, or **FWA_PRIVATE_LINE** | *FWA* | UEs running FWA services do not switch to a sparse SSSG. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Ultimate Cloud X experience | **gNBQciBearer**.***ServiceType*** set to **EMBB_CLOUD_VR**, **EMBB_CLOUD_GAMING**, or **EMBB_CLOUD_PHONE** | *Ultimate Cloud X Experience* | UEs with ultimate Cloud X experience in effect do not switch to a sparse SSSG. |
| Low-frequency TDD | PDCCH skipping | **PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter | *UE Power Saving* | Assume that both PDCCH skipping and RedCap SSSG switching are enabled. If a UE reports both **pdcch-SkippingWithoutSSSG-r17** and **SSSG-Switching-1BitInd-r17** but not **pdcch-SkippingWithSSSG-r17**, only RedCap SSSG switching takes effect on the UE. |

#### 5.3.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 5.3.8.3.4 Cells

For a TDD cell, the cell slot assignment can only be **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

### 5.3.8.3.5 Others

RedCap SSSG switching requires that UEs must support the capabilities indicated by **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17**.

## 5.3.8.4 Operation and Maintenance

### 5.3.8.4.1 Data Configuration

## Data Preparation

Table 5-15 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-15** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if RedCap SSSG switching is required. |
| UE Power Saving Switch | **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** | REDCAP_SSSG_SWITCHING_SW | Select this option if RedCap SSSG switching is required for 1RX RedCap UEs. |
| UE Power Saving Switch | **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** | REDCAP_SSSG_SWITCH_EXT_SW | Select this option if RedCap SSSG switching is required for non-1RX RedCap UEs. |
| Whitelist Control Extension Switch | **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** | REDCAP_SSSG_SWITCH_EXT_SW_ON | Select this option if RedCap SSSG switching is required for specified non-1RX RedCap UEs. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| SSSG Switching Level | **NRDUCellUePwrSaving**.***SssgSwitchingLevel*** | None | Use the recommended value. |
| RedCap SSSG SW Throughput Threshold | **NRDUCellUePwrSaving**.***RedCapSssgSwThptThld*** | None | Use the recommended value. |
| RedCap SSSG SW Volume Threshold Slot Num | **NRDUCellUePwrSaving**.***RedCapSssgSwVolThldSlotNum*** | None | Use the recommended value. |

## Using MML Commands

Before using MML commands, refer to **5.3.8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Turning on the RedCap SSSG switching switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
UePowerSavingSwitch=REDCAP_SSSG_SWITCHING_SW-1&REDCAP_SSSG_SWITCH_EXT_SW-1;
//(Optional) Enabling whitelist control over RedCap SSSG switching if RedCap SSSG switching is required for
specified non-1RX RedCap UEs (UE information needs to be added based on live network conditions.)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL;
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_SSSG_SWITCH_EXT_SW_ON-1;
//Configuring the SSSG switching level
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, SssgSwitchingLevel=LEVEL0;
//Configuring the thresholds related to SSSG switching for RedCap UEs
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,RedCapSssgSwThptThld
=1000,RedCapSssgSwVolThldSlotNum=4;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling RedCap SSSG switching for specified non-1RX RedCap UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_SSSG_SWITCH_EXT_SW_ON-0;
//Turning off the RedCap SSSG switching switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
UePowerSavingSwitch=REDCAP_SSSG_SWITCHING_SW-0&REDCAP_SSSG_SWITCH_EXT_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.8.4.2 Activation Verification

**Step 1** Enable a UE that supports **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17** to initially access the network or initiate a handover.

**Step 2** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 3** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 4** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the RRC_RECFG message contains "SearchSpace > searchSpaceGroupIDList-r17", RedCap SSSG switching has taken effect. Otherwise, RedCap SSSG switching has not taken effect.

**----End**

### 5.3.8.4.3 Network Monitoring

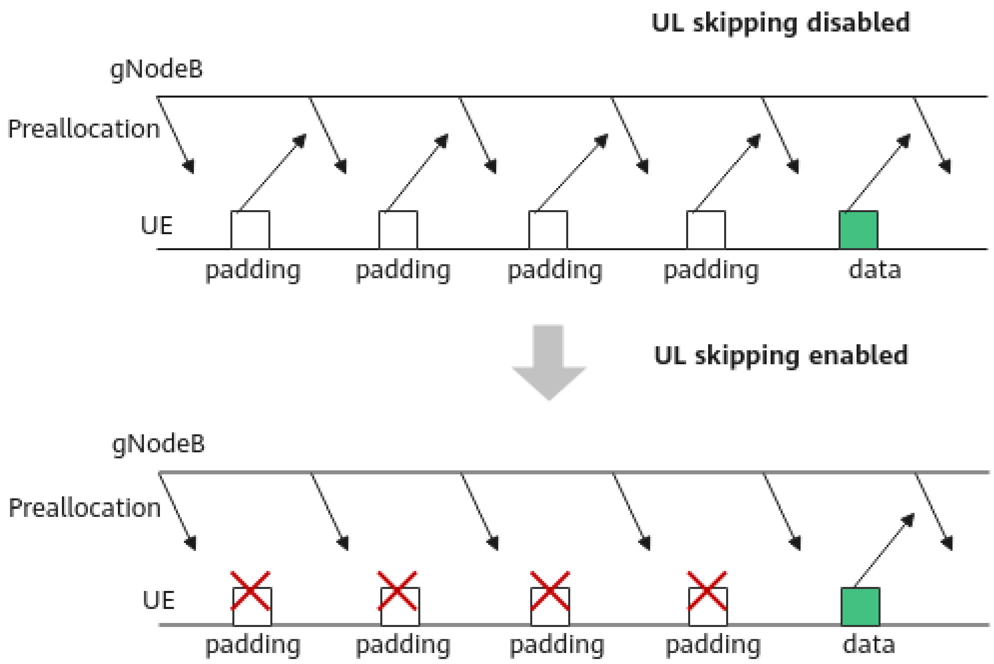
The performance indicators described in **5.3.8.2.2 Impacts** can be used for network monitoring of this function.

## 5.3.9 RedCap UL Skipping

### 5.3.9.1 Principles

DRX saves UE power but increases service delay, affecting user experience. To reduce the impact of power saving on delay, uplink RedCap preallocation is usually enabled. However, after uplink RedCap preallocation is enabled, a UE must respond to the base station with padding packets upon receiving UL grants even if the UE has no data to transmit, consuming excessive UE power. As such, 3GPP specifications proposed RedCap UL skipping for UE power saving. **For a UE supporting UL skipping, sending padding packets is not required if the UE has no data to transmit after it receives UL grants from the base station**, as shown in **Figure 5-10**. **In this way, power consumption is reduced**.

**Figure 5-10** RedCap UL skipping




The process of this function is as follows:

1. The UE and base station exchange capability information.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The UE indicates the UL skipping capability by using the **SkipUplinkTxDynamic**, **enhancedSkipUplinkTxDynamic-r16**, or **enhancedSkipUplinkTxDynamic-v1660** field.
2. The base station sends an RRC message to the UE for which preallocation can take effect, notifying the UE that the UE is allowed to perform UL skipping. In the RRC message, the **SkipUplinkTxDynamic** field is set to **True** or the **enhancedSkipUplinkTxDynamic-r16** field is carried.
3. When the base station sends a UL grant to the UE after delivering RRC configurations to the UE, the UE does not send any padding packets if it has no data to transmit. In this way, power saving is achieved.

This function is controlled by the following parameters:

- For 1RX RedCap UEs, this function is controlled by the **REDCAP_UPLINK_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter and is applicable to 1RX RedCap UEs that newly access or are newly handed over to the cell. To disable this function for specified 1RX RedCap UEs, select the **UPLINK_SKIP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter. For details about the blacklist mechanism, see "UL skipping" in *UE Power Saving*.

- For non-1RX RedCap UEs, this function is controlled by the **REDCAP_UPLINK_SKIP_EXT_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter and is applicable to 1RX RedCap UEs that newly access or are newly handed over to the cell. If whitelist control over RedCap UL skipping is required for specified non-1RX RedCap UEs, select the **REDCAP_UPLINK_SKIP_EXT_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter.

## 5.3.9.2 Network Analysis

### 5.3.9.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

### 5.3.9.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

Enabling RedCap UL skipping may have the following impacts:

- The proportion of padding packets in a cell and the data rate at the MAC layer decrease.
- For UEs for which UL skipping takes effect, BWP switching can be performed only through RRC connection reconfiguration. In addition, the decrease in padding packets may cause false or missing DTX detection. Therefore, UL skipping may increase the service delay.
- The number of uplink padding packets sent by UEs decreases. As a result, the number of DTX detections increases, and the number of times PUSCH RSRP values fall into the range of index 0 indicated by **N.UL.PUSCH.RSRP.Index0** increases.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The service drop rate increases.

The involved indicators are as follows:

| Counter Description | Formula |
|---|---|
| Service delay | **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num** |

| Counter Description | Formula |
|---|---|
| Uplink IBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink IBLER | (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |
| Uplink RBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink RBLER | (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |
| Service drop rate | **N.PDUSession.AbnormRel**/ (**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) x 100% |

### 5.3.9.3 Requirements

#### 5.3.9.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UL skipping depends on RedCap Introduction. |
| Low-frequency TDD | RedCap Power Saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap UL skipping requires that RedCap Power Saving be enabled. |
| Low-frequency TDD | Uplink RedCap preallocation | **UL_REDCAP_PREALLOCATION_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter | *RedCap* | RedCap UL skipping takes effect only after uplink RedCap preallocation is enabled. |

### Mutually Exclusive Functions

None

### Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Time-domain adaptive filtering | **PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter | *Turbo Uplink (Low-Frequency TDD)* | Time-domain adaptive filtering does not take effect on UEs with RedCap UL skipping in effect. |
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** | *High Speed Mobility* | When High-speed Railway Superior Experience is enabled, RedCap UL skipping does not take effect. |
| Low-frequency TDD | Basic VoNR functions | **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter | *VoNR* | RedCap UL skipping does not take effect on UEs running VoNR services. Once a UE with RedCap UL skipping in effect initiates a VoNR service, RedCap UL skipping will be deactivated for that UE. |

### 5.3.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 5.3.9.3.4 Others

RedCap UL skipping requires that UEs support **SkipUplinkTxDynamic**, **enhancedSkipUplinkTxDynamic-r16**, or **enhancedSkipUplinkTxDynamic-v1660**.

## 5.3.9.4 Operation and Maintenance

### 5.3.9.4.1 Data Configuration

## Data Preparation

Table 5-16 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-16** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option when the RedCap UL skipping function is required. |
| Scheduling Power Saving Switch | **NRDUCellUePwrSaving**.***SchPwrSavingSw*** | REDCAP_UPLINK_SKIP_SW | Select this option if the RedCap UL skipping function is required for 1RX RedCap UEs. |
| Scheduling Power Saving Switch | **NRDUCellUePwrSaving**.***SchPwrSavingSw*** | REDCAP_UPLINK_SKIP_EXT_SW | Select this option if the RedCap UL skipping function is required for non-1RX RedCap UEs. |
| Whitelist Control Extension Switch | **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** | REDCAP_UPLINK_SKIP_EXT_SW_ON | Select this option if the RedCap UL skipping function is required for specified non-1RX RedCap UEs. |

## Using MML Commands

Before using MML commands, refer to **5.3.9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Turning on the RedCap UL Skipping switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
SchPwrSavingSw=REDCAP_UPLINK_SKIP_SW-1&REDCAP_UPLINK_SKIP_EXT_SW-1;
//(Optional) Enabling whitelist control over RedCap UL skipping if RedCap UL skipping is required for
specified non-1RX RedCap UEs (UE information needs to be added based on live network conditions.)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL;
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_UPLINK_SKIP_EXT_SW_ON-1;
```

## Deactivation Command Examples

```
//(Optional) Disabling RedCap UL skipping for specified non-1RX RedCap UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_UPLINK_SKIP_EXT_SW_ON-0;
//Turning off the RedCap UL Skipping switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
SchPwrSavingSw=REDCAP_UPLINK_SKIP_SW-0&REDCAP_UPLINK_SKIP_EXT_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

## Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.9.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** Create a Uu interface tracing task on the MAE-Access. Enable a UE with the **SkipUplinkTxDynamic** capability defined in 3GPP Release 15 or the

**enhancedSkipUplinkTxDynamic-r16** or **enhancedSkipUplinkTxDynamic-v1660** capability defined in 3GPP Release 16 to initially access the network.

**Step 4** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the value of the **SkipUplinkTxDynamic** field under **MAC-CellGroupConfig** in the RRC_RECFG message is **True** or the **enhancedSkipUplinkTxDynamic-r16** field is present under **MAC-CellGroupConfig** in the RRC_RECFG message, the RedCap UL skipping function has taken effect. Otherwise, this function has not taken effect.

**----End**

### 5.3.9.4.3 Network Monitoring

After this function is enabled, the value of **N.ThpVol.UL.Cell** (total volume of uplink data received at the MAC layer in a cell) decreases and that of **N.ThpVol.UL** (total volume of uplink data received at the RLC layer in a cell) remains basically unchanged.

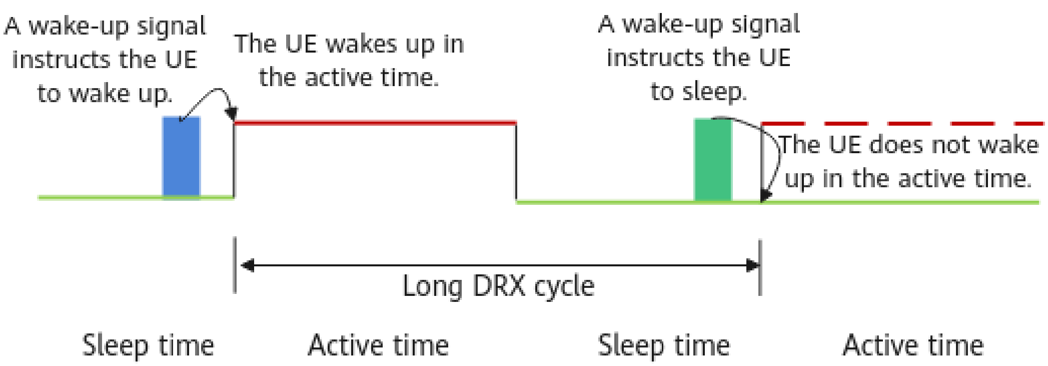
## 5.3.10 RedCap DRX Adaptation

### 5.3.10.1 Principles

When DRX takes effect, UEs do not monitor the PDCCH during the DRX sleep time. This mechanism significantly reduces the power consumption of UEs in the RRC_CONNECTED state by minimizing the frequency of PDCCH monitoring. However, in each DRX active time, the UE has to wake up to monitor the PDCCH. When the traffic volume of a UE is small, the scheduling probability is low. Waking up in each cycle consumes excessive UE power. To address this issue, RedCap DRX adaptation is introduced.

After the function takes effect, **a WUS DCI format 2_6 is sent to the UE before the UE enters the On Duration. This signal indicates whether the UE needs to monitor PDCCH scheduling information while the long-cycle DRX On Duration Timer is running. In this way, the UE can wake up when required**. Because the length of a WUS in the time domain is shorter than the On Duration Timer, as shown in **Figure 5-11**, this method can save UE power.

**Figure 5-11** RedCap DRX adaptation




The process of this function is as follows:

1. The UE and base station exchange capability information.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The **drx-Adaptation-r16** field in the message indicates that the UE supports RedCap DRX adaptation.

2. The base station delivers the DCI format 2_6 configuration.

   If the RedCap DRX adaptation function is enabled, the base station delivers the DCI format 2_6 configuration to UEs that support the capability indicated by **drx-Adaptation-r16**. Such configuration includes:

   – ps-Offset-r16: indicates the interval between the start position of a DCI format 2_6 and the start of the On Duration Timer corresponding to the long DRX cycle. Based on this information, the UE can calculate when to detect the DCI format 2_6.
   – sizeDCI-2-6-r16: indicates the length of a DCI format 2_6.
   – ps-PositionDCI-2-6: indicates the specific bit occupied by a DCI format 2_6 for a UE.
   – ps-WakeUp-r16: indicates that a UE needs to wake up to monitor the PDCCH during the On Duration of the next long DRX cycle if the UE does not receive a DCI format 2_6. The value is always **TRUE**.
   – ps-TransmitOtherPeriodicCSI-r16: indicates that the UE needs to report periodic CSI measurement results within the On Duration. This field is carried only when the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected.

3. DCI format 2_6 is used to indicate whether a UE needs to wake up or can remain to sleep in the next On Duration. This applies only to the On Duration of a long DRX cycle, not that of a short DRX cycle.

   When there is no data to be sent in the RLC buffer, the base station instructs the UE to sleep through the DCI format 2_6 with a bit value of **0**. When there is data to be transmitted in the RLC buffer, the base station instructs the UE to wake up through the DCI format 2_6 with a bit value of **1**. If the UE is instructed to sleep, it does not need to wake up to monitor the PDCCH in the On Duration of the next long DRX cycle, and does not send the SRS in the On Duration period.

This function is controlled by the following parameters:

- For 1RX RedCap UEs, this function is controlled by the **REDCAP_DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter and is applicable to 1RX RedCap UEs that newly access or are newly handed over to the cell. To disable this function for specified 1RX RedCap UEs, select the **DRX_ADAPTATION_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter. For details about the blacklist mechanism, see "WUS" in *UE Power Saving*.
- For 1RX RedCap UEs, this function is controlled by the **REDCAP_DRX_ADAPTATION_EXT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter and is applicable to 1RX RedCap UEs that newly access or are newly handed over to the cell. If whitelist control over RedCap DRX adaptation is required for specified non-1RX RedCap UEs, select the **REDCAP_DRX_ADAPT_EXT_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter.

### 5.3.10.2 Network Analysis

#### 5.3.10.2.1 Benefits

The RedCap DRX adaptation function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

When the RedCap DRX adaptation function is enabled and the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, the delay may decrease and the downlink user-perceived rate may increase, compared with when this option is deselected.

#### 5.3.10.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

RedCap DRX adaptation may result in:

- Decrease in average uplink and downlink cell throughput
- Significant decrease in average uplink and downlink UE throughput
- Increase in delay
- Increase or decrease in the uplink and downlink initial block error rates (IBLERs)/residual block error rates (RBLERs)
- Increase in the service drop rate

After the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, UE power consumption increases and the downlink user-perceived rate increases. However, if the CSI measurement results are inaccurate, the downlink user-perceived rate decreases.

The involved indicators are as follows:

| Counter Description | Formula |
|---|---|
| Average uplink cell throughput | **Cell Uplink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average uplink UE throughput | **User Uplink Average Throughput (DU)** |
| Average downlink UE throughput | **User Downlink Average Throughput (DU)** |
| Delay | **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num** |

| Counter Description | Formula |
|---|---|
| Uplink IBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink IBLER | (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |
| Uplink RBLER | (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100% |
| Downlink RBLER | (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100% |
| Service drop rate | **N.PDUSession.AbnormRel**/ (**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) x 100% |

### 5.3.10.3 Requirements

#### 5.3.10.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.3.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

**Prerequisite Functions**

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap DRX adaptation depends on RedCap Introduction. |
| Low-frequency TDD | RedCap power saving | **REDCAP_ENERGY_SAVING_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | RedCap DRX adaptation requires that RedCap Power Saving be enabled. |
| Low-frequency TDD | DRX | **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter | *DRX* | RedCap DRX adaptation requires that DRX be enabled. |

## Mutually Exclusive Functions

None

## Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PEI | **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter | *UE Power Saving* | The transmission of the DCI format 2_7 may affect the transmission of the WUS DCI format 2_6 and fallback DCI scheduling. As a result, the proportion of occasions where RedCap DRX adaptation takes effect decreases. |
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | When RedCap UEs are using BWP1 with the NCD-SSB, RedCap DRX adaptation does not take effect on these RedCap UEs. |
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** | *High Speed Mobility* | When High-speed Railway Superior Experience is enabled, RedCap DRX adaptation does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Basic VoNR functions | **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter | *VoNR* | RedCap DRX adaptation does not take effect on UEs running VoNR services. Once a UE with RedCap DRX adaptation in effect initiates a VoNR service, RedCap DRX adaptation will be deactivated for that UE. |

### 5.3.10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 5.3.10.3.4 Cells

For a TDD cell, the **NRDUCell**.***SlotAssignment*** parameter cannot be set to **3_7_DSUUUUSUUU**.

#### 5.3.10.3.5 Others

RedCap DRX adaptation requires that UEs support "longDRX-Cycle" and 3GPP-R16-defined "drx-Adaptation-r16".

### 5.3.10.4 Operation and Maintenance

#### 5.3.10.4.1 Data Configuration

##### Data Preparation

Table 5-17 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-17** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UE Capability-based Function Switch | **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** | REDCAP_ENERGY_SAVING_SW | Select this option if RedCap DRX adaptation is required. |
| NR DU Cell DRX Algorithm Switch | **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** | REDCAP_DRX_ADAPTATION_SW | Select this option if RedCap DRX adaptation is required for 1RX RedCap UEs. |
| NR DU Cell DRX Algorithm Switch | **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** | REDCAP_DRX_ADAPTATION_EXT_SW | Select this option if RedCap DRX adaptation is required for non-1RX RedCap UEs. |
| NR DU Cell DRX Algorithm Switch | **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** | SLEEP_DRXOD_CSI_RPT_SW | Select this option if CSI measurement result reporting during the DRX sleep time specified by the On Duration timer is required. |
| Whitelist Control Extension Switch | **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** | REDCAP_DRX_ADAPT_EXT_SW_ON | Select this option if RedCap DRX adaptation is required for specified non-1RX RedCap UEs. |

##### Using MML Commands

Before using MML commands, refer to 5.3.10.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-1;
//Turning on the RedCap DRX adaptation switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=REDCAP_DRX_ADAPTATION_SW-1&REDCAP_DRX_ADAPTATION_EXT_SW-1;
//(Optional) Turning on SLEEP_DRXOD_CSI_RPT_SW if CSI measurement result reporting during the DRX
sleep time specified by the On Duration timer is required
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=SLEEP_DRXOD_CSI_RPT_SW-1;
//(Optional) Enabling whitelist control over RedCap DRX adaptation if RedCap DRX adaptation is required
for specified non-1RX RedCap UEs (UE information needs to be added based on live network conditions.)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL;
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_DRX_ADAPT_EXT_SW_ON-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling RedCap DRX adaptation for specified non-1RX RedCap UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=REDCAP_DRX_ADAPT_EXT_SW_ON-0;
//(Optional) Turning off SLEEP_DRXOD_CSI_RPT_SW
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=SLEEP_DRXOD_CSI_RPT_SW-0;
//Turning off the RedCap DRX adaptation switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,
NrDuCellDrxAlgoSwitch=REDCAP_DRX_ADAPTATION_SW-0&REDCAP_DRX_ADAPTATION_EXT_SW-0;
//Disabling RedCap Power Saving
MOD NRDUCELLFEATURESW: NrDuCellId=0, UeCapbBasedFunSw=REDCAP_ENERGY_SAVING_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.3.10.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** Create a Uu interface tracing task on the MAE-Access. Enable a UE that supports the **drx-Adaptation-r16** capability defined in 3GPP Release 16 to initially access the network.

**Step 4** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If the RRC_RECFG message contains **physicalCellGroupConfig->dcp-Config-r16**, the RedCap DRX adaptation function has taken effect. Otherwise, this function has not taken effect.

**----End**

#### 5.3.10.4.3 Network Monitoring

After this function is enabled, monitor the following changes described in **5.3.10.2.2 Impacts**:

- Decrease in average uplink and downlink cell throughput
- Significant decrease in average uplink and downlink UE throughput
- Increase in delay
- Possible increase or decrease in the uplink and downlink initial block error rates (IBLERs)/residual block error rates (RBLERs)
- Increase in the service drop rate

Check the values of **N.Cdrx.Sleep.Dur.Total** (total duration of RRC_CONNECTED UEs in the sleep time in a cell) and **N.Cdrx.Active.Dur.Total** (total duration of RRC_CONNECTED UEs in the active time in a cell) to evaluate the power saving effect of UEs.

# 5.4 RedCap Mobility Management Enhancement

## 5.4.1 Uplink-Experience-based Inter-Frequency Handover

### 5.4.1.1 Principles

The uplink experience of RedCap UEs on a frequency varies from that of eMBB UEs. While eMBB UEs can exploit large bandwidths to keep their uplink user-perceived rates hardly affected by subframe configurations, RedCap UEs' uplink user-perceived rates are susceptible to these configurations. This is due to the limited maximum bandwidth supported by RedCap UEs, which is only 20 MHz—much less than the 100 MHz available to eMBB UEs.

However, key RedCap services, such as security surveillance and vehicle-mounted surveillance, require support for high uplink user-perceived rates. If RedCap UEs have low uplink user-perceived rates due to limited uplink frames under certain subframe configurations, delivering high-resolution high-bitrate uplink video services becomes challenging. To address this issue, uplink-experience-based inter-frequency handover is introduced. **This function hands over target RedCap UEs with poor uplink experience on an NR TDD network to an inter-frequency NR network to increase their uplink user-perceived rates.**

This function is controlled by the **UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

**Figure 5-12** Uplink-experience-based inter-frequency handover

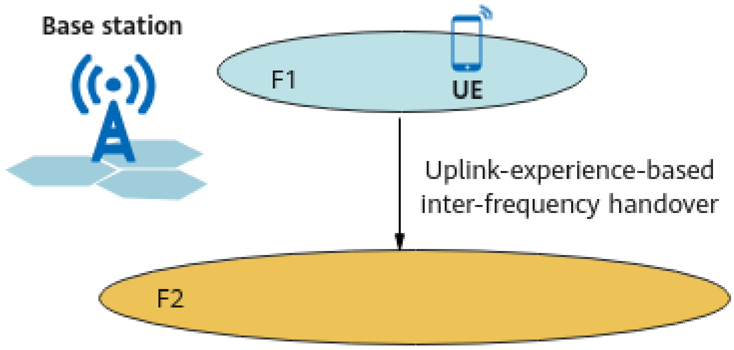
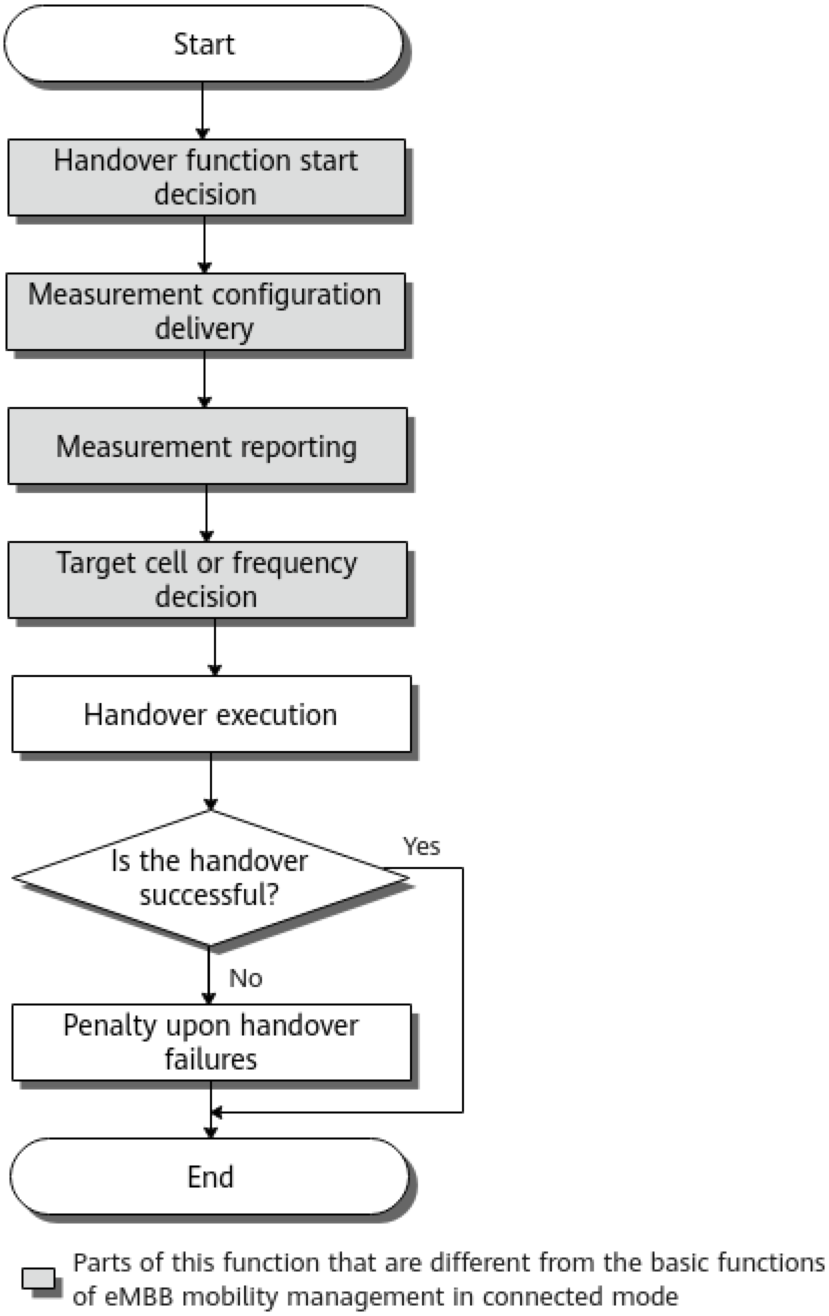



**Figure 5-13** shows the procedure for this function. This section describes only the parts of this function that are different from the basic functions of eMBB mobility management in connected mode. For details about these basic functions, see the corresponding descriptions in *SA Mobility Management in Connected Mode*.

**Figure 5-13** Procedure for uplink-experience-based inter-frequency handover




The differences lie in the following parts: handover function start decision, measurement configuration delivery, measurement reporting, and target cell or frequency decision. In addition, uplink-experience-based inter-frequency handover supports measurement-based mode but not blind mode, and supports handover but not redirection.

- Handover function start decision

  Uplink-experience-based inter-frequency handover is started in a cell when all of the following conditions are met:

  – The **UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter is selected.

  - There are target RedCap UEs in the cell. A UE is considered to be a target RedCap UE if it meets both of the following conditions:
    - The UE is identified as an uplink large-packet UE. That is, it meets the following condition at least five times within 10s: The uplink traffic volume of the UE in 1s is greater than or equal to the threshold specified by the **gNBRedCapServIdenCfg**.***UlBigPktDataVolThld*** parameter in the **gNBRedCapServIdenCfg** MO identified by the **NRCellRedCap**.***UlExpIfHoServParamGroupId*** parameter for the cell.
    - The UE is at the cell edge. That is, the SRS SINR of the UE is less than the threshold specified by the **NRCellRedCap**.***UlExpBasedInterFHoSinrThld*** parameter minus a system-defined hysteresis.
  - The UEs support inter-frequency measurements and handovers, and can work in the frequency bands that contain to-be-measured neighboring frequencies.
- Measurement configuration delivery

  The gNodeB delivers measurement configurations to a target RedCap UE.

  The gNodeB selects a target RedCap UE in the serving cell every 10s and delivers measurement configurations to the UE, so that the UE can start measurements for an uplink-experience-based inter-frequency handover. The measurement configurations can be customized for RedCap UEs. These configurations include the target frequencies for the handover and event A5 configurations. Specifically:
  - Target frequencies for uplink-experience-based inter-frequency handover: The frequencies included in the measurement configurations are those for which the RedCap uplink-experience-based frequency priority (specified by **NRCellFreqRelation**.***RedCapUlExpFreqPriority***) is set to a value other than **0**. The gNodeB selects frequencies based on these priorities in descending order until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see the section "Frequency Selection for Measurement" in *SA Mobility Management in Connected Mode*. If multiple frequencies have the same priority, the NR frequency with a smaller **NRCellFreqRelation**.***SsbFreqPos*** value is preferentially selected.
  - Event A5: Event A5 indicates that the signal quality of the serving cell is lower than RSRP threshold 1 for event A5 related to RedCap uplink-experience-based inter-frequency and the signal quality of a neighboring cell is higher than RSRP threshold 2 for this event. RSRP threshold 1 is specified by the **NRCellRedCap**.***RedCapUlExpInterFA5RsrpTh1*** parameter, and RSRP threshold 2 is specified by the **NRCellRedCap**.***RedCapUlExpInterFA5RsrpTh2*** parameter.
- Measurement reporting

  After receiving the measurement configurations from the gNodeB, the target RedCap UE measures frequencies in descending order of uplink-experience-based frequency priorities (specified by the **NRCellFreqRelation**.***RedCapUlExpFreqPriority*** parameter) based on the event configurations, and then reports measurement results. If no neighboring cell has been reported when the inter-frequency measurement wait time (specified by the **NRCellMobilityConfig**.***InterFreqMeasWaitTime*** parameter) elapses, the gNodeB deletes the measurement configurations and instructs the target RedCap UE to stop the event A5 measurements.

- Target cell or frequency decision

  After receiving event A5 measurement reports, the gNodeB selects the first reported cell as the target cell and checks whether this target cell meets certain conditions. If the conditions are all met, a handover is initiated. Otherwise, no handover is performed and the corresponding measurements are deleted. These conditions are as follows:

  - The target cell allows RedCap UEs to access.
  - The target cell is not in the high-load state. That is, the cell meets all the following conditions: (1) Uplink PRB usage ≤ **NRCellRedCap**.***UlExpIfHoUlPrbUsageThld***; (2) Uplink CCE allocation failure rate ≤ **NRCellRedCap**.***UlExpIfHoCceAllocUlFailTh***; (3) **Downlink Resource Block Utilizing Rate (DU)** ≤ 50%.

  If the target RedCap UE fails to be handed over, another procedure for uplink-experience-based inter-frequency handover can start again only after the UE's RRC connection is released and the UE accesses the network again.

If RedCap UEs performing uplink large-packet services each need to continuously camp on the target inter-frequency cell after an inter-frequency handover, the uplink-experience-based inter-frequency handover collaboration function can be enabled for the target inter-frequency cell (by selecting the **UL_EXP_INTER_FREQ_HO_COLLAB_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter).

If uplink-experience-based inter-frequency handover is enabled, it is recommended that the following functions be enabled as well:

- Subscription to uplink interference information about inter-frequency neighboring cells

  This function enables the serving cell to obtain information about uplink interference in intra- or inter-base-station neighboring cells. The serving cell then uses the obtained information to determine whether to instruct a UE to initiate a handover. Specifically:

  - If the serving cell determines that the target cell is a cell with low uplink interference, the gNodeB instructs the UE to be handed over to the target cell.
  - Otherwise, the gNodeB does not instruct the UE to perform a handover.

  This function is controlled by the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch**.***NCellInfoCtrlSwitch*** parameter.

- Inter-frequency handover collaboration

  This function enables the serving cell to exchange load information with neighboring cells. If the serving cell determines that the target cell is heavily loaded based on the obtained load information, the gNodeB does not instruct a UE to perform a handover to the target cell. This prevents ping-pong handovers. This function is controlled by the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter.

### 5.4.1.2 Network Analysis

#### 5.4.1.2.1 Benefits

This function hands over target RedCap UEs with poor uplink experience on an NR TDD network to an inter-frequency NR network to increase their uplink user-perceived rates. Use this function when all of the following conditions are met:

- Multiple NR TDD and NR FDD frequencies are deployed, and NR FDD cells provide contiguous coverage.
- NR FDD cells are under a light or medium load (that is, the uplink PRB usage of a cell is less than or equal to 40% and the uplink CCE allocation failure rate of the cell is less than or equal to 20%).
- The uplink bandwidths of NR FDD cells are not 5 MHz or 10 MHz.

NOTE

Before enabling this function, consider the networking of the live network and how this function will collaborate with other handover functions. If these factors are not considered, ping-pong handovers may occur, affecting user experience. For details about networking requirements, see **5.4.1.3.5 Networking**.

#### 5.4.1.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

- Service interruption occurs when a RedCap UE synchronizes with the target cell and initiates random access during a handover, causing delay and affecting throughput.
- Measurement gaps are used for inter-frequency measurement, leading to throughput decrease.
- The number of intra-NR inter-frequency outgoing handover executions and the number of successful intra-NR inter-frequency outgoing handovers increase, and the **Inter-Frequency Handover Out Success Rate (CU)** changes.

### 5.4.1.3 Requirements

#### 5.4.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Uplink-experience-based inter-frequency handover can take effect only when RedCap Introduction has been enabled. |

### Mutually Exclusive Functions

None

### Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Connected mode MLB | **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter | *Mobility Load Balancing in Connected Mode* | Uplink-experience-based inter-frequency handover does not take effect on RedCap UEs that are handed over to the local cell through connected mode mobility load balancing (MLB). |
| Low-frequency TDD | Multi-frequency beam coordination | **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter | *Multi-Frequency Smart Selection* | Uplink-experience-based inter-frequency handover does not take effect on RedCap UEs that are handed over to the local cell through multi-frequency beam coordination. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Coverage-based inter-frequency handover | **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter | *SA Mobility Management in Connected Mode* | If (1) both uplink-experience-based inter-frequency handover and coverage-based inter-frequency handover are enabled, and (2) the value of **NRCellRedCap**.***RedCapUlExpInterFA5RsrpTh2*** is less than or equal to that of **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld***, then ping-pong between uplink-experience-based inter-frequency handovers and coverage-based inter-frequency handovers will occur. |
| Low-frequency TDD | Frequency-priority-based inter-frequency handover | **FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter | *SA Mobility Management in Connected Mode* | Frequency-priority-based inter-frequency handovers will not be triggered for RedCap UEs if they have completed uplink-experience-based inter-frequency handovers, thereby preventing ping-pong handovers. |
| Low-frequency TDD | Service-based inter-frequency handover | **SERVICE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter | *SA Mobility Management in Connected Mode* | Uplink-experience-based inter-frequency handover is not triggered for UEs for which service-based inter-frequency handover has taken effect. |
| Low-frequency TDD | Low-speed UE outmigration | **NRCellMobilityConfig**.***LowSpeedUeOutgoingPolicy*** | *High Speed Mobility* | Uplink-experience-based inter-frequency handover is not triggered for low-speed UEs that have been migrated out of the high-speed railway dedicated network. |
| Low-frequency TDD | High-speed UE return | **HIGH_SPEED_UE_REDIRECT_SW** option of the **NRCellAlgoSwitch**.***HighSpeedDedPolicySw*** parameter | *High Speed Mobility* | Uplink-experience-based inter-frequency handover does not take effect on high-speed UEs. |

#### 5.4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

For NR TDD: All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

#### 5.4.1.3.4 Cells

None

#### 5.4.1.3.5 Networking

During network planning, if unnecessary handovers (handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers) need to be enabled for multiple cells, avoid the situation where a RedCap UE is handed over back to the source cell through the same or different unnecessary handovers. Otherwise, ping-pong handovers occur, affecting user experience. For example:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an uplink-experience-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through an uplink-experience-based inter-frequency handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an uplink-experience-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through an operator-specific-priority-based handover.

#### 5.4.1.3.6 Others

None

### 5.4.1.4 Operation and Maintenance

#### 5.4.1.4.1 Data Configuration

## Data Preparation

Before activating this function, ensure that the NR cell's frequency relationships (including frequency priorities) and neighbor relationships have been configured. For details about the configurations of the NR cell's frequency relationships, see **Table 5-18**. For details about the configurations of neighbor relationships, see the section "Data Preparation" for the basic functions of mobility management in connected mode in *SA Mobility Management in Connected Mode*.

**Table 5-19** describes the parameters used for function activation.

**Table 5-18** Parameters used for activation (parameters related to the NR cell's frequency relationships)

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| NR Cell ID | **NRCellFreqRelation**.***NrCellId*** | Set this parameter based on the network plan. |
| SSB Frequency Position | **NRCellFreqRelation**.***SsbFreqPos*** | Set this parameter based on the network plan. |
| SSB Frequency Position Describe Method | **NRCellFreqRelation**.***SsbDescMethod*** | Use the recommended value. |
| Frequency Band | **NRCellFreqRelation**.***FrequencyBand*** | Set this parameter based on the network plan. |
| Subcarrier Spacing | **NRCellFreqRelation**.***SubcarrierSpacing*** | Set this parameter based on the network plan. |
| RedCap UL-Exp-based Frequency Priority | **NRCellFreqRelation**.***RedCapUlExpFreqPriority*** | If uplink-experience-based inter-frequency handover is required, set the priority of an NR FDD frequency to be higher than that of an NR TDD frequency. |

**Table 5-19** Parameters used for activation (function-related parameters)

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | To enable uplink-experience-based inter-frequency handover, select the **UL_EXP_BASED_INTER_FREQ_HO_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap Algorithm Switch | **NRCellRedCap**.***RedCapAlgoSwitch*** | To enable RedCap UEs performing large-packet services each to continuously camp on the target inter-frequency cell after an inter-frequency handover, select the **UL_EXP_INTER_FREQ_HO_COLLAB_SW** option for the target inter-frequency cell. |
| UL Big Packet Data Volume Threshold | **gNBRedCapServIdenCfg**.***UlBigPktDataVolThld*** | Retain the default value. |
| UL-Exp-based Inter-Freq HO SINR Threshold | **NRCellRedCap**.***UlExpBasedInterFHoSinrThld*** | Use the recommended value. |
| UL Exp Inter-Freq HO CCE Alloc UL Fail Rate Thld | **NRCellRedCap**.***UlExpIfHoCceAllocUlFailTh*** | Use the recommended value. |
| UL Exp Inter-Freq HO UL PRB Usage Thld | **NRCellRedCap**.***UlExpIfHoUlPrbUsageThld*** | Use the recommended value. |
| RedCap UL Exp Inter-Freq A5 RSRP Thld1 | **NRCellRedCap**.***RedCapUlExpInterFA5RsrpTh1*** | Use the recommended value. |
| RedCap UL Exp Inter-Freq A5 RSRP Thld2 | **NRCellRedCap**.***RedCapUlExpInterFA5RsrpTh2*** | Use the recommended value. |
| UL Exp Inter-Freq HO Service Parameter Group ID | **NRCellRedCap**.***UlExpIfHoServParamGroupId*** | Use the recommended value.<br>Ensure that the **NRCellRedCap**.***UlExpIfHoServParamGroupId*** parameter value is the same as the value of the **gNBRedCapServIdenCfg**.***ServParamGroupId*** parameter in a **gNBRedCapServIdenCfg** MO. |
| Neighboring Cell Information Control Switch | **NRCellAlgoSwitch**.***NCellInfoCtrlSwitch*** | You are advised to select the **INTER_FREQ_UL_INTRF_SW** option when uplink-experience-based inter-frequency handover is enabled. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Inter-Frequency Handover Switch | **NRCellAlgoSwitch**.*InterFreqHoSwitch* | You are advised to select the **INTER_FREQ_HO_COLLABORATION_SW** option when uplink-experience-based inter-frequency handover is enabled. |

## Using MML Commands

Before using MML commands, refer to **5.4.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Adding a frequency relationship with the NR cell (using handover from an NR TDD cell to an NR FDD cell as an example)
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=1840, SsbDescMethod=SSB_DESC_TYPE_GSCN, FrequencyBand=N12, SubcarrierSpacing=15KHZ, RedCapUlExpFreqPriority=1;
//Adding an external cell (using handover from an NR TDD cell to an NR FDD cell as an example)
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0, Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=1840, FrequencyBand=N12, NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Enabling uplink-experience-based inter-frequency handover
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=UL_EXP_BASED_INTER_FREQ_HO_SW-1;
//(Optional) To enable RedCap UEs performing large-packet services each to camp on the target inter-frequency cell after an inter-frequency handover, turning on UL_EXP_INTER_FREQ_HO_COLLAB_SW for the target inter-frequency cell
MOD NRCELLREDCAP: NrCellId=1, RedCapAlgoSwitch=UL_EXP_INTER_FREQ_HO_COLLAB_SW-1;
//Configuring UlBigPktDataVolThld
MOD GNBREDCAPSERVIDENCFG: ServParamGroupId=0, UlBigPktDataVolThld=3M;
MOD NRCELLREDCAP: NrCellId=0, UlExpIfHoServParamGroupId=0;
//Configuring measurement parameter groups for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1, InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=2, InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=3, InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2;
//Setting RSRP thresholds 1 and 2 for event A5 related to RedCap uplink-experience-based inter-frequency handover
MOD NRCELLREDCAP: NrCellId=0, RedCapUlExpInterFA5RsrpTh1=-43, RedCapUlExpInterFA5RsrpTh2=-102;
//Modifying the parameter settings for target RedCap UE identification (using NR TDD as an example)
MOD NRCELLREDCAP: NrCellId=0, UlExpBasedInterFHoSinrThld=60, UlExpIfHoServParamGroupId=0;
//Modifying the thresholds related to the load of the target cell
MOD NRCELLREDCAP: NrCellId=0, UlExpIfHoCceAllocUlFailTh=20, UlExpIfHoUlPrbUsageThld=40;
//Modifying the inter-frequency measurement wait time
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasWaitTime=6;
```

```
//(Optional, recommended when uplink-experience-based inter-frequency handover is enabled for the serving cell and uplink-interference-based inter-frequency handover is enabled for the target cell) Turning on the switch for "subscription to uplink interference information about inter-frequency neighboring cells" for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
//(Optional, recommended when uplink-experience-based inter-frequency handover is enabled for the serving cell) Enabling inter-frequency handover collaboration for the serving cell to avoid ping-pong handovers
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
```

### Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Turning off UL_EXP_INTER_FREQ_HO_COLLAB_SW
MOD NRCELLREDCAP: NrCellId=1, RedCapAlgoSwitch=UL_EXP_INTER_FREQ_HO_COLLAB_SW-0;
//Disabling uplink-experience-based inter-frequency handover
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=UL_EXP_BASED_INTER_FREQ_HO_SW-0;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 5.4.1.4.2 Activation Verification

If the number of intra-NR inter-frequency outgoing handover executions and the number of successful intra-NR inter-frequency outgoing handovers increase, uplink-experience-based inter-frequency handover has taken effect.

#### 5.4.1.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)**.

# 5.5 BWP Load Balancing

## 5.5.1 UE-Number-based Periodic BWP Load Balancing

### 5.5.1.1 Principles

In RedCap hybrid service scenarios, UEs are unevenly distributed on different RedCap BWPs. As a result, the experience of some RedCap UEs is poor. The UE-number-based periodic BWP load balancing function is introduced to address this issue. **After this function takes effect, the gNodeB transfers RedCap UEs in the RRC_CONNECTED state from heavy-load RedCap BWPs to light-load BWPs every 10 minutes to achieve load balancing.**

This function is controlled by the **UE_NUM_BASED_PRDC_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. It can take effect only when all of the following conditions are met:

- There are RedCap UEs in a cell.
- More than one RedCap BWP is activated in the cell.
- The following is true: (maxLoad – expectLoad)/maxLoad > 10%. Specifically:
  - maxLoad = Number of RRC_CONNECTED UEs on the RedCap BWP with the heaviest load
  - expectLoad = Total number of RRC_CONNECTED UEs on all RedCap BWPs/Number of RedCap BWPs

### 5.5.1.2 Network Analysis

#### 5.5.1.2.1 Benefits

After the UE-number-based periodic BWP load balancing function is enabled, UEs become more evenly distributed across RedCap BWPs, enabling efficient resource utilization across multiple BWP1s.

#### 5.5.1.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

The UE-number-based periodic BWP load balancing function has the following impacts on the network:

- The average uplink/downlink cell throughput fluctuates because the RedCap UE distribution on BWP1s changes in the cell.
- As user-number-based load balancing is periodically performed for RedCap UEs between BWPs in a cell, the load of each BWP tends to be close to each other, the number of RRC connection reconfigurations for RedCap UEs increases, and the service drop rate of the cell may increase.

**Table 5-20** lists the related indicators.

**Table 5-20** Indicators affected by UE-number-based periodic BWP load balancing

| Indicator | Formula |
|---|---|
| Average uplink cell throughput | **Cell Uplink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Service drop rate of a cell | **N.QosFlow.AbnormRel**/(**N.QosFlow.AbnormRel** + **N.QosFlow.NormRel**) x 100% |

### 5.5.1.3 Requirements

#### 5.5.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

**Prerequisite Functions**

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | UE-number-based periodic BWP load balancing requires RedCap Introduction. |

**Mutually Exclusive Functions**

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | High-speed Railway Superior Experience | **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED** | *High Speed Mobility* | UE-number-based periodic BWP load balancing cannot be enabled in high-speed cells. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PRB-and-CCE-resource-based BWP1 adaptation | **RESOURCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | UE-number-based periodic BWP load balancing cannot be enabled together with PRB-and-CCE-resource-based BWP1 adaptation. |

## Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | QCI-specific UE-number-based periodic BWP load balancing | **QCI_UE_NUM_BASED_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter and **QCI_UE_NUM_BASED_PRDC_BLB_SW** option of the **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** parameter | *RedCap* | When QCI-specific UE-number-based periodic BWP load balancing and UE-number-based periodic BWP load balancing are both enabled, the former takes precedence, followed by the later. Together, they help maintain a near-balanced distribution of UEs across RedCap BWPs. |
| Low-frequency TDD | BWP scheduling optimization for RedCap UEs | **REDCAP_UE_BWP_SCH_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter | *RedCap* | When both BWP scheduling optimization for RedCap UEs and UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect. |

#### 5.5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

#### 5.5.1.3.4 Cells

None

#### 5.5.1.3.5 Others

None

### 5.5.1.4 Operation and Maintenance

#### 5.5.1.4.1 Data Configuration

## Data Preparation

Table 5-21 describes the parameters used for function activation.

**Table 5-21** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | UE_NUM_BASED_PRDC_BLB_SW | Select this option if UE-number-based load balancing is required in a cell. |

## Using MML Commands

Before using MML commands, refer to **5.5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling UE-number-based periodic BWP load balancing
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=UE_NUM_BASED_PRDC_BLB_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling UE-number-based periodic BWP load balancing
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=UE_NUM_BASED_PRDC_BLB_SW-0;
```

## Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.5.1.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. Observe the *locationAndBandwidth* IE in "bwp-Common" under "downlinkBWP-ToAddModList" to check the RedCap UE distribution on BWPs in the frequency domain. If RedCap UEs are evenly distributed on multiple BWP1s, UE-number-based periodic load balancing has taken effect.

**----End**

### 5.5.1.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **5.5.2.2.2 Impacts**.

# 5.5.2 QCI-specific UE-Number-based Periodic BWP Load Balancing

## 5.5.2.1 Principles

Due to service differences in RedCap hybrid service scenarios, RedCap UEs performing a certain type of services may be concentrated on a BWP1, resulting in poor user experience. To address this issue, the QCI-specific UE-number-based periodic BWP load balancing function is introduced to perform load balancing for RedCap UEs running services of a specific QCI. This function consists of access load balancing and periodic connected mode load balancing.

### Access load balancing

After access load balancing takes effect, the gNodeB by default allocates a RedCap UE to a BWP1 that serves the minimum number of RedCap UEs for which bearers of a specific QCI are set up in either of the following scenarios: (1) The RedCap UE initially accesses the network and a bearer with the specific QCI needs to be set up for the UE. (2) The RedCap UE is already online and a bearer with the specific QCI needs to be set up for it. (The RedCap UEs for which bearers of the specific QCI are set up are referred to as BLB UEs in the following sections. BLB refers to BWP Load Balance.) This function takes effect only when more than one RedCap BWP1 is activated in a cell.

In addition, QCI-specific UE-number-based periodic BWP load balancing supports the access load balancing weighting mechanism to achieve different balancing effects.

For NR TDD, CD-SSB BWP load balancing weighting is used. **Table 5-22** lists the load balancing methods in different scenarios.

**Table 5-22** CD-SSB BWP load balancing weighting (TDD)

| Value of the Downlink BWP Load Balancing Weight Factor[a] | Access Load Balancing Method |
|---|---|
| 0 | The gNodeB allocates the RedCap UE to a RedCap BWP1 that does not contain the CD-SSB in either of the preceding scenarios. |
| 100 (default value) | The gNodeB allocates the RedCap UE to a RedCap BWP1 that serves the minimum number of BLB UEs in either of the preceding scenarios. |

| Value of the Downlink BWP Load Balancing Weight Factor[a] | Access Load Balancing Method |
|---|---|
| *K* (values other than 0 and 100) | The gNodeB allocates the RedCap UE to a RedCap BWP1 that has the minimum load in either of the preceding scenarios. Specifically:<br>● Load of the BWP1 with the CD-SSB = Number of BLB UEs on this BWP1/(*K*/100)<br>● Load of the BWP1 without the CD-SSB = Number of BLB UEs on this BWP1 |
| a: The **downlink BWP load balancing weight factor** is configured using the **gNBDURedCapBlbGrp**.***DLBwpLbWeightFactor*** parameter. | |

The access load balancing function is controlled by the **QCI_UE_NUM_BASED_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter. The **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** parameter is used to bind a specific QCI with the RedCap BWP load balancing parameter group corresponding to the **gNBDURedCapBlbGrp**.***RedCapBlbParamGroupId*** parameter.

NOTE

- After a specific QCI is bound to the RedCap BWP load balancing parameter group corresponding to the **gNBDURedCapBlbGrp**.***RedCapBlbParamGroupId*** using the **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** parameter, RedCap UEs that have bearers of this QCI set up before the relationship binding are not considered as BLB UEs. They are identified as **BLB UEs** only after the execution of the **STR NRCELLSERVICEPARAMEFFECT** command. In addition, the **UeType** parameter can be set to **REDCAP** to select only RedCap UEs (including eRedCap UEs) for executing this MML command.
- When the **QCI_UE_NUM_BASED_BLB_SW** option of **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** is selected, **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** must be set to a value other than **255** in at least one **NRDUCellQciBearer** MO associated with **NRDUCellRedCap**.***NrDuCellId***.

## Periodic connected mode load balancing

After periodic connected mode load balancing takes effect, the gNodeB transfers RRC_CONNECTED BLB UEs from RedCap BWPs serving a large number of BLB UEs to RedCap BWPs serving a small number of BLB UEs every 10 minutes. This achieves a balance on BLB UE numbers among BWPs.

In addition, QCI-specific UE-number-based periodic BWP load balancing supports the weighting mechanism for connected mode load balancing to achieve different balancing effects.

For NR TDD, CD-SSB BWP load balancing weighting is used. **Table 5-23** lists the load balancing methods in different scenarios.

**Table 5-23** CD-SSB BWP load balancing weighting (TDD)

| Value of the Downlink BWP Load Balancing Weight Factor[a] | Method of Periodic Connected Mode Load Balancing |
|---|---|
| 0 | The number of BLB UEs is 0 on each RedCap BWP1 with the CD-SSB, and is close to each other across RedCap BWP1s with the NCD-SSBs. |
| 100 (default value) | The number of BLB UEs is close to each other on each RedCap BWP1. |
| *K* (values other than 0 and 100) | The load is close to each other on each RedCap BWP1. For example, if five BWP1s are activated, the ratio of the BLB UE number on the RedCap BWP with the CD-SSB to the BLB UE numbers on the other four RedCap BWP1s with NCD-SSBs is close to (*K*/100):1:1:1:1. |
| a: The **downlink BWP load balancing weight factor** is configured using the **gNBDURedCapBlbGrp**.***DLBwpLbWeightFactor*** parameter. | |

The periodic connected mode load balancing function is controlled by the **QCI_UE_NUM_BASED_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter and the **QCI_UE_NUM_BASED_PRDC_BLB_SW** option of the **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** parameter. This function can take effect only when all the following conditions are met:

- There are RedCap UEs in a cell.
- More than one RedCap BWP is activated in the cell.
- The following is true: (maxLoad – expectLoad)/maxLoad ≥ 10%. Specifically:
  - maxLoad = Number of RRC_CONNECTED UEs on the RedCap BWP with the heaviest load/Weight factor of the RedCap BWP
  - expectLoad = Total number of RRC_CONNECTED UEs on all RedCap BWPs/Sum of the weight factors of all RedCap BWPs

The periodic connected mode load balancing function uses the **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** parameter to bind a specific QCI with the RedCap BWP load balancing parameter group corresponding to the **gNBDURedCapBlbGrp**.***RedCapBlbParamGroupId*** parameter.

### 5.5.2.2 Network Analysis

#### 5.5.2.2.1 Benefits

The QCI-specific UE-number-based periodic BWP load balancing function meets differentiated balancing requirements and maximizes resource utilization across multiple BWP1s.

#### 5.5.2.2.2 Impacts

The impacts between this function and other functions are described in **Function Impacts**.

The QCI-specific UE-number-based periodic BWP load balancing function has the following impacts on the network:

- The average uplink/downlink cell throughput fluctuates because the RedCap UE distribution on BWP1s changes in the cell.
- When the weighting mechanism for load balancing for RRC_CONNECTED UEs takes effect, RRC connection reconfigurations are periodically initiated for BLB UEs for them to switch to other BWP1s. As a result, the service drop rate of the cell may increase.

**Table 5-24** lists the related indicators.

**Table 5-24** Indicators affected by QCI-specific UE-number-based periodic BWP load balancing

| Indicator | Formula |
| --- | --- |
| Average uplink cell throughput | **Cell Uplink Average Throughput (DU)** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Service drop rate of a cell | **N.QosFlow.AbnormRel**/(**N.QosFlow.AbnormRel** + **N.QosFlow.NormRel**) x 100% |

### 5.5.2.3 Requirements

#### 5.5.2.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- |
| FOFD-091206 | Premium RedCap | NR0S00RCCE00 | per Cell |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 5.5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

## Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | QCI-specific UE-number-based periodic BWP load balancing requires RedCap Introduction. |

## Mutually Exclusive Functions

Periodic connected mode load balancing:

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | High-reliability services | **gNBQciBearer**.***ServiceType*** set to **URLLC_HIGH_RELIABILITY_SERVICE** | *URLLC* | The **QCI_UE_NUM_BASED_PRDC_BLB_SW** option of the **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** parameter cannot be selected in a parameter group bound to a QCI indicating high-reliability services (the **gNBQciBearer**.***ServiceType*** parameter is set to **URLLC_HIGH_RELIABILITY_SERVICE** for this QCI). |

## Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | UE-number-based BWP1 adaptation | **USER_NUM_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | For an NR TDD cell, the base station reserves at least one RedCap BWP1 with the NCD-SSB when the **gNBDURedCapBlbGrp**.***DLBwpLbWeightFactor*** parameter is set to **0** and any of the following functions is enabled: UE-number-based BWP1 adaptation, PDCCH-CCE-load-based BWP1 adaptation, and PRB-and-CCE-resource-based BWP1 adaptation. This is because RedCap UEs running services of the corresponding 5QI are not expected to access the CD-SSB BWP. |
| Low-frequency TDD | PDCCH-CCE-load-based BWP1 adaptation | **PDCCH_CCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | |
| Low-frequency TDD | PRB-and-CCE-resource-based BWP1 adaptation | **RESOURCE_BASED_BWP_ACTV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | |
| Low-frequency TDD | UE-number-based periodic BWP load balancing | **UE_NUM_BASED_PRDC_BLB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | When QCI-specific UE-number-based periodic BWP load balancing and UE-number-based periodic BWP load balancing are both enabled, the former takes precedence, followed by the later. Together, they help maintain a near-balanced distribution of UEs across RedCap BWPs. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | BWP Scheduling Optimization for RedCap UEs | **REDCAP_UE_BWP_SCH_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter | *RedCap* | When both BWP scheduling optimization for RedCap UEs and QCI-specific UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect. |

### 5.5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

### 5.5.2.3.4 Cells

- The bandwidth of an NR TDD cell must be greater than or equal to 40 MHz. (That is, the **NRDUCell**.***UlBandwidth*** and **NRDUCell**.***DlBandwidth*** parameters must be set to **CELL_BW_40M** or a larger value).
- For an NR TDD cell, the **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- The **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to a value greater than **20M**.

#### 5.5.2.3.5 Others

None

### 5.5.2.4 Operation and Maintenance

#### 5.5.2.4.1 Data Configuration

## Data Preparation

Table 5-25 describes the parameters used for function activation.

**Table 5-25** Parameters used for activation

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| RedCap BWP Load Balancing Parameter Group ID | **gNBDURedCapBlbGrp**.***RedCapBlbParamGroupId*** | None | Set this parameter based on the network plan. |
| RedCap BWP Load Balancing Parameter Group ID | **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** | None | Set this parameter based on the network plan.<br>If the **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** parameter is set to a value other than **255**, the **gNBDURedCapBlbGrp** MO corresponding to this **NRDUCellQciBearer**.***RedCapBlbParamGroupId*** parameter must be already configured. |
| RedCap Algorithm Extension Switch | **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** | QCI_UE_NUM_BASED_BLB_SW | Select both options if differentiated load balancing for RedCap UEs is required. |
| RedCap BWP Load Balancing Algorithm Switch | **gNBDURedCapBlbGrp**.***RedCapBlbAlgoSwitch*** | QCI_UE_NUM_BASED_PRDC_BLB_SW | |
| DL BWP Load Balancing Weight Factor | **gNBDURedCapBlbGrp**.***DLBwpLbWeightFactor*** | None | Set this parameter based on the network plan. |

| Parameter Name | Parameter ID | Option | Setting Notes |
|---|---|---|---|
| UL BWP Load Balancing Weight Factor | **gNBDURedCapBlbGrp.*ULBwpLbWeightFactor*** | None | Set this parameter based on the network plan. |

## Using MML Commands

Before using MML commands, refer to **5.5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Adding RedCap BWP load balancing parameter group 1. Access load balancing is performed for UEs involved in this group.
ADD GNBDUREDCAPBLBGRP: RedCapBlbParamGroupId=1;
//Adding RedCap BWP load balancing parameter group 2. Access load balancing and periodic connected mode load balancing are performed for UEs involved in this group. The number of UEs accessing the CD-SSB BWP is half the number accessing a single NCD-SSB BWP1.
ADD GNBDUREDCAPBLBGRP: RedCapBlbParamGroupId=2, RedCapBlbAlgoSwitch=QCI_UE_NUM_BASED_PRDC_BLB_SW-1, DLBwpLbWeightFactor=50, ULBwpLbWeightFactor=100;
//Binding a QCI (QCI 100 used as an example) to parameter group 1 so that the corresponding load balancing policy is applied (If QCI 100 has not been added, run the ADD command. Otherwise, run the MOD command.)
ADD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, RedCapBlbParamGroupId=1;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, RedCapBlbParamGroupId=1;
//Binding another QCI (QCI 101 used as an example) to parameter group 2 so that the corresponding load balancing policy is applied (If QCI 101 has not been added, run the ADD command. Otherwise, run the MOD command.)
ADD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=101, RedCapBlbParamGroupId=2;
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=101, RedCapBlbParamGroupId=2;
//Turning on QCI_UE_NUM_BASED_BLB_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=QCI_UE_NUM_BASED_BLB_SW-1;
//(Optional) Enabling QCI-specific UE-number-based periodic BWP load balancing for UEs that remain online (using 5QI 9 as an example)
STR NRCELLSERVICEPARAMEFFECT: NrCellId=0, NrNetworkingOption=SA, First5qi=9, UeType=REDCAP;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off QCI_UE_NUM_BASED_BLB_SW
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoExtSwitch=QCI_UE_NUM_BASED_BLB_SW-0;
//Removing the binding relationships between QCI 100 and RedCap BWP load balancing parameter group 1 and between QCI 101 and RedCap BWP load balancing parameter group 2
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=100, RedCapBlbParamGroupId=255;
```

```
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=101, RedCapBlbParamGroupId=255;
//Turning off QCI_UE_NUM_BASED_PRDC_BLB_SW
MOD GNBDUREDCAPBLBGRP: RedCapBlbParamGroupId=2,
RedCapBlbAlgoSwitch=QCI_UE_NUM_BASED_PRDC_BLB_SW-0;
//Removing RedCap BWP load balancing parameter groups 1 and 2
RMV GNBDUREDCAPBLBGRP: RedCapBlbParamGroupId=1;
RMV GNBDUREDCAPBLBGRP: RedCapBlbParamGroupId=2;
```

### Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 5.5.2.4.2 Activation Verification

**Step 1** On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

**Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

**Step 3** On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. Observe the *locationAndBandwidth* IE in "bwp-Common" under "downlinkBWP-ToAddModList" to check the BLB UE distribution on BWPs in the frequency domain. If BLB UEs are evenly distributed on multiple BWP1s, QCI-specific UE-number-based periodic BWP load balancing has taken effect.

**----End**

### 5.5.2.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **5.5.2.2.2 Impacts**.

# 6 RedCap Downlink Experience Improvement (NR TDD)

The RedCap Downlink Experience Improvement (NR TDD) feature further increases the average downlink throughput of RedCap UEs using the following means: resource scheduling avoidance, MU pairing enhancement, refined scheduling granularity, flexible BWP1 selection in the frequency domain, and inter-BWP power sharing.

## 6.1 Start RB Avoidance

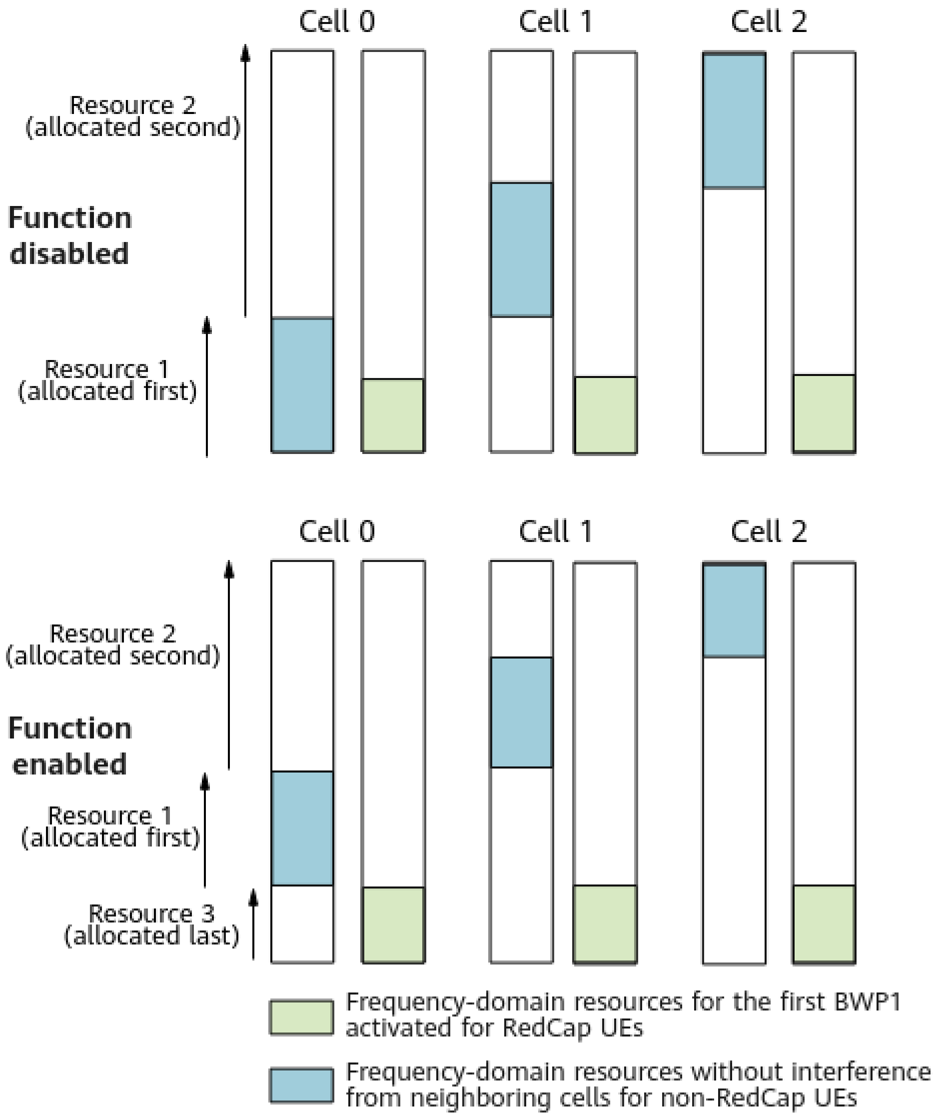
### 6.1.1 Principles

Start RB avoidance aims to increase the RedCap downlink spectral efficiency and average downlink throughput of RedCap UEs. This function is recommended when **NRDUCell**.***DlBandwidth*** is set to **CELL_BW_60M** or a larger value and the **DL_INTRF_RANDOM_SW** option of **NRDUCellAlgoswitch**.***AdaptiveEdgeExpEnhSwitch*** is selected.

With this function, **downlink scheduling of non-RedCap UEs does not start from the 20 MHz spectrum at the lower-frequency end (the end that starts with RB 0). Instead, these UEs preferentially use frequency-domain resources other than that 20 MHz spectrum**. For example, in cell 0 in **Figure 6-1**, non-RedCap UEs are allocated first resource 1, and then resources 2 and 3 in sequence. This ensures that the 20 MHz spectrum at the lower-frequency end is idle, so that RedCap UEs can use this spectrum when it serves as the first BWP1 activated for RedCap UEs. As a result, the average downlink throughput of RedCap UEs will increase.

This function is controlled by the **START_RB_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

**Figure 6-1** Start RB avoidance (example: a 100 MHz cell where the first BWP1 activated for RedCap UEs is on the 20 MHz spectrum at the lower-frequency end)




## 6.1.2 Network Analysis

### 6.1.2.1 Benefits

This function applies to medium- or light-load networks where multiple cells offer contiguous coverage. It significantly increases the average downlink throughput of RedCap UEs. This is because downlink scheduling of non-RedCap UEs does not start from the first BWP1 activated for RedCap UEs, allowing RedCap UEs to use idle frequency-domain resources on this BWP1.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap – N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

#### 6.1.2.2 Impacts

The impacts between this function and other functions are described in **6.1.3.2.3 Function Impacts**.

As downlink scheduling of non-RedCap UEs does not start from the first BWP1 activated for RedCap UEs, the following impacts may occur on the network:

- On a network with a medium or light load, non-RedCap UEs are left with fewer frequency-domain resources that are free from interference from neighboring cells. As a result, the gains of downlink interference randomization-based scheduling decrease for non-RedCap UEs and their average downlink throughput slightly decreases.
- When frequency-domain resources on the first BWP1 activated for RedCap UEs are allocated to non-RedCap UEs, the average downlink throughput of RedCap UEs decreases because non-RedCap UEs have a higher scheduling priority.

### 6.1.3 Requirements

#### 6.1.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |
| For the license control item NR0S00RDEE00:<br>• If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>• If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the **START_RB_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter cannot be modified, preventing cell activation failures. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.1.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Start RB avoidance can be enabled only when RedCap Introduction is enabled. |
| Low-frequency TDD | Downlink interference randomization-based scheduling | **DL_INTRF_RANDOM_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter | *Downlink Scheduling* | Downlink interference-randomization-based scheduling is required for start RB avoidance to take effect. |

#### 6.1.3.2.2 Mutually Exclusive Functions

None

#### 6.1.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink interference randomization-based scheduling | **DL_INTRF_RANDOM_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter | *Downlink Scheduling* | Assume that start RB avoidance is enabled and the downlink RBs allocated to non-RedCap UEs are not on the 20 MHz spectrum at the lower-frequency end after non-RedCap UEs avoid the frequency-domain resources on BWP1s activated for RedCap UEs. If downlink interference randomization-based scheduling is also enabled, the benefits of both functions will decrease. Therefore, the benefits of start RB avoidance vary among cells with different PCIs. |
| Low-frequency TDD | Downlink multi-BWP interference avoidance | **DL_BWP_HYBRID_INTRF_RANDOM_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | Downlink multi-BWP interference avoidance does not take effect when enabled together with start RB avoidance. |
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | *RedCap* | When the function "multiple BWP1s for RedCap" is in effect, start RB avoidance does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Avoidance of RedCap downlink scheduling | **AVOID_REDCAP_DL_SCH_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter | *RedCap* | When avoidance of RedCap downlink scheduling is in effect, start RB avoidance does not take effect. |

### 6.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

### 6.1.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- The cell bandwidth must be greater than or equal to 60 MHz (that is, **NRDUCell**.***UlBandwidth*** and **NRDUCell**.***DlBandwidth*** must be set to **CELL_BW_60M** or a larger value).

### 6.1.3.5 Others

None

# 6.1.4 Operation and Maintenance

## 6.1.4.1 Data Configuration

### 6.1.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation.

**Table 6-1** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | On a medium- or light-load network, select the **START_RB_AVOID_SW** option if the RedCap downlink spectral efficiency and average downlink throughput of RedCap UEs need to be increased.<br>It is recommended that this function be enabled when multiple cells offer contiguous coverage. |

### 6.1.4.1.2 Using MML Commands

Before using MML commands, refer to **6.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling start RB avoidance
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=START_RB_AVOID_SW-1;
```

## Deactivation Command Examples

```
//Disabling start RB avoidance
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=START_RB_AVOID_SW-0;
```

### 6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 6.1.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

#### 6.1.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.1.2.2 Impacts**.

## 6.2 Downlink Adaptive MU Pairing for RedCap UEs

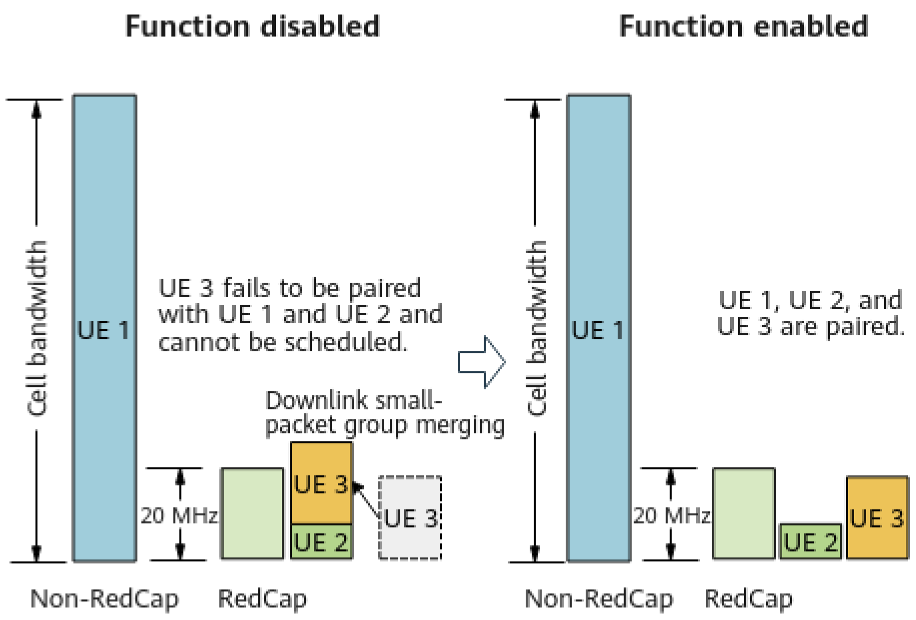
### 6.2.1 Principles

RedCap UEs with high downlink traffic, such as RedCap FWA UEs, need a large amount of scheduling resources. During downlink MU-MIMO in a medium- or heavy-load cell, RedCap UEs cannot be scheduled if merging downlink small-packet groups results in a demand for more RBs than are available on a BWP1 for RedCap UEs. This will cause poor experience of RedCap UEs. To address this issue, downlink adaptive MU pairing for RedCap UEs has been introduced.

**This function allows UEs using different downlink DMRS types to be paired. In addition, if downlink small-packet group merging is enabled and RedCap UEs in two small-packet groups use the same BWP1, the gNodeB checks against certain conditions to determine whether to merge the two groups**. (Downlink small-packet group merging is enabled if the **DL_SMALL_PACKET_MERGE_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter is selected. For details about this function, see *MIMO (TDD)*.) Specifically:

- If the RedCap UEs in the two groups are estimated to require more downlink RBs than are available on the BWP1 for RedCap UEs, the groups will not be merge.
- Otherwise, the small-packet groups will be merged.

**Figure 6-2** Downlink adaptive MU pairing for RedCap UEs




This function is controlled by the **REDCAP_DL_MU_ADAPT_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

## 6.2.2 Network Analysis

### 6.2.2.1 Benefits

On a medium- or heavy-load network with downlink small-packet group merging enabled, two groups are no longer merged if the RedCap UEs in the groups are estimated to require more downlink RBs than are available on a BWP1 for RedCap UEs. This increases the pairing success rate of RedCap UEs and brings the following benefits:

- The average downlink throughput of RedCap UEs and the average downlink cell throughput increase. The specific increases depend on UE distribution and traffic volume.
- The average number of paired layers for downlink MU may increase.

**Table 6-2** lists the related indicators.

**Table 6-2** Indicators related to the benefits of downlink adaptive MU pairing for RedCap UEs

| Indicator | Definition |
| --- | --- |
| Average downlink throughput of RedCap UEs | (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/ **N.ThpTime.DL.RmvLastSlot.RedCap** |

| Indicator | Definition |
|---|---|
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average number of paired layers for downlink MU pairing | **N.ChMeas.MIMO.DL.Pair.Layer/ N.ChMeas.MIMO.DL.Pair.PRB** |

### 6.2.2.2 Impacts

The impacts between this function and other functions are described in **6.2.3.2.3 Function Impacts**.

On a medium- or heavy-load network with downlink small-packet group merging enabled, two groups are no longer merged if the RedCap UEs in the groups are estimated to require more downlink RBs than those available on a BWP1 for RedCap UEs. This leads to power allocation to more paired RedCap UEs and generates inter-layer interference. As a result, the average downlink throughput of non-RedCap UEs decreases, and the downlink RB usage and the average number of used PDCCH CCEs change.

**Table 6-3** lists the related indicators.

**Table 6-3** Indicators related to the network impacts of downlink adaptive MU pairing for RedCap UEs

| Indicator | Definition |
|---|---|
| Downlink RB usage | **Downlink Resource Block Utilizing Rate (DU)** |
| Average number of used PDCCH CCEs | **N.CCE.Used.Avg** |

## 6.2.3 Requirements

### 6.2.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the **REDCAP_DL_MU_ADAPT_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter cannot be modified, preventing cell activation failures. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.2.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Downlink adaptive MU pairing for RedCap UEs requires RedCap Introduction to be enabled. |
| Low-frequency TDD | Downlink MU-MIMO for RedCap UEs | **REDCAP_DL_MU_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | Downlink adaptive MU pairing for RedCap UEs can be enabled only when downlink MU-MIMO for RedCap UEs is enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink small-packet group merging | **DL_SMALL_PACKET_MERGE_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter | *MIMO (TDD)* | Downlink adaptive MU pairing for RedCap UEs requires downlink small-packet group merging to be enabled. |

#### 6.2.3.2.2 Mutually Exclusive Functions

None

#### 6.2.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Power saving BWP | **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | Downlink adaptive MU pairing for RedCap UEs increases the spectral efficiency of a cell. This may enable more UEs to work on BWP2s, on condition that power saving BWP is enabled. In this case, if scheduling in common PDCCH symbols for RedCap UEs (controlled by the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter) is also enabled, the uplink CCE allocation success rate of BWP2 UEs decreases and the cell-level uplink CCE allocation success rate also decreases. |

### 6.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

### 6.2.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

### 6.2.3.5 Others

None

## 6.2.4 Operation and Maintenance

### 6.2.4.1 Data Configuration

#### 6.2.4.1.1 Data Preparation

Table 6-4 describes the parameters used for function activation.

**Table 6-4** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | Select the **REDCAP_DL_MU_ADAPT_PAIR_SW** option for a medium- or heavy-load network. |

#### 6.2.4.1.2 Using MML Commands

Before using MML commands, refer to 6.2.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling downlink adaptive MU pairing for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_DL_MU_ADAPT_PAIR_SW-1;
```

## Deactivation Command Examples

```
//Disabling downlink adaptive MU pairing for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=REDCAP_DL_MU_ADAPT_PAIR_SW-0;
```

#### 6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 6.2.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.2.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.2.2.2 Impacts**.

# 6.3 Avoidance of RedCap Downlink Scheduling

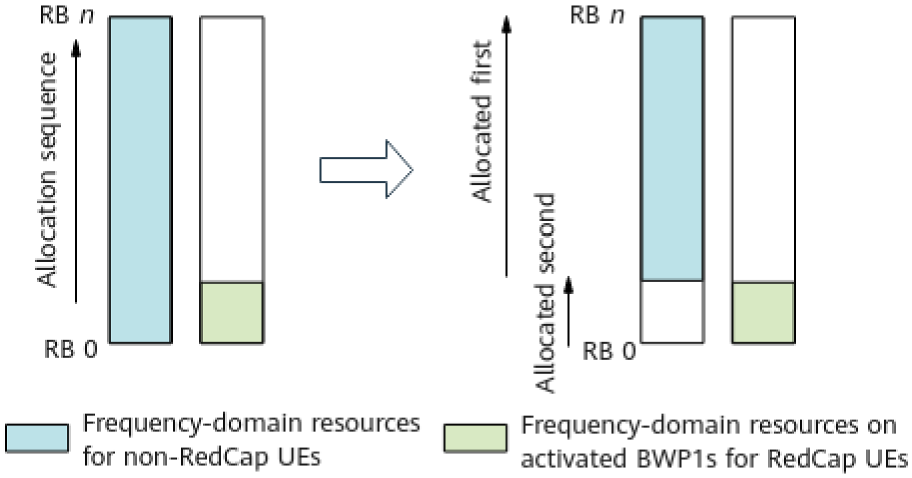
## 6.3.1 Principles

When there are both RedCap UEs and non-RedCap large-bandwidth UEs, they contend for resources. To increase the average downlink throughput of RedCap UEs, avoidance of RedCap downlink scheduling has been introduced.

With this function, non-RedCap UEs avoid the frequency-domain resources on all BWP1s activated for RedCap UEs. That is, in downlink scheduling, **the gNodeB preferentially allocates frequency-domain resources outside these BWP1s to non-RedCap UEs; only after exhausting these resources, it allocates resources within the BWP1s to non-RedCap UEs**. This reduces resource contention between non-RedCap UEs and RedCap UEs and increases the average downlink throughput of RedCap UEs.

**Figure 6-3** shows how avoidance of RedCap downlink scheduling works.

**Figure 6-3** Avoidance of RedCap downlink scheduling




This function is controlled by the **AVOID_REDCAP_DL_SCH_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.

## 6.3.2 Network Analysis

### 6.3.2.1 Benefits

The average downlink throughput of RedCap UEs will increase significantly when there are a large number of small-packet services on non-RedCap UEs. The increase depends on the scheduling priority of RedCap UEs. A higher scheduling priority of RedCap UEs will result in a smaller increase.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.3.2.2 Impacts

The impacts between this function and other functions are described in **6.3.3.2.3 Function Impacts**.

Non-RedCap UEs will experience greater interference from neighboring cells because they avoid all frequency-domain resources on the activated BWP1s for RedCap UEs. As a result, the average downlink throughput of non-RedCap UEs slightly decreases.

More UEs will be scheduled in a slot after non-RedCap UEs avoid resources for RedCap UEs. This affects a medium- or heavy-load network as follows: the uplink and downlink CCE allocation success rates change; low-priority non-RedCap UEs have fewer available scheduling resources; the average uplink and downlink throughputs of non-RedCap UEs may slightly decrease; the average number of downlink PRBs used by RedCap UEs increases; the service delay decreases; and the total number of paired RBs for downlink MU-MIMO changes.

The involved indicators are defined as follows:

- Average number of downlink PRBs used by RedCap UEs: **N.PRBUsed.DL.RedCap.Avg**
- Service delay: (**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**)/1000 + (**N.QoS.DL.PktDelaygNBDU.Time**/**N.QoS.DL.PktDelaygNBDU.Num**)/1000
- Total number of paired RBs for downlink MU-MIMO: **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + ... + **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**

## 6.3.3 Requirements

### 6.3.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td>FOFD-091204</td><td>RedCap Downlink Experience Improvement (NR TDD)</td><td>NR0S00RDEE00</td><td>per Cell</td></tr>
<tr><td colspan="4">For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the AVOID_REDCAP_DL_SCH_SW option of the NRDUCellRedCap.RedCapSchAlgoSwitch parameter cannot be modified, preventing cell activation failures.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 6.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.3.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Avoidance of RedCap downlink scheduling can be enabled only when RedCap Introduction is enabled. |

#### 6.3.3.2.2 Mutually Exclusive Functions

None

#### 6.3.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink interference randomization-based scheduling | **DL_INTRF_RANDOM_SW** option of the **NRDUCellAlgoswitch**.***AdaptiveEdgeExpEnhSwitch*** parameter | *Downlink Scheduling* | Assume that avoidance of RedCap downlink scheduling and downlink interference randomization-based scheduling are both enabled and the downlink RBs allocated to non-RedCap UEs do not overlap with those allocated to RedCap UEs. The benefits of both downlink interference randomization-based scheduling for non-RedCap UEs and avoidance of RedCap downlink scheduling will decrease after non-RedCap UEs avoid the frequency-domain resources on BWP1s activated for RedCap UEs. In such scenarios, the benefits of avoidance of RedCap downlink scheduling vary with PCIs. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Inter-TRP interference coordinated scheduling for high-speed railways | **NRDUCellAlgoSwitch**.***HighSpeedIntrfRandomSchSw*** | *High Speed Mobility* | When inter-TRP interference coordinated scheduling for high-speed railways is in effect, avoidance of RedCap downlink scheduling does not take effect. |
| Low-frequency TDD | Start RB avoidance | **START_RB_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | When avoidance of RedCap downlink scheduling is in effect, start RB avoidance does not take effect. |

### 6.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

### 6.3.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).

- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

#### 6.3.3.5 Others

None

## 6.3.4 Operation and Maintenance

### 6.3.4.1 Data Configuration

#### 6.3.4.1.1 Data Preparation

Table 6-5 describes the parameters used for function activation.

**Table 6-5** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** | It is recommended that the **AVOID_REDCAP_DL_SCH_SW** option be selected if the average downlink throughput of RedCap UEs needs to be increased when there are both RedCap UEs and non-RedCap large-bandwidth UEs. |

#### 6.3.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

```
//Enabling avoidance of RedCap downlink scheduling
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=AVOID_REDCAP_DL_SCH_SW-1;
```

### Deactivation Command Examples

```
//Disabling avoidance of RedCap downlink scheduling
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=AVOID_REDCAP_DL_SCH_SW-0;
```

#### 6.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 6.3.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.3.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.3.2.2 Impacts**.

# 6.4 FWA RedCap Downlink Type 1 Scheduling

## 6.4.1 Principles

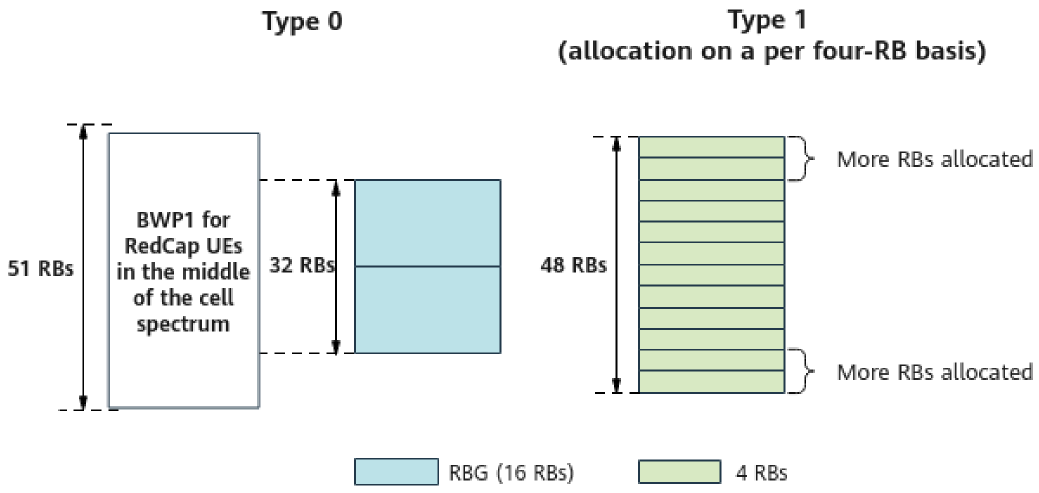
In downlink resource allocation type 0, resources are allocated on a per RBG basis. For example, in a 100 MHz cell, one RBG (16 RBs) is allocated to a UE even if the UE needs fewer resources. Because a BWP1 for RedCap UEs is 20 MHz (containing 51 RBs), at most two or three RBGs can be allocated to a UE to avoid resource waste (at most two RBGs can be allocated in the case of a BWP1 in the middle of the cell spectrum). This restriction severely affects the downlink experience of RedCap UEs. In contrast, type 1 enables resource allocation at the granularity of RBs, ensuring that more RBs are allocated to RedCap UEs without wasting resources.

Considering that RedCap FWA users mainly perform large-packet services in the downlink, FWA RedCap downlink type 1 scheduling has been introduced to increase their average downlink throughputs. This function also prevents SU-MIMO of RedCap UEs from affecting the average downlink cell throughput and the average downlink throughput of non-RedCap FWA users.

This function targets RedCap FWA UEs that support dynamic indication of PDSCH type 0 or 1 (the support is indicated by the **dynamicSwitchRA-Type0-1-PDSCH** field). When the function is enabled, **the base station adaptively selects type 0 or type 1** for such a UE based on the estimated maximum number of available RBs on the corresponding BWP and the estimated traffic volume of all to-be-scheduled UEs in the current slot. **If type 1 is selected, resources are allocated to the UE on a per four-RB basis. This function increases the number of RBs that can be allocated** and consequently increases the average downlink throughput of SU-MIMO RedCap UEs.

**Figure 6-4** shows how FWA RedCap downlink type 1 scheduling works.

**Figure 6-4** FWA RedCap downlink type 1 scheduling




In scenarios where all services of a single UE are carried on only one bearer, this function is controlled by the **FWA_TYPE1_4RB_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter. In scenarios where all services of a single UE are carried on multiple bearers, this function is controlled by the **FWA_TYPE1_4RB_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter and the **RSVDSWPARAM3_BIT30** option of the **NRDUCellRsvdExt01**.***RsvdSwParam3*** parameter.

NOTE

FWA UEs are identified if the service type of their QCIs (specified by **gNBQciBearer**.***ServiceType***) is **FWA_INTERNET**, **FWA_VIDEO**, or **FWA_PRIVATE_LINE**. For details, see *FWA*.

## 6.4.2 Network Analysis

### 6.4.2.1 Benefits

The average downlink throughput of RedCap UEs may increase in lightly loaded cells. In scenarios where all services of a single UE are carried on multiple bearers, the proportion of type 1 scheduling and the function benefits decrease when there are burst services.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.4.2.2 Impacts

The impacts between this function and other functions are described in **6.4.3.2.3 Function Impacts**.

This function has no impact on the network.

## 6.4.3 Requirements

#### 6.4.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |
| For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the **FWA_TYPE1_4RB_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter cannot be modified, preventing cell activation failures. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 6.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

##### 6.4.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | FWA RedCap downlink type 1 scheduling requires that RedCap Introduction be enabled. |

#### 6.4.3.2.2 Mutually Exclusive Functions

None

#### 6.4.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap downlink scheduling policy | **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter deselected | *RedCap* | When FWA RedCap downlink type 1 scheduling is enabled (indicating that resources are allocated on a per four-RB basis) and the **REDCAP_DL_SCH_POL_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter is deselected (indicating that resources are allocated on a per RB basis), resources are preferentially allocated on a per RB basis in downlink resource allocation type 1. |
| Low-frequency TDD | Supplementary type 1 scheduling for UEs performing small-packet services | **SMALL_PKT_TYPE1_SUPPL_SCH_SW** option of the **NRDUCellDlSchRes**.***DlSchResAlgoSw*** parameter | *Downlink Scheduling* | FWA RedCap downlink type 1 scheduling is not applicable to UEs that meet the conditions of "supplementary type 1 scheduling for UEs performing small-packet services". |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | UE bandwidth adaptation | **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter | *Scalable Bandwidth* | In a single-user peak rate test, when UE bandwidth adaptation and FWA RedCap downlink type 1 scheduling are both enabled for a cell, the average downlink throughput of RedCap UEs and the average downlink throughput of all UEs in the cell decrease. |
| Low-frequency TDD | User PRB control for FWA | **PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter | *FWA* | When user PRB control for FWA and FWA RedCap downlink type 1 scheduling are both enabled and the **NRDUCellRes**.***DlRbMaxRatio*** parameter is set to a value other than **255**, downlink PRB resources are more likely to be insufficient because more downlink RBs are allocated at a time to RedCap UEs. As a result, scheduling of RedCap UEs is more likely to be suspended, and the average downlink throughput of RedCap UEs changes. |

#### 6.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

#### 6.4.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

#### 6.4.3.5 Others

None

## 6.4.4 Operation and Maintenance

### 6.4.4.1 Data Configuration

#### 6.4.4.1.1 Data Preparation

**Table 6-6** describes the parameters used for function activation.

**Table 6-6** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **FWA_TYPE1_4RB_SCH_SW** option be selected in FWA scenarios. |
| Reserved Switch Parameter 3 | **NRDUCellRsvdExt01**.***RsvdSwParam3*** | If type 1 scheduling is required for FWA RedCap UEs in the downlink in scenarios where all services of a single UE are carried on multiple bearers, the **RSVDSWPARAM3_BIT30** option also needs to be selected. |

### 6.4.4.1.2 Using MML Commands

Before using MML commands, refer to **6.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling FWA RedCap downlink type 1 scheduling
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=FWA_TYPE1_4RB_SCH_SW-1;
//(Optional) Turning on RSVDSWPARAM3_BIT30 as well if type 1 scheduling is required for FWA RedCap UEs in the downlink in scenarios where all services of a single UE are carried on multiple bearers
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT30-1;
```

## Deactivation Command Examples

```
//(Optional) Turning off RSVDSWPARAM3_BIT30
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT30-0;
//Disabling FWA RedCap downlink type 1 scheduling
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=FWA_TYPE1_4RB_SCH_SW-0;
```

### 6.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 6.4.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap – N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

#### 6.4.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.4.2.2 Impacts**.

## 6.5 Downlink Large-Packet Alignment Pairing for RedCap UEs

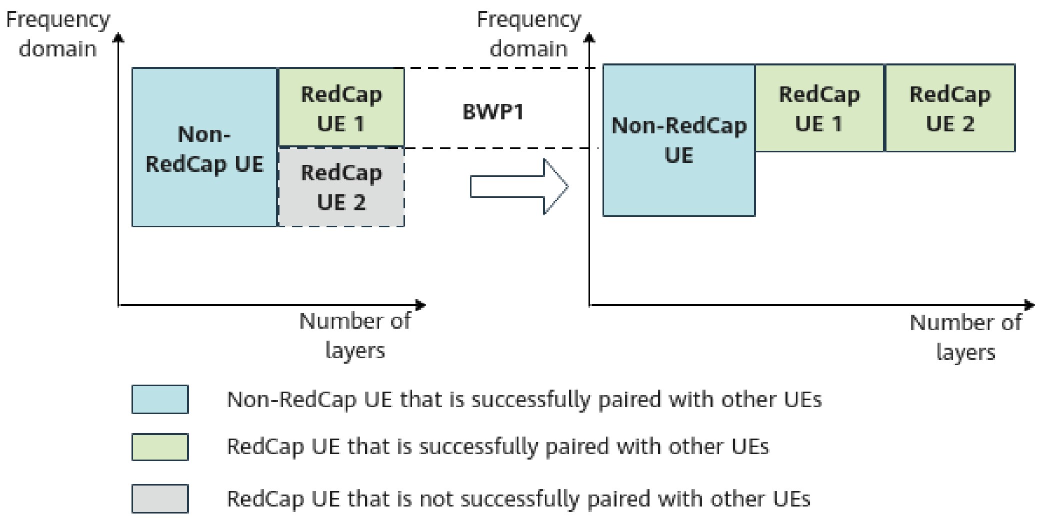
### 6.5.1 Principles

In FWA scenarios, RedCap services in the downlink are mainly large-packet services. To increase the average downlink throughput of RedCap UEs when RedCap UEs are paired with non-RedCap UEs in such scenarios, the "downlink large-packet alignment pairing for RedCap UEs" function is introduced.

After this function is enabled, **the base station determines whether RedCap UEs with downlink large-packet services can be divided into several MU groups based on the frequency-domain positions of the BWPs for the UEs and the correlation between the UEs. If the conditions for the division are met, the base station divides the UEs into several groups and pairs UEs on a per MU group basis.** This increases the MU pairing probability, which in turn increases cell spectral efficiency and average downlink throughput of RedCap UEs. The conditions for division are as follows:

- The estimated total number of downlink RBs required by RedCap UEs is greater than the number of RBs available in the RedCap BWP1 bandwidth.
- The correlation between UEs meets the correlation requirement.

**Figure 6-5** Downlink large-packet alignment pairing for RedCap UEs




This function is controlled by the **DL_BIG_PKT_ALIGN_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.

# 6.5.2 Network Analysis

## 6.5.2.1 Benefits

- The average downlink throughput of RedCap UEs and the average downlink cell throughput increase. The specific increases depend on UE distribution and traffic volume.
- The average number of paired layers for downlink MU-MIMO may increase.

**Table 6-7** lists the related indicators.

**Table 6-7** Indicators related to the benefits of downlink large-packet alignment pairing for RedCap UEs

| Indicator | Definition |
| --- | --- |
| Average downlink throughput of RedCap UEs | (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/ **N.ThpTime.DL.RmvLastSlot.RedCap** |
| Average downlink cell throughput | **Cell Downlink Average Throughput (DU)** |
| Average number of paired layers for downlink MU-MIMO | **N.ChMeas.MIMO.DL.Pair.Layer**/ **N.ChMeas.MIMO.DL.Pair.PRB** |

## 6.5.2.2 Impacts

The impacts between this function and other functions are described in **6.5.3.2.3 Function Impacts**.

After "downlink large-packet alignment pairing for RedCap UEs" takes effect, more RedCap UEs are paired, and power allocation to these UEs and inter-layer interference due to pairing cause the following impacts: The average downlink throughput of non-RedCap UEs decreases, the downlink RB usage, average number of used PDCCH CCEs, PDCCH CCE usage, uplink and downlink CCE allocation success rates, and average downlink rank fluctuate.

**Table 6-8** lists the related indicators.

**Table 6-8** Indicators related to the impacts of downlink large-packet alignment pairing for RedCap UEs

| Indicator | Definition |
| --- | --- |
| Downlink RB usage | **Downlink Resource Block Utilizing Rate (DU)** |
| Average number of used PDCCH CCEs | **N.CCE.Used.Avg** |
| PDCCH CCE usage | **N.CCE.Used.Avg**/**N.CCE.Avail.Avg** x 100% |
| Uplink CCE allocation success rate | (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/ **N.CCE.UL.AllocReq.Num** x 100% |
| Downlink CCE allocation success rate | (**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/ **N.CCE.DL.AllocReq.Num** x 100% |
| Average downlink rank | SUM(*k* x **N.PDSCH.InitTbDl.Rank.***k*)/ SUM(**N.PDSCH.InitTbDl.Rank.***k*)<br>*k* = 1 to 4 |

## 6.5.3 Requirements

### 6.5.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- |
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |

| Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- |
| For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the **DL_BIG_PKT_ALIGN_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter cannot be modified, preventing cell activation failures. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 6.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.5.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Downlink large-packet alignment pairing for RedCap UEs requires that RedCap Introduction be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
| --- | --- | --- | --- | --- |
| Low-frequency TDD | Downlink MU-MIMO for RedCap UEs | **REDCAP_DL_MU_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | Downlink large-packet alignment pairing for RedCap UEs can be enabled only when downlink MU-MIMO for RedCap UEs is enabled. |
| Low-frequency TDD | Downlink MU pairing optimization in UE gathering scenarios | **NRDUCellPdsch**.***DlMuGatherOptSw*** | *MIMO (TDD)* | Downlink large-packet alignment pairing for RedCap UEs requires "downlink MU pairing optimization in UE gathering scenarios" to take effect. |
| Low-frequency TDD | Downlink small-packet group merging | **DL_SMALL_PACKET_MERGE_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter | *MIMO (TDD)* | Downlink large-packet alignment pairing for RedCap UEs requires "downlink small-packet group merging" to take effect. |

#### 6.5.3.2.2 Mutually Exclusive Functions

None

#### 6.5.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Power saving BWP | **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter | *UE Power Saving* | Downlink large-packet alignment pairing for RedCap UEs increases the spectral efficiency of a cell. If power saving BWP is enabled, more UEs may use BWP2. In this case, if scheduling in common PDCCH symbols for RedCap UEs (controlled by the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter) is also enabled, the uplink CCE allocation success rate of BWP2 UEs decreases and the cell-level uplink CCE allocation success rate also decreases. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Downlink small-packet pairing enhancement | **DL_SMALL_PACKET_PAIR_ENH_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter | *MIMO (TDD)* | When downlink small-packet pairing enhancement is enabled for a cell, more small-packet UEs can be paired for MU-MIMO. If downlink large-packet alignment pairing for RedCap UEs is also enabled for the cell and both non-RedCap UEs and RedCap UEs are scheduled in the cell, large-packet RedCap UEs are less likely to be paired. As a result, there will be a smaller increase and even a decrease in the average downlink throughput of RedCap UEs.<br>Therefore, it is not recommended that downlink small-packet pairing enhancement be enabled when downlink large-packet alignment pairing for RedCap UEs has been enabled. |

### 6.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

No requirements

### 6.5.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

### 6.5.3.5 Others

None

## 6.5.4 Operation and Maintenance

### 6.5.4.1 Data Configuration

#### 6.5.4.1.1 Data Preparation

**Table 6-9** describes the parameters used for function activation.

**Table 6-9** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** | It is recommended that the **DL_BIG_PKT_ALIGN_PAIR_SW** option be selected in medium- and heavy-load scenarios. |

#### 6.5.4.1.2 Using MML Commands

Before using MML commands, refer to **6.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling downlink large-packet alignment pairing for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=DL_BIG_PKT_ALIGN_PAIR_SW-1;
```

## Deactivation Command Examples

```
//Disabling downlink large-packet alignment pairing for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapSchAlgoSwitch=DL_BIG_PKT_ALIGN_PAIR_SW-0;
```

### 6.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

**NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 6.5.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap – N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.5.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.5.2.2 Impacts**.

# 6.6 BWP Frequency-Domain Adaptation for RedCap UEs

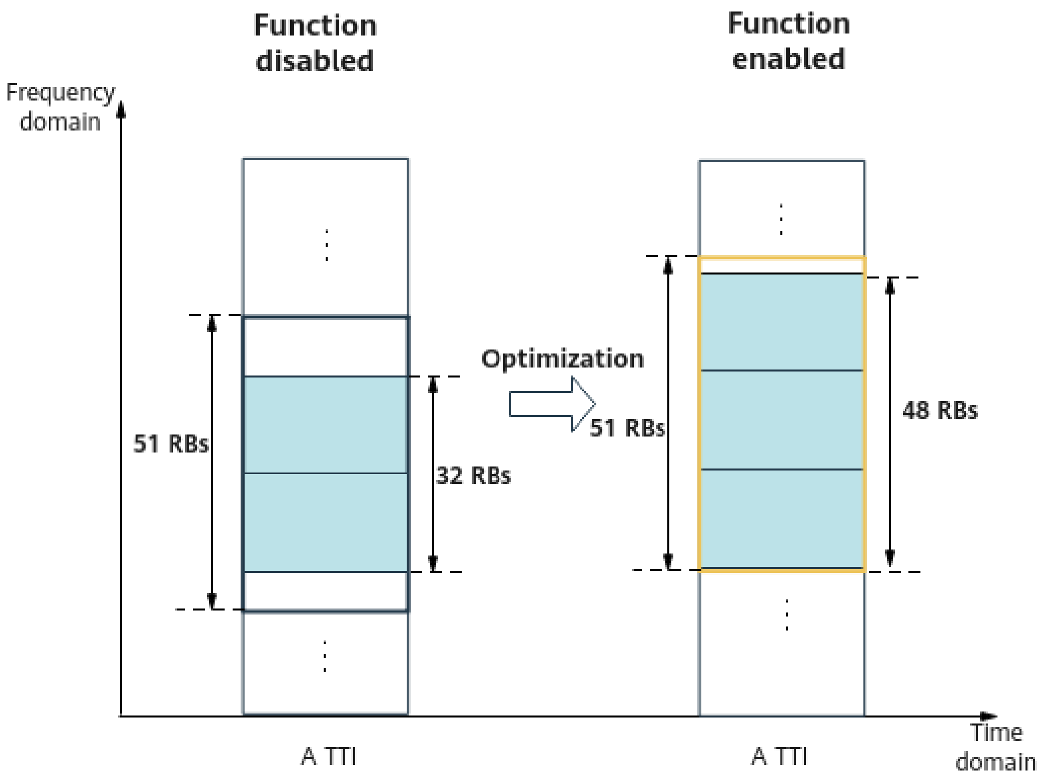
## 6.6.1 Principles

In RedCap MU-MIMO scenarios, resource allocation type 0 (with a resource granularity of RBGs) is used. Considering the overall system performance, only two complete RBGs can be scheduled on the middle RedCap BWP1, which severely affects the downlink experience of RedCap UEs. To address this issue, BWP frequency-domain adaptation for RedCap UEs is introduced.

After this function is enabled, **the base station selects the optimal 20 MHz spectrum (51 RBs) from the cell bandwidth as the first BWP for RedCap (that**

**is, RedCap BWP1 with the CD-SSB) based on the configured SSB frequency and the optimal comprehensive performance of data and control channels. If multiple BWPs need to be activated (a maximum of four RedCap BWP1s can be activated), the base station selects a second optimal 20 MHz spectrum from the remaining bandwidth as an extension BWP for RedCap (that is, a RedCap BWP with an NCD-SSB)**.

**Figure 6-6** BWP frequency-domain adaptation for RedCap UEs




This function is controlled by the **BWP_FREQ_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter.

# 6.6.2 Network Analysis

## 6.6.2.1 Benefits

The average downlink throughput of RedCap UEs may increase. The specific increase depends on the traffic model of UEs. The higher the proportion of large-packet UEs and the higher the proportion of RedCap UEs paired for MU-MIMO, the larger the increase.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

## 6.6.2.2 Impacts

The impacts between this function and other functions are described in **6.6.3.2.3 Function Impacts**.

- This function selects optimal resources in the frequency domain as RedCap BWP1s for RedCap UEs. As a result, the number of RBs that can be scheduled for RedCap UEs increases, the average number of downlink PRBs used by RedCap UEs (**N.PRBUsed.DL.RedCap.Avg**) increases, the number of RBs that can be scheduled for non-RedCap UEs decreases when resources are insufficient, and the average downlink throughput of non-RedCap UEs changes. Because the radio performance of RedCap UEs is lower than that of common NR UEs, the values of related indicators including the average downlink MCS index, downlink BLER, PDCCH CCE usage, and average cell throughput will change. For details about the affected indicators, see **network impacts** of **RedCap Introduction**, where there are descriptions of the network impacts present when RedCap UEs and NR UEs coexist.
- When multiple RedCap BWP1s are activated, their frequency bands become discrete and more RBs can be scheduled on BWP1s in the middle. As a result, the probability of RB truncation during scheduling for non-RedCap UEs with uplink large-packet services increases, leading to slight fluctuations in their average uplink throughput.
- This function adjusts the frequency-domain positions of RedCap BWP1s. When the CD-SSBs are located in the middle of the cell frequency band, the number of BWP1s that actually take effect for RedCap UEs increases or decreases in certain scenarios. The specific impact is the same as that of the "multiple BWP1s for RedCap" function. For details, see **4.2.2 Impacts**. When the cell bandwidth is less than 60 MHz, the number of BWP1s that take effect for RedCap UEs is small. To prevent this function from affecting the number of BWP1s and subsequently affecting user experience, you are advised not to enable this function.
- This function adjusts the frequency-domain positions of RedCap BWP1s, and the corresponding SSB positions also change. If this function is enabled only in some non-continuous cells, inter-cell interference will be introduced due to the NCD-SSB frequency position change, affecting the downlink performance of UEs. Therefore, it is recommended that this function be enabled on the entire network or in a number of contiguous cells.
- This function requires cell reactivation, which causes service drops and re-access of admitted UEs. Therefore, it is recommended that this function be enabled when there are a small number of UEs.

## 6.6.3 Requirements

### 6.6.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the **BWP_FREQ_ADAPT_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter cannot be modified, preventing cell activation failures. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

#### 6.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

##### 6.6.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | BWP frequency-domain adaptation for RedCap UEs requires that RedCap Introduction be enabled. |
| Low-frequency TDD | CD-SSB inclusion in the RedCap BWP0 | **REDCAP_BWP0_INCLUDE_CD_SSB_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter | *RedCap* | BWP frequency-domain adaptation for RedCap UEs requires that CD-SSB inclusion in the RedCap BWP0 be enabled. |

#### 6.6.3.2.2 Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | Multiple BWP1s for RedCap | **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** | *RedCap* | If the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter is set to **100M**, BWP frequency-domain adaptation for RedCap UEs must not be enabled. |

#### 6.6.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap BWP frequency-domain staggering | **REDCAP_BWP_NON_OVERLAP_SW** option of the **NRDUCellRedCap**.***RedCapAlgoExtSwitch*** parameter | *RedCap* | When RedCap BWP frequency-domain staggering and "BWP frequency-domain adaptation for RedCap UEs" are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, RedCap BWP frequency-domain adaptation takes effect based on the new activation sequence of RedCap BWP1s. |

#### 6.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

## Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

No requirements

#### 6.6.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

#### 6.6.3.5 Others

None

## 6.6.4 Operation and Maintenance

### 6.6.4.1 Data Configuration

#### 6.6.4.1.1 Data Preparation

**Table 6-10** describes the parameters used for function activation.

**Table 6-10** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| RedCap Schedule Algorithm Switch | **NRDUCellRedCap**.***RedCapAlgoSwitch*** | It is recommended that the **BWP_FREQ_ADAPT_SW** option be selected when the frequency-domain resources on the middle BWP1 have been activated for RedCap UEs. |

### 6.6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

A cell automatically restarts when the status of the **BWP_FREQ_ADAPT_SW** option is changed.

## Activation Command Examples

```
//Enabling BWP frequency-domain adaptation for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=BWP_FREQ_ADAPT_SW-1;
```

## Deactivation Command Examples

```
//Disabling BWP frequency-domain adaptation for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0, RedCapAlgoSwitch=BWP_FREQ_ADAPT_SW-0;
```

### 6.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 6.6.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect.

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.6.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.6.2.2 Impacts**.

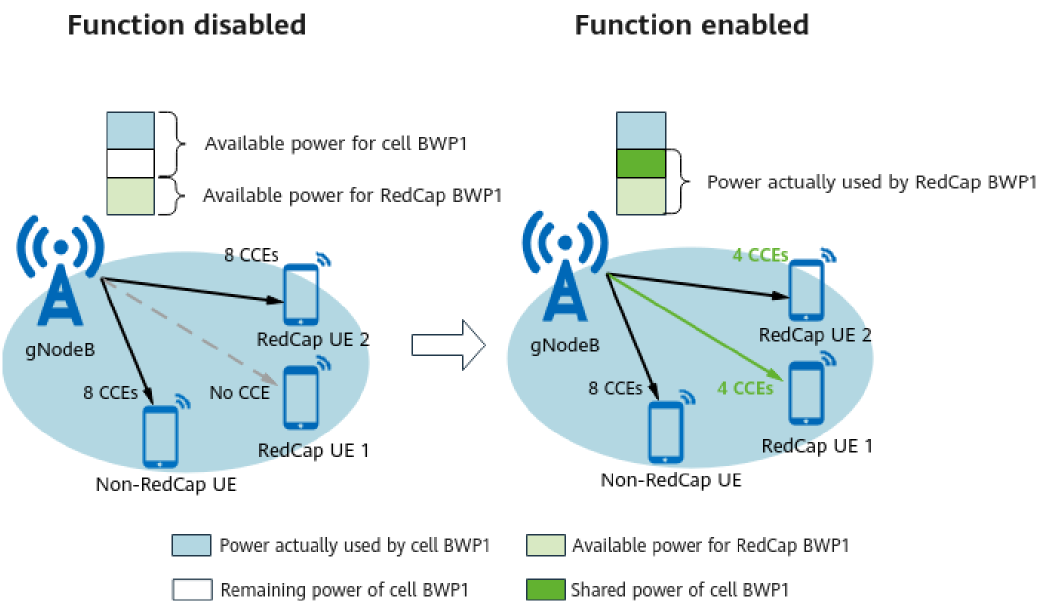
# 6.7 Inter-BWP Power Sharing

## 6.7.1 Principles

A RedCap BWP1, which has a small bandwidth (20 MHz), is more likely to be affected by CCE allocation failures caused by insufficient PDCCH CCE resources than the cell BWP1, which has a large bandwidth (up to 100 MHz for non-RedCap). Inter-BWP power sharing is introduced to increase the number of PDCCH CCEs that can be allocated to RedCap UEs and increase the downlink CCE allocation success rate for RedCap UEs.

After this function is enabled, **the gNodeB checks the types of UEs to be scheduled in a downlink slot and the remaining power of cell BWP1 in each TTI. If there are only RedCap UEs to be scheduled in a downlink slot and cell BWP1 has remaining power, the gNodeB enables the remaining power of the cell BWP1 to be used by the RedCap BWP1 and reduces the PDCCH CCE aggregation level for the RedCap UEs,** thereby increasing the downlink CCE allocation success rate for these UEs.

**Figure 6-7** Inter-BWP power sharing




This function is controlled by the **INTER_BWP_POWER_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter.

## 6.7.2 Network Analysis

### 6.7.2.1 Benefits

As the downlink CCE allocation success rate increases for RedCap UEs, their average downlink throughput increases accordingly. The definitions of the indicators involved are described as follows:

- Downlink CCE allocation success rate = (**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num** x 100%
- Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap** – **N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

### 6.7.2.2 Impacts

The impacts between this function and other functions are described in **6.7.3.2.3 Function Impacts**.

This function has no impact on the network.

## 6.7.3 Requirements

### 6.7.3.1 Licenses

| Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|
| FOFD-091204 | RedCap Downlink Experience Improvement (NR TDD) | NR0S00RDEE00 | per Cell |
| For the license control item NR0S00RDEE00:<br>● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.<br>● If the feature is not enabled for an activated cell, the setting of the **INTER_BWP_POWER_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter cannot be modified when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. This prevents the cell from activation failures due to license insufficiency. | | | |

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 6.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 6.7.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | RedCap Introduction | **REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter | *RedCap* | Inter-BWP power sharing requires that RedCap Introduction be enabled. |

#### 6.7.3.2.2 Mutually Exclusive Functions

None

#### 6.7.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PDCCH power control for CEUs | **NRDUCellPdcchAlgo**.***PdcchEdgePwrCtrlPolicy*** set to **BASED_ON_AGG_LVL** or **BASED_ON_CSI_RPT** | *Power Control* | When PDCCH power control for CEUs and inter-BWP power sharing are both enabled, the power used by a single UE increases, leading to power insufficiency in the cell. This, in turn, decreases the CCE allocation success rate and may reduce overall benefits. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency TDD | PDCCH aggregation level reduction | **PDCCH_AGG_LVL_COMPR_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter | *Massive MIMO iBeam (Low-Frequency TDD)* | When PDCCH aggregation level reduction and inter-BWP power sharing are both enabled, the overall benefits provided by these functions decrease. |

### 6.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

### RF Modules

All RF modules whose power aggregation capability is greater than or equal to 3 dB support this function.

### 6.7.3.4 Cells

- The cell must be an NR TDD cell (that is, **NRDUCell**.***DuplexMode*** must be set to **CELL_TDD**).
- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N39**, **N40**, **N41**, **N77**, **N78**, or **N79**.

### 6.7.3.5 Others

None

# 6.7.4 Operation and Maintenance

## 6.7.4.1 Data Configuration

### 6.7.4.1.1 Data Preparation

Table 6-11 describes the parameters used for function activation.

**Table 6-11** Parameters used for activation

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| PDCCH Algorithm Extension Switch | **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** | It is recommended that the **INTER_BWP_POWER_SHARING_SW** option be selected. |

### 6.7.4.1.2 Using MML Commands

Before using MML commands, refer to **6.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

```
//Enabling inter-BWP power sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=INTER_BWP_POWER_SHARING_SW-1;
```

## Deactivation Command Examples

```
//Disabling inter-BWP power sharing
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=INTER_BWP_POWER_SHARING_SW-0;
```

### 6.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

#### 6.7.4.2 Activation Verification

After this function is enabled, observe the increase in the average downlink throughput of RedCap UEs to check whether this function has taken effect. Where

Average downlink throughput of RedCap UEs = (**N.ThpVol.DL.RedCap – N.ThpVol.DL.LastSlot.RedCap**)/**N.ThpTime.DL.RmvLastSlot.RedCap**

#### 6.7.4.3 Network Monitoring

After this function is enabled, monitor the network impacts of this function using the performance indicators listed in **6.7.2 Network Analysis**.

# 7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

**NOTE**

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

**Step 1** Open the EXCEL file of parameter reference.

**Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All parameters related to the feature are displayed.

**----End**

**FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

**Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

**----End**

# 8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

**NOTE**

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ: How do I find the counters related to a certain feature from performance counter reference?**

**Step 1** Open the EXCEL file of performance counter reference.

**Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

**Step 3** Click **OK**. All counters related to the feature are displayed.

**----End**

# 9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 10 Reference Documents

- 3GPP TS 38.331: "NR;Radio Resource Control (RRC) protocol specification(Release 17)"
- 3GPP TS 38.304: "NR;User Equipment (UE) procedures in Idle mode and RRC Inactive state(Release 17)"
- 3GPP TS 38.133: "NR;Requirements for support of radio resource management(Release 17)"
- 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception(Release 17)"
- 3GPP TS 38.423: "NG-RAN;Xn application protocol (XnAP)(Release 18)"
- 3GPP TS 38.306: "NR;User Equipment (UE) radio access capabilities(Release 18)"
- *5G Networking and Signaling*
- *CoMP*
- *DRX*
- *Hyper Cell*
- *SA Mobility Management in Inactive Mode*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *Random Access Control*
- *URLLC*
- *Beam Management*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Multi-Frequency Smart Selection*
- *Interference Avoidance*
- *High Speed Mobility*
- *Energy Conservation and Emission Reduction*
- *Scalable Bandwidth*
- *Network Slicing*

- *Cell Combination*
- *Channel Management*
- *UE Power Saving*
- *Multi-RAT Coordinated Energy Saving*
- *Virtual Grid-based Multi-Frequency Coordination*
- *MRO*
- *Super Uplink*
- *UL and DL Decoupling*
- *License Control Item Lists*
- *QoS Management*
- *FWA*
- *3D Networking Experience Improvement*
- *MIMO (TDD)*
- *VoNR*
- *Power Control*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Cell Management*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Standards Compliance*
- *Mobility Load Balancing in Connected Mode*
- *Co-MM (Low-Frequency TDD)*
- *Ultimate Cloud X Experience*
- *Uplink Flexible Spectrum Scheduling*
- *Extended Cell Range*
- *Remote Interference Management (Low-Frequency TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *Service-differentiated Experience Guarantee*
- *Device-Pipe Synergy*
- *SAI-based Service Experience Improvement*